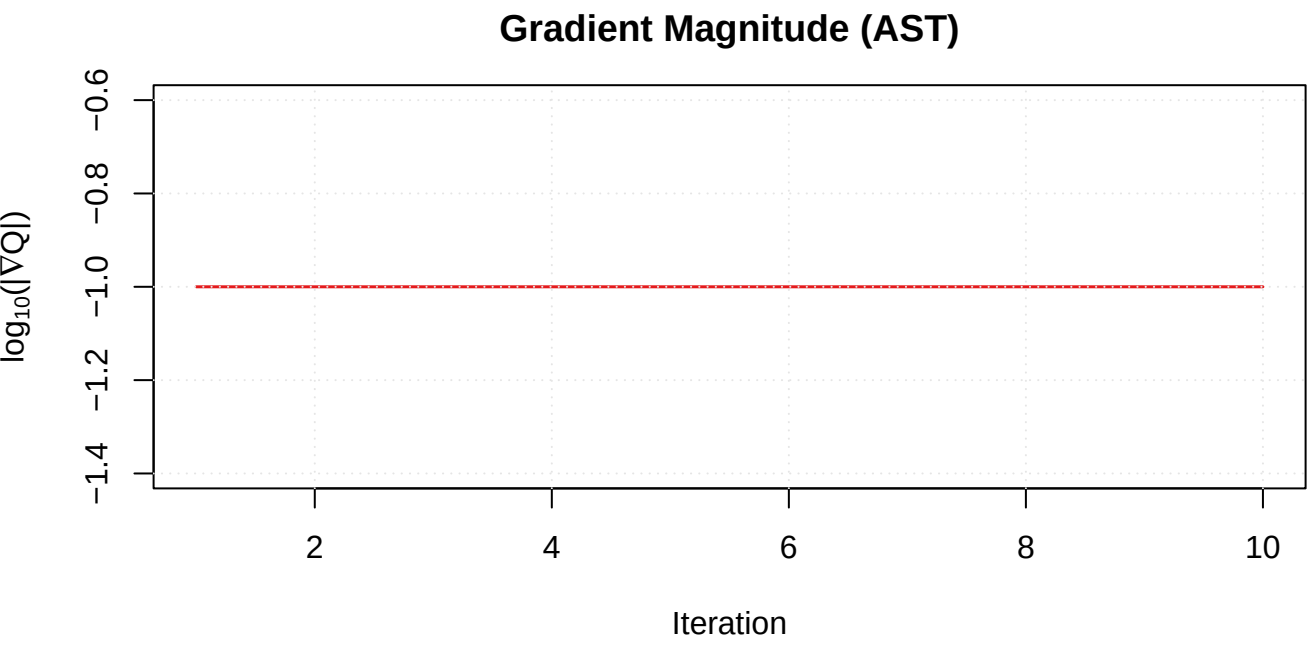
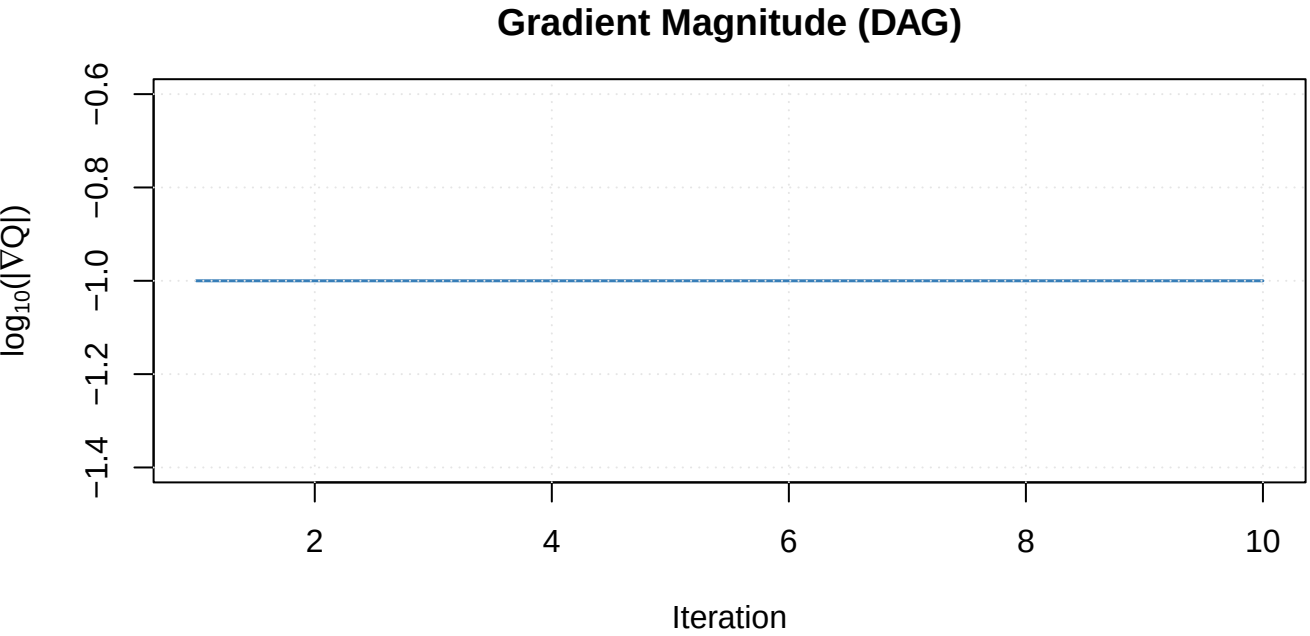
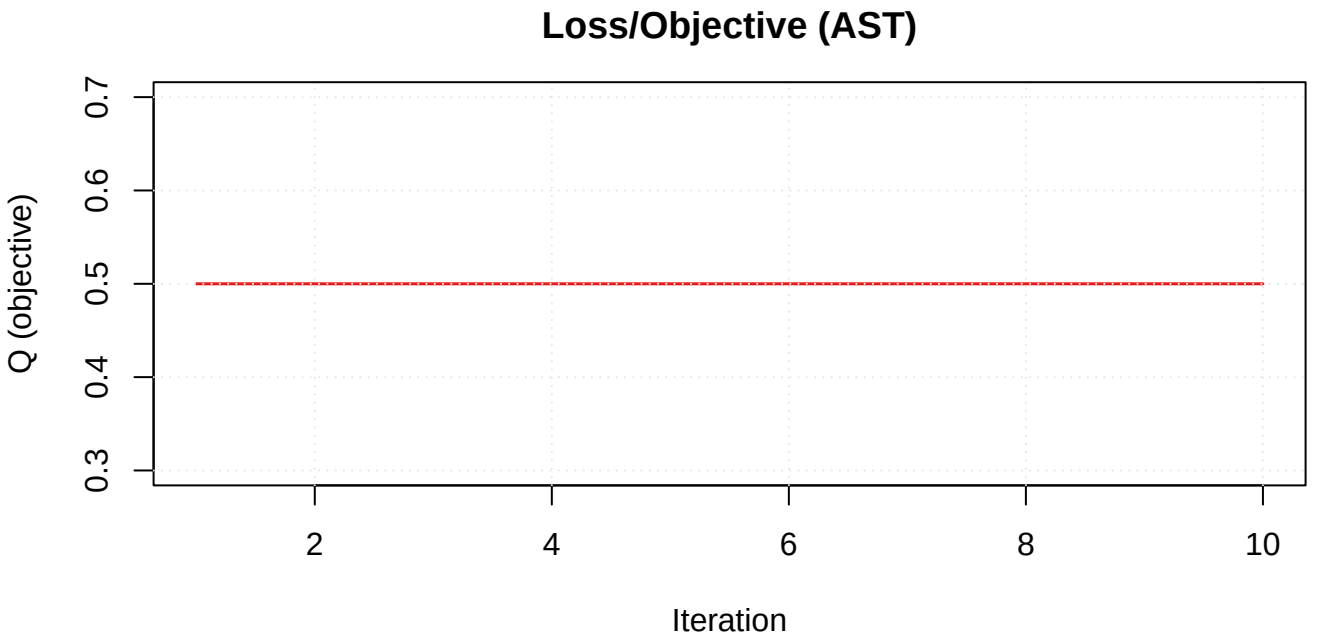
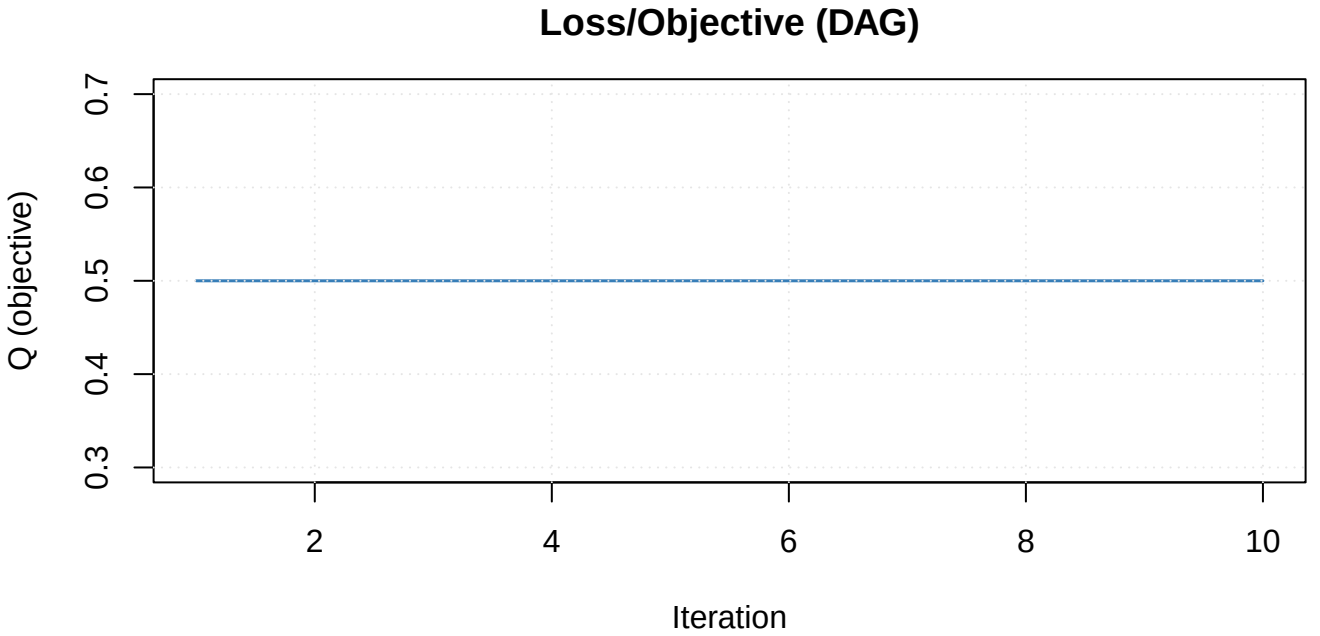
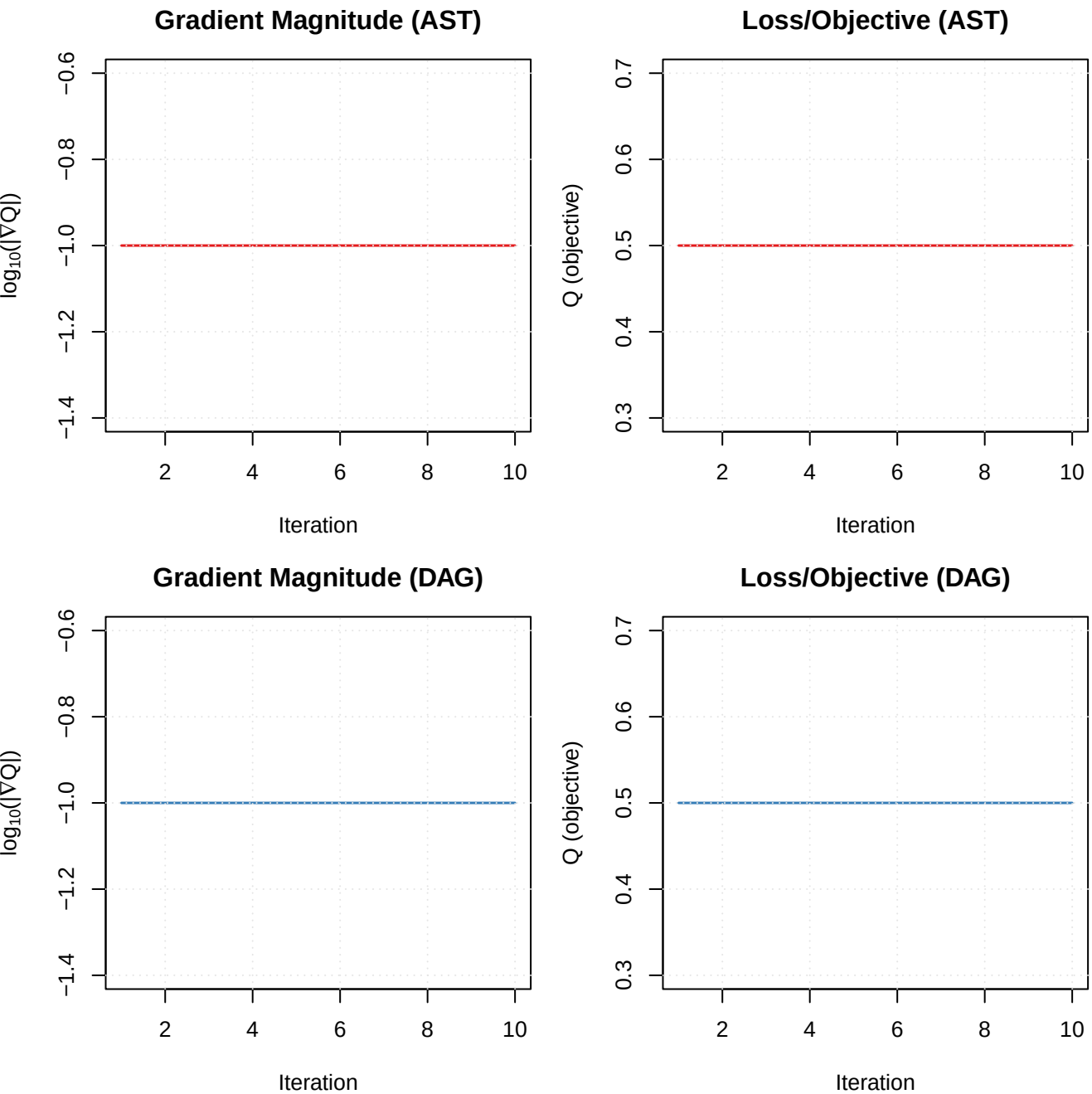
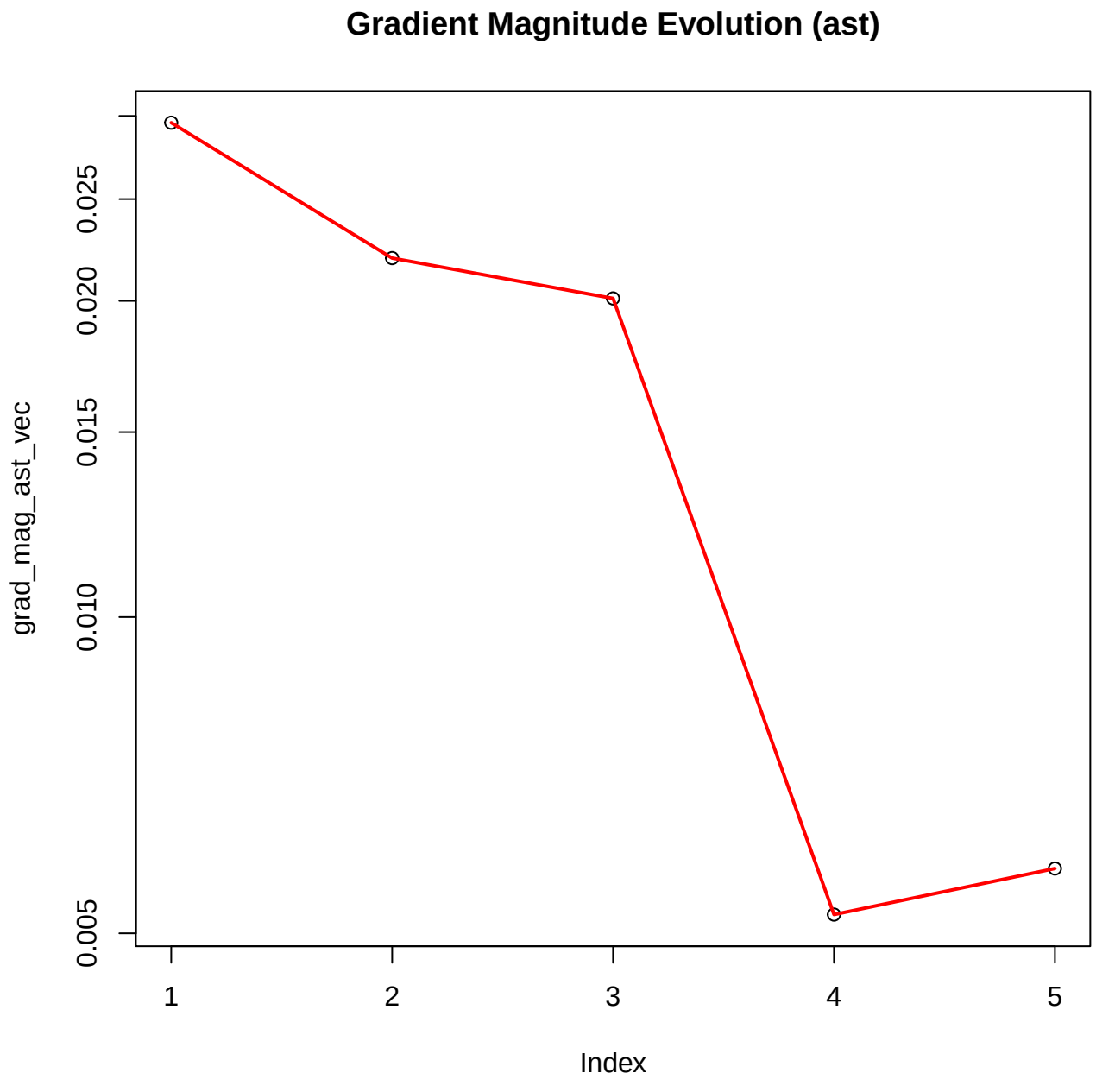

Gradient Magnitude (AST)
log10(|∇Q|)
−0.6
−0.8
−1.0
−1.2
−1.4
2
4
6
8
10
Iteration

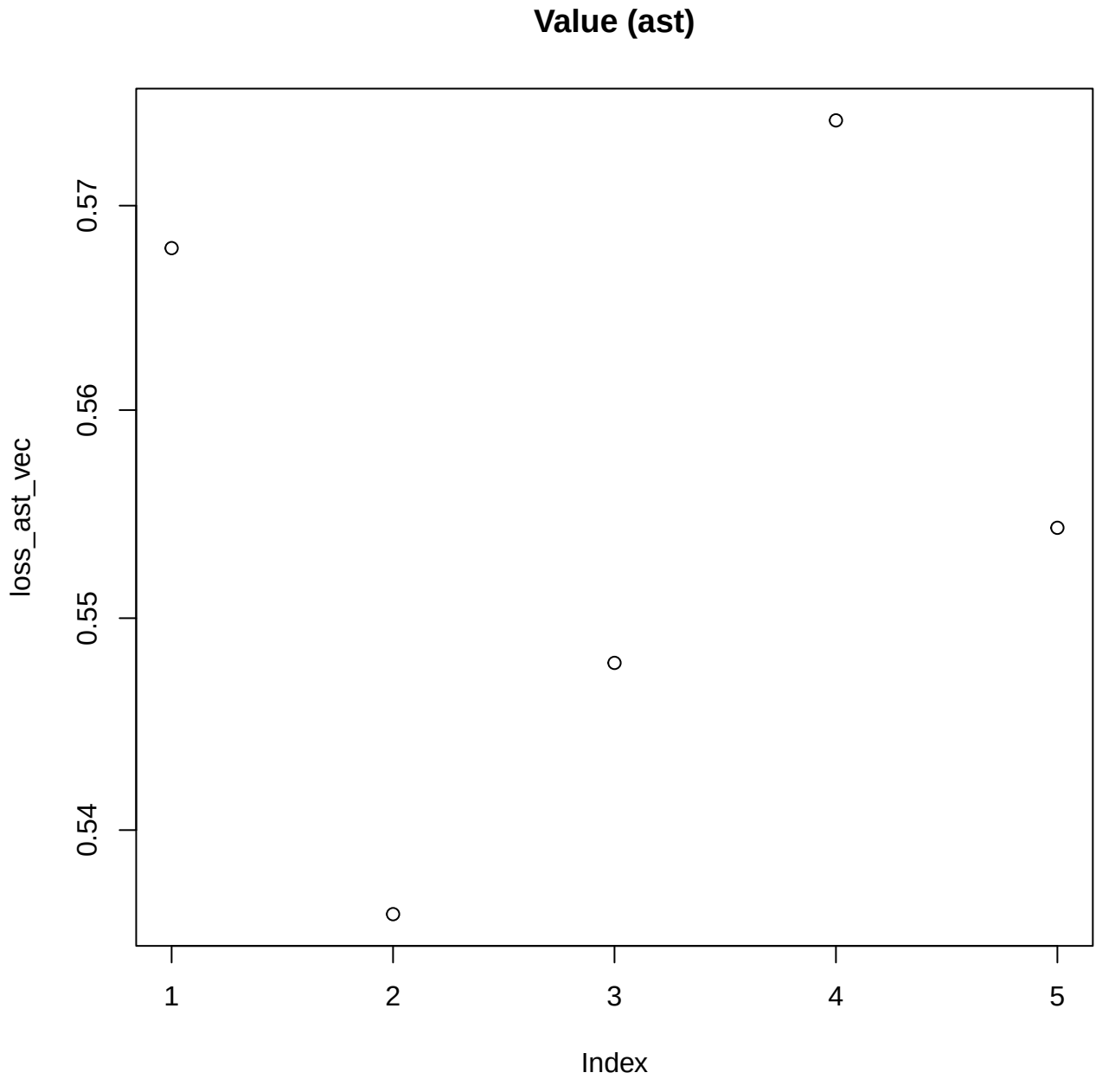

Gradient Magnitude (DAG)
log10(|∇Q|)
−0.6
−0.8
−1.0
−1.2
−1.4
2
4
6
8
10
Iteration

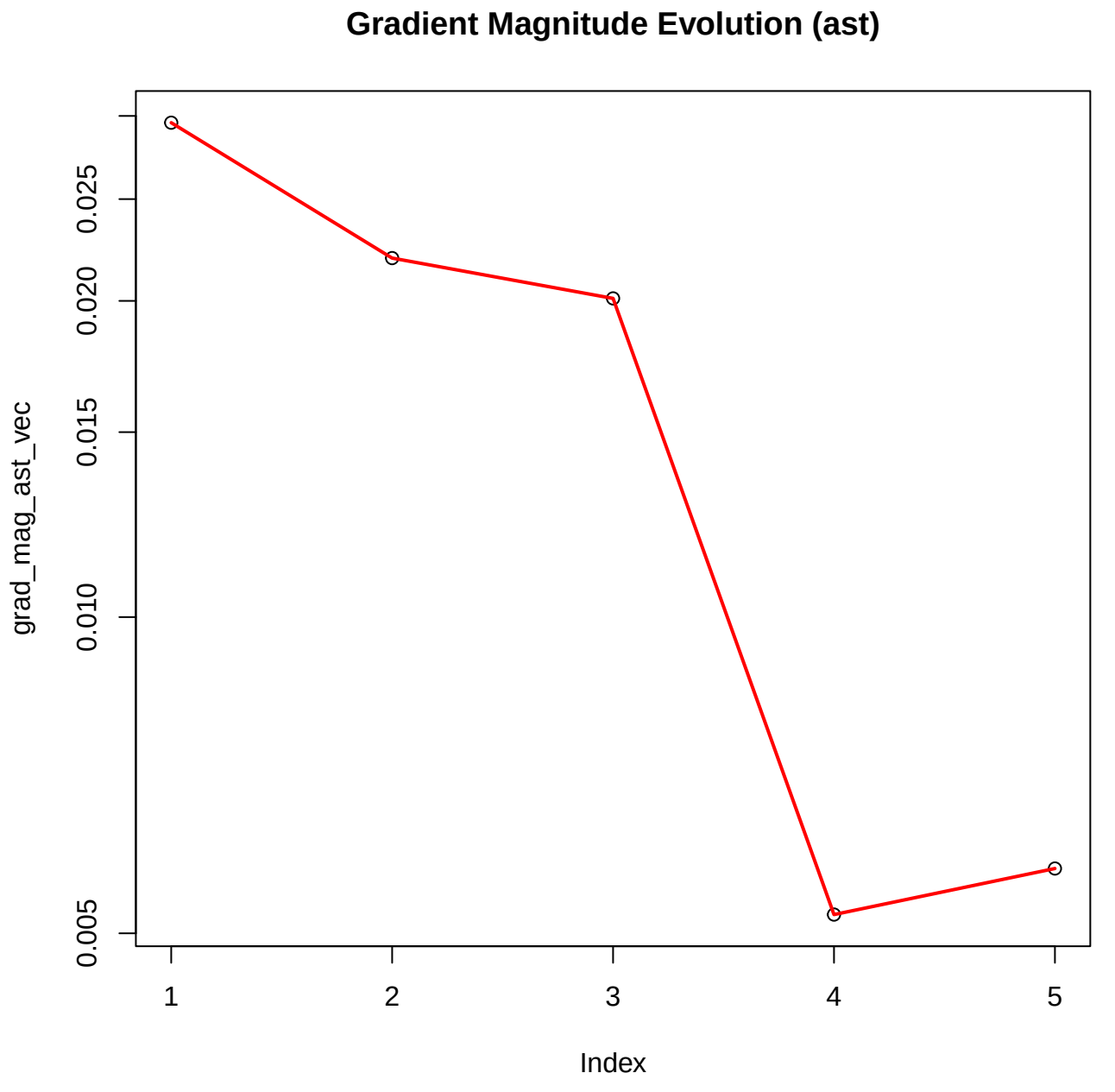

Loss/Objective (AST)
Q (objective)
0.3
0.4
0.5
0.6
0.7
2
4
6
8
10
Iteration

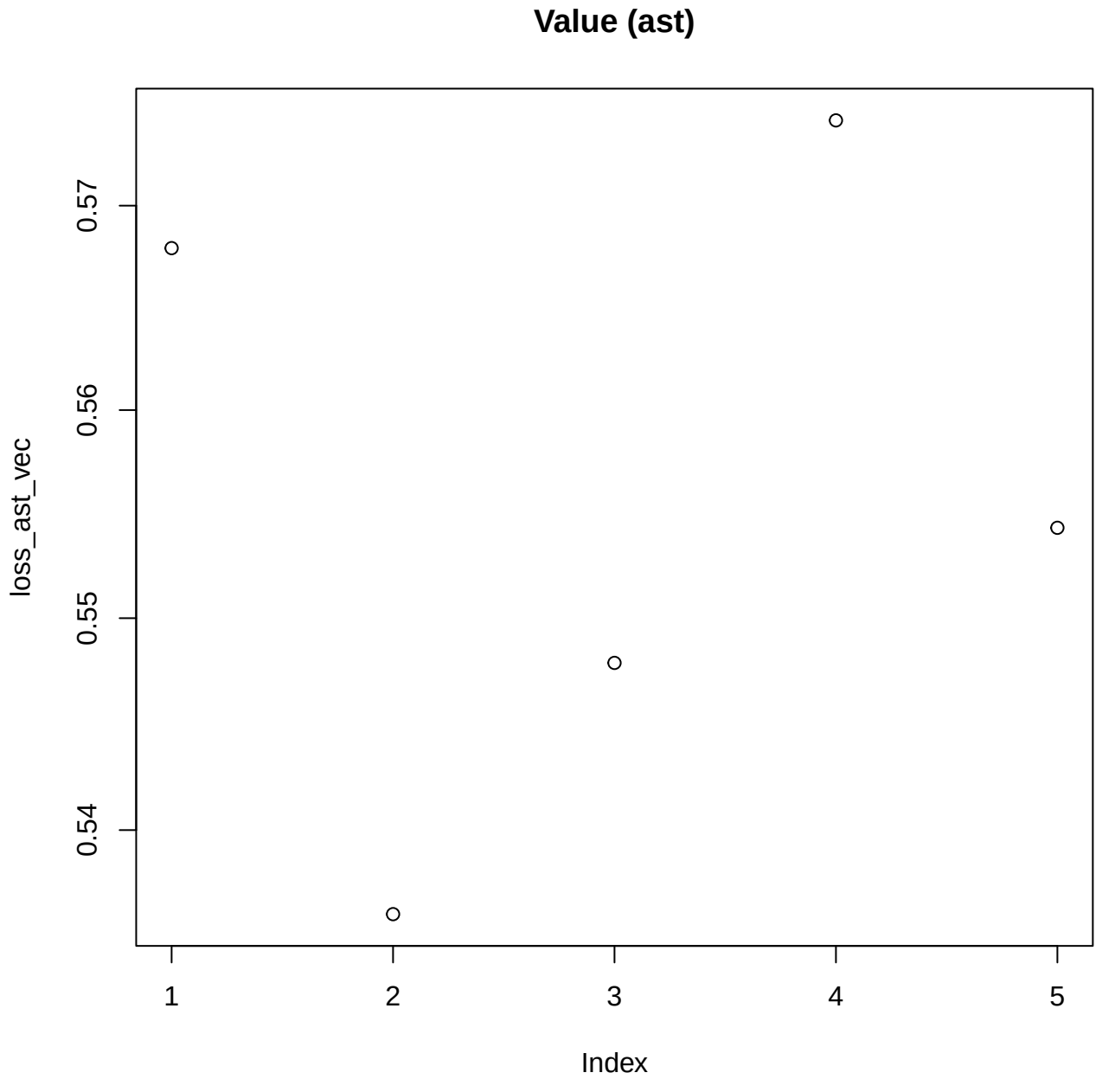

Loss/Objective (DAG)
Q (objective)
0.3
0.4
0.5
0.6
0.7
2
4
6
8
10
Iteration

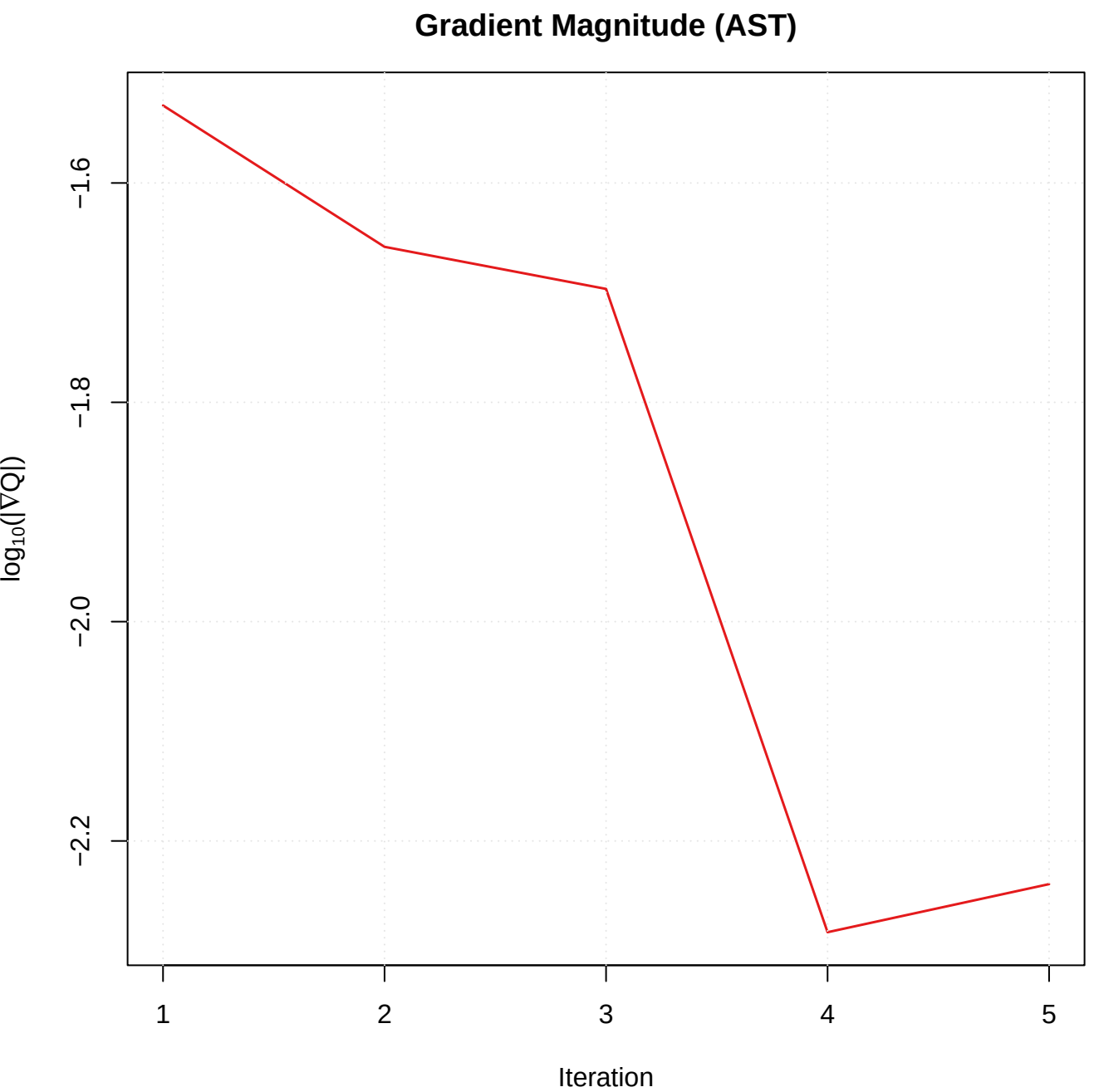

Gradient Magnitude (AST)
log10(|∇Q|)
−0.6
−0.8
−1.0
−1.2
−1.4
2
4
6
8
10
Iteration
Loss/Objective (AST)
Q (objective)
0.7
0.6
0.5
0.4
0.3
2
4
6
8
10
Iteration
Gradient Magnitude (DAG)
log10(|∇Q|)
−0.6
−0.8
−1.0
−1.2
−1.4
2
4
6
8
10
Iteration
Loss/Objective (DAG)
Q (objective)
0.7
0.6
0.5
0.4
0.3
2
4
6
8
10
Iteration

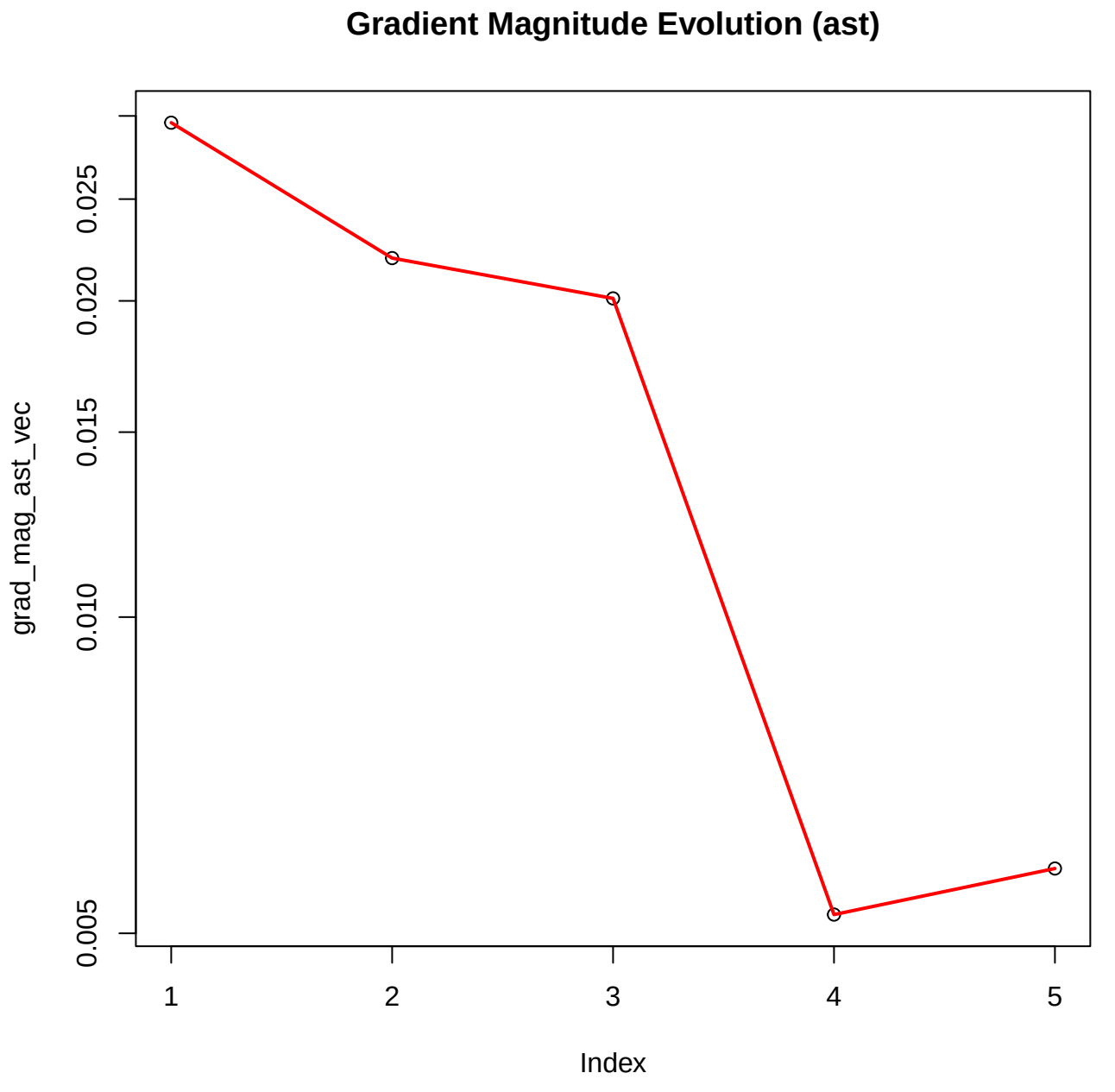

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.005
0.010
0.015
0.020
0.025
1
2
3
4
5
Index

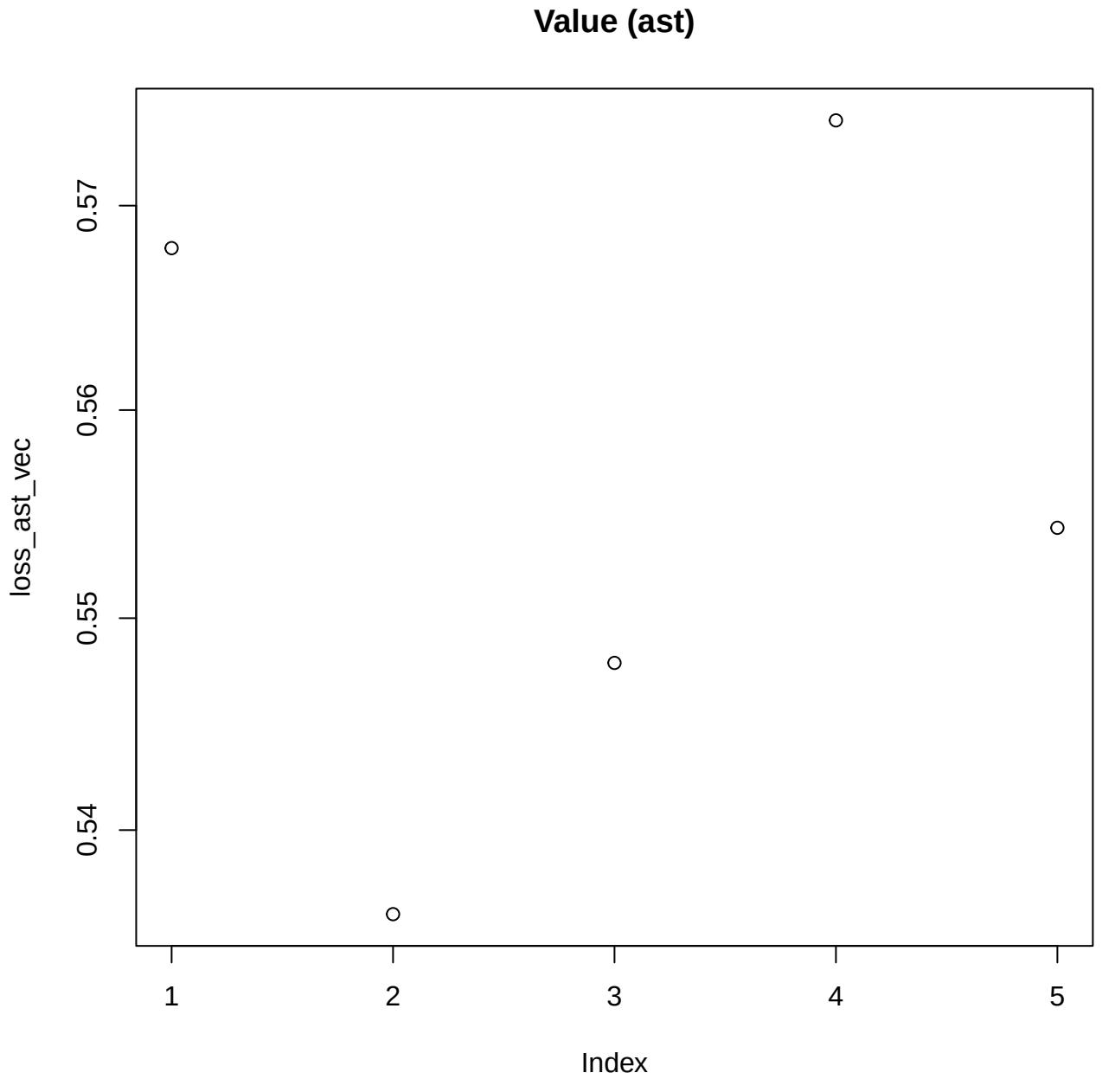

Value (ast)
loss_ast_vec
0.54
0.55
0.56
0.57
1
2
3
4
5
Index

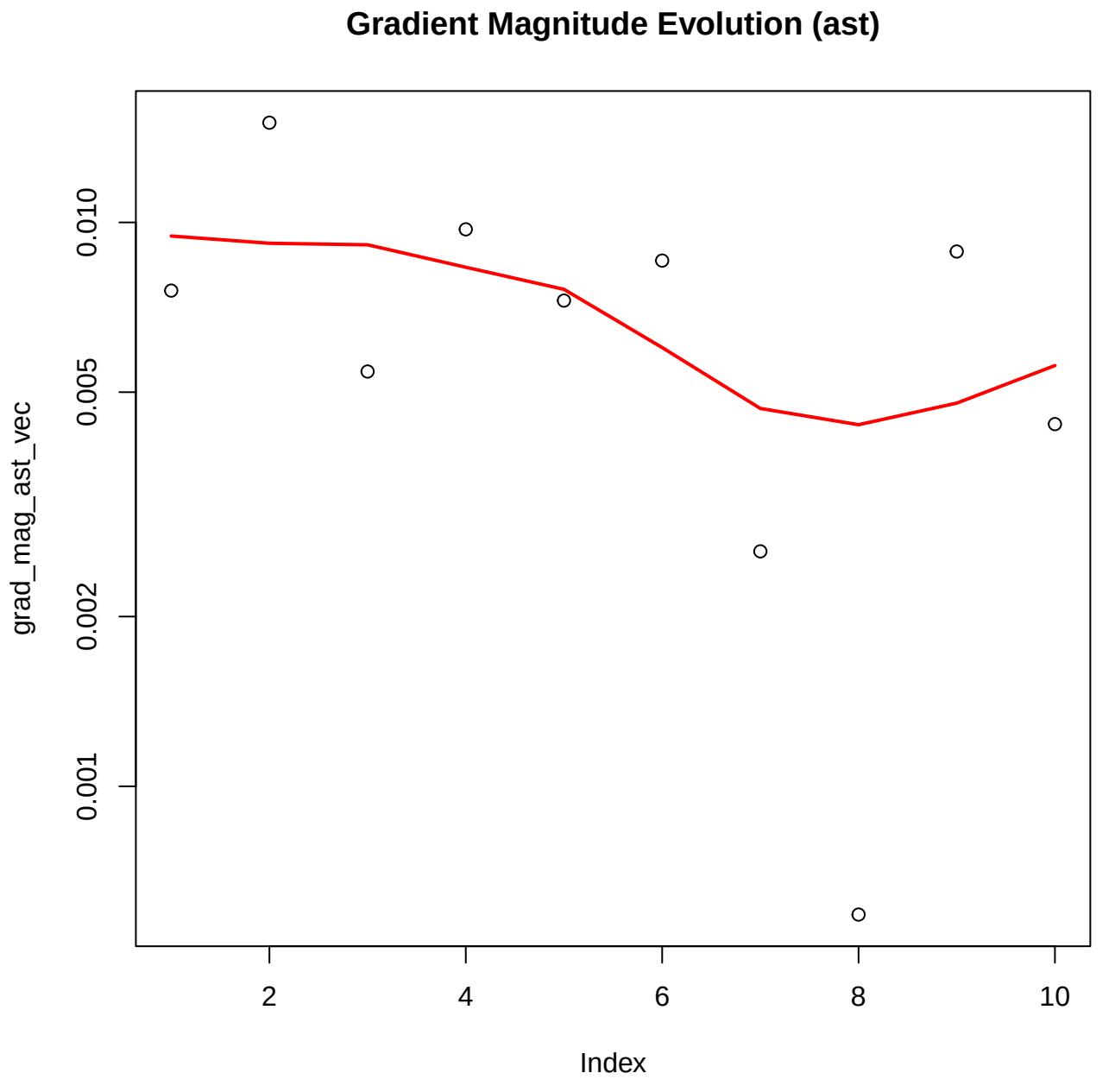

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.005
0.010
0.015
0.020
0.025
1
2
3
4
5
Index

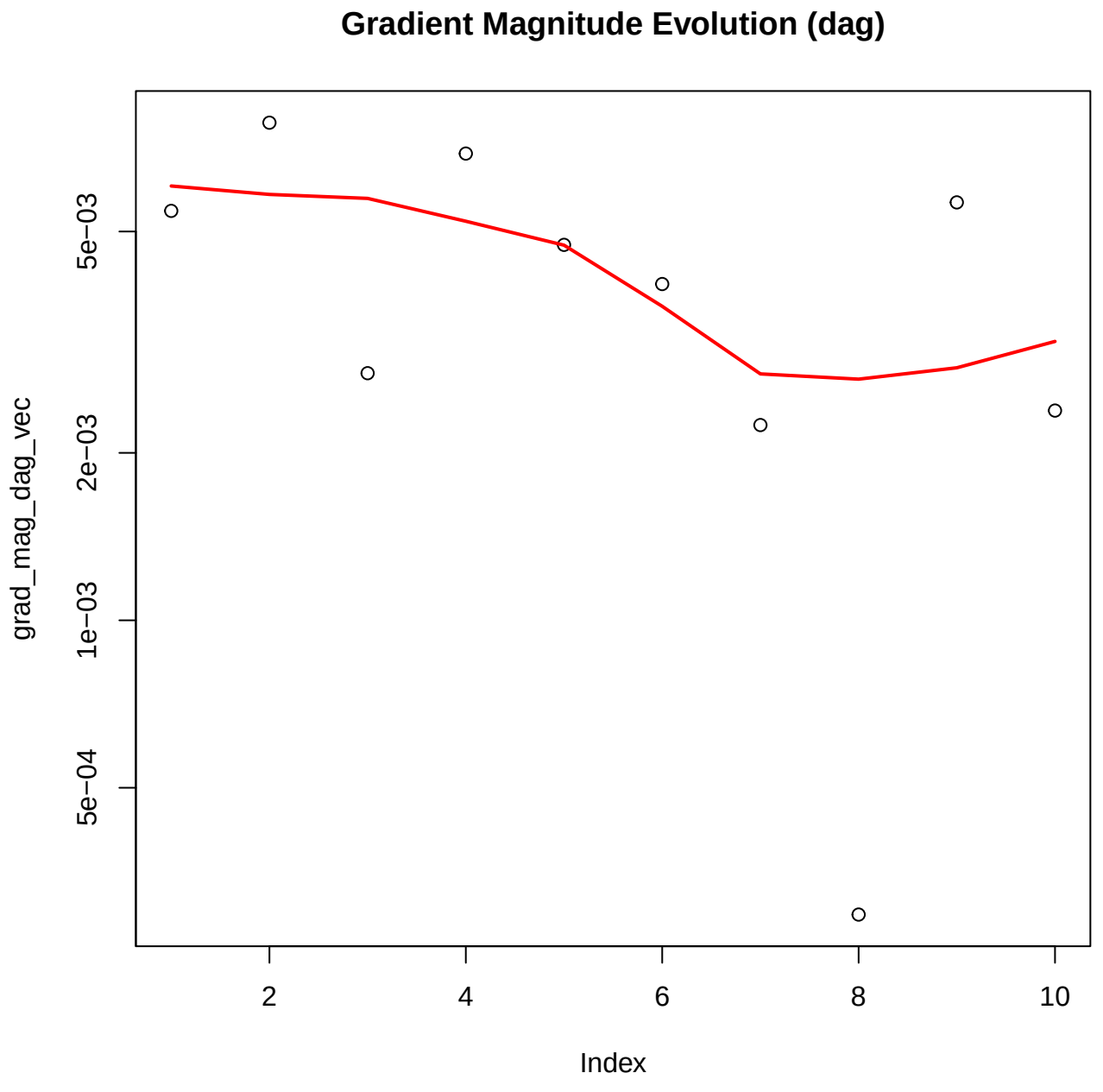

Value (ast)
loss_ast_vec
0.54
0.55
0.56
0.57
1
2
3
4
5
Index

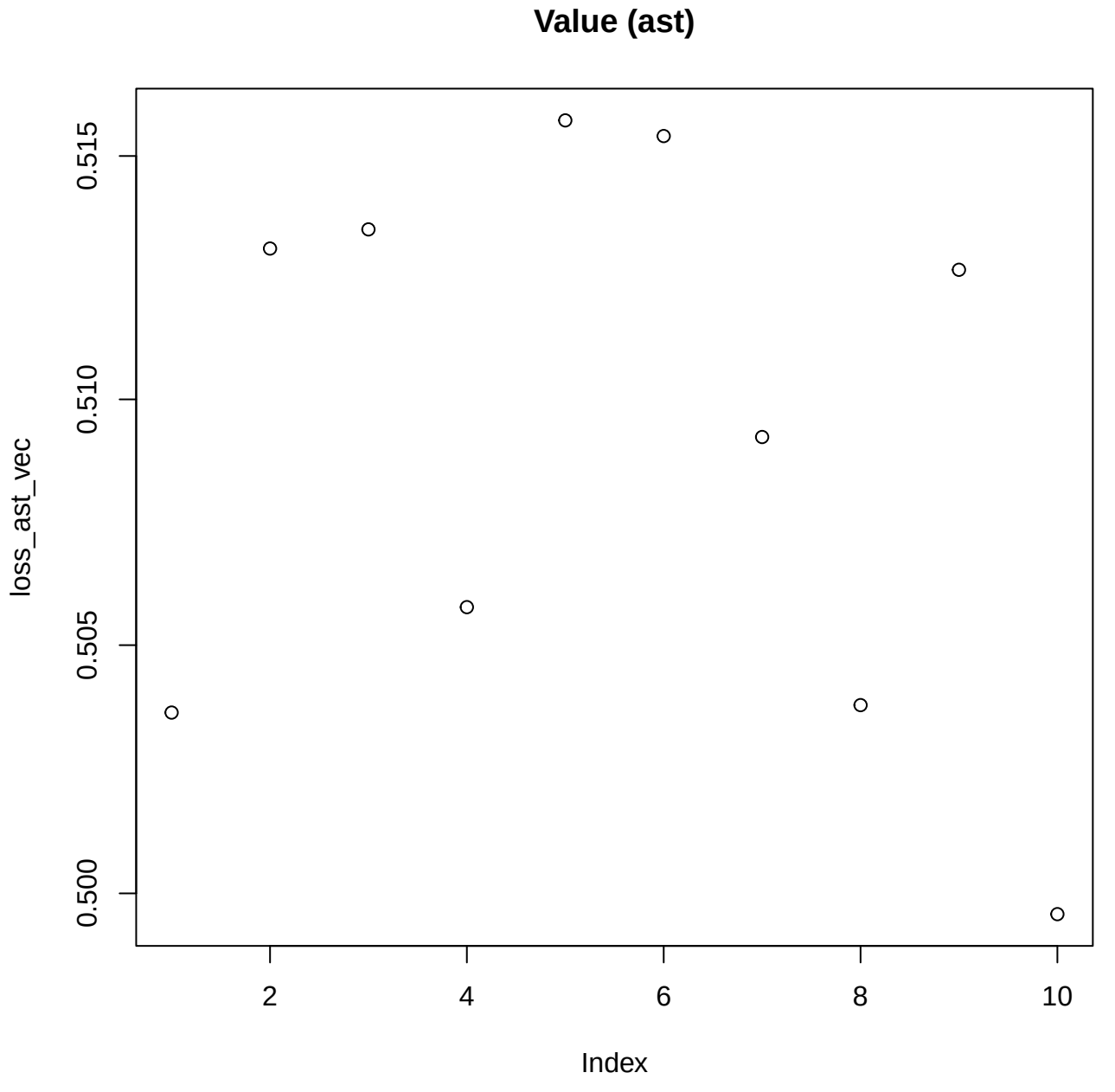

Gradient Magnitude (AST)
−1.6
−1.8
−2.0
−2.2
log10(|∇Q|)
1
2
3
4
5
Iteration

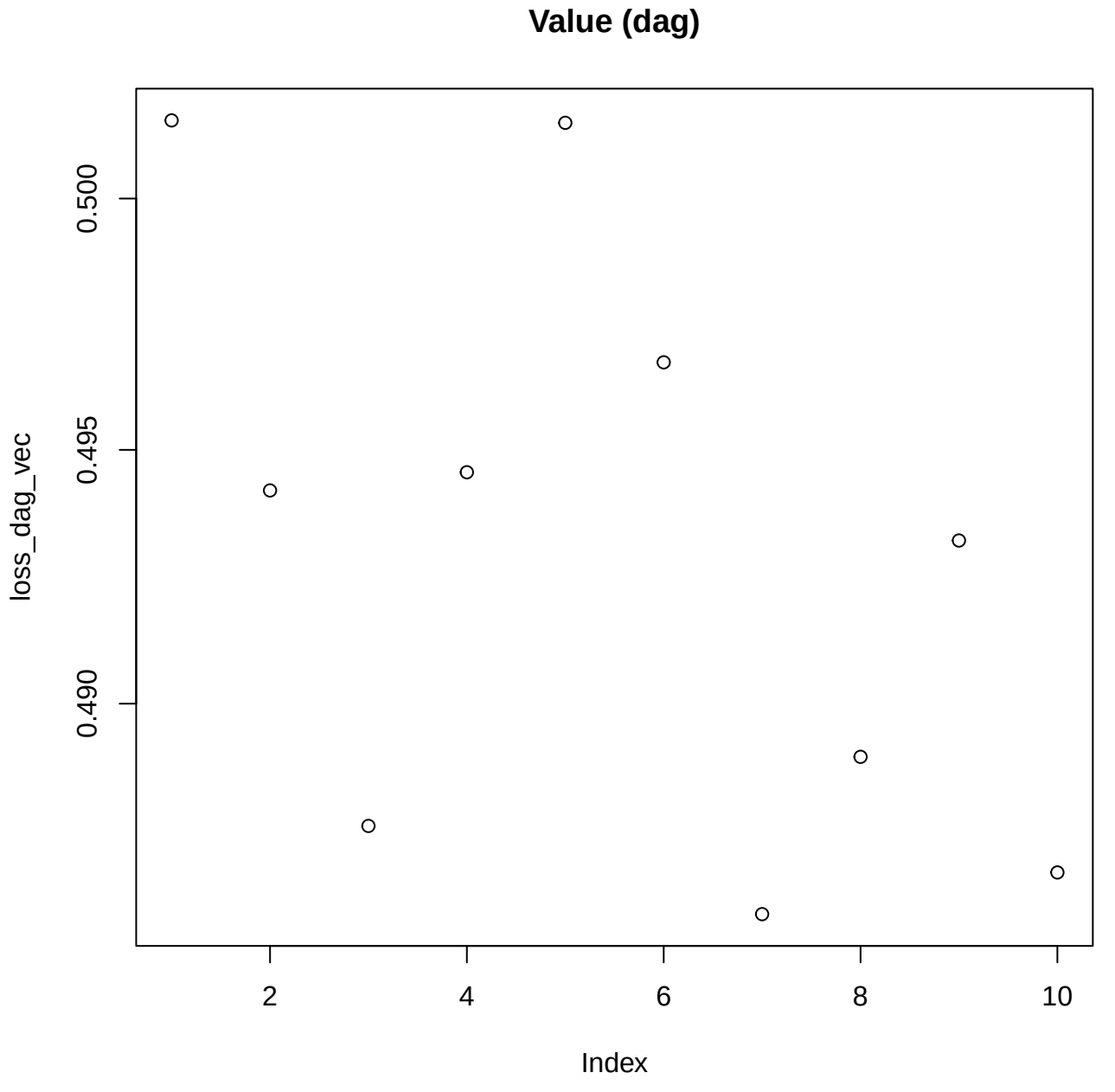

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.005
0.010
0.015
0.020
0.025
1
2
3
4
5
Index

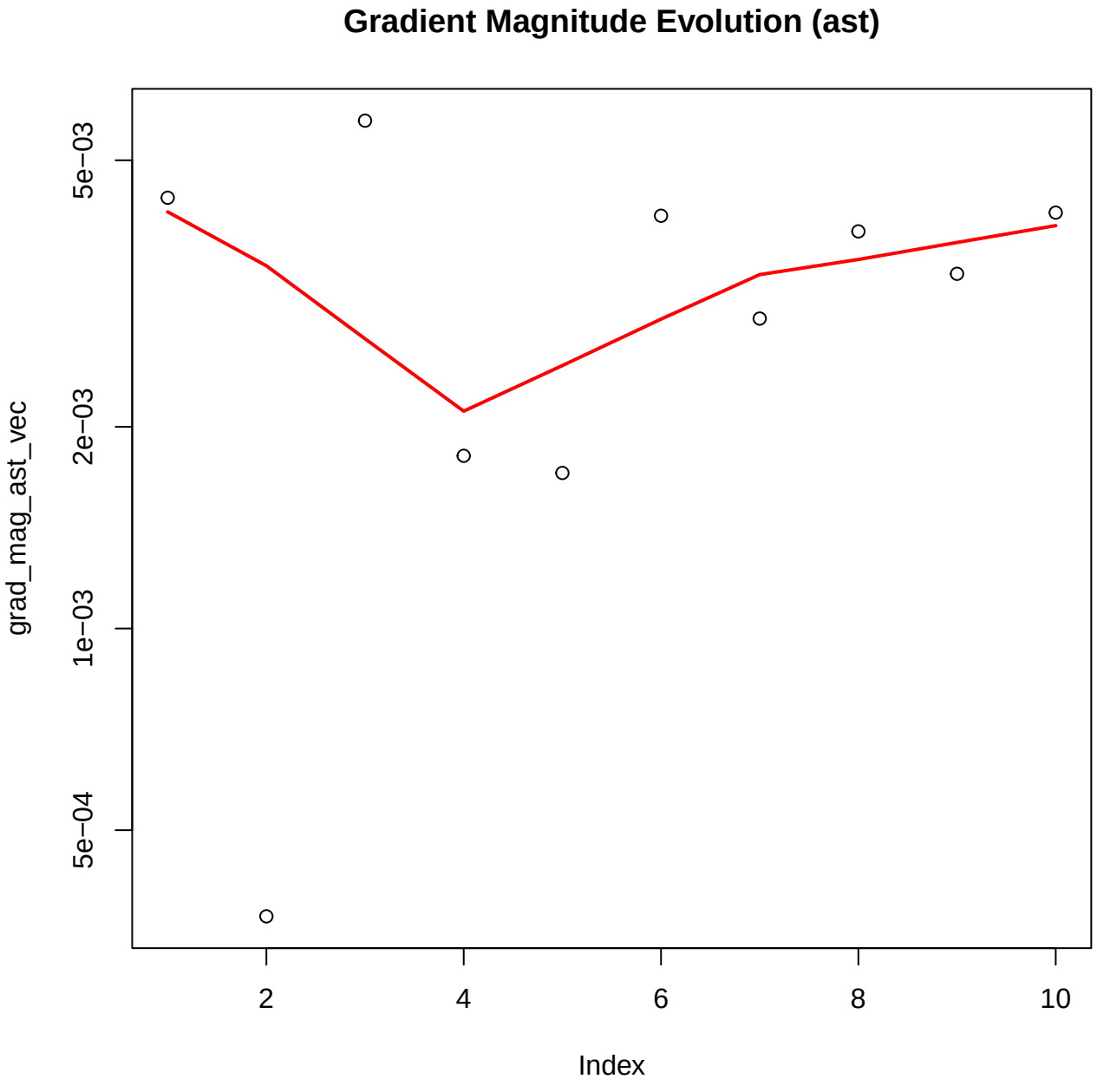

Value (ast)
loss_ast_vec
0.54
0.55
0.56
0.57
1
2
3
4
5
Index

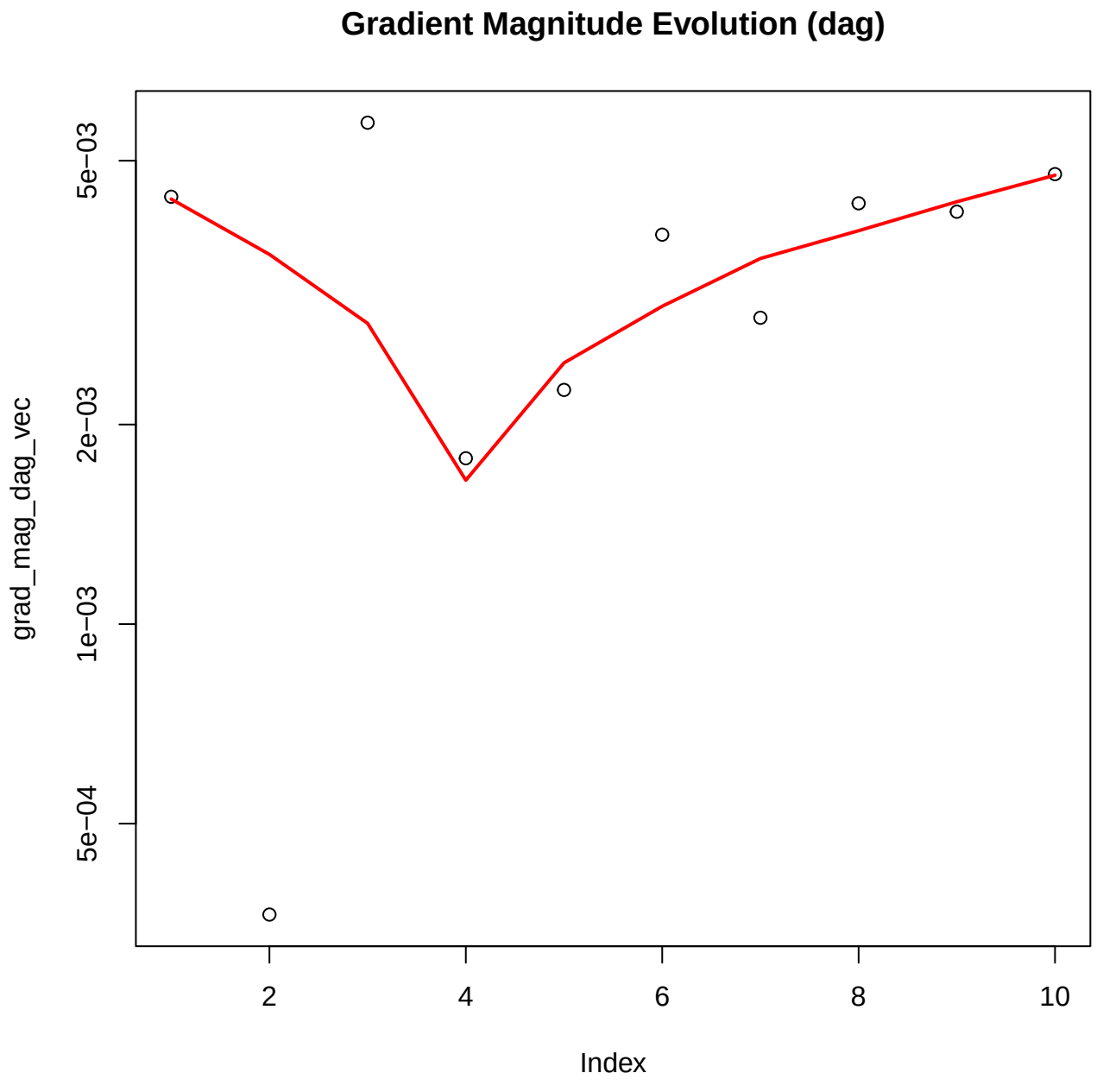

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.001
0.002
0.005
0.010
2
4
6
8
10
Index

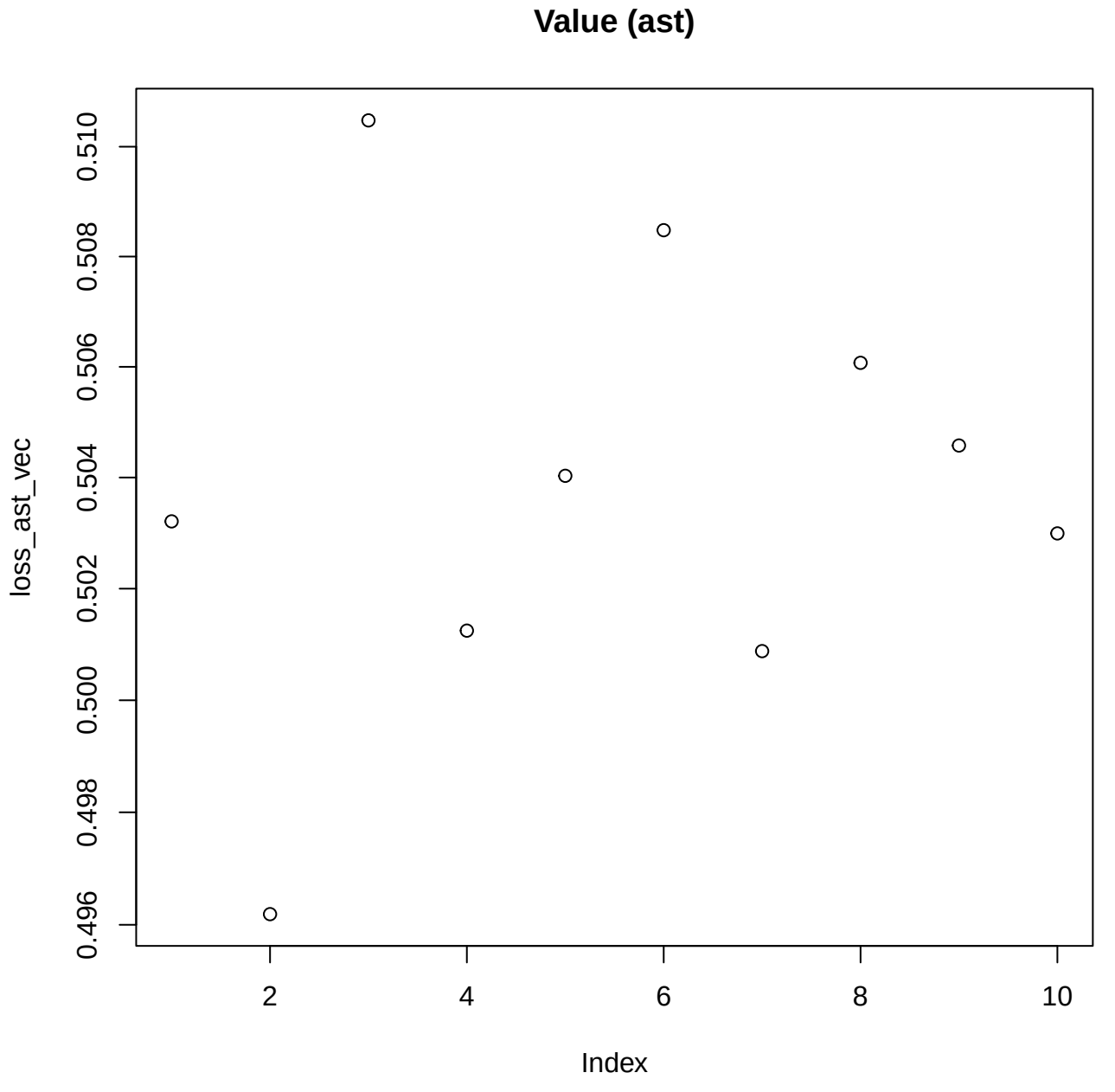

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
5e−03
2e−03
1e−03
5e−04
2
4
6
8
10
Index

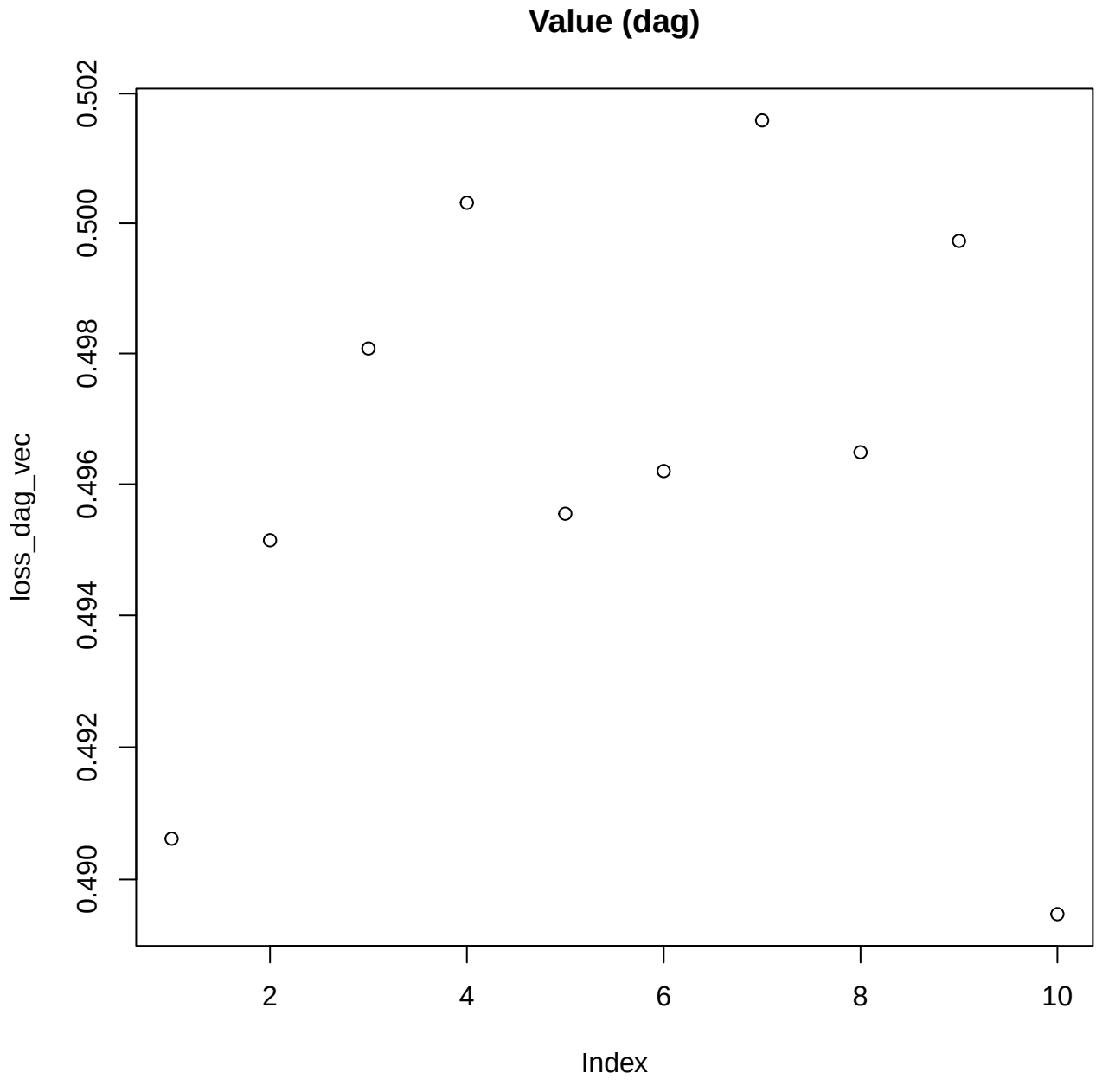

Value (ast)
loss_ast_vec
0.500
0.505
0.510
0.515
2
4
6
8
10
Index

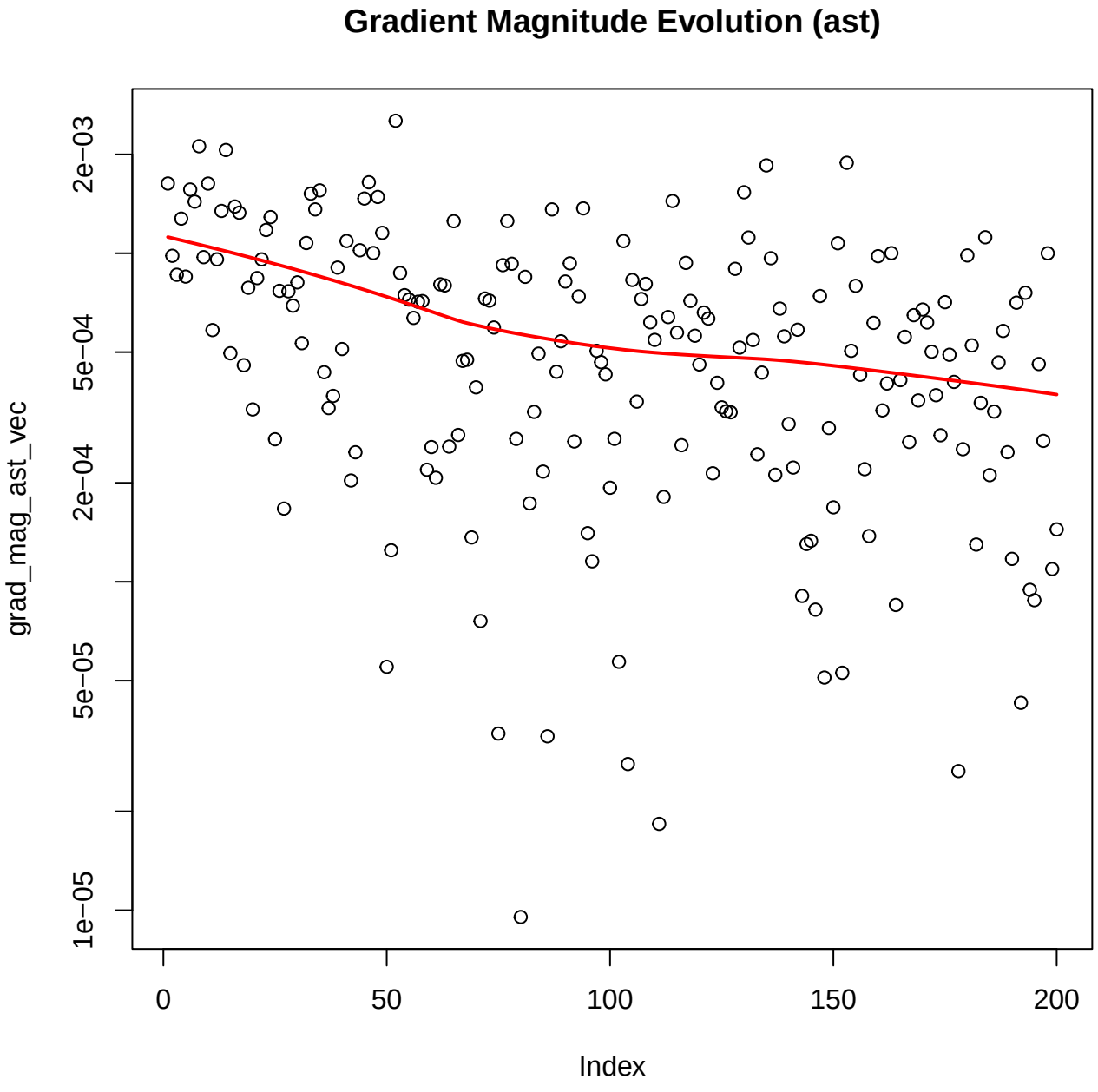

Value (dag)
loss_dag_vec
0.490
0.495
0.500
2
4
6
8
10
Index

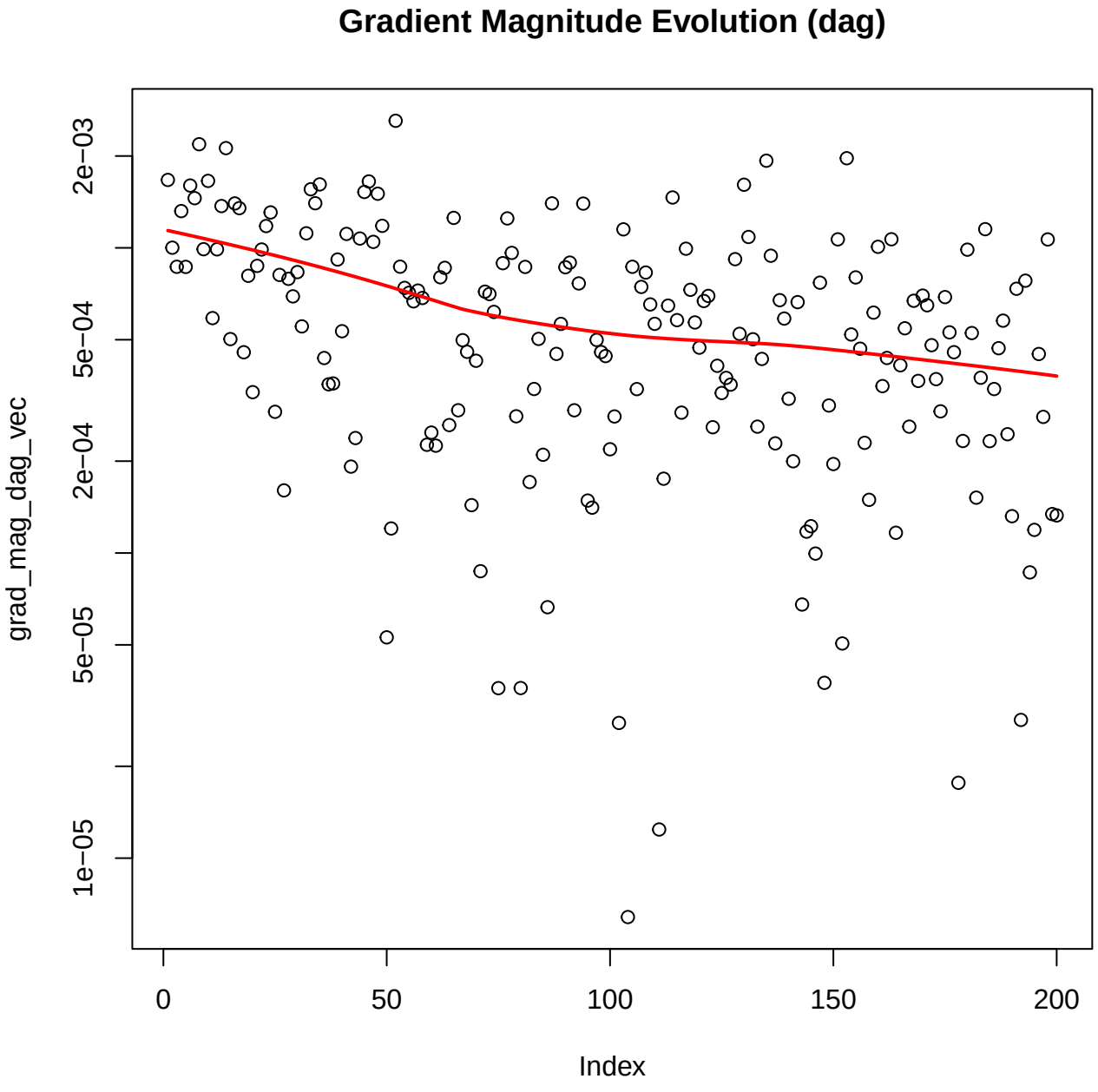

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
5e−03
2e−03
1e−03
5e−04
2
4
6
8
10
Index

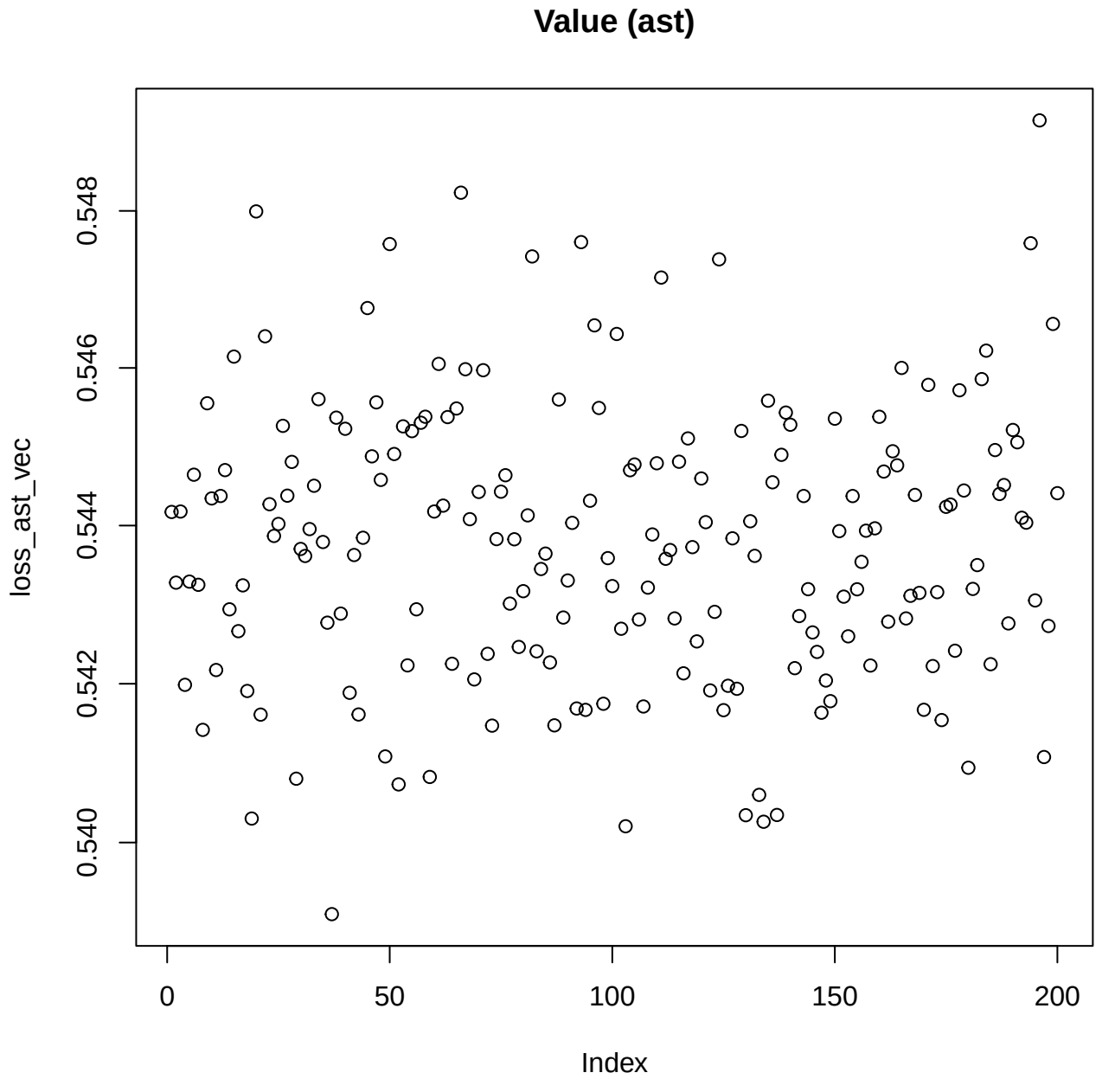

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
5e−03
2e−03
1e−03
5e−04
2
4
6
8
10
Index

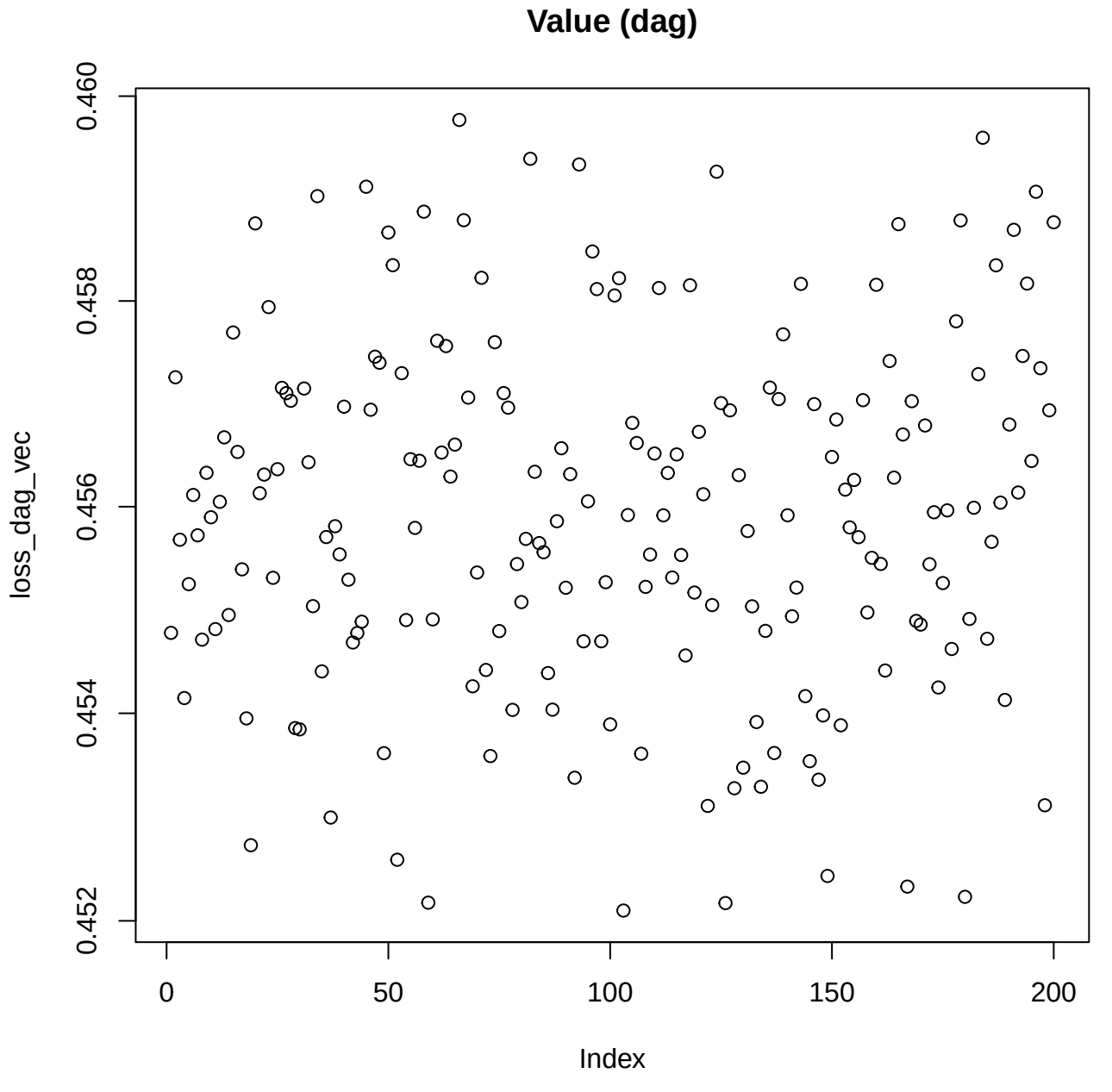

Value (ast)
loss_ast_vec
0.496
0.498
0.500
0.502
0.504
0.506
0.508
0.510
2
4
6
8
10
Index

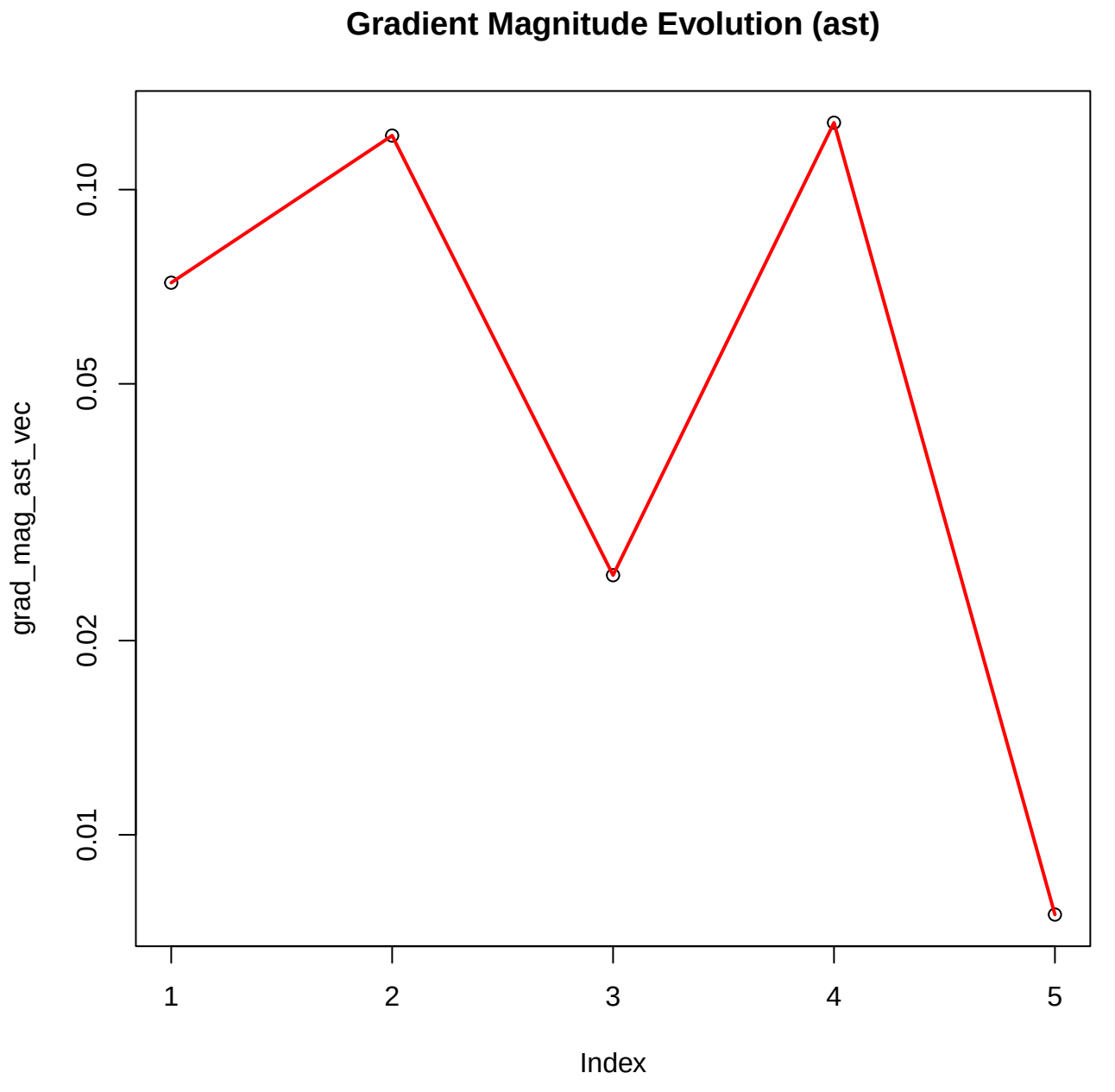

Value (dag)
loss_dag_vec
0.490
0.492
0.494
0.496
0.498
0.500
0.502
2
4
6
8
10
Index

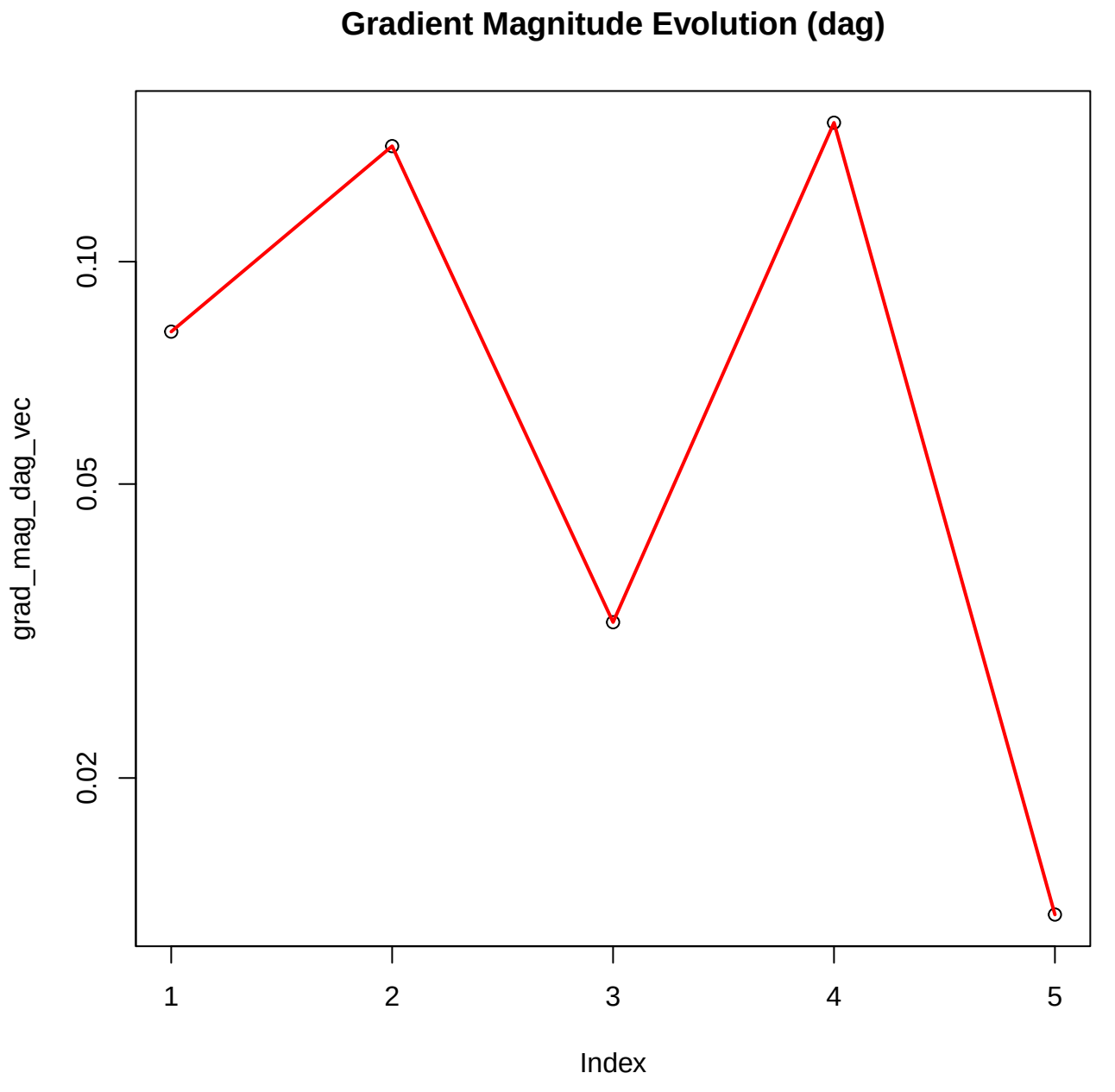

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
2e−03
5e−04
2e−04
5e−05
1e−05
0
50
100
150
200
Index

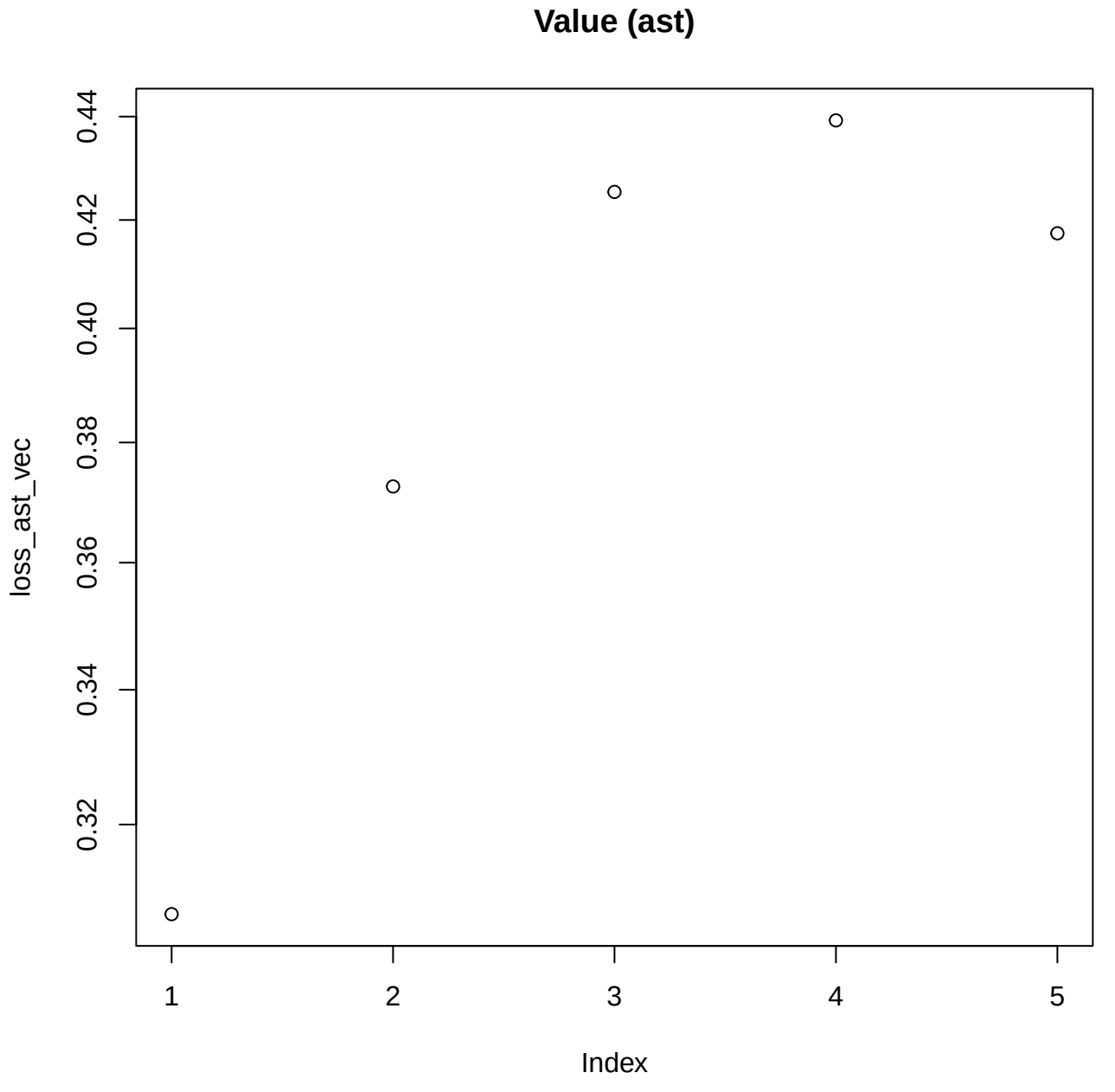

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
2e−03
5e−04
2e−04
5e−05
1e−05
0
50
100
150
200
Index

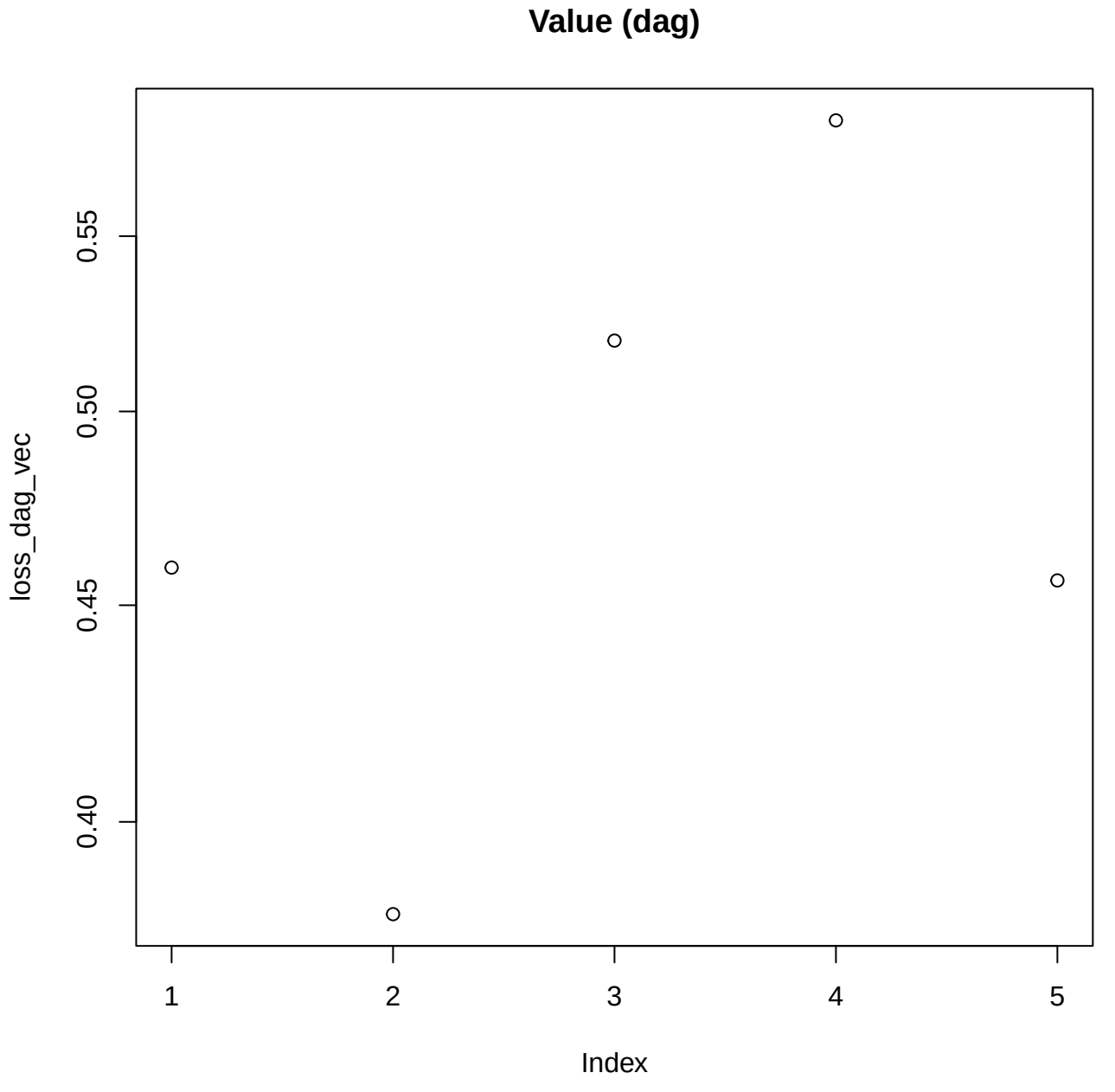

Value (ast)
loss_ast_vec
0.540
0.542
0.544
0.546
0.548
0
50
100
150
200
Index

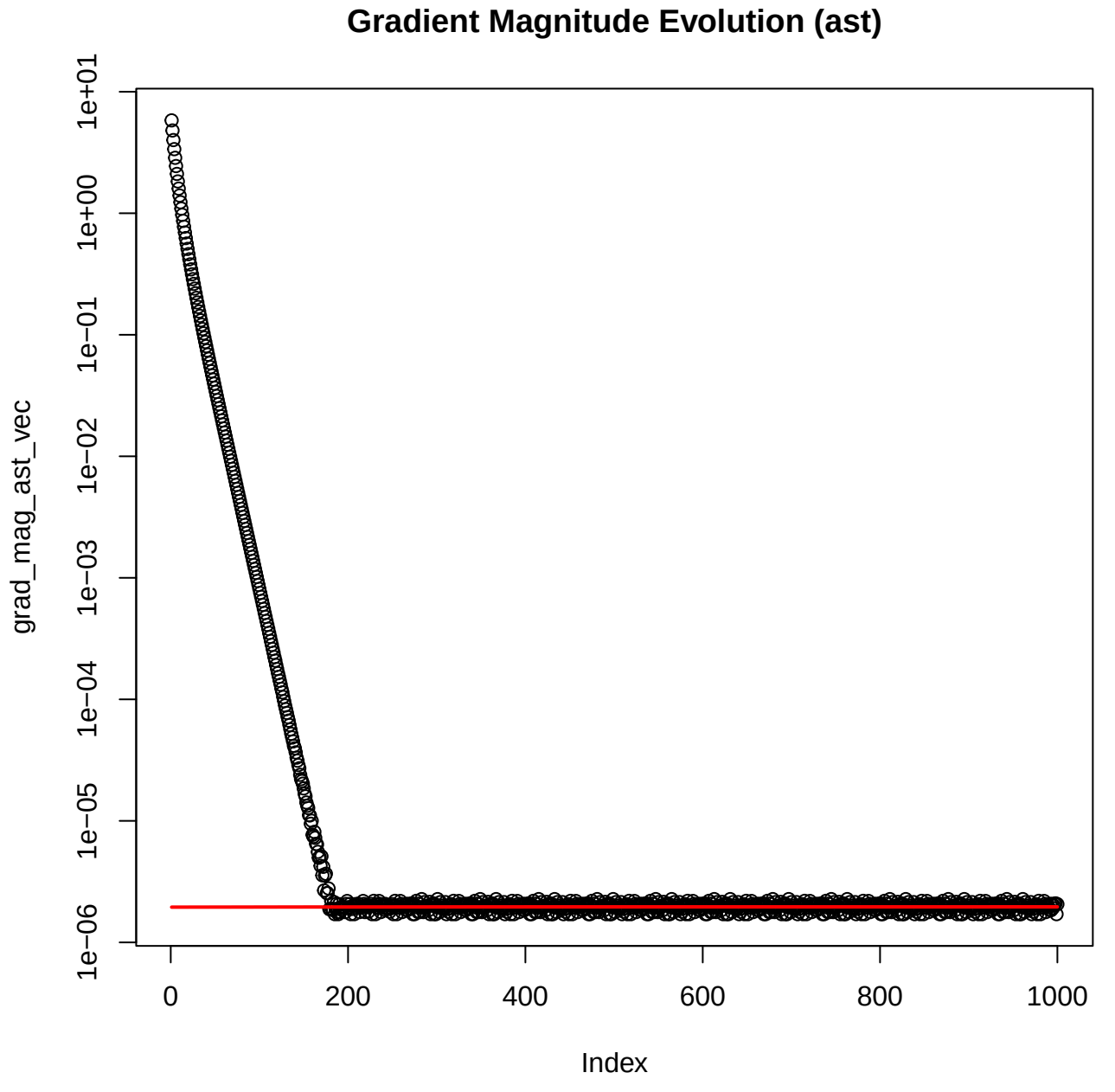

Value (dag)
loss_dag_vec
0.452
0.454
0.456
0.458
0.460
0
50
100
150
200
Index

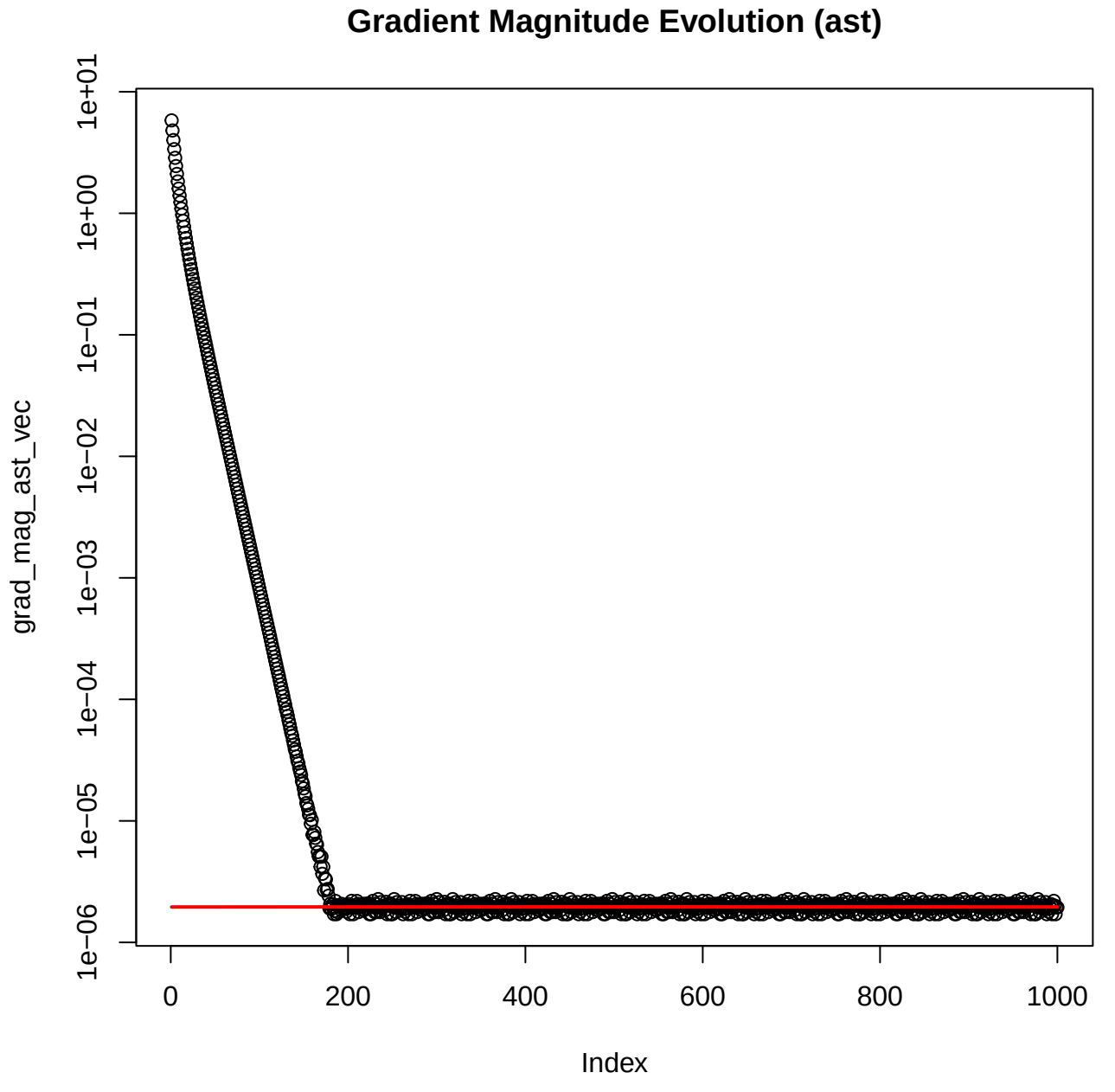

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.01
0.02
0.05
0.10
1
2
3
4
5
Index

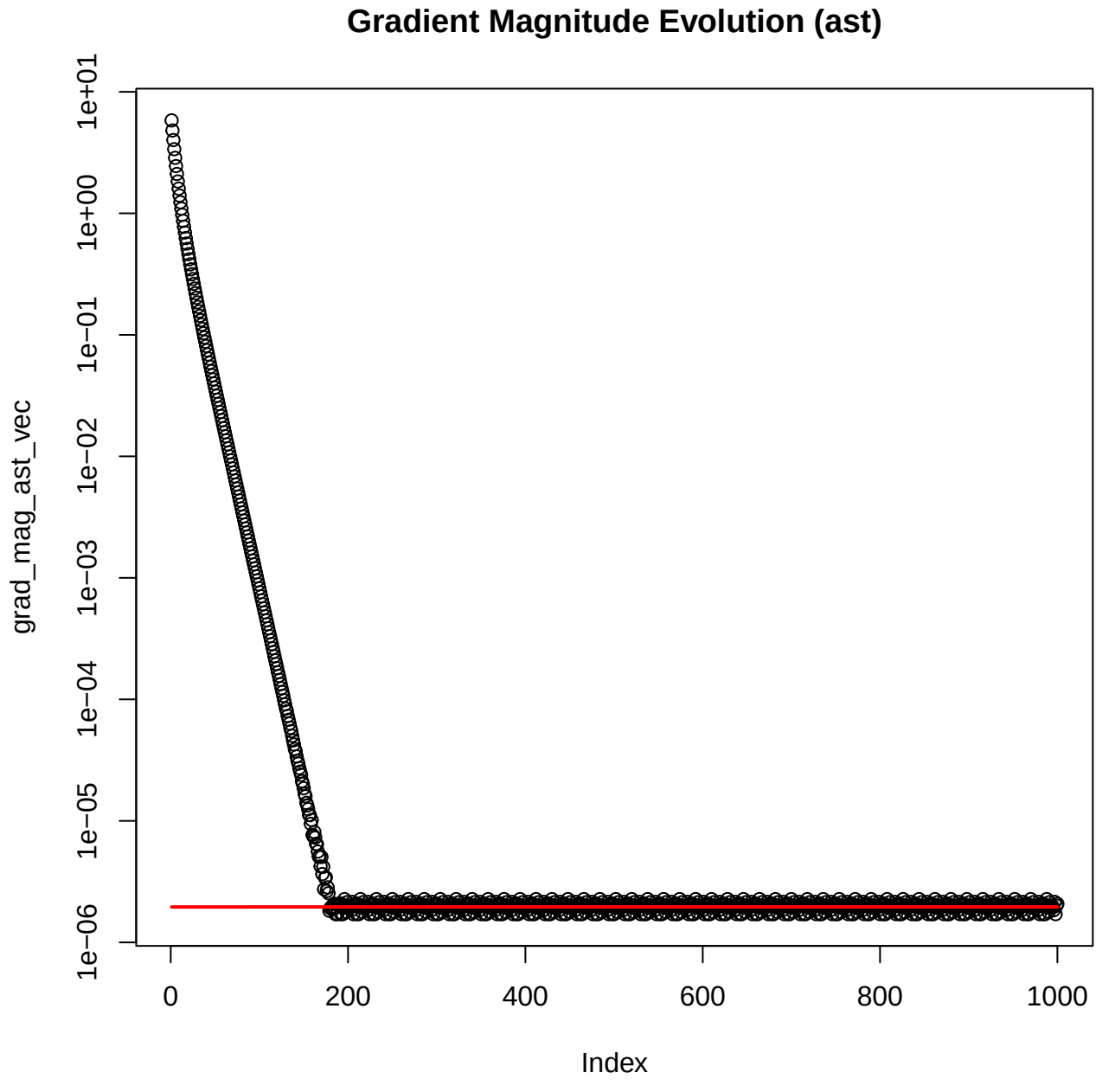

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.02
0.05
0.10
1
2
3
4
5
Index

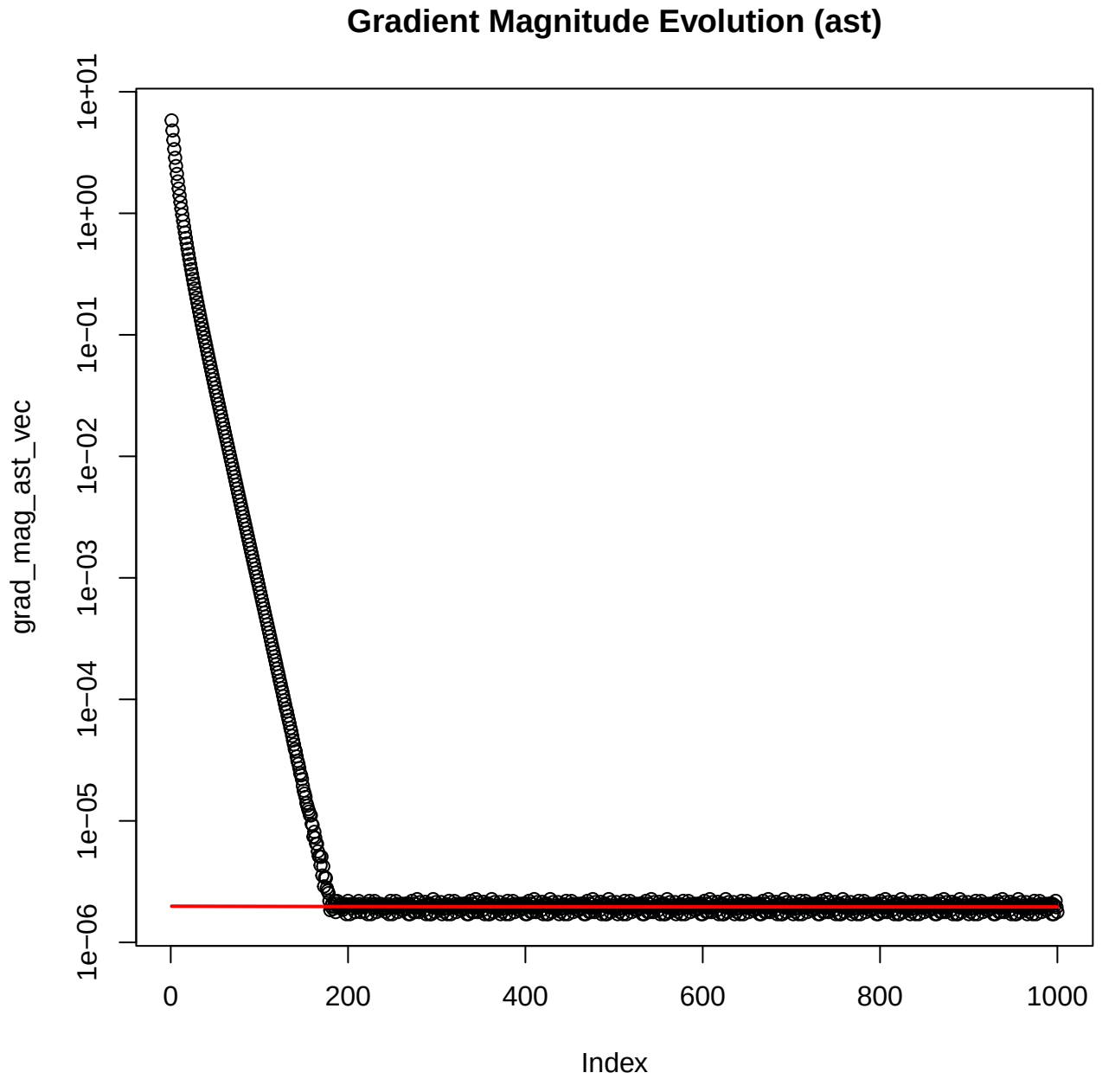

Value (ast)
loss_ast_vec
0.32
0.34
0.36
0.38
0.40
0.42
0.44
1
2
3
4
5
Index

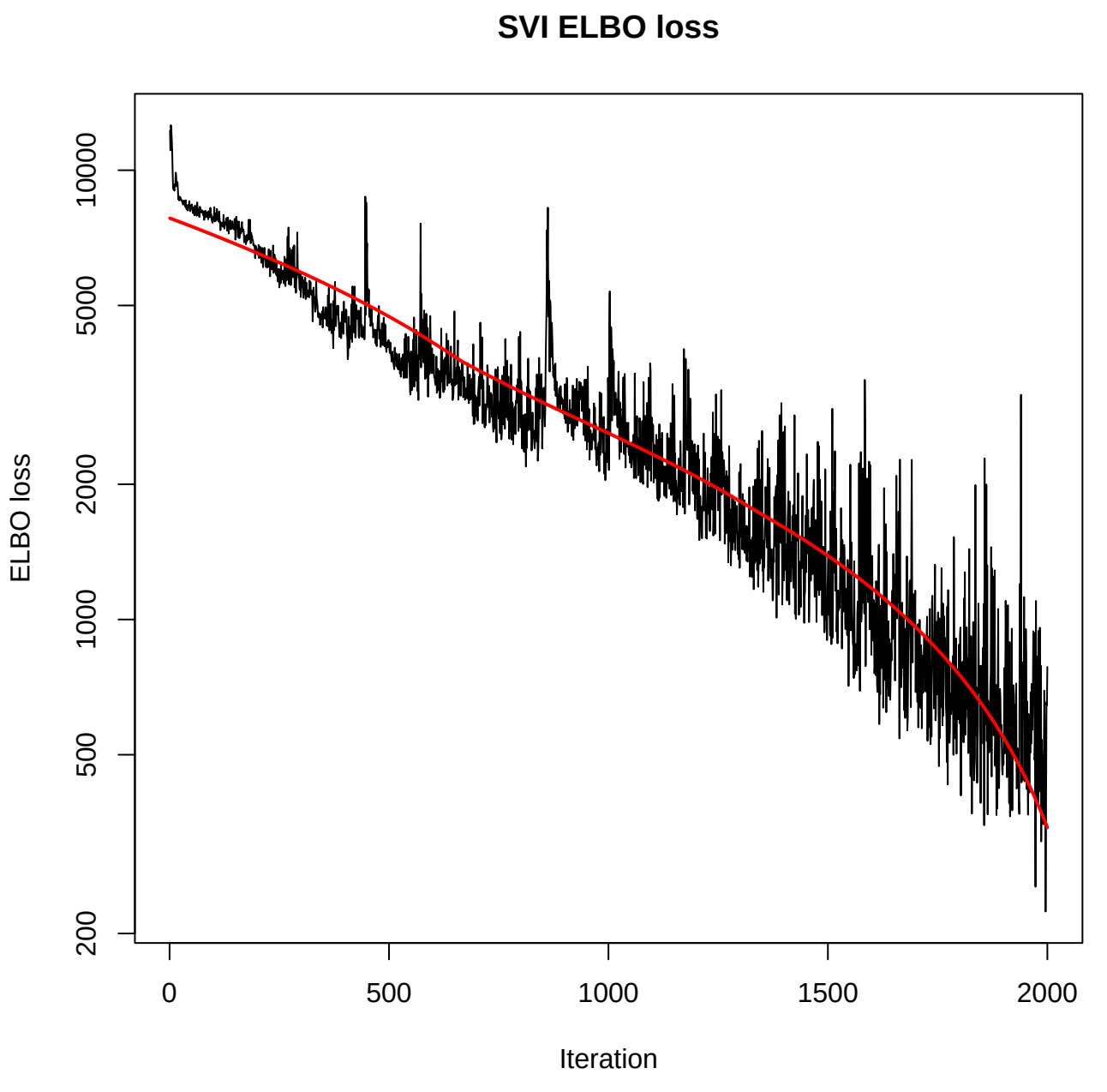

Value (dag)
loss_dag_vec
0.40
0.45
0.50
0.55
1
2
3
4
5
Index

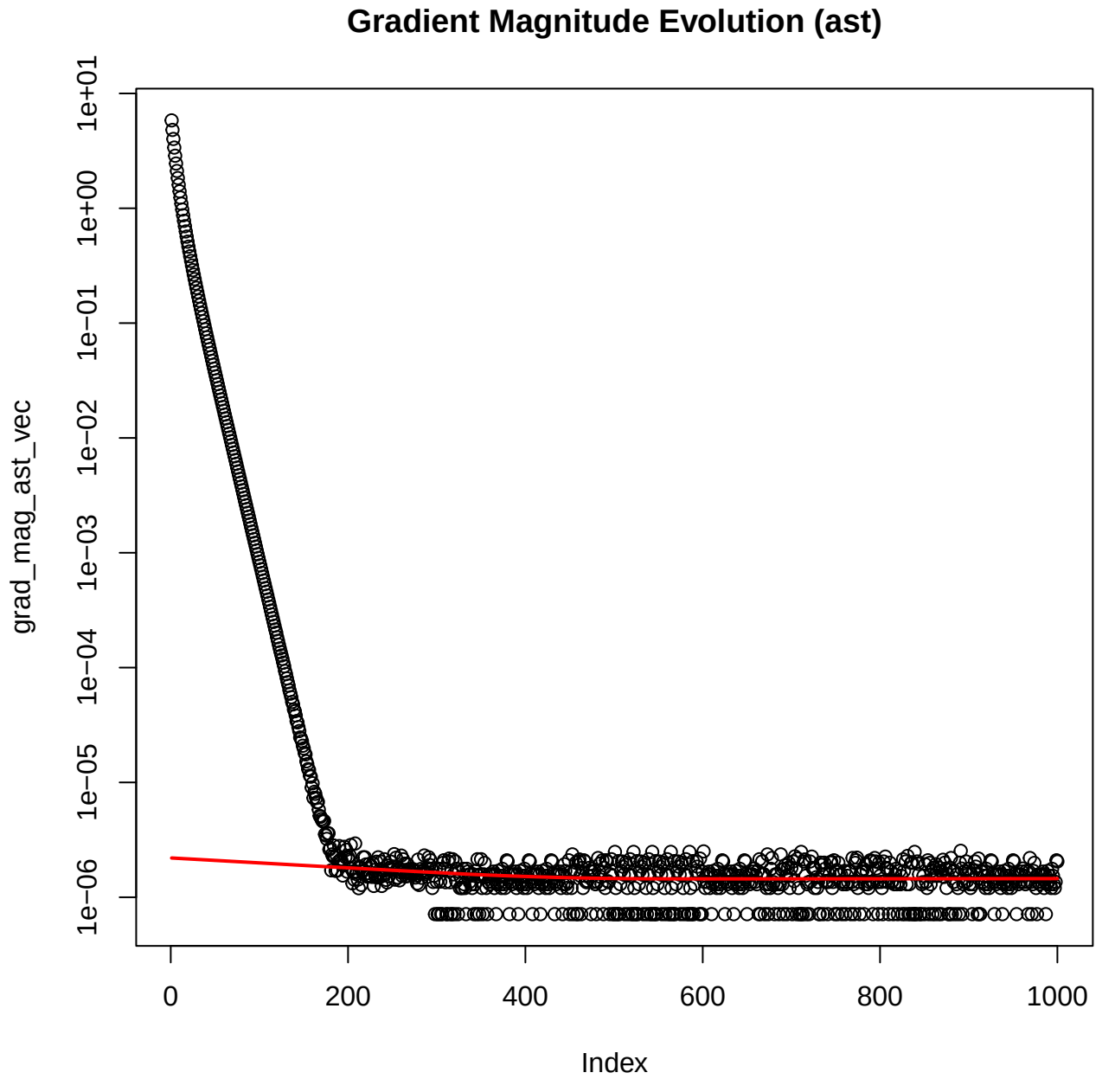

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

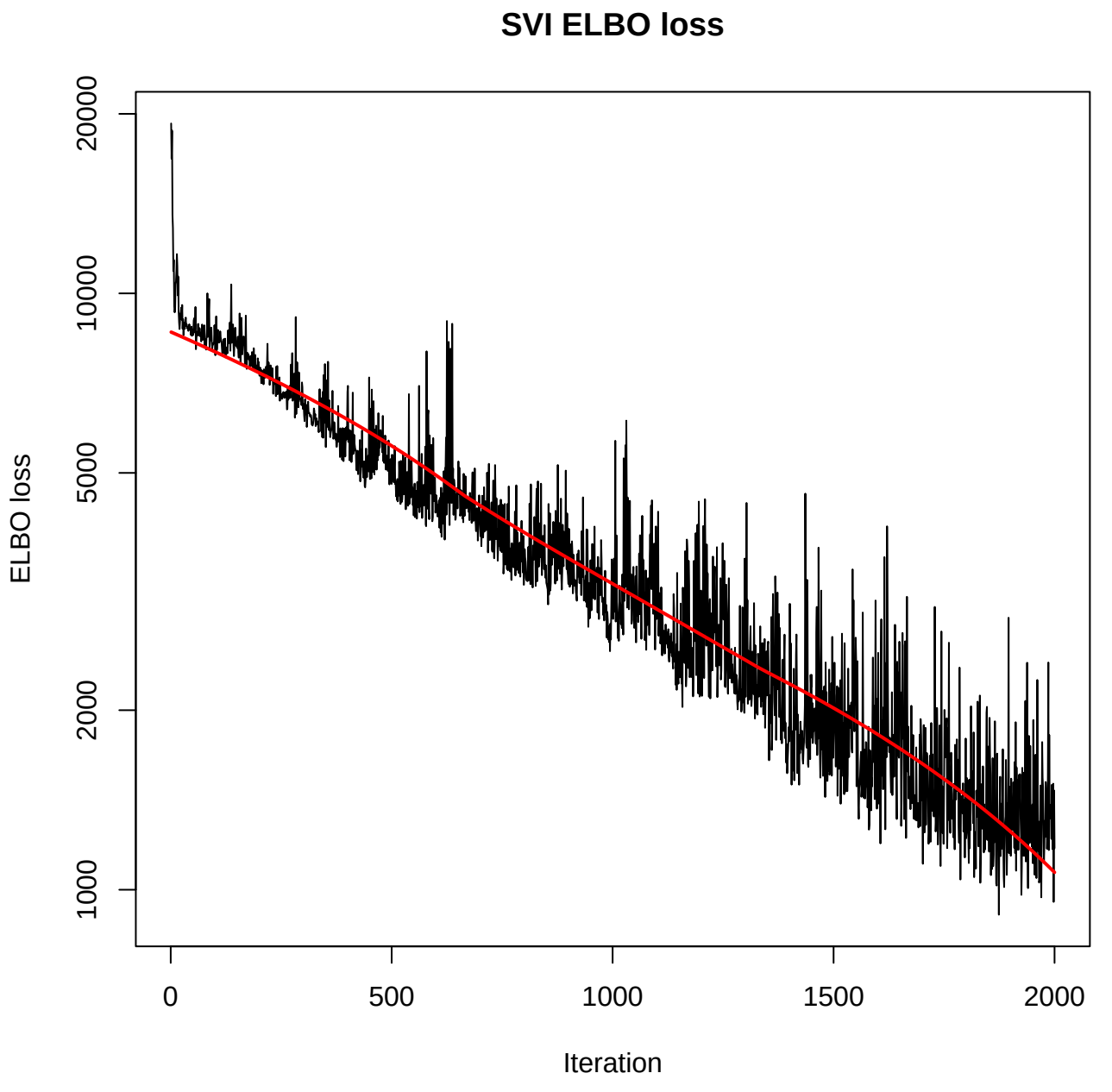

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

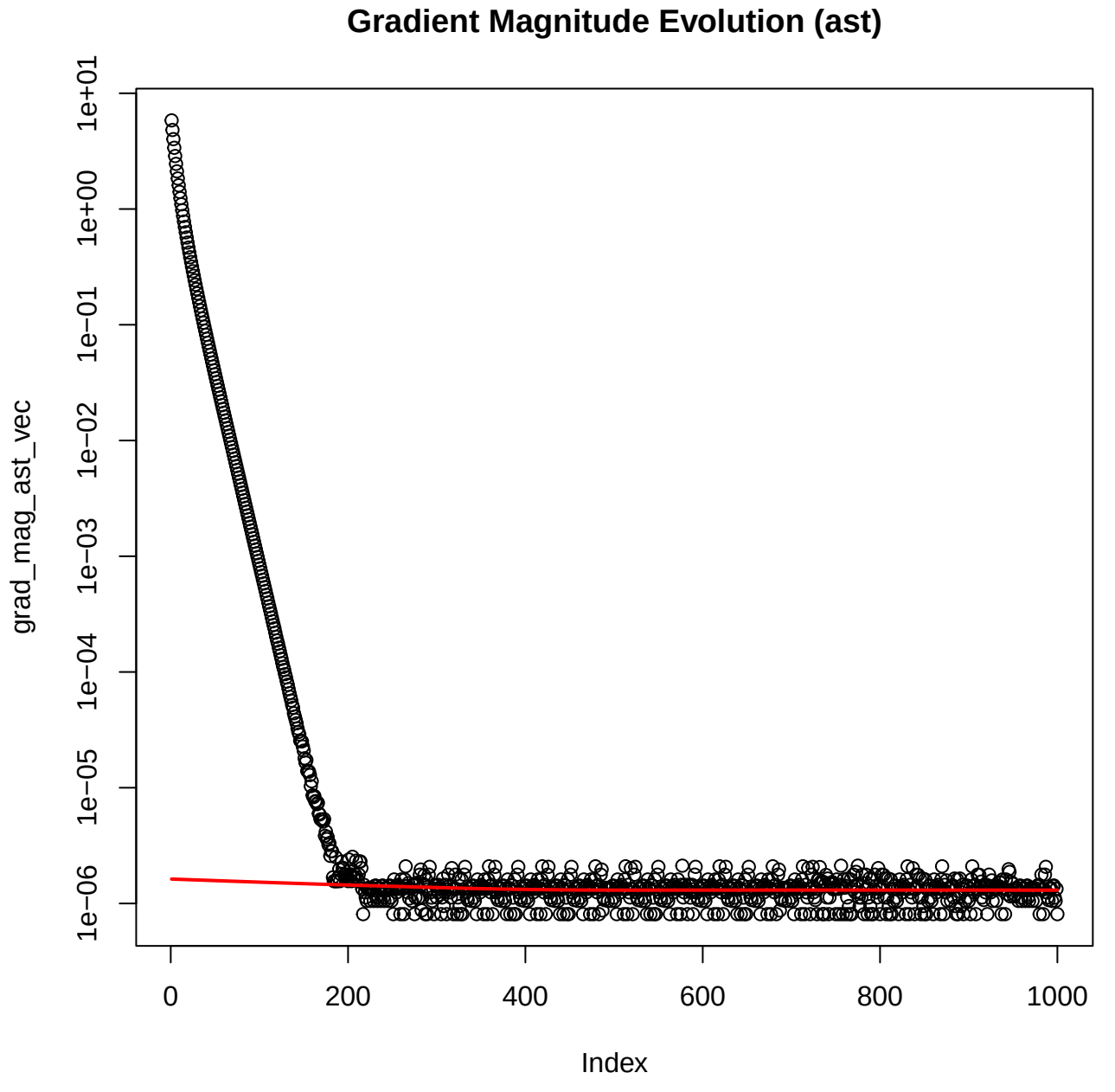

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

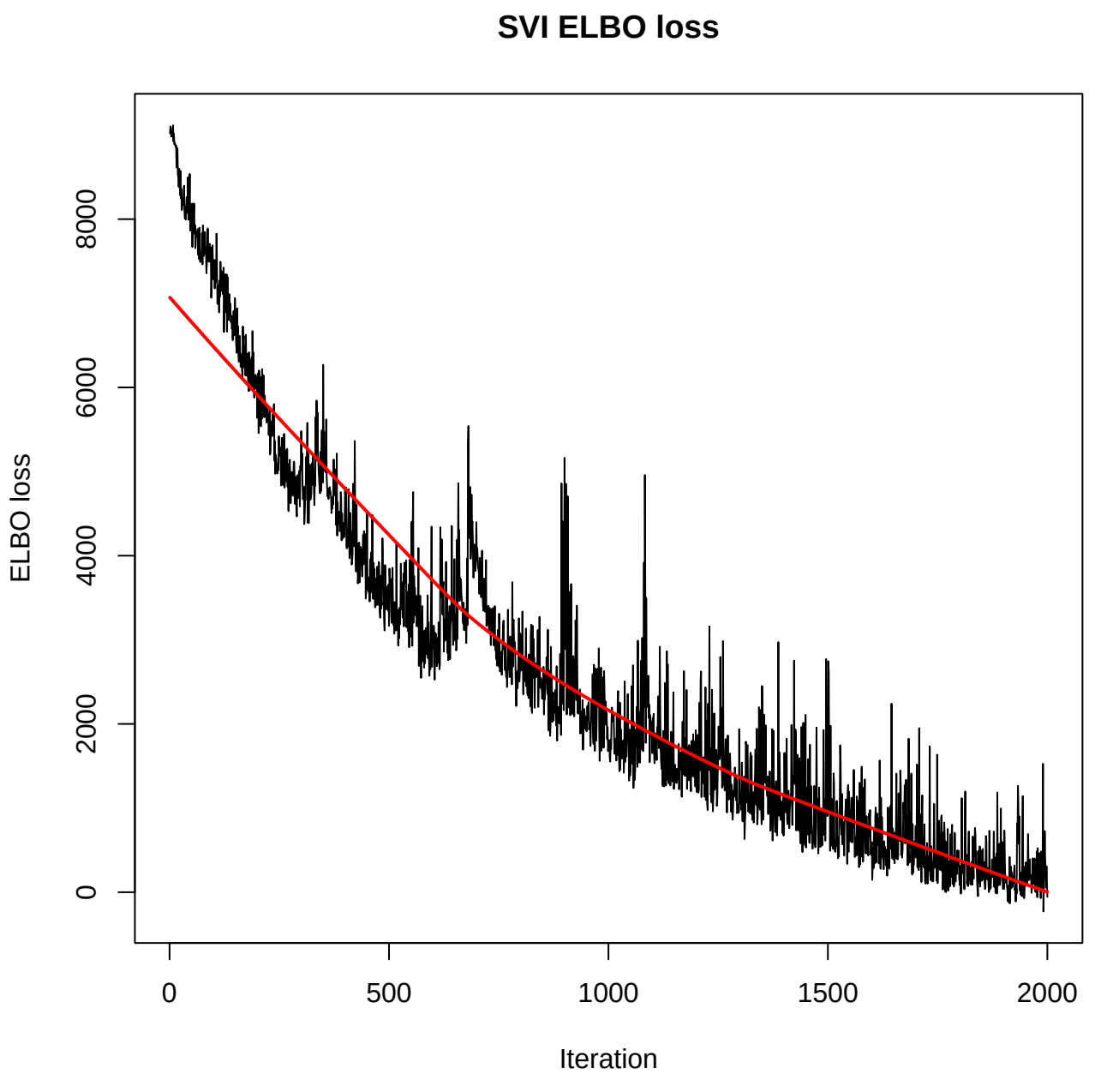

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

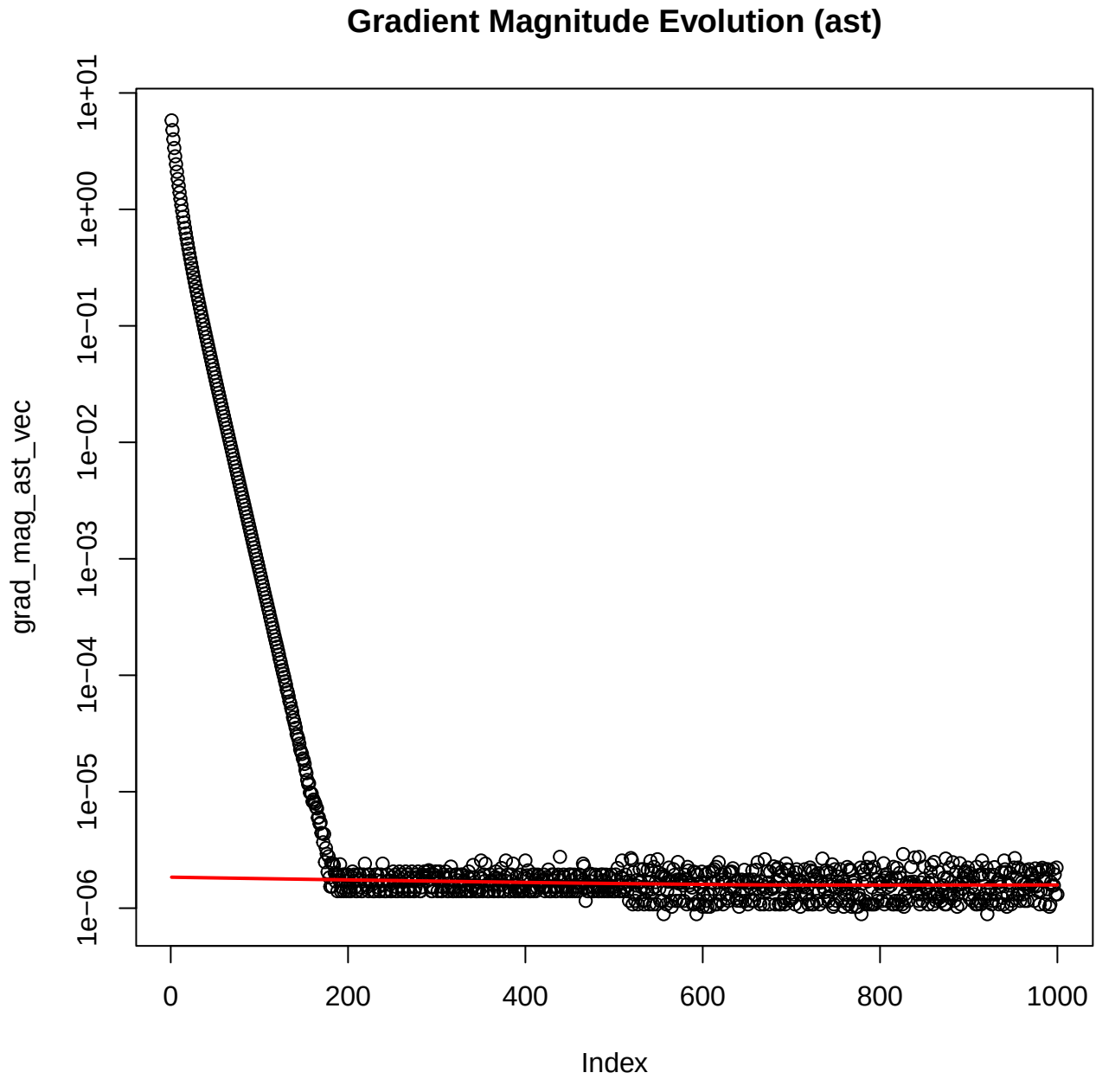

SVI ELBO loss
ELBO loss
10000
5000
2000
1000
500
200
0
500
1000
1500
2000
Iteration

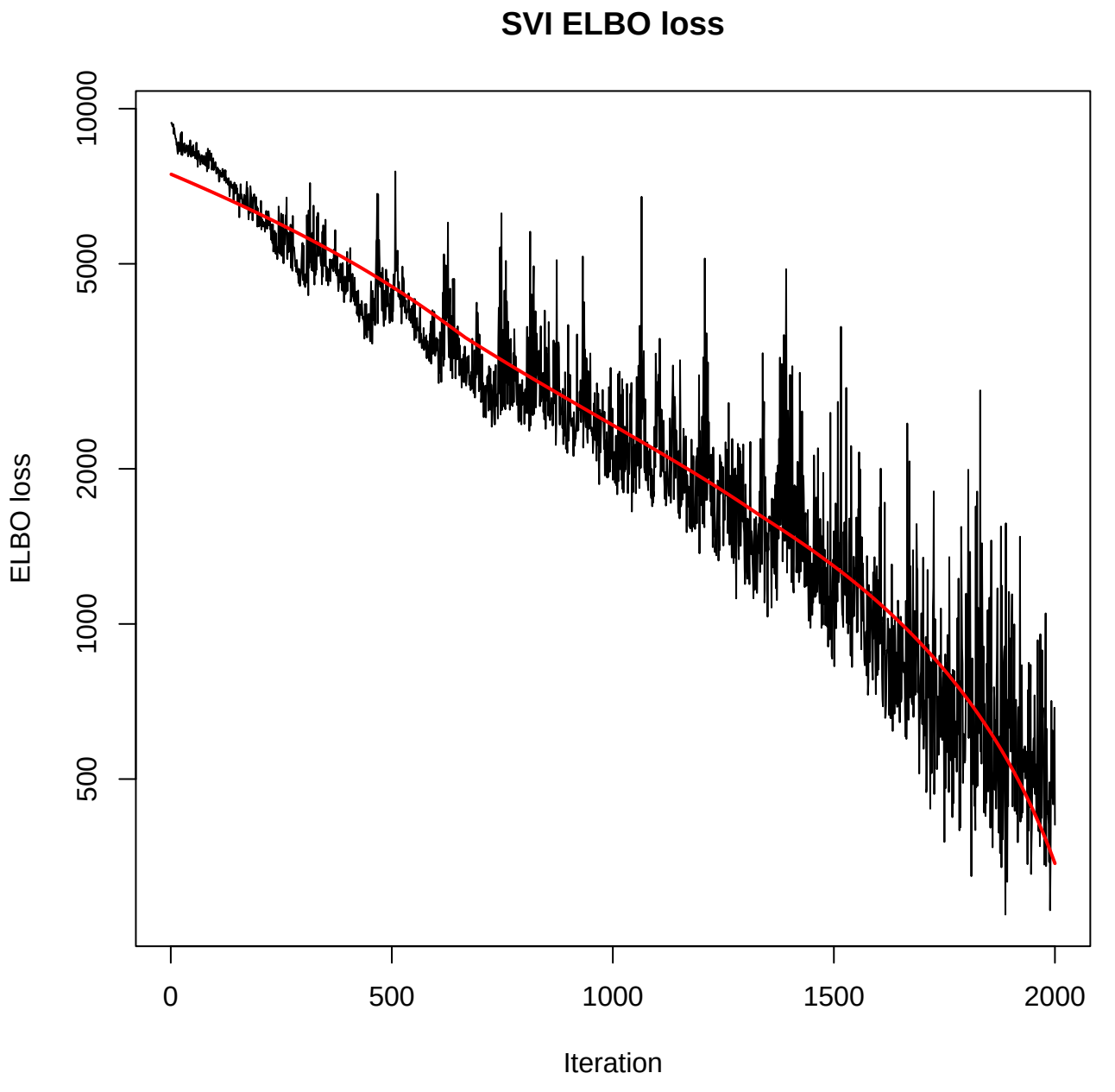

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

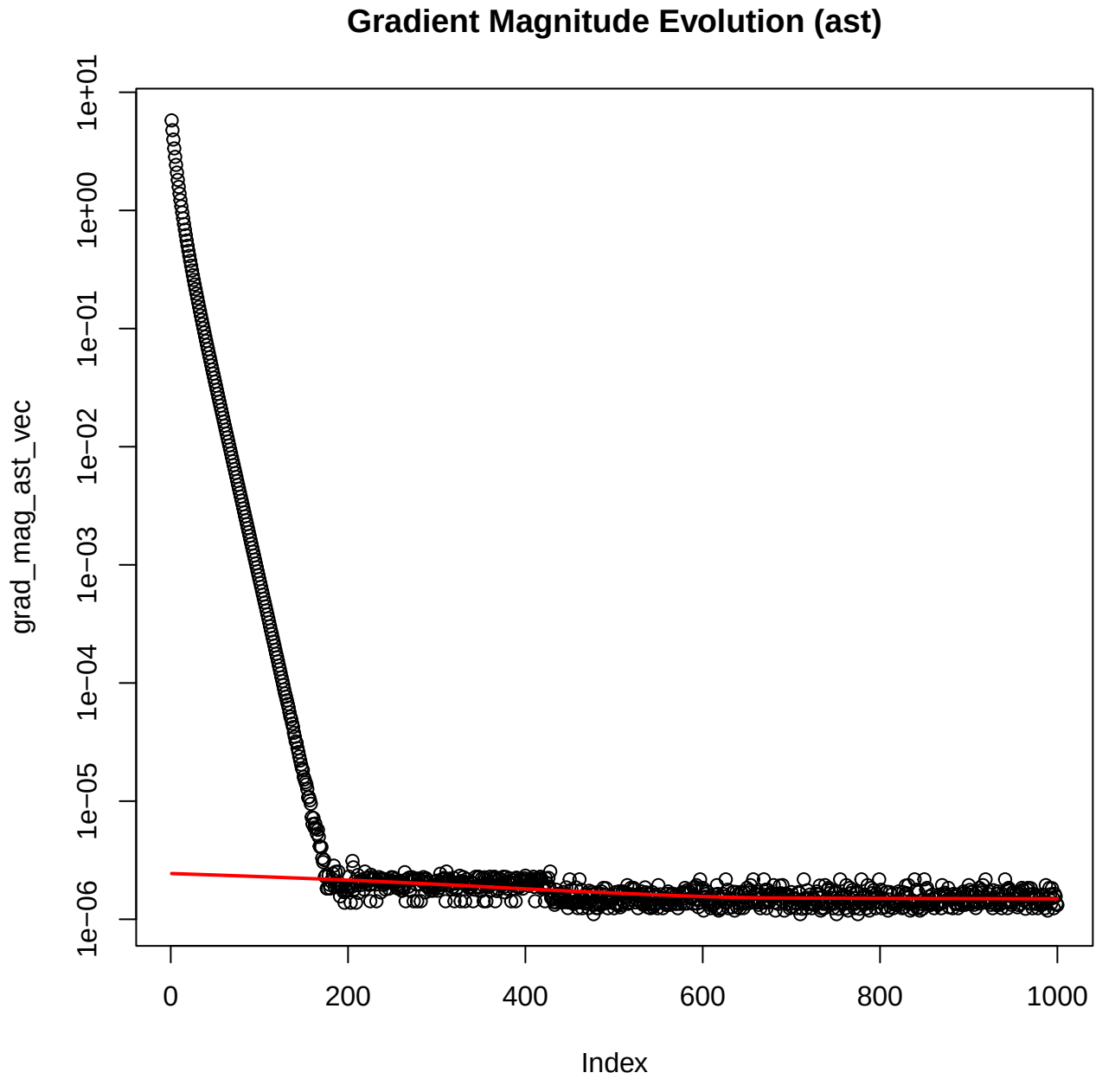

SVI ELBO loss
ELBO loss
20000
10000
5000
2000
1000
0
500
1000
1500
2000
Iteration

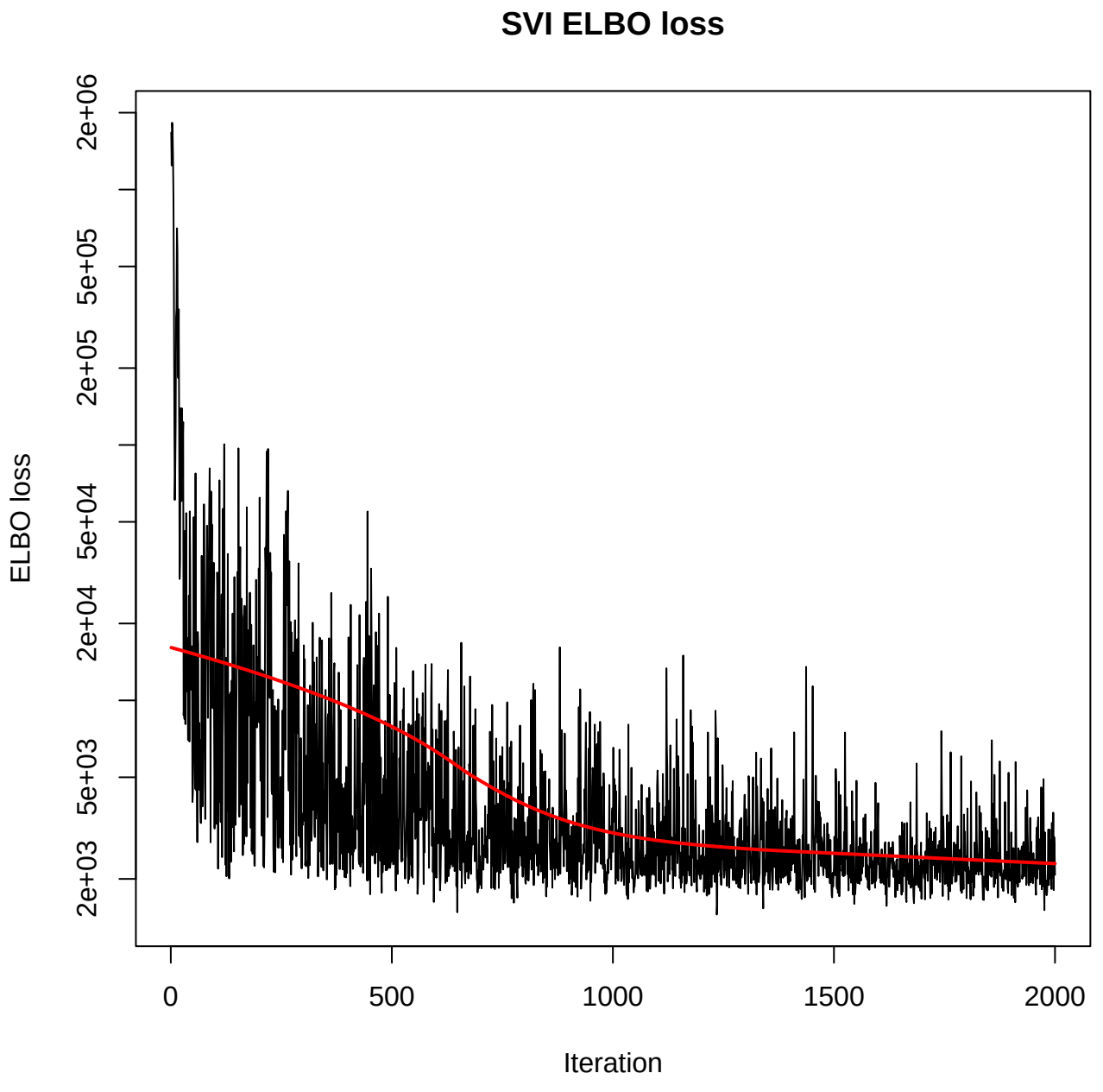

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

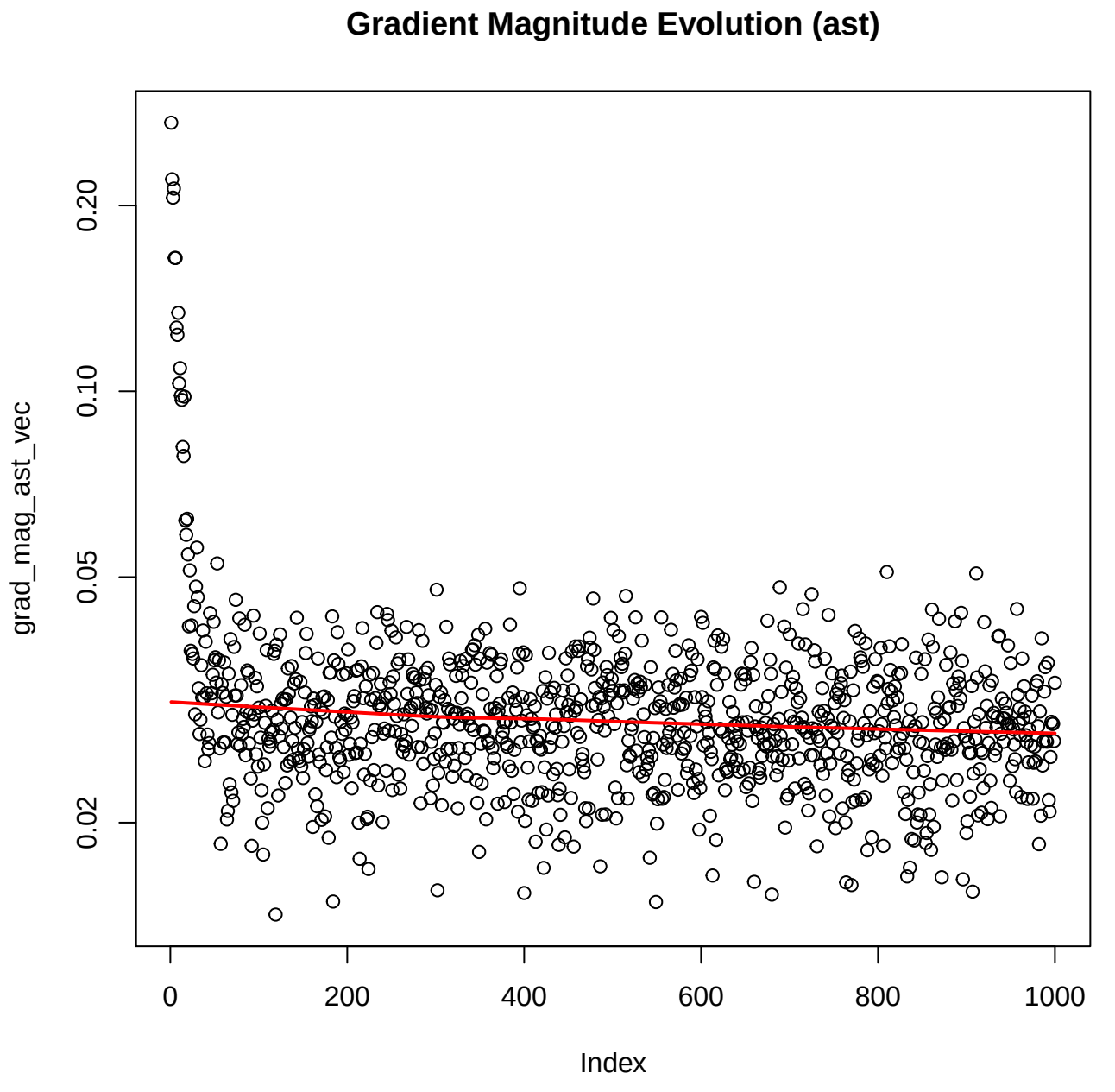

SVI ELBO loss
ELBO loss
0
2000
4000
6000
8000
0
500
1000
1500
2000
Iteration

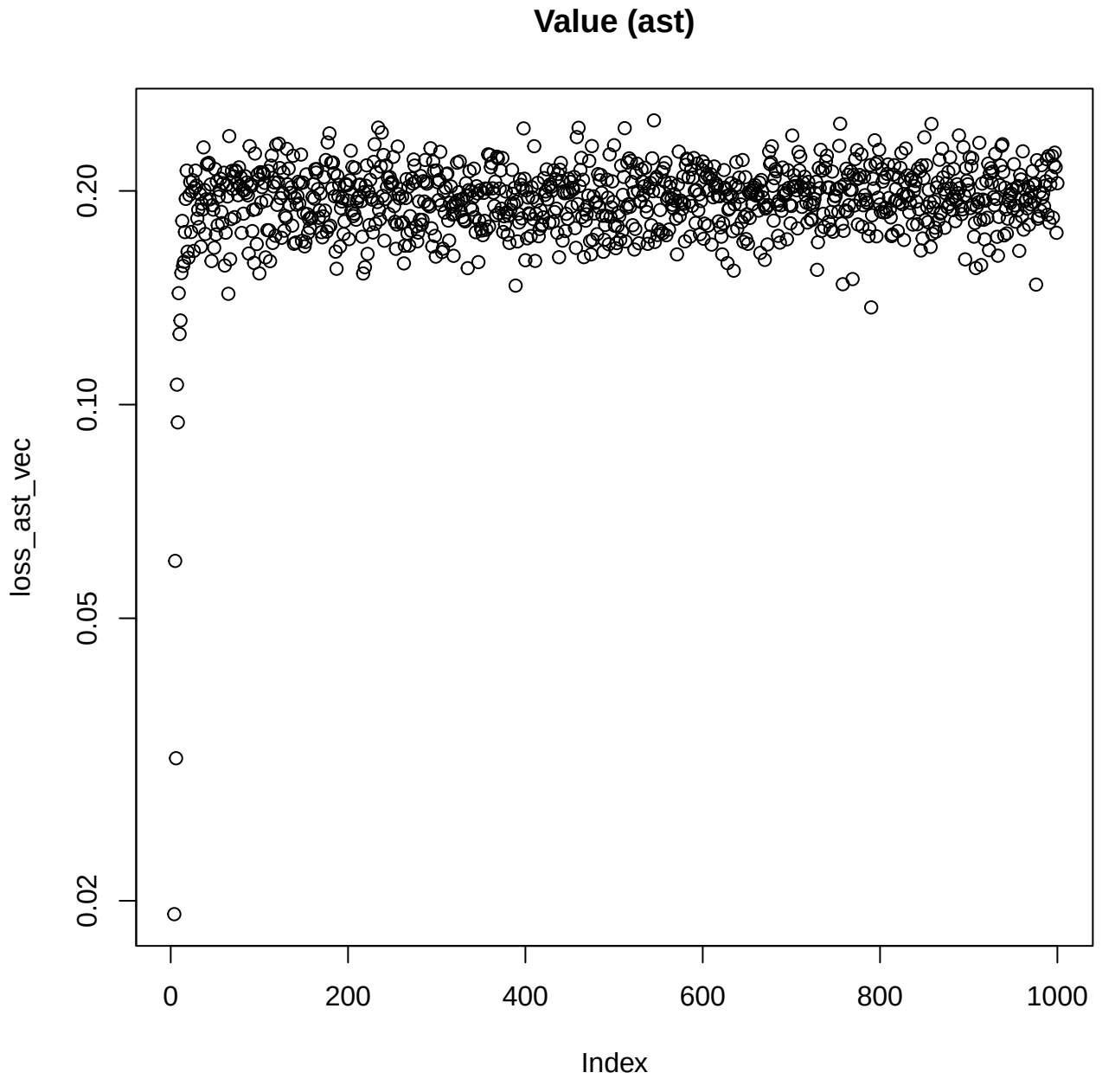

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

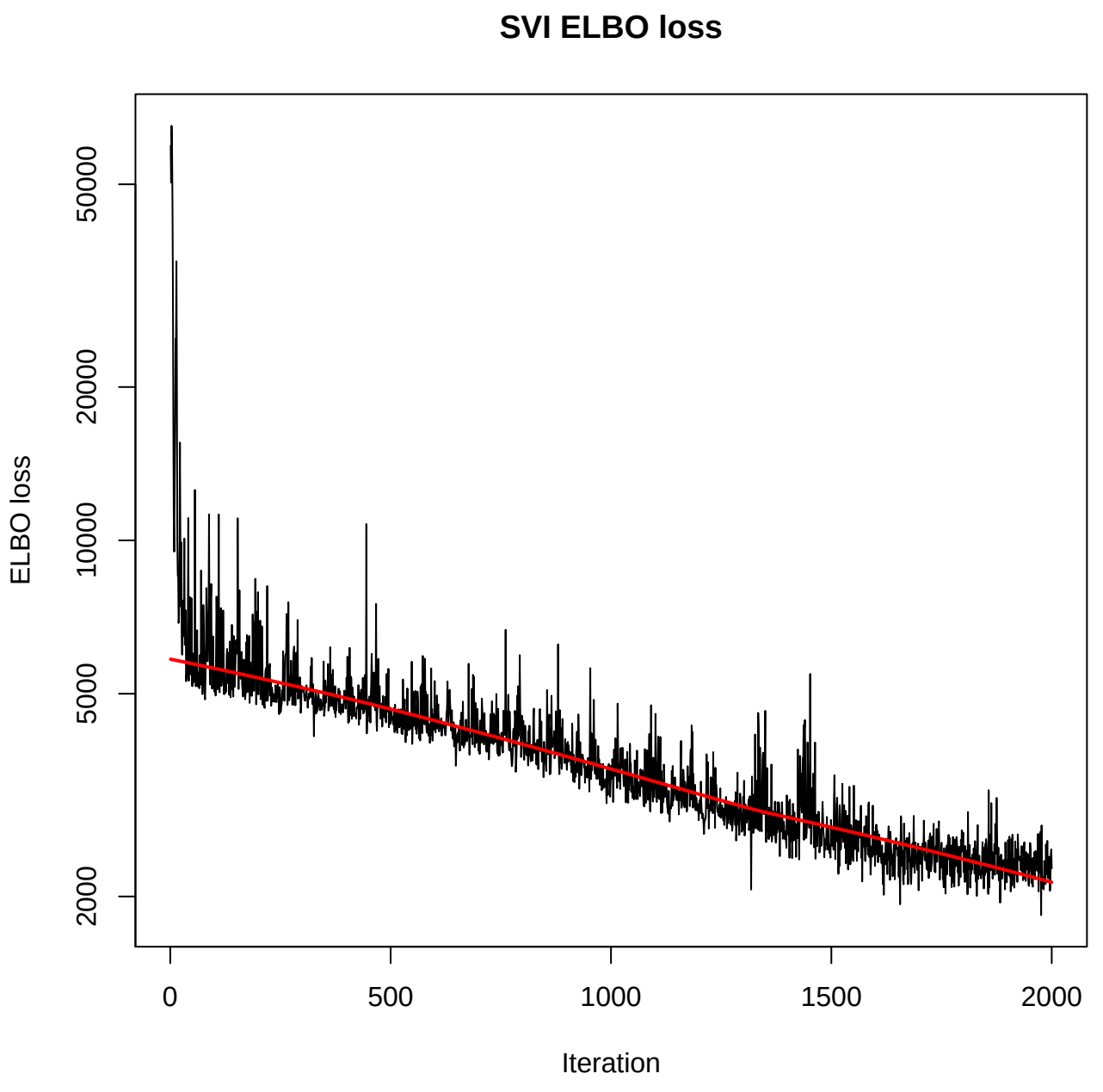

SVI ELBO loss
ELBO loss
10000
5000
2000
1000
500
0
500
1000
1500
2000
Iteration

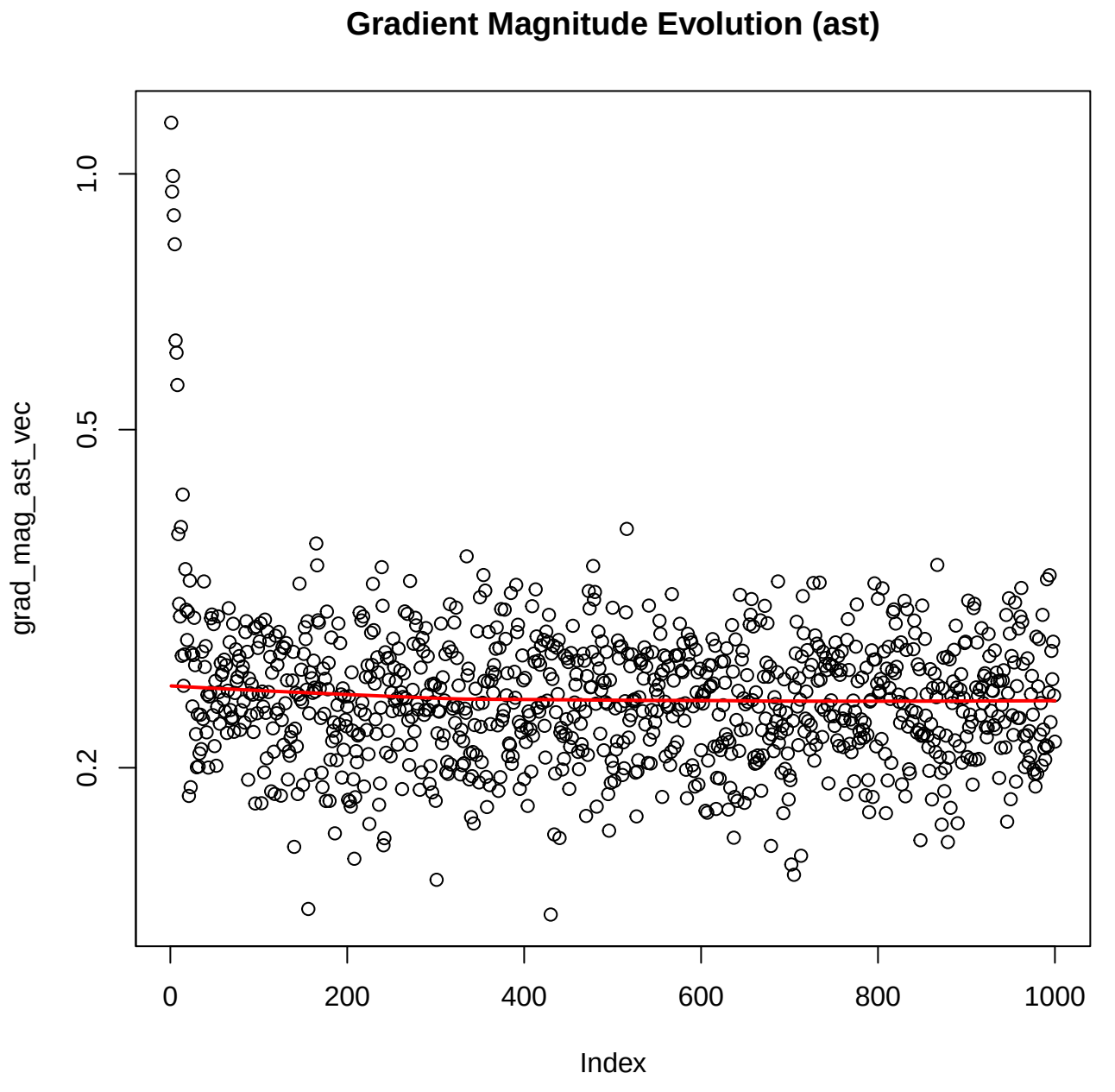

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

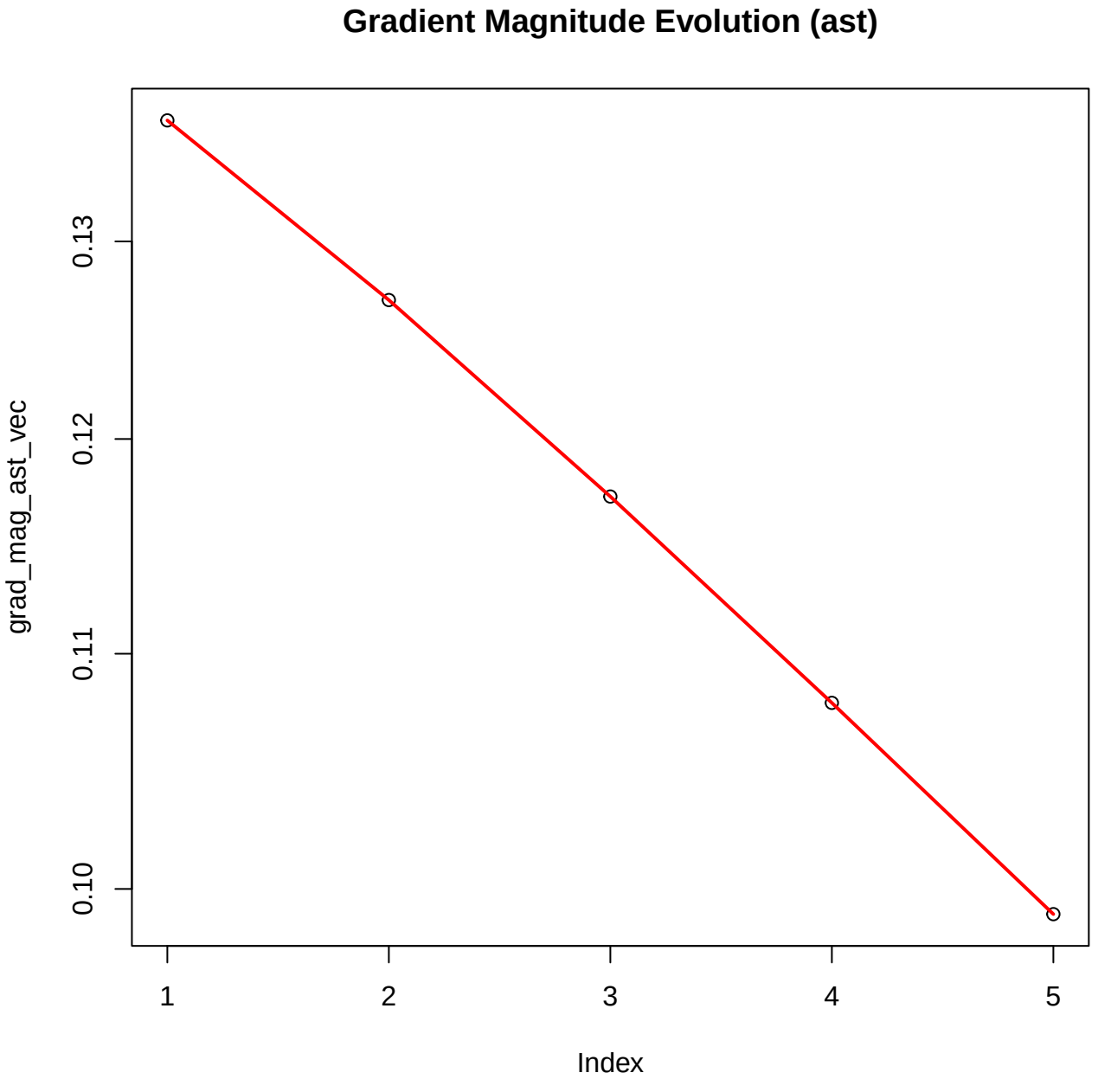

SVI ELBO loss
ELBO loss
2e+06
5e+05
2e+05
5e+04
2e+04
5e+03
2e+03
0
500
1000
1500
2000
Iteration

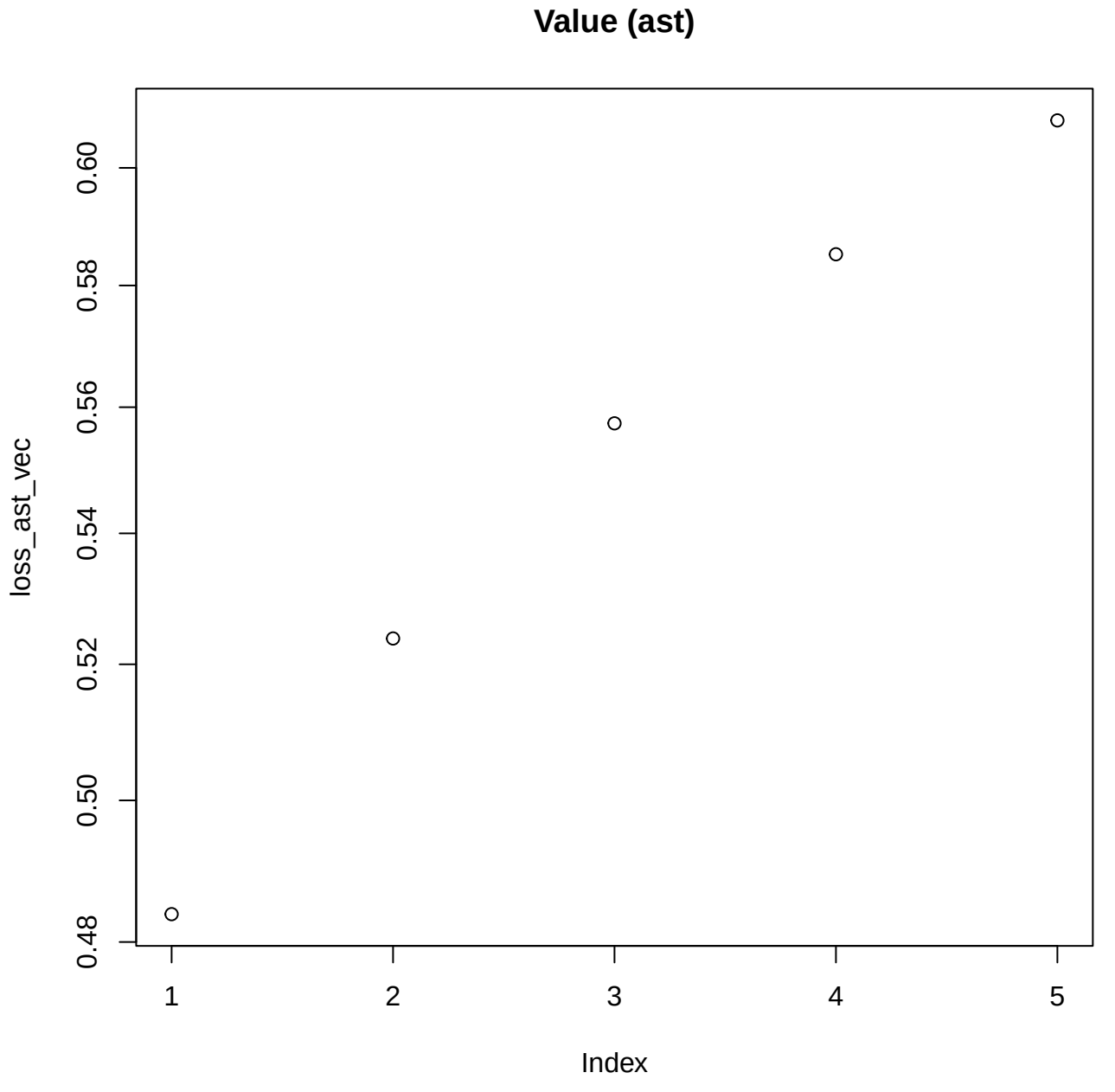

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.02
0.05
0.10
0.20
0
200
400
600
800
1000
Index

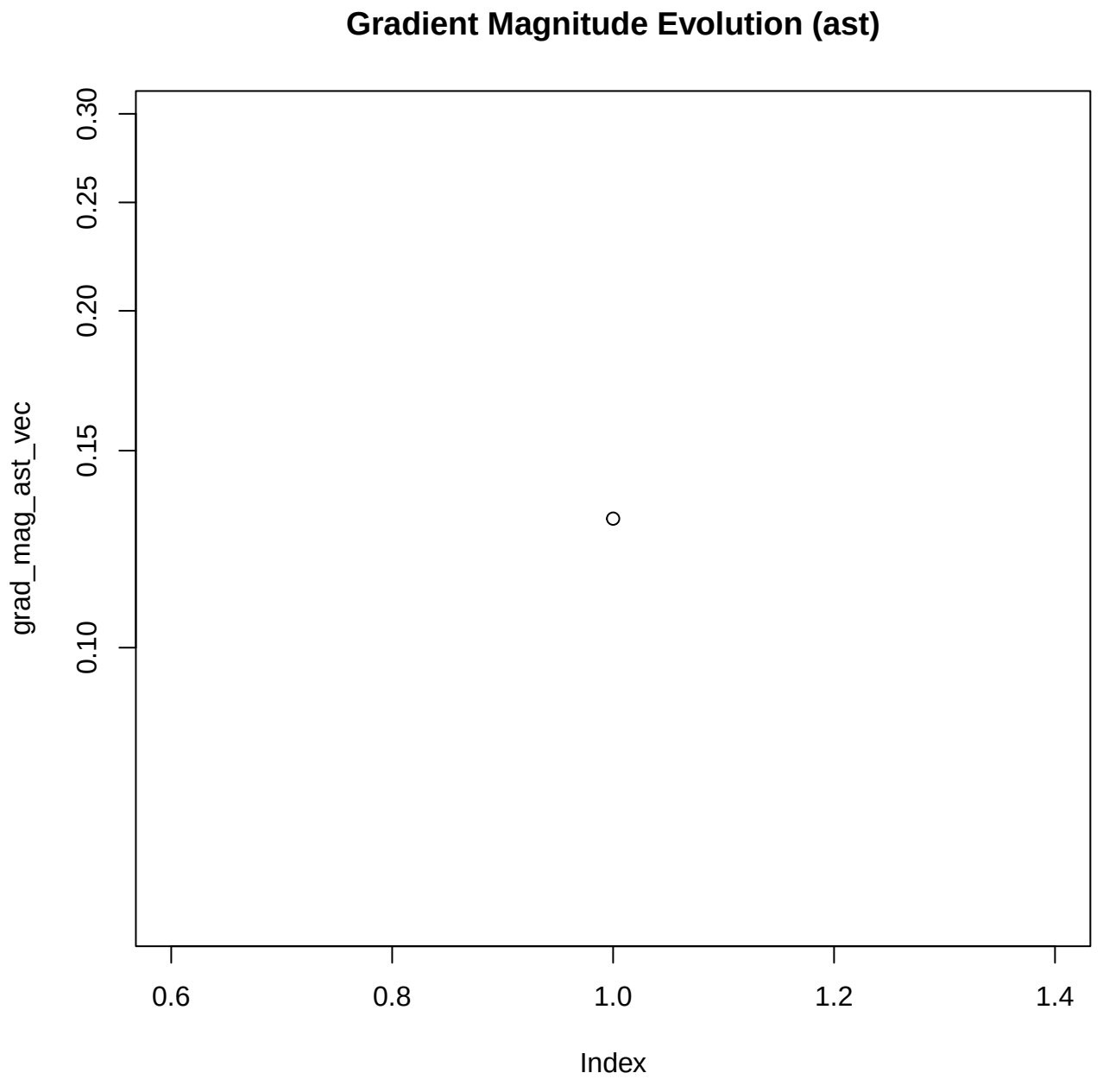

Value (ast)
loss_ast_vec
0.02
0.05
0.10
0.20
0
200
400
600
800
1000
Index

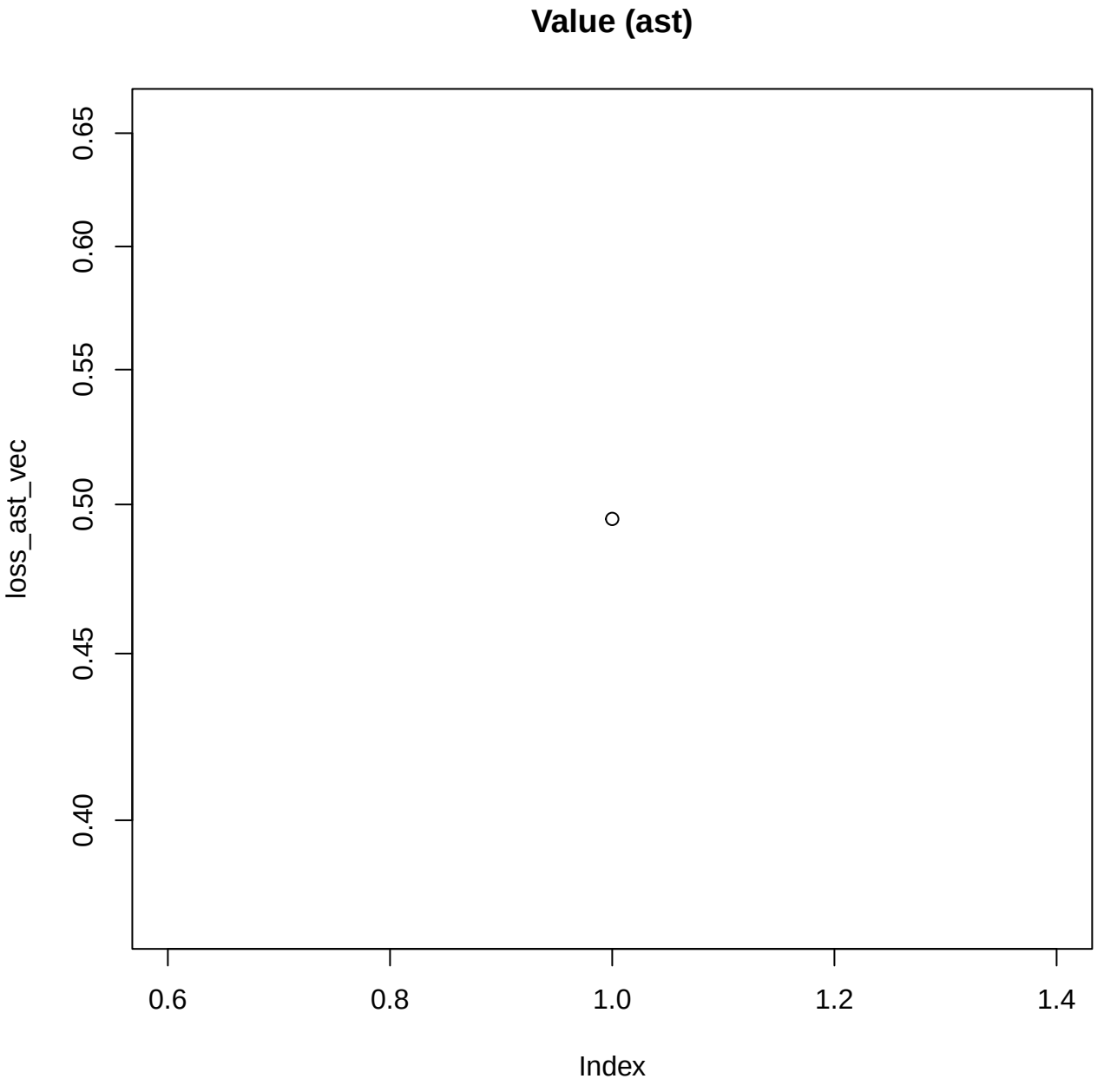

SVI ELBO loss
ELBO loss
50000
20000
10000
5000
2000
0
500
1000
1500
2000
Iteration

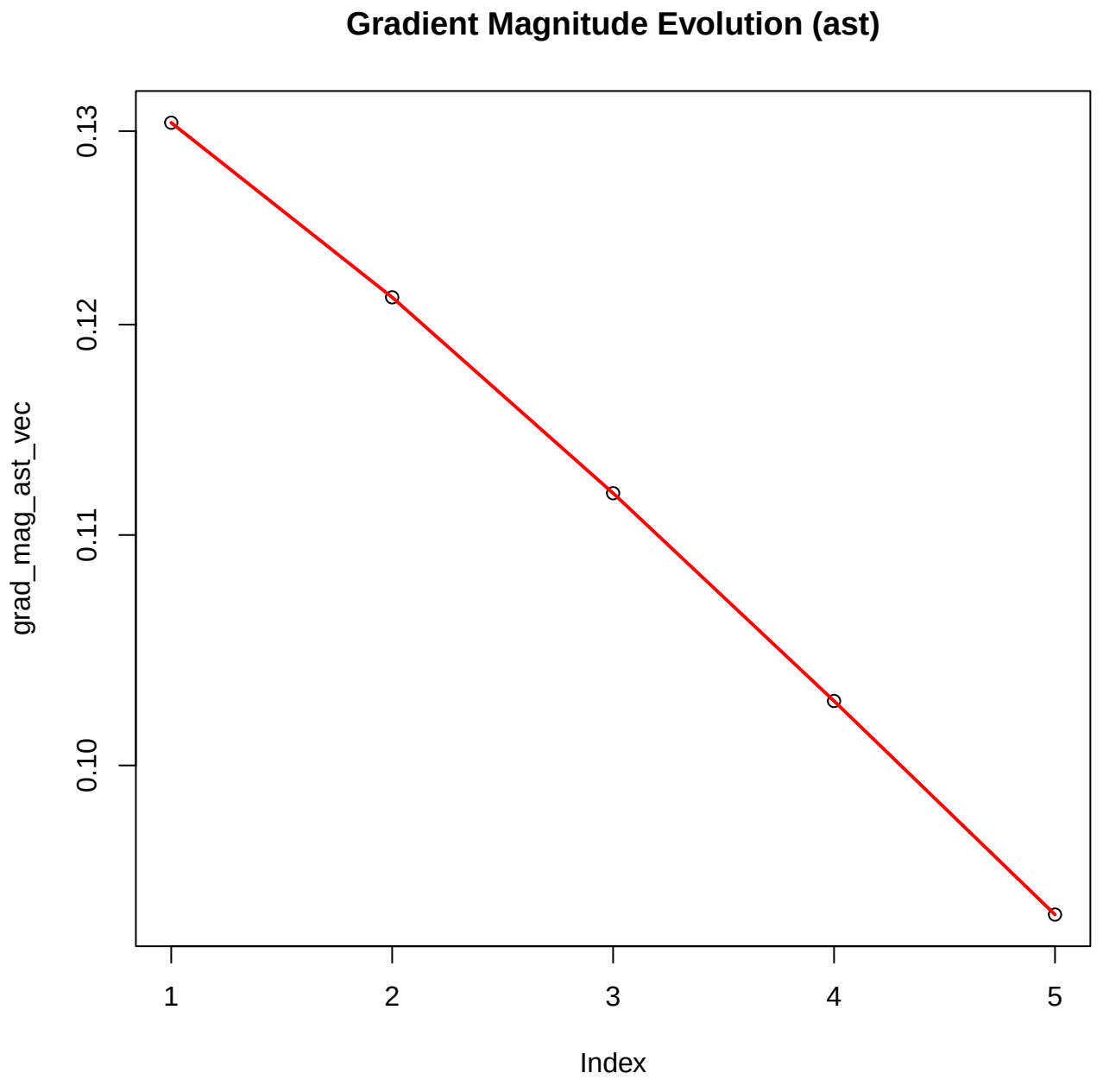

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1.0
0.5
0.2
0
200
400
600
800
1000
Index

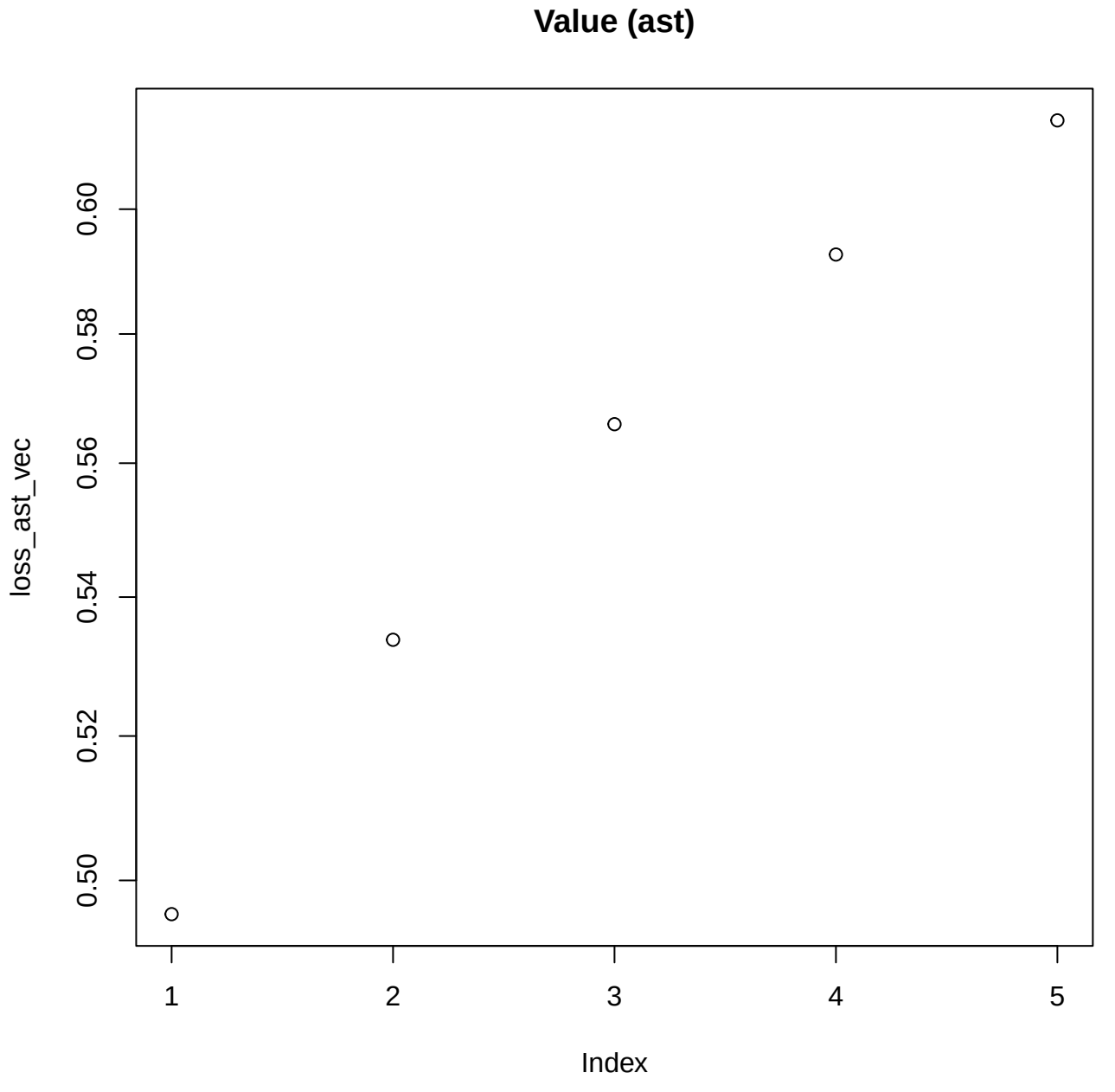

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index

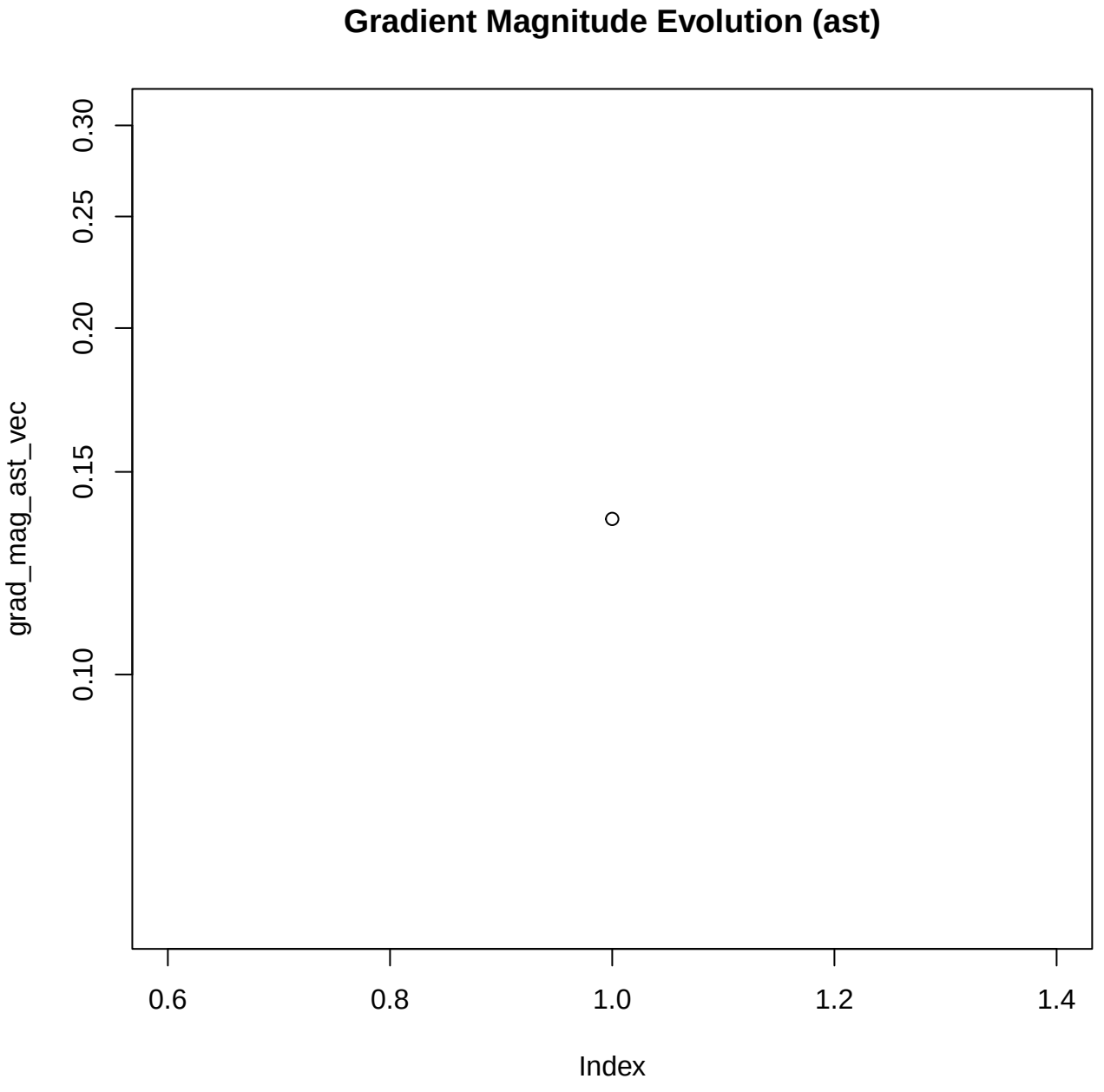

Value (ast)
loss_ast_vec
0.48
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index

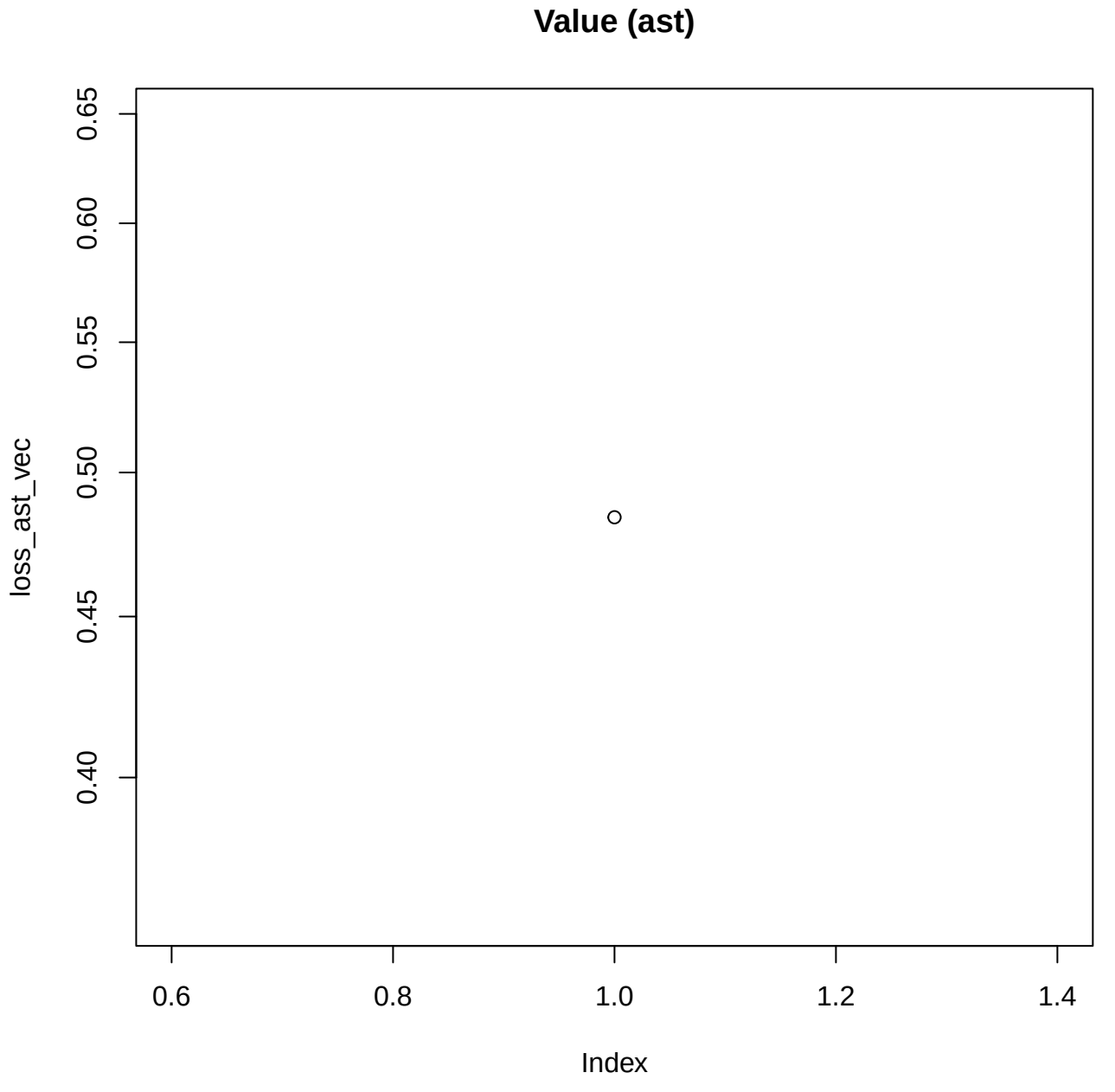

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.15
0.20
0.25
0.30
0.6
0.8
1.0
1.2
1.4
Index

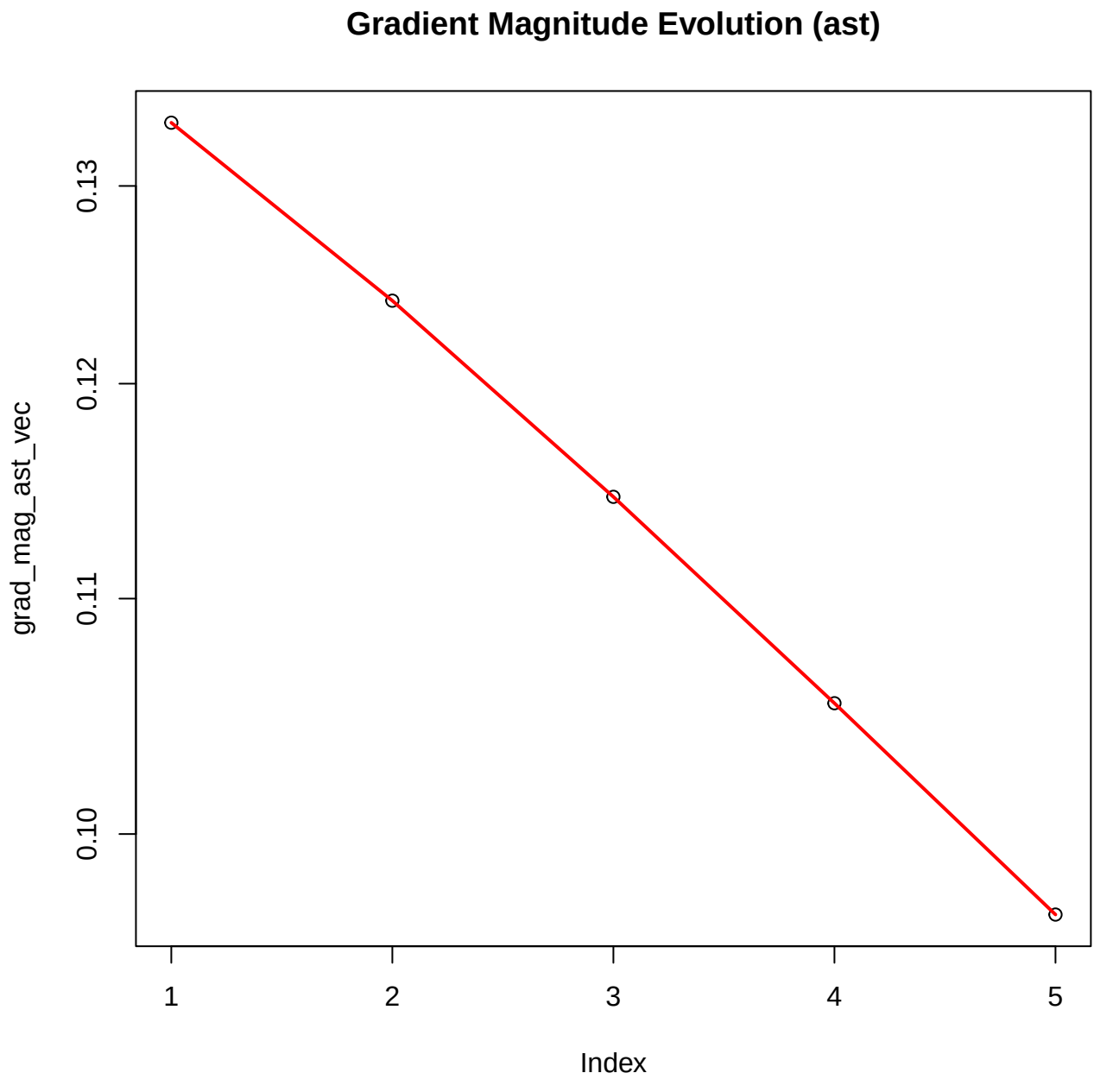

Value (ast)
loss_ast_vec
0.65
0.60
0.55
0.50
0.45
0.40
0.6
0.8
1.0
1.2
1.4
Index

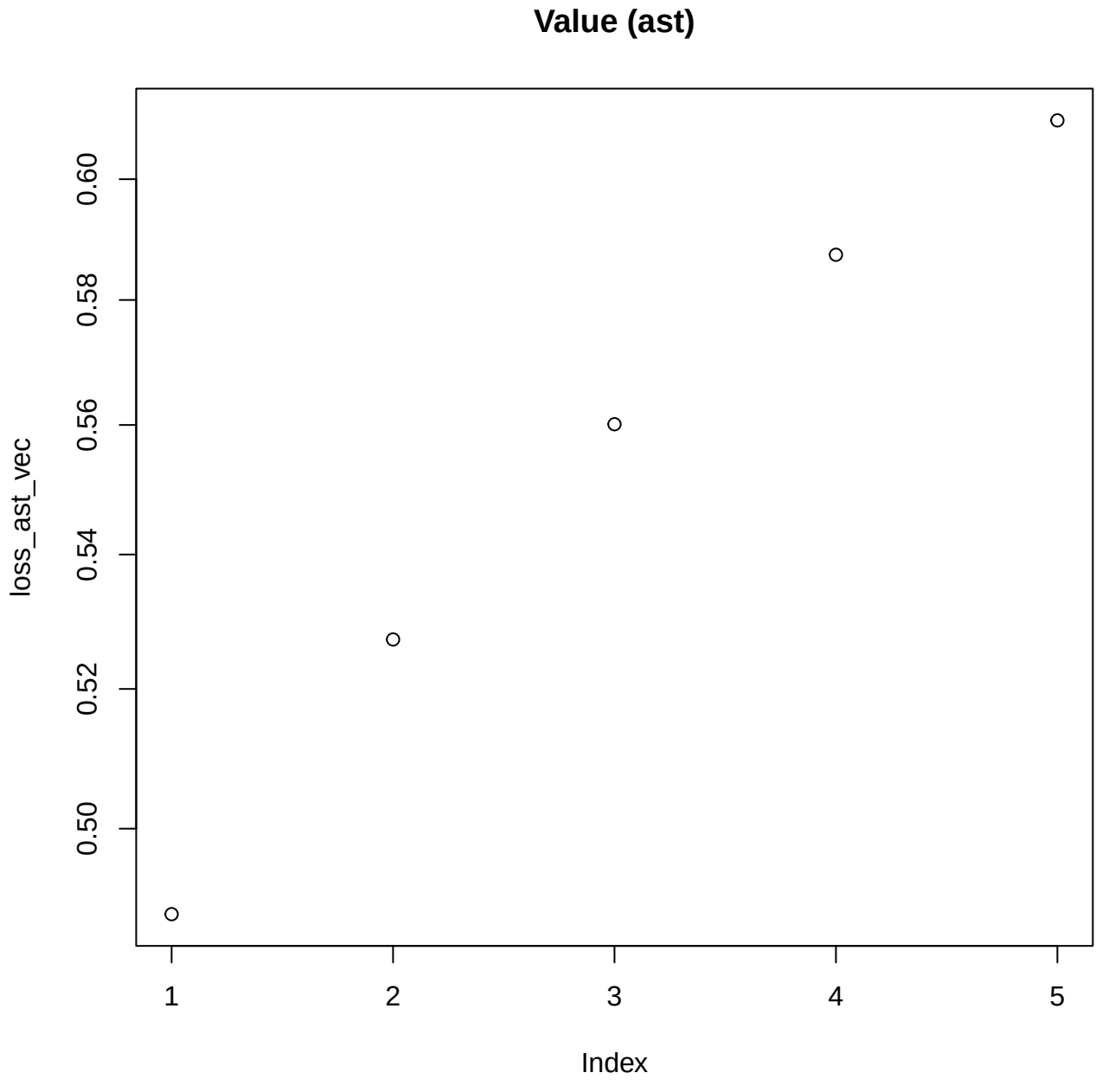

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index

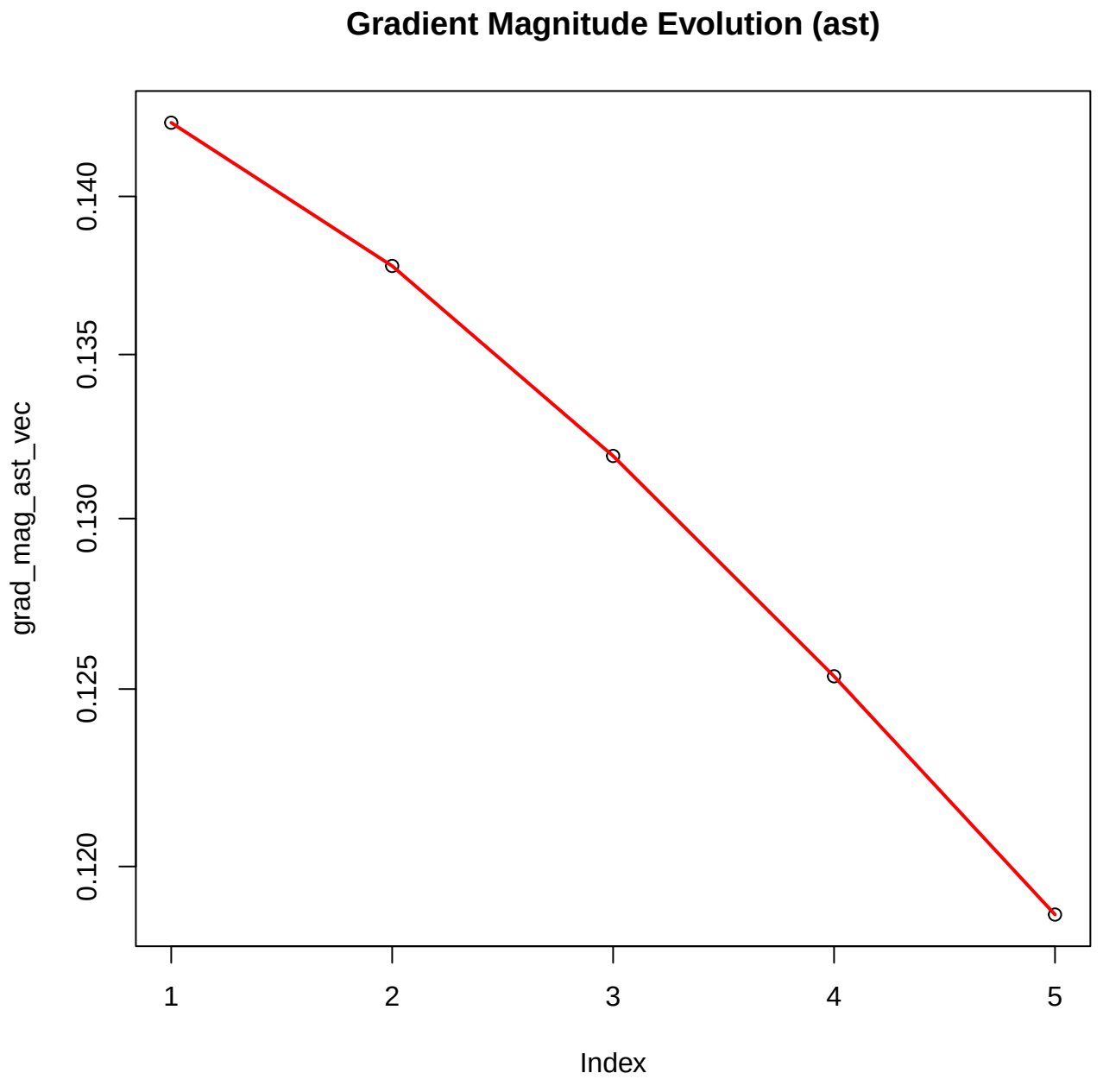

Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index

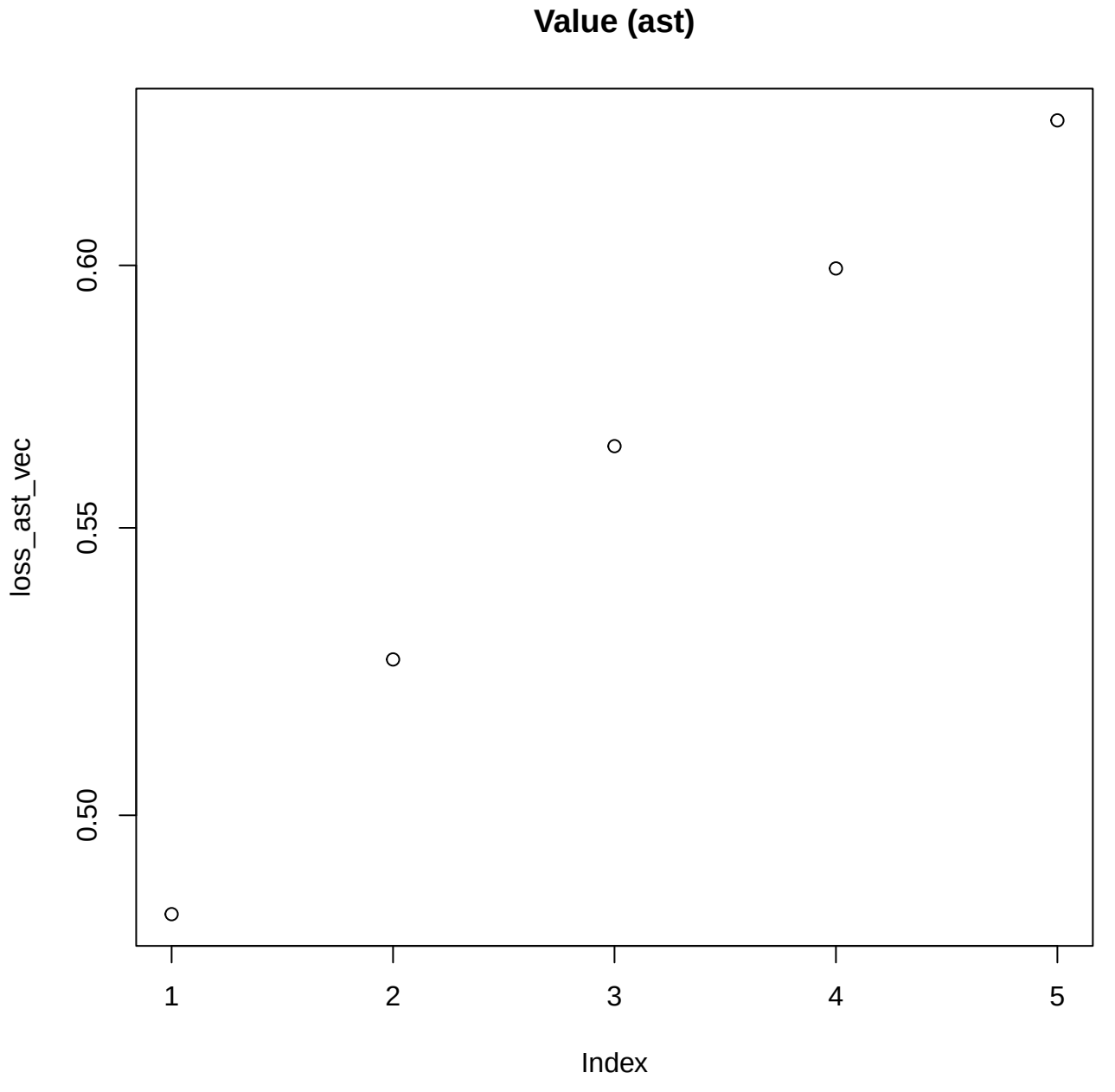

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.15
0.20
0.25
0.30
0.6
0.8
1.0
1.2
1.4
Index

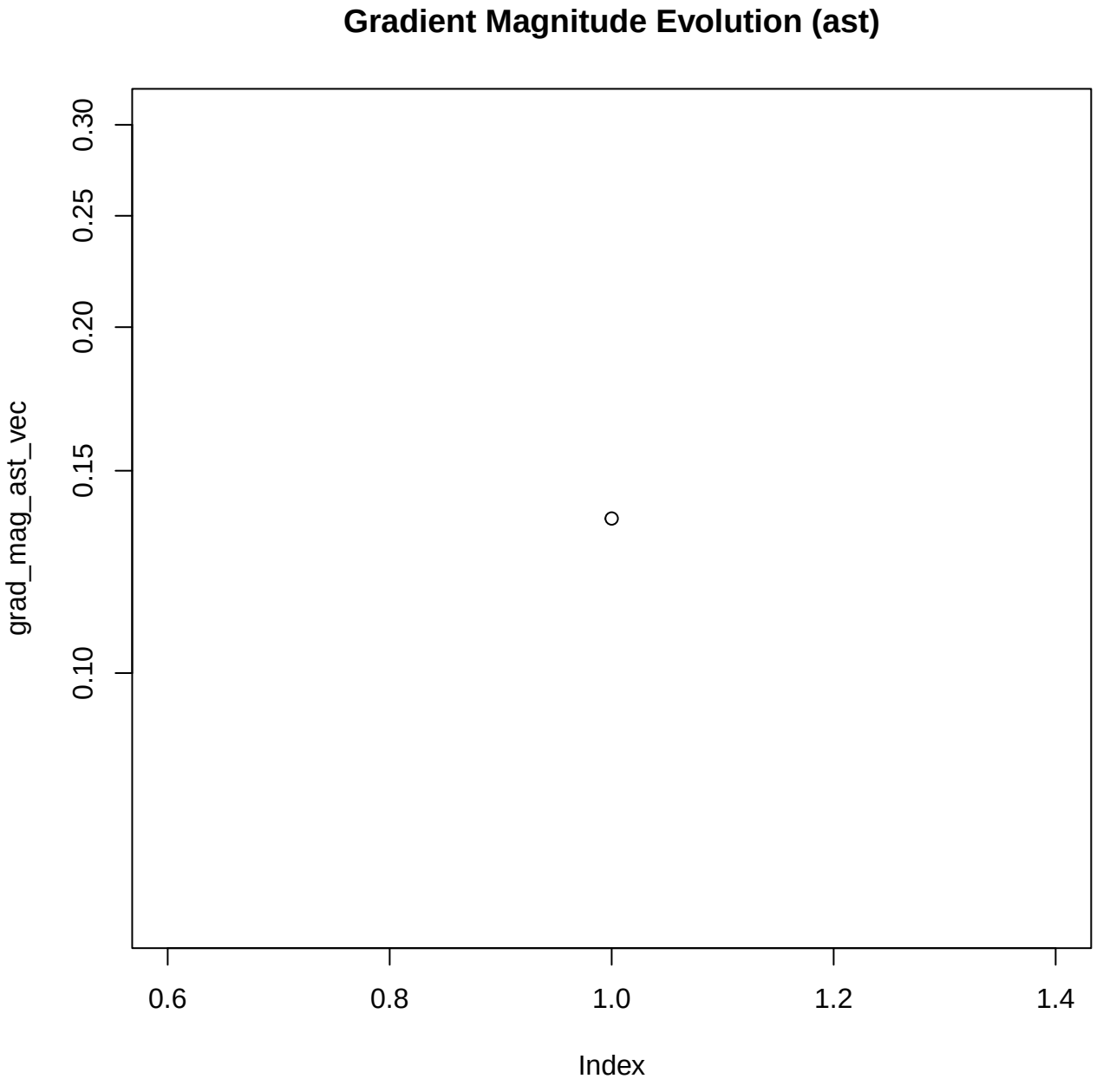

Value (ast)
loss_ast_vec
0.65
0.60
0.55
0.50
0.45
0.40
0.6
0.8
1.0
1.2
1.4
Index

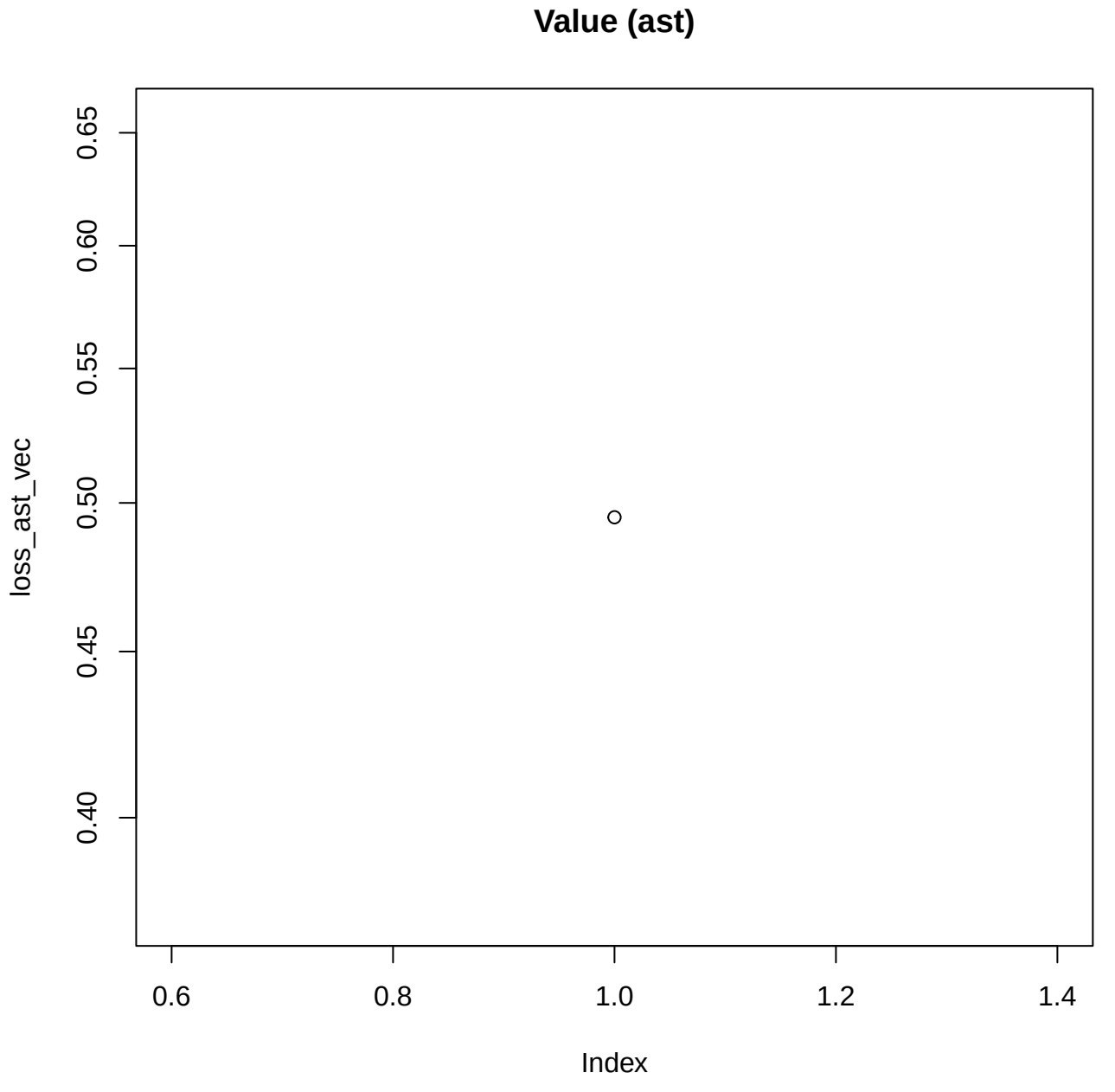

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index

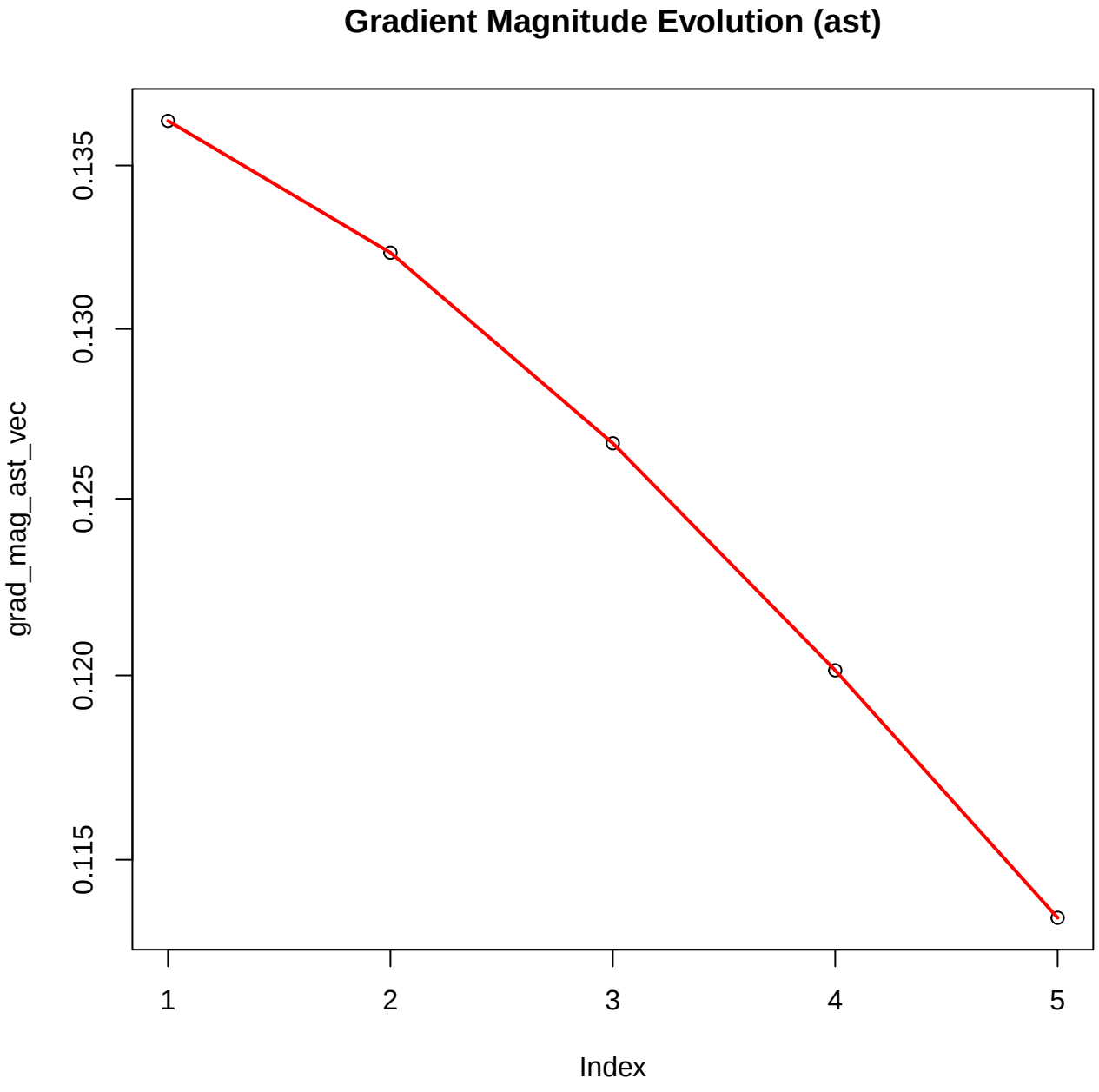

Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index

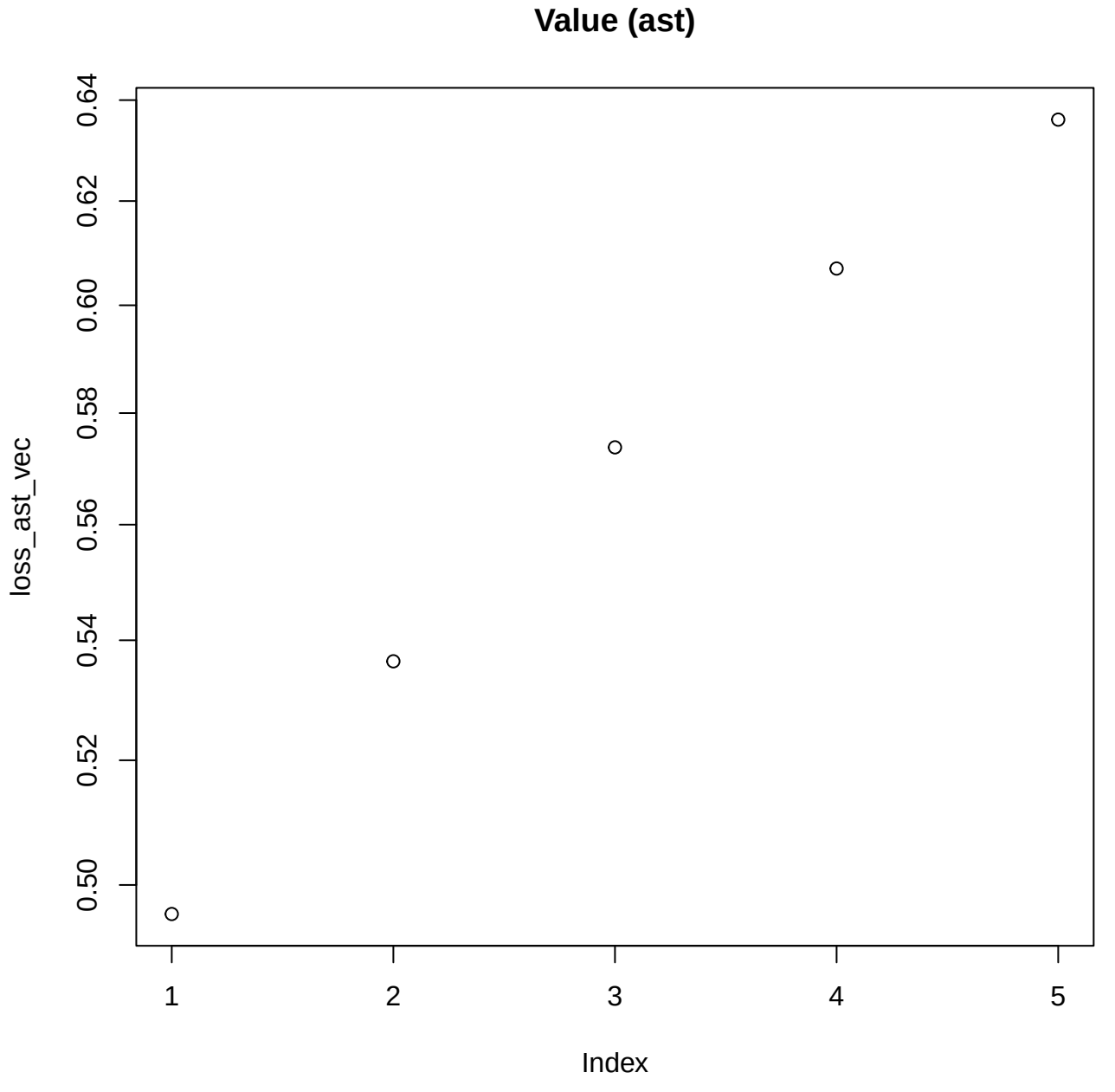

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.120
0.125
0.130
0.135
0.140
1
2
3
4
5
Index

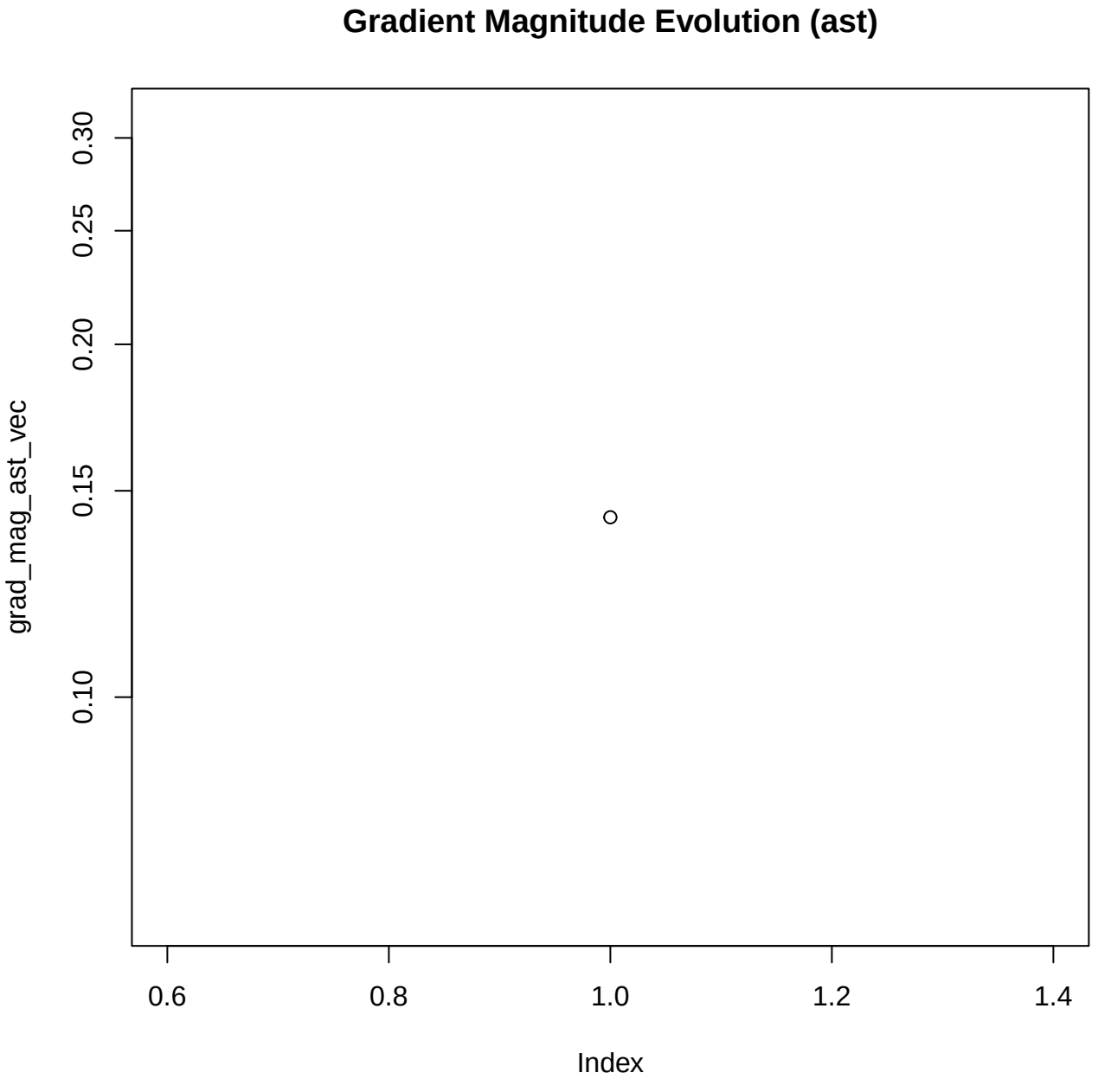

Value (ast)
loss_ast_vec
0.50
0.55
0.60
1
2
3
4
5
Index

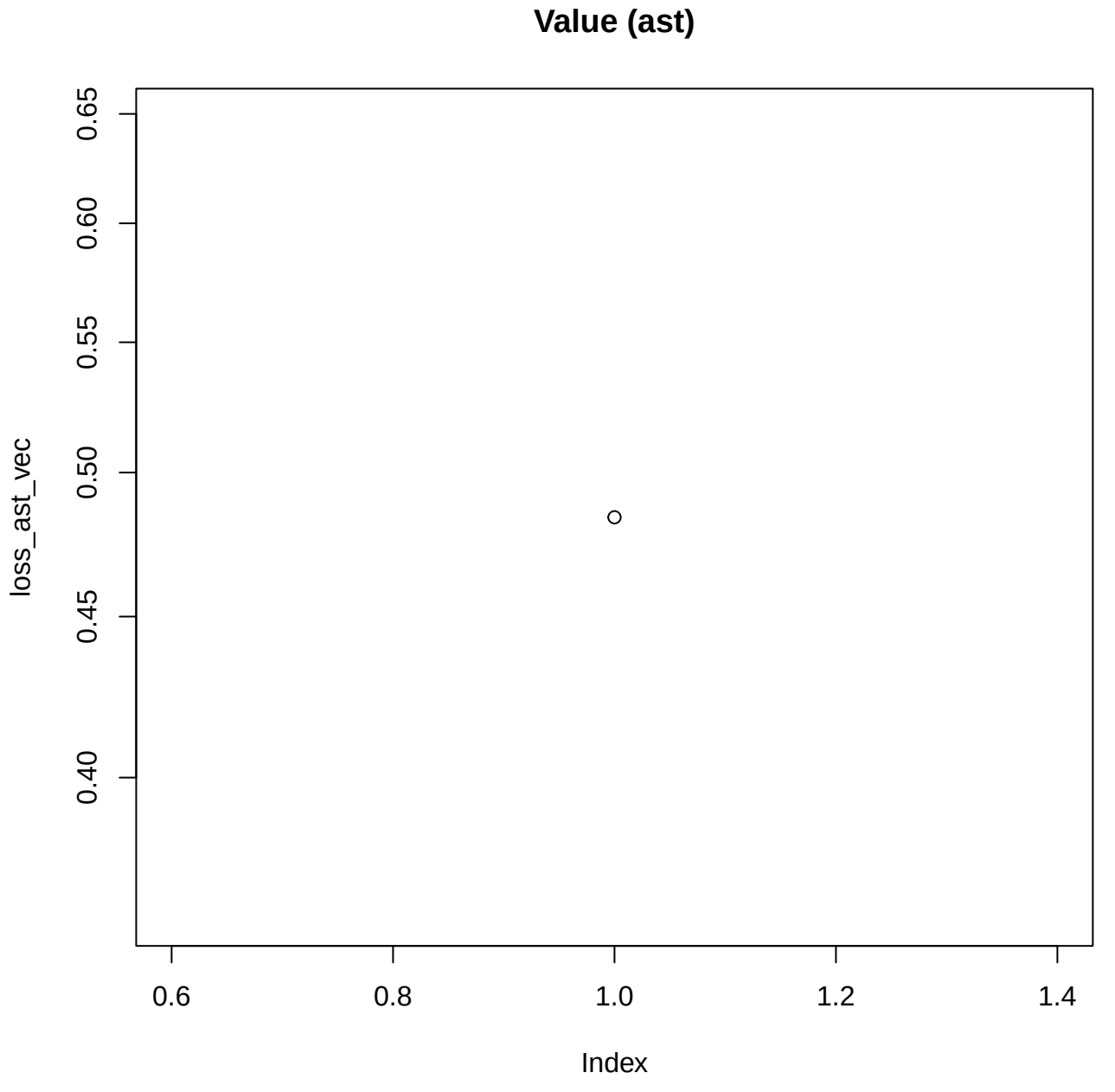

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.15
0.20
0.25
0.30
0.6
0.8
1.0
1.2
1.4
Index

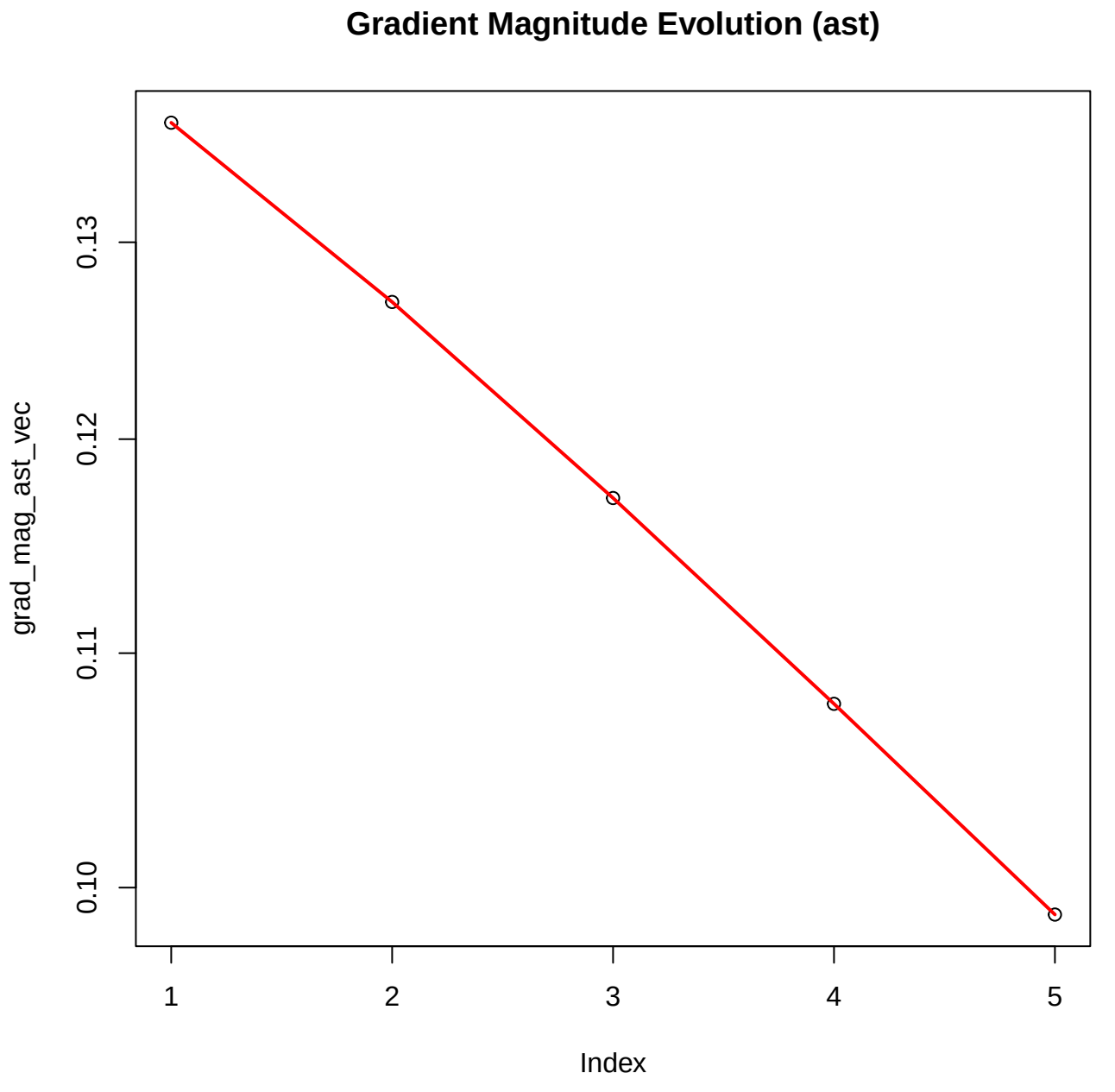

Value (ast)
loss_ast_vec
0.65
0.60
0.55
0.50
0.45
0.40
0.6
0.8
1.0
1.2
1.4
Index

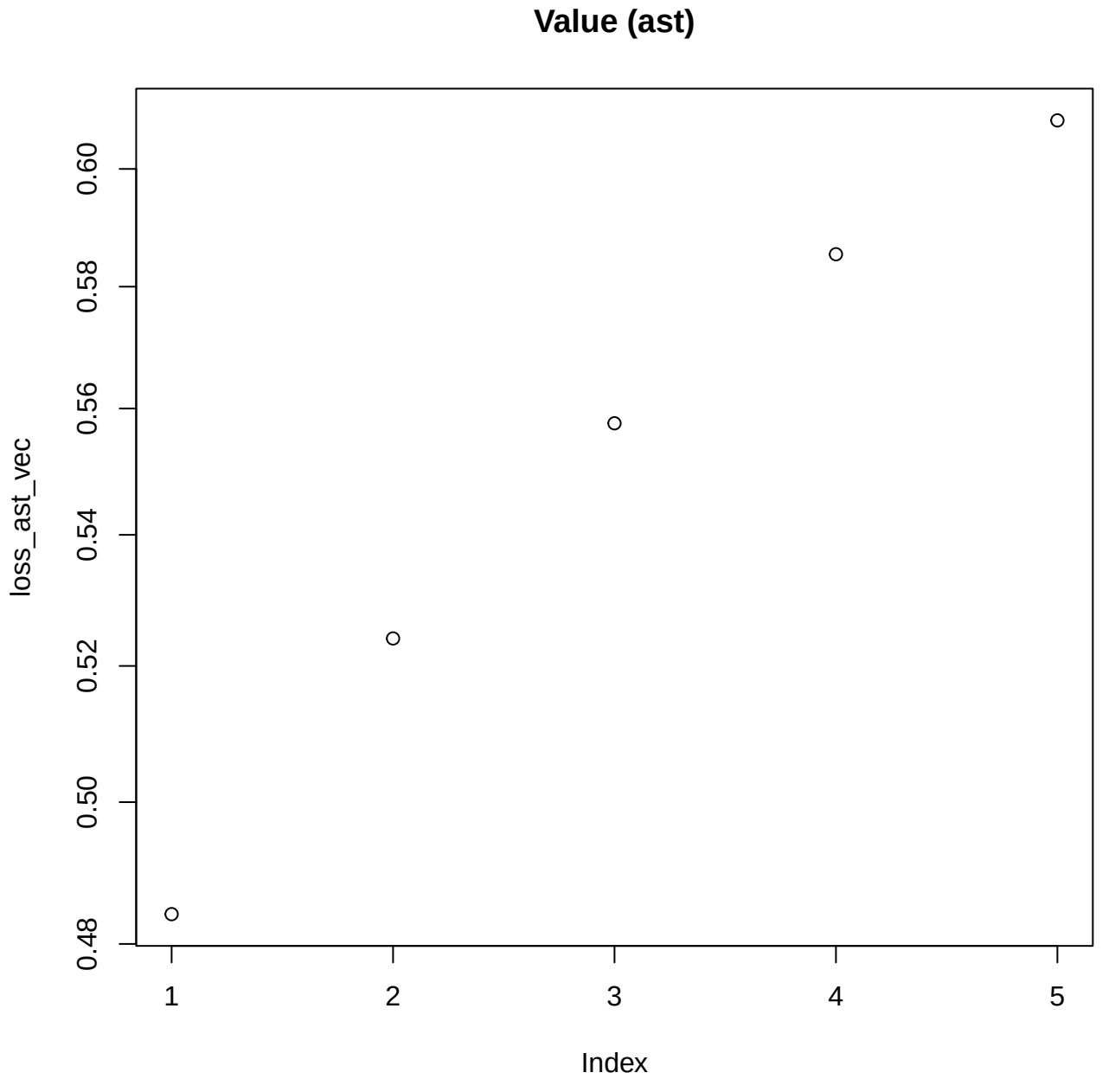

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.115
0.120
0.125
0.130
0.135
1
2
3
4
5
Index

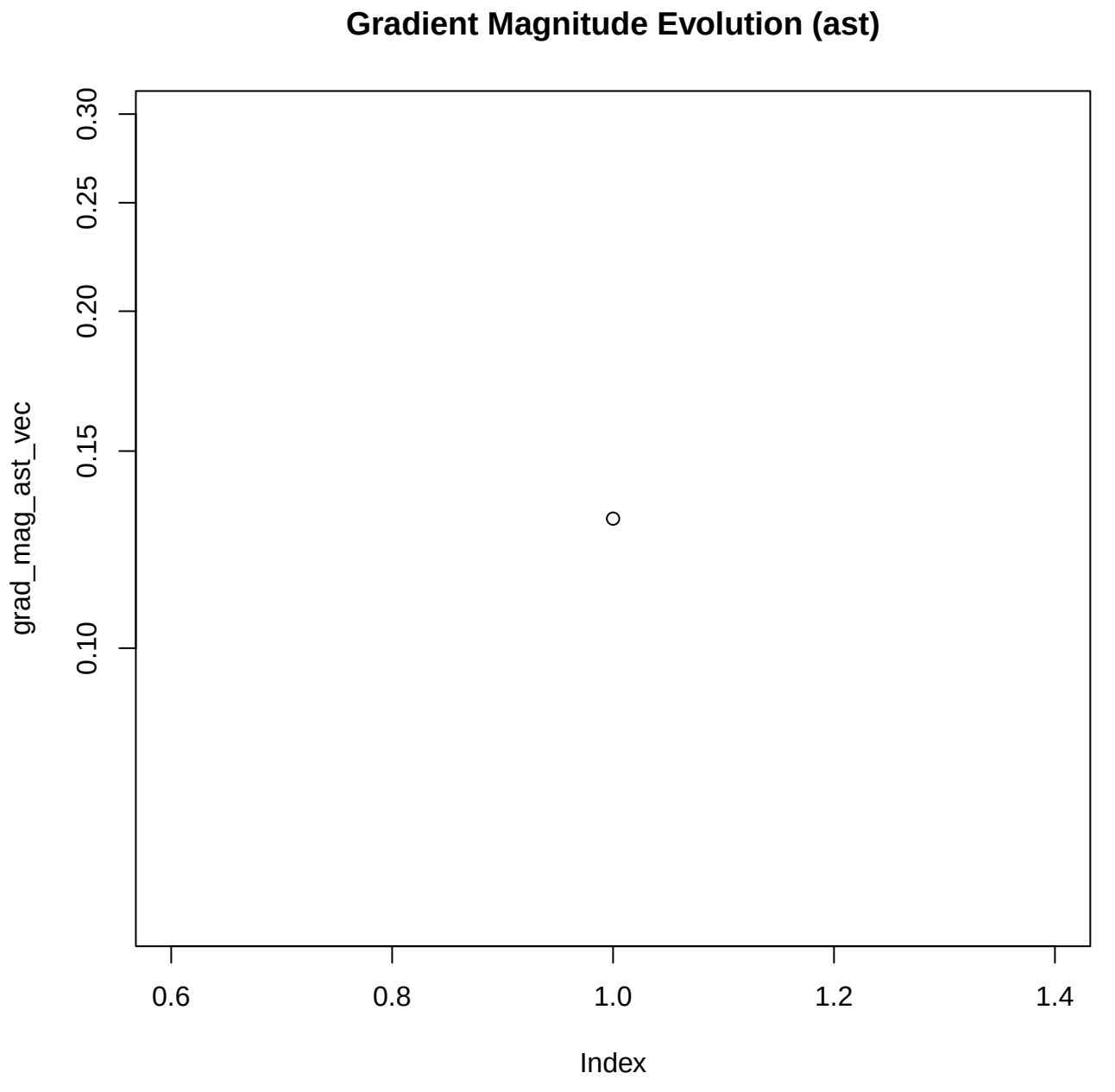

Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
0.62
0.64
1
2
3
4
5
Index

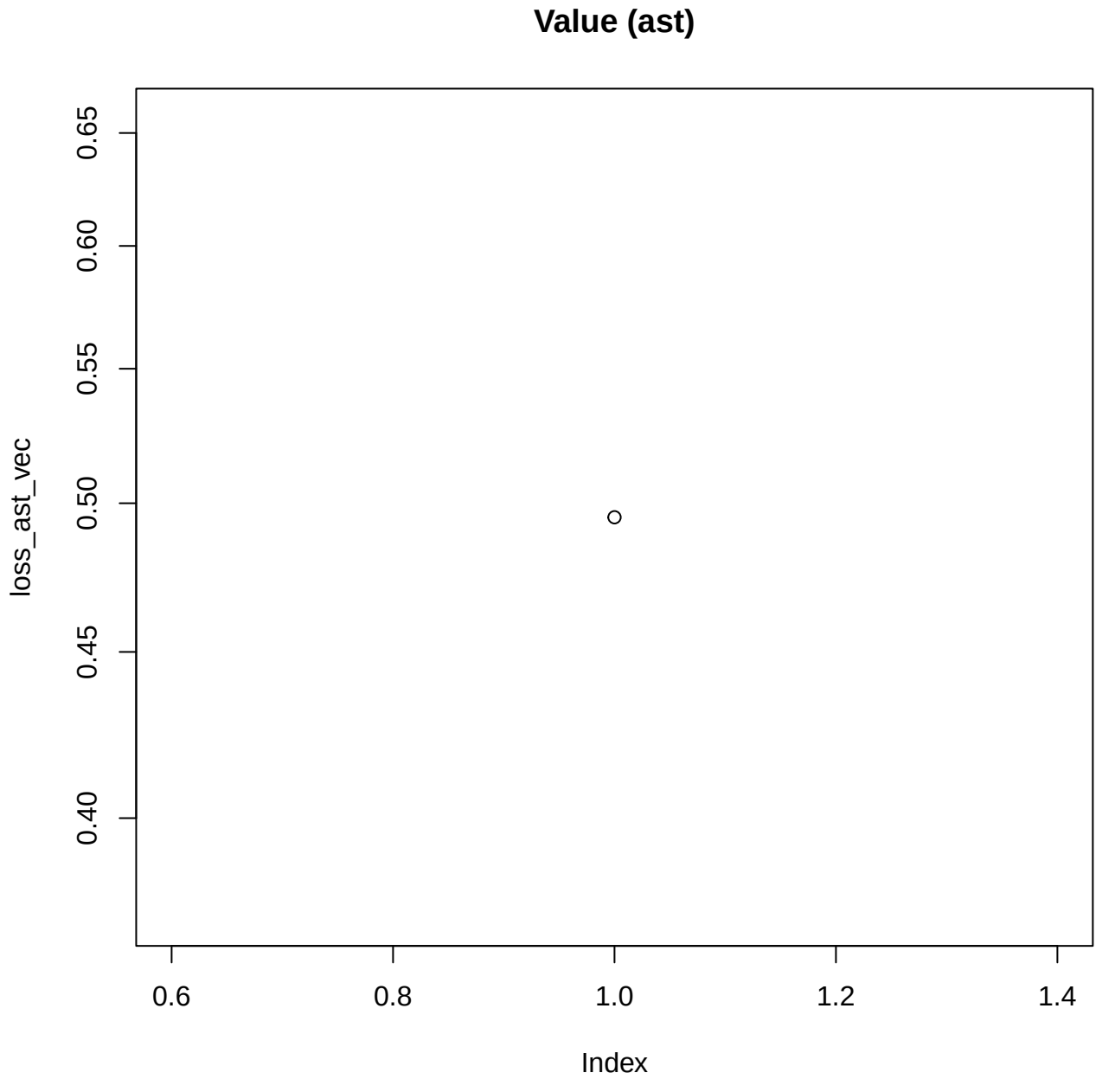

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.15
0.20
0.25
0.30
0.6
0.8
1.0
1.2
1.4
Index

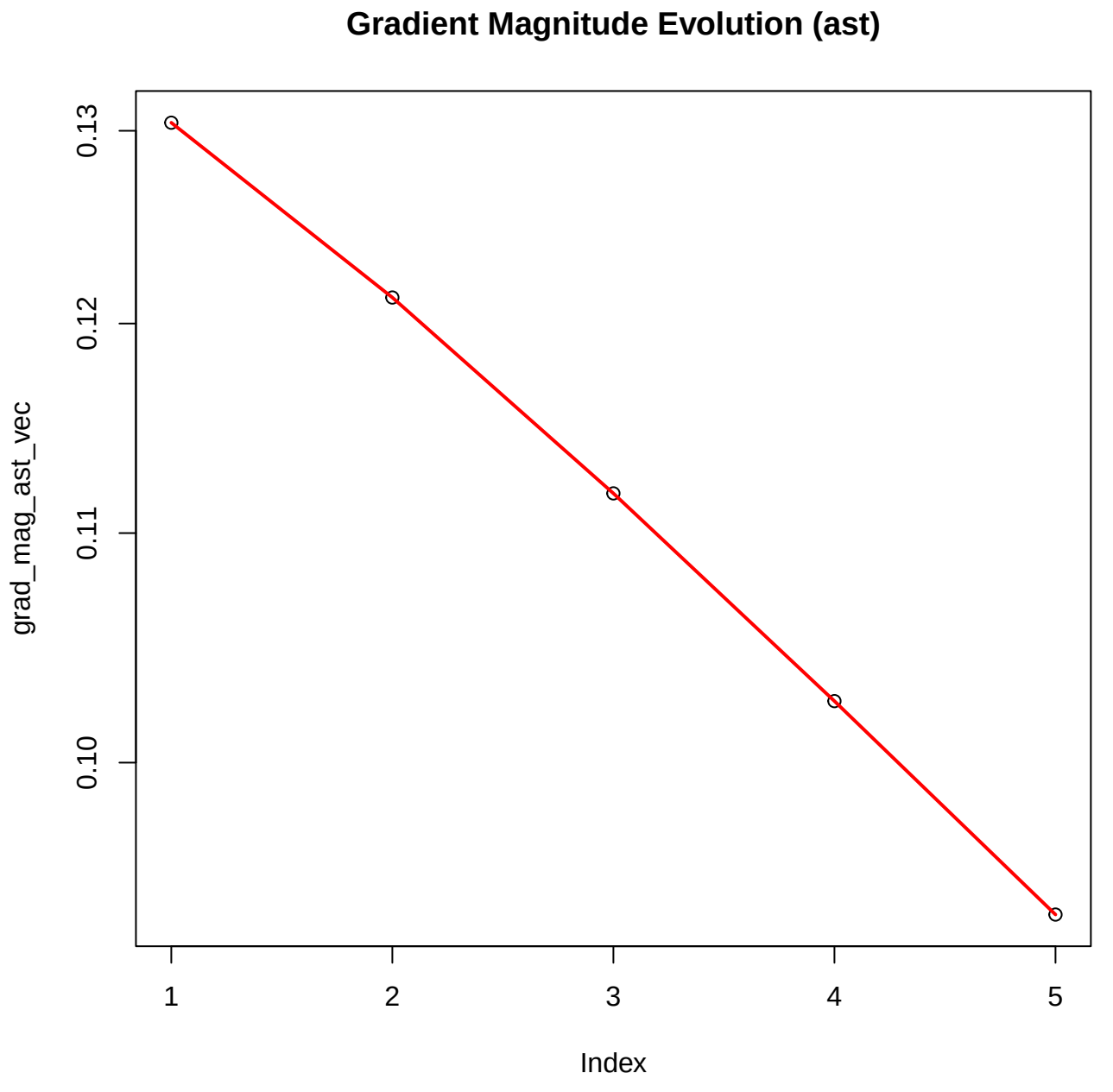

Value (ast)
loss_ast_vec
0.65
0.60
0.55
0.50
0.45
0.40
0.6
0.8
1.0
1.2
1.4
Index

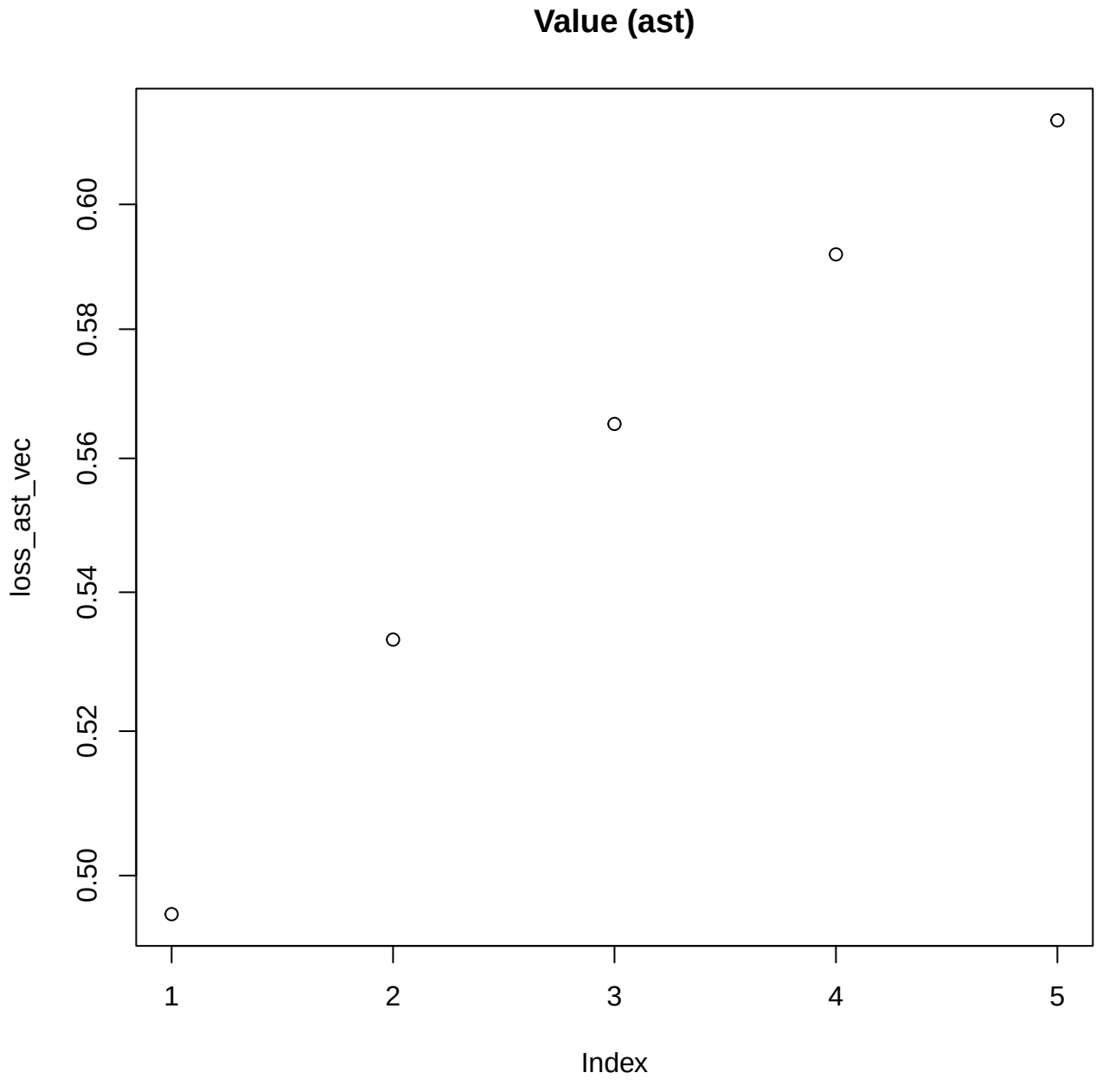

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index

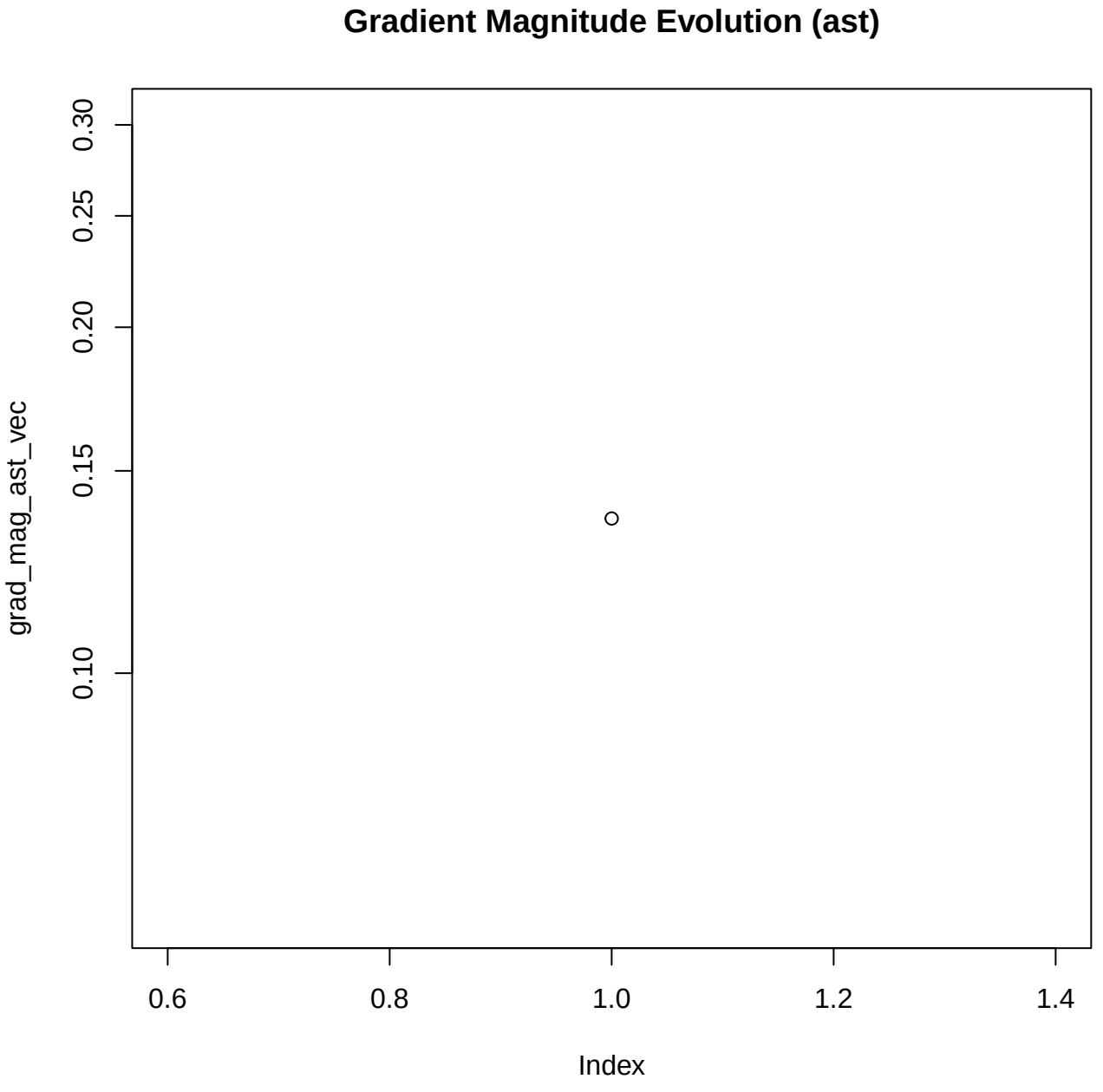

Value (ast)
loss_ast_vec
0.48
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index

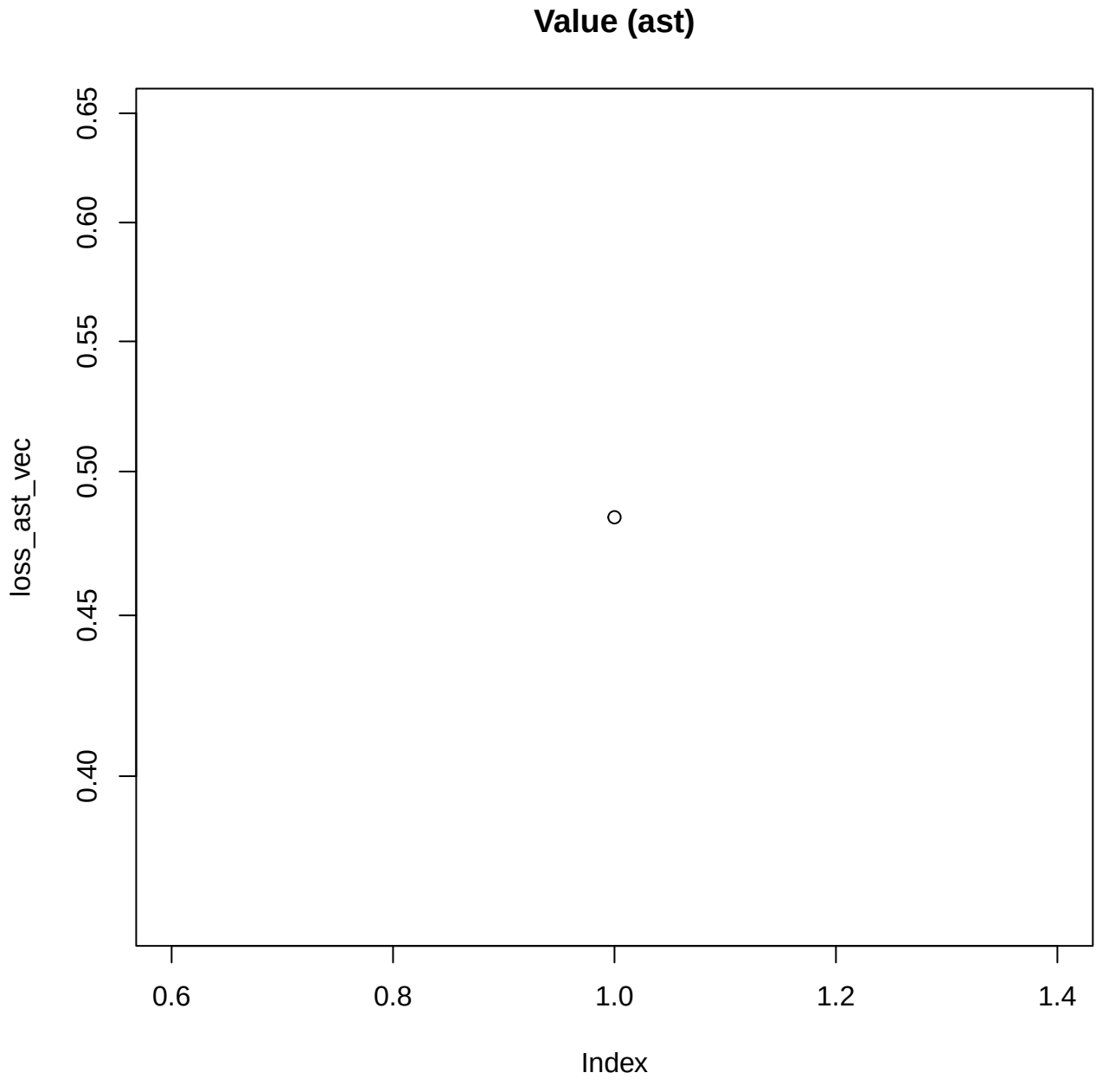

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.15
0.20
0.25
0.30
0.6
0.8
1.0
1.2
1.4
Index

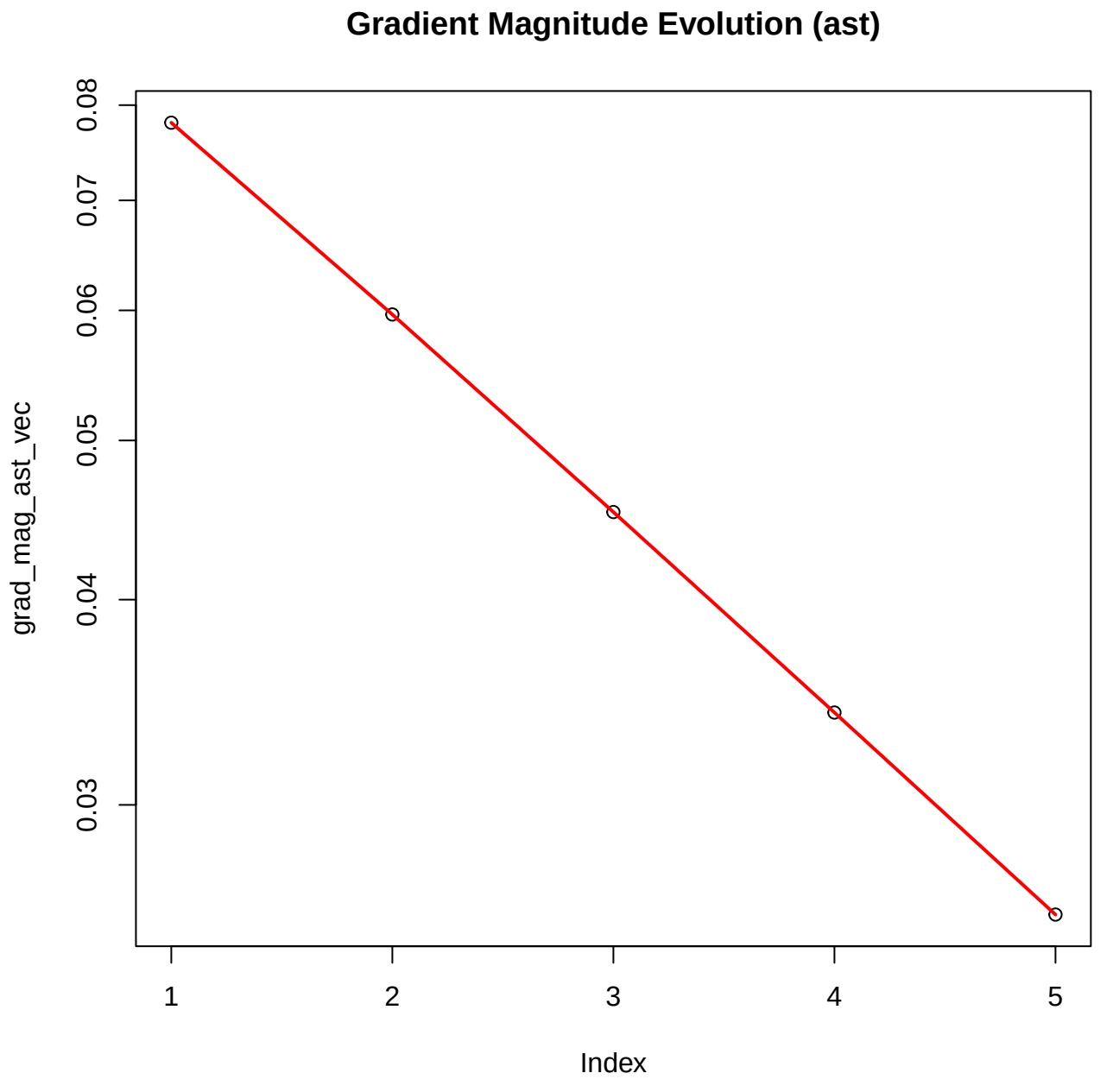

Value (ast)
loss_ast_vec
0.40
0.45
0.50
0.55
0.60
0.65
0.6
0.8
1.0
1.2
1.4
Index

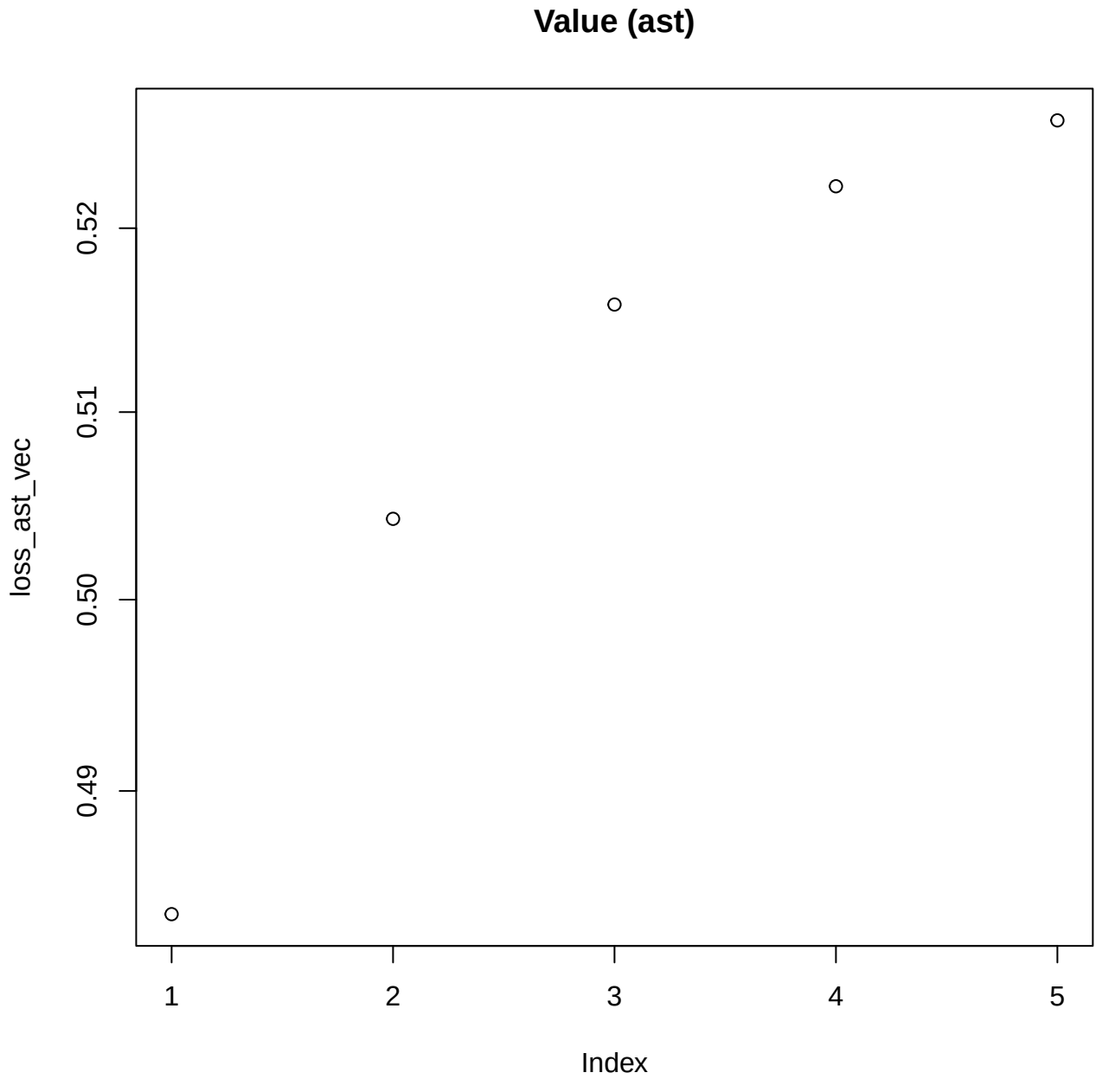

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index

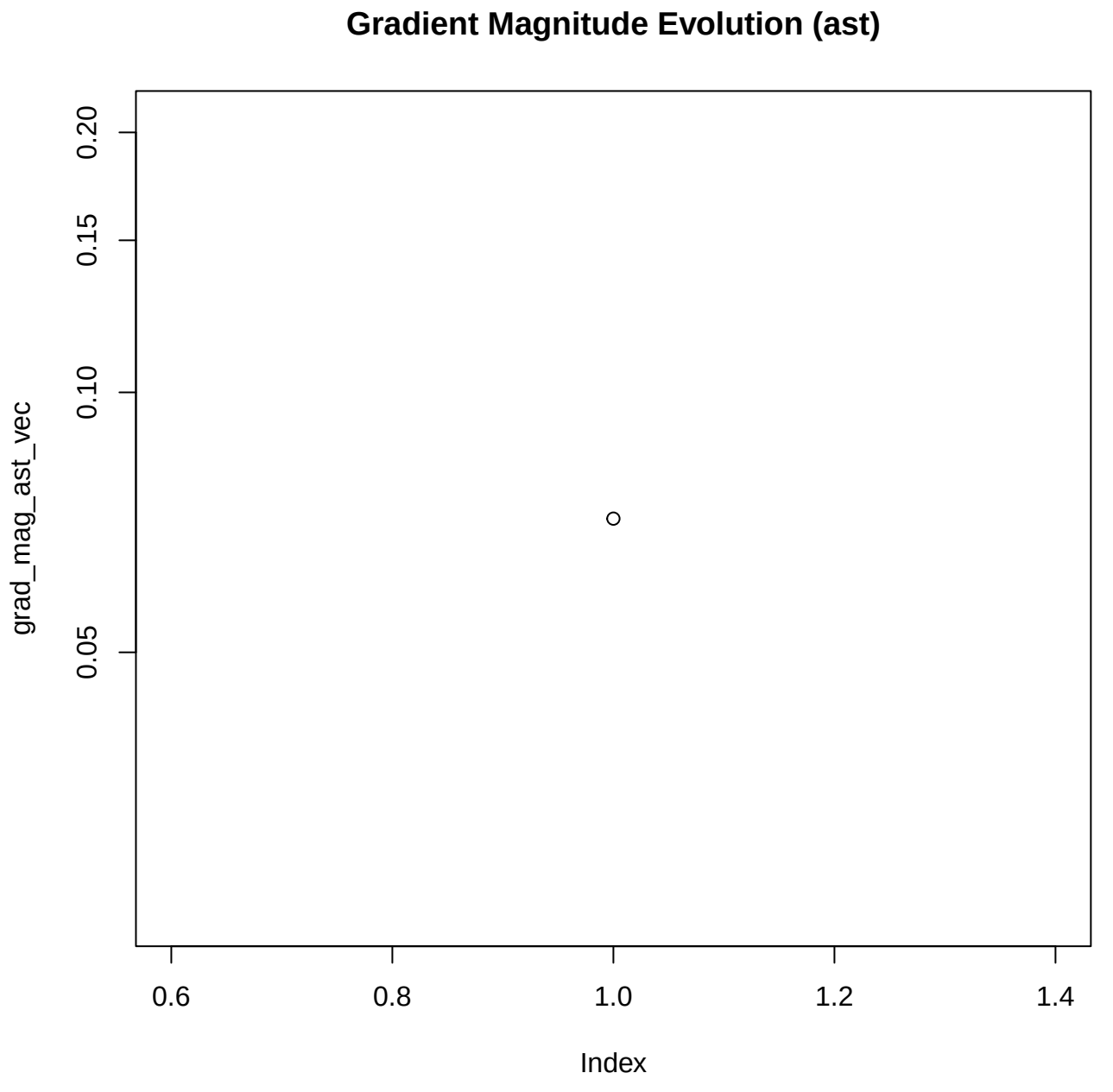

Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index

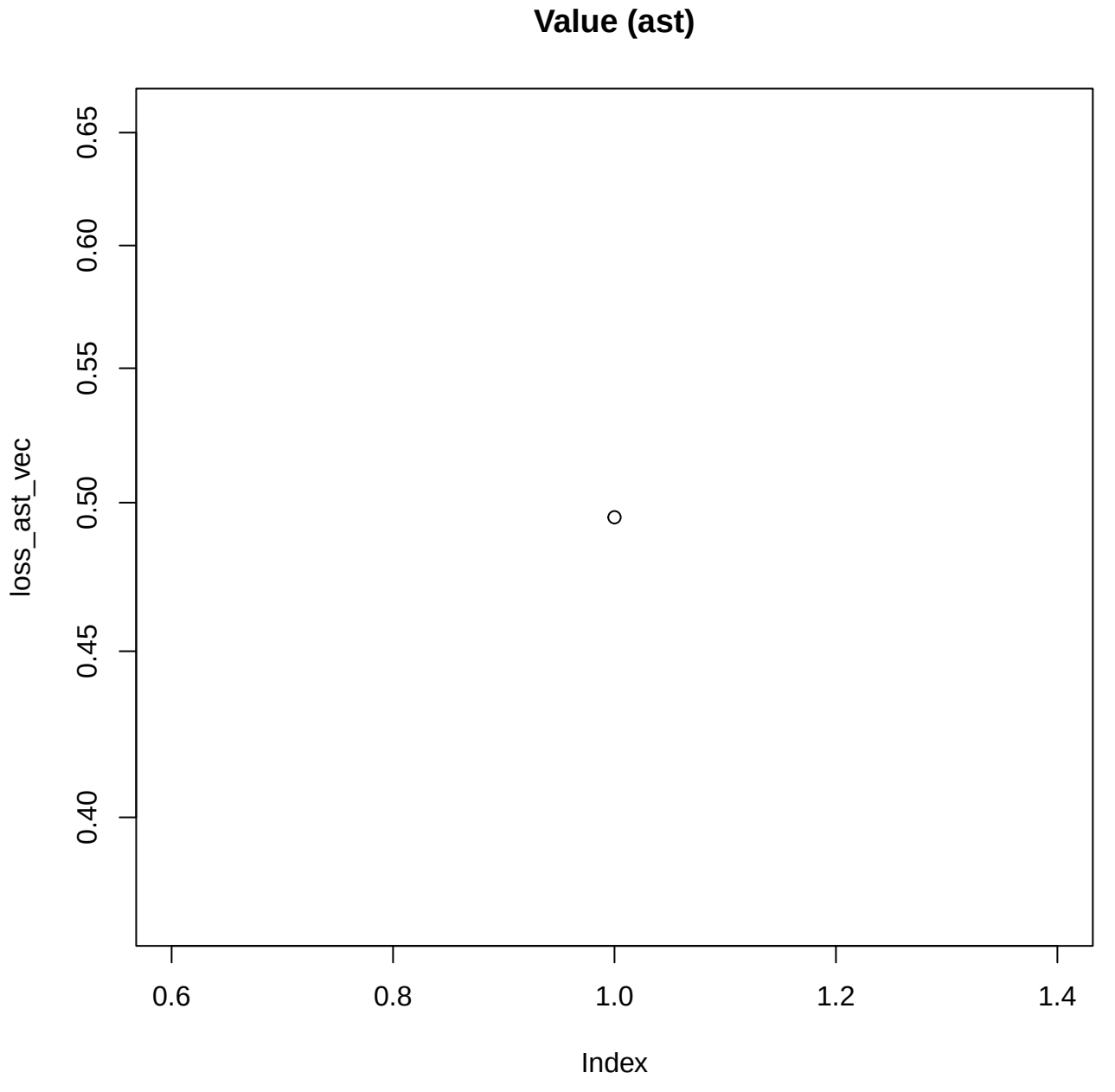

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.15
0.20
0.25
0.30
0.6
0.8
1.0
1.2
1.4
Index

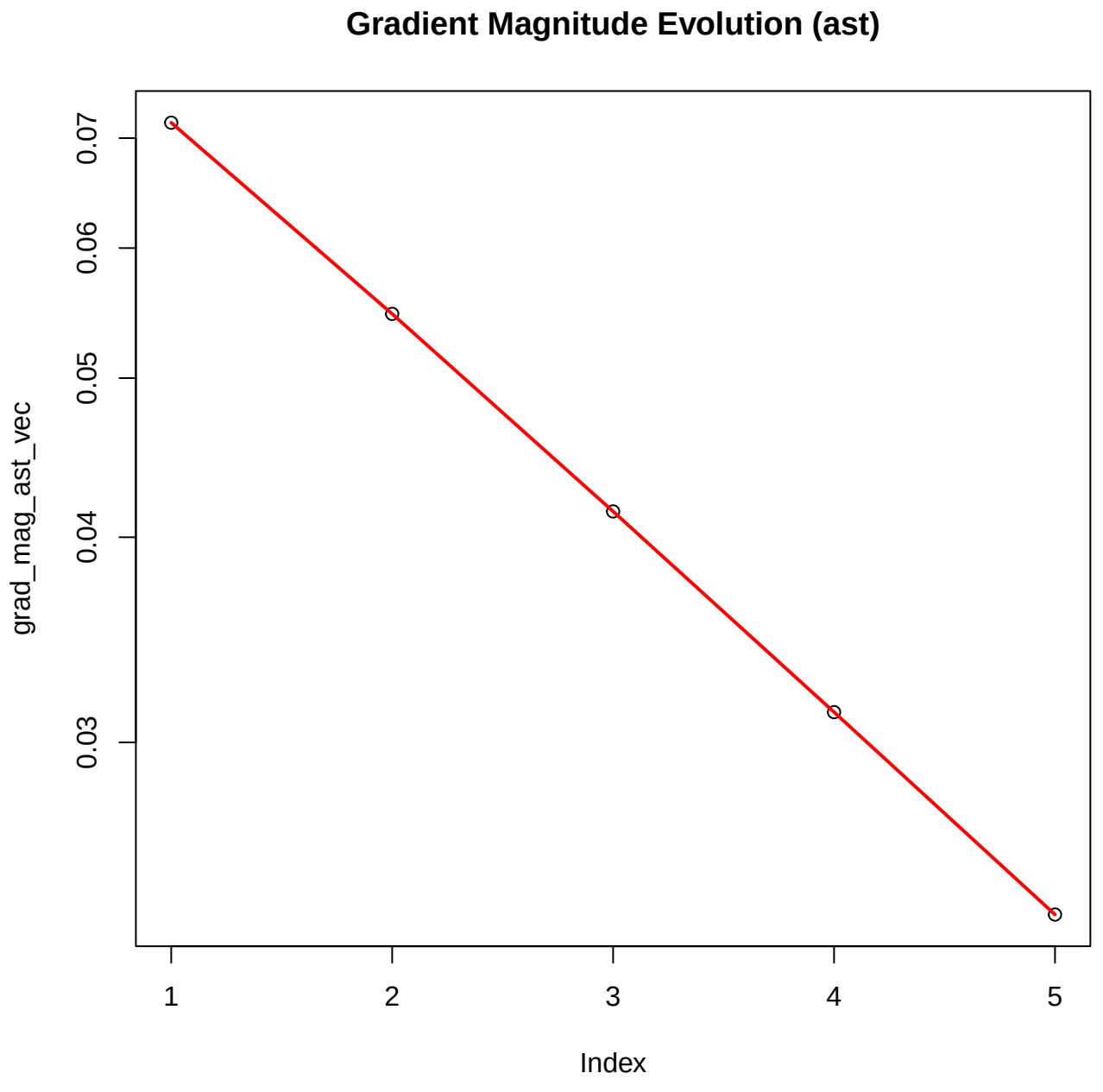

Value (ast)
loss_ast_vec
0.65
0.60
0.55
0.50
0.45
0.40
0.6
0.8
1.0
1.2
1.4
Index

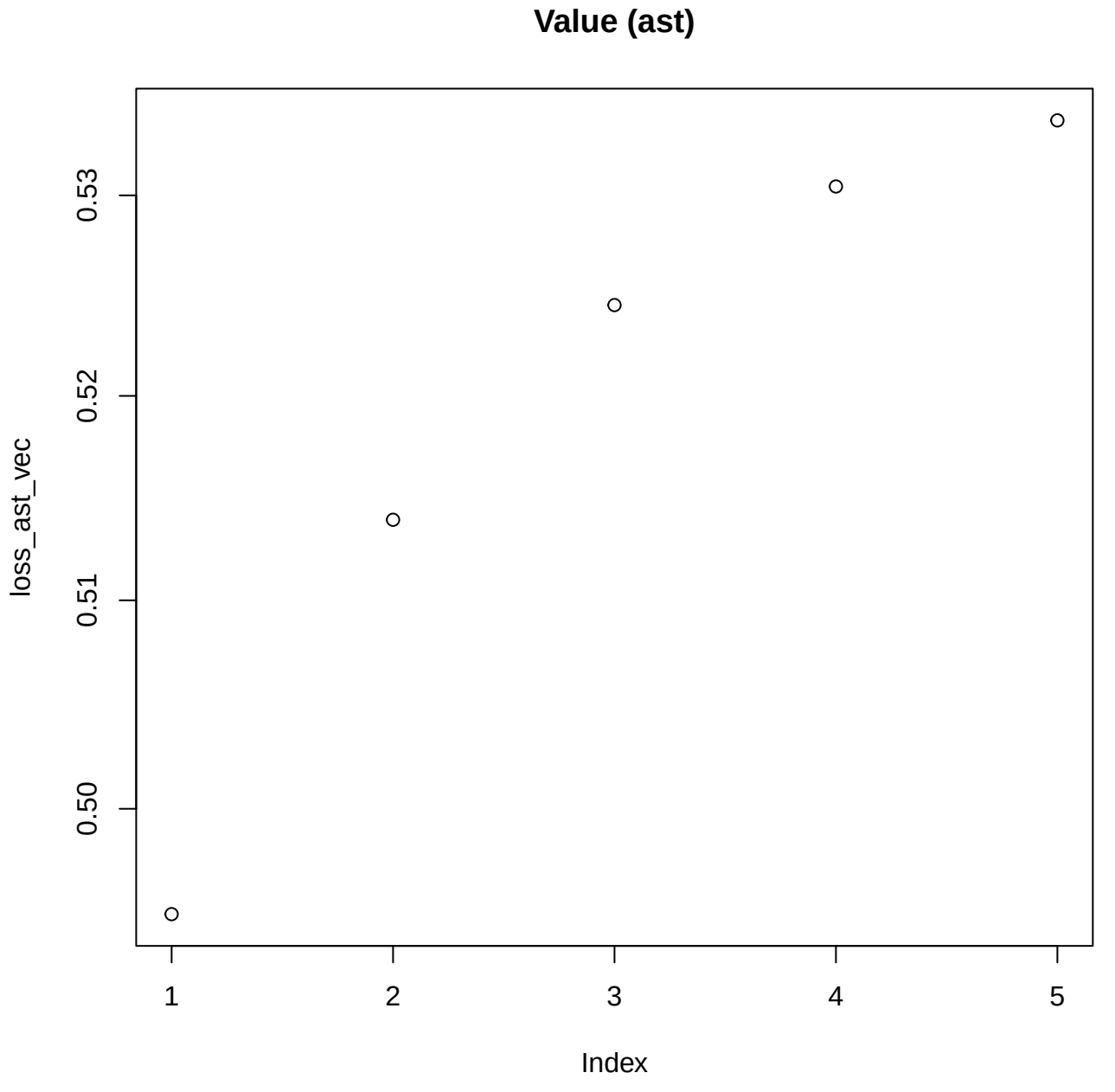

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.08
0.07
0.06
0.05
0.04
0.03
1
2
3
4
5
Index

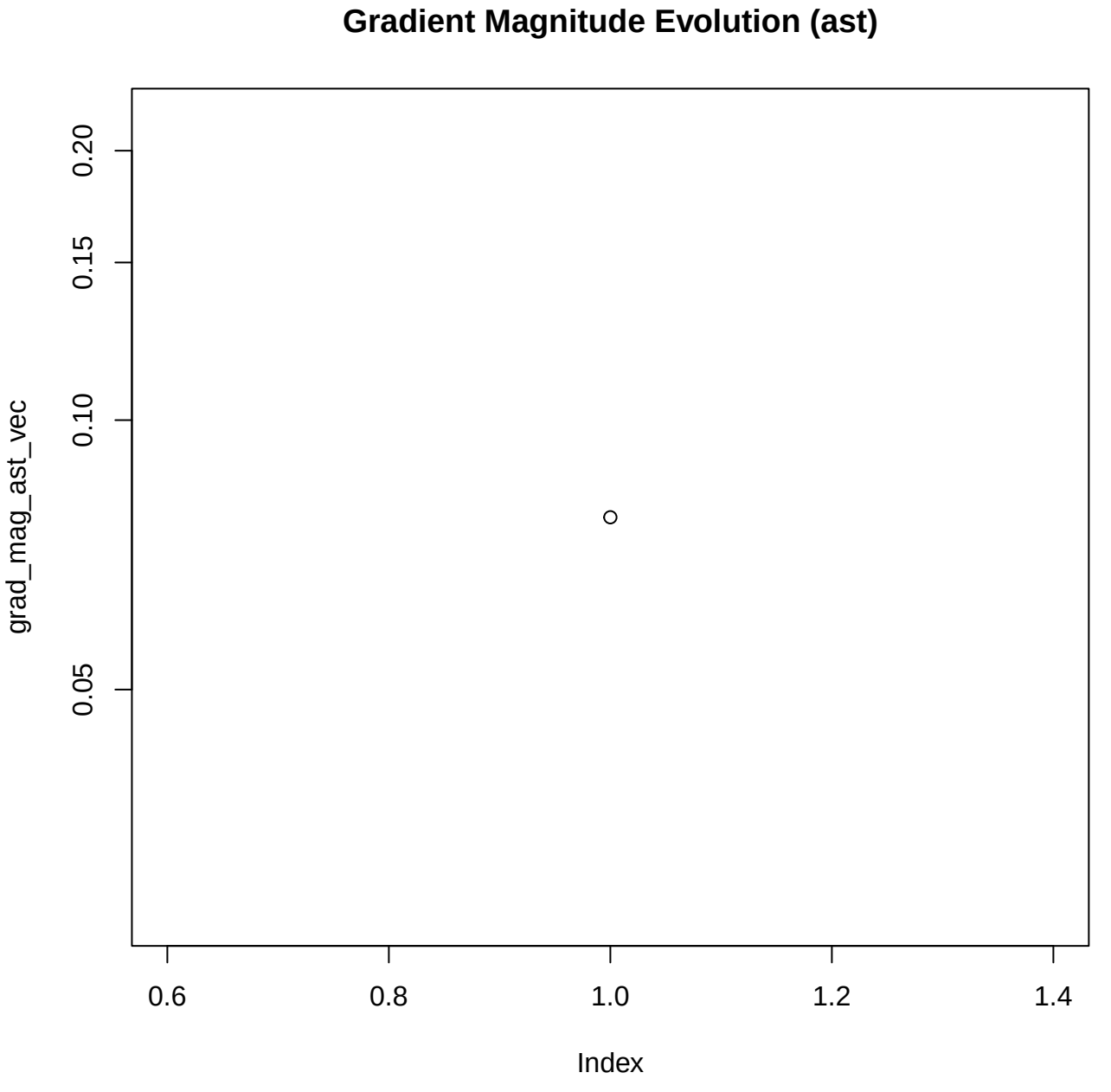

Value (ast)
loss_ast_vec
0.49
0.50
0.51
0.52
1
2
3
4
5
Index

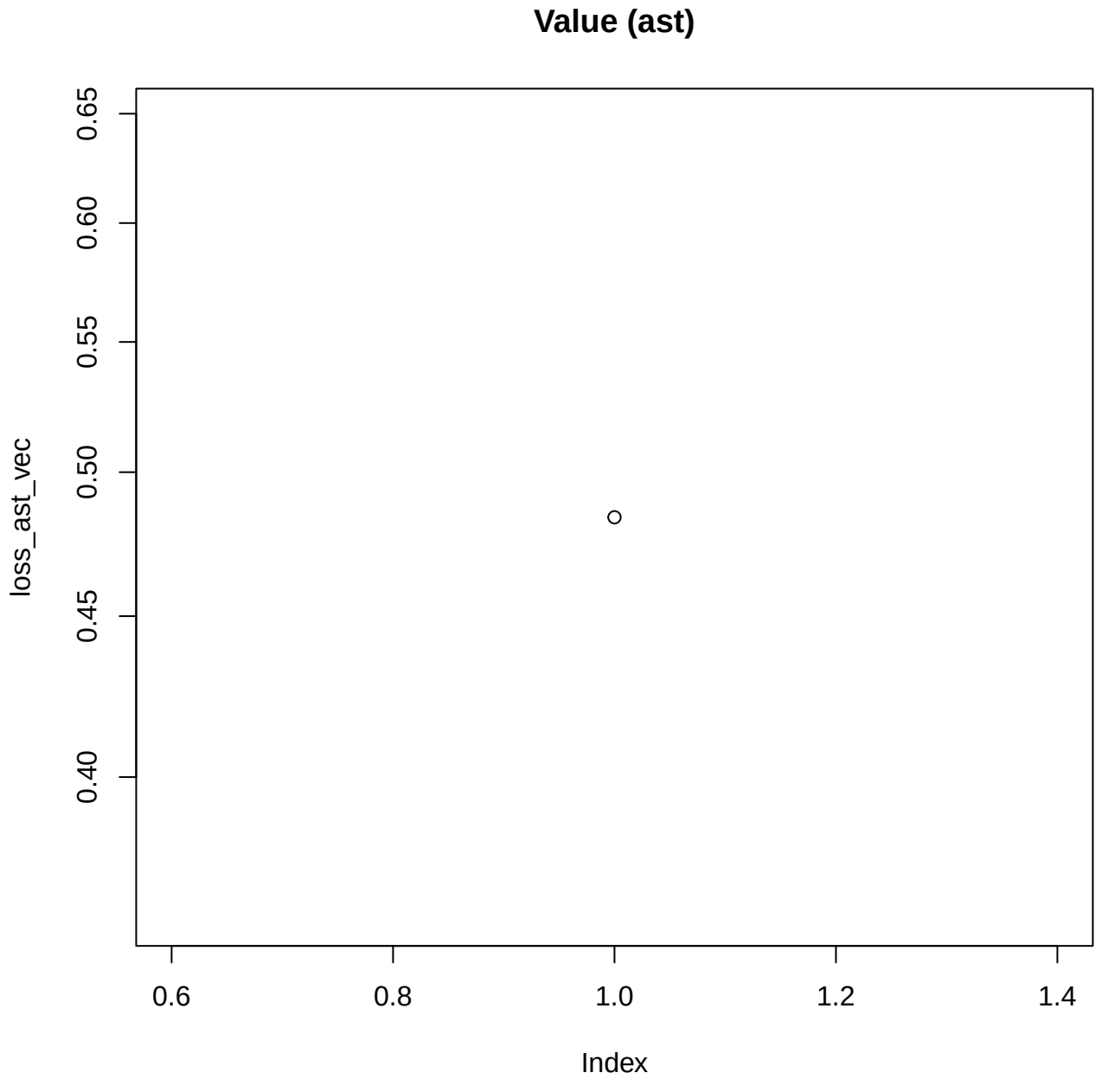

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.05
0.10
0.15
0.20
0.6
0.8
1.0
1.2
1.4
Index

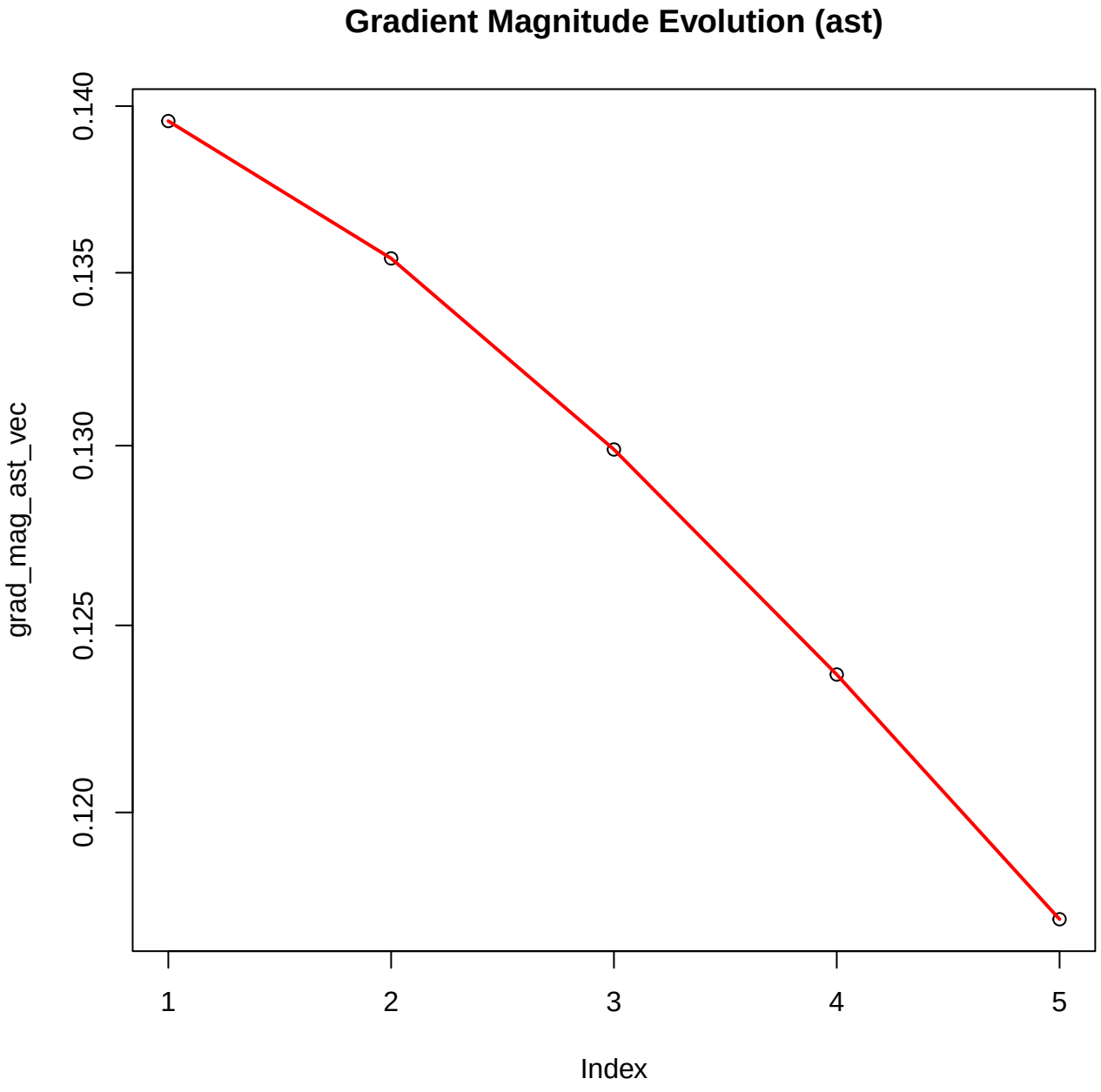

Value (ast)
loss_ast_vec
0.65
0.60
0.55
0.50
0.45
0.40
0.6
0.8
1.0
1.2
1.4
Index

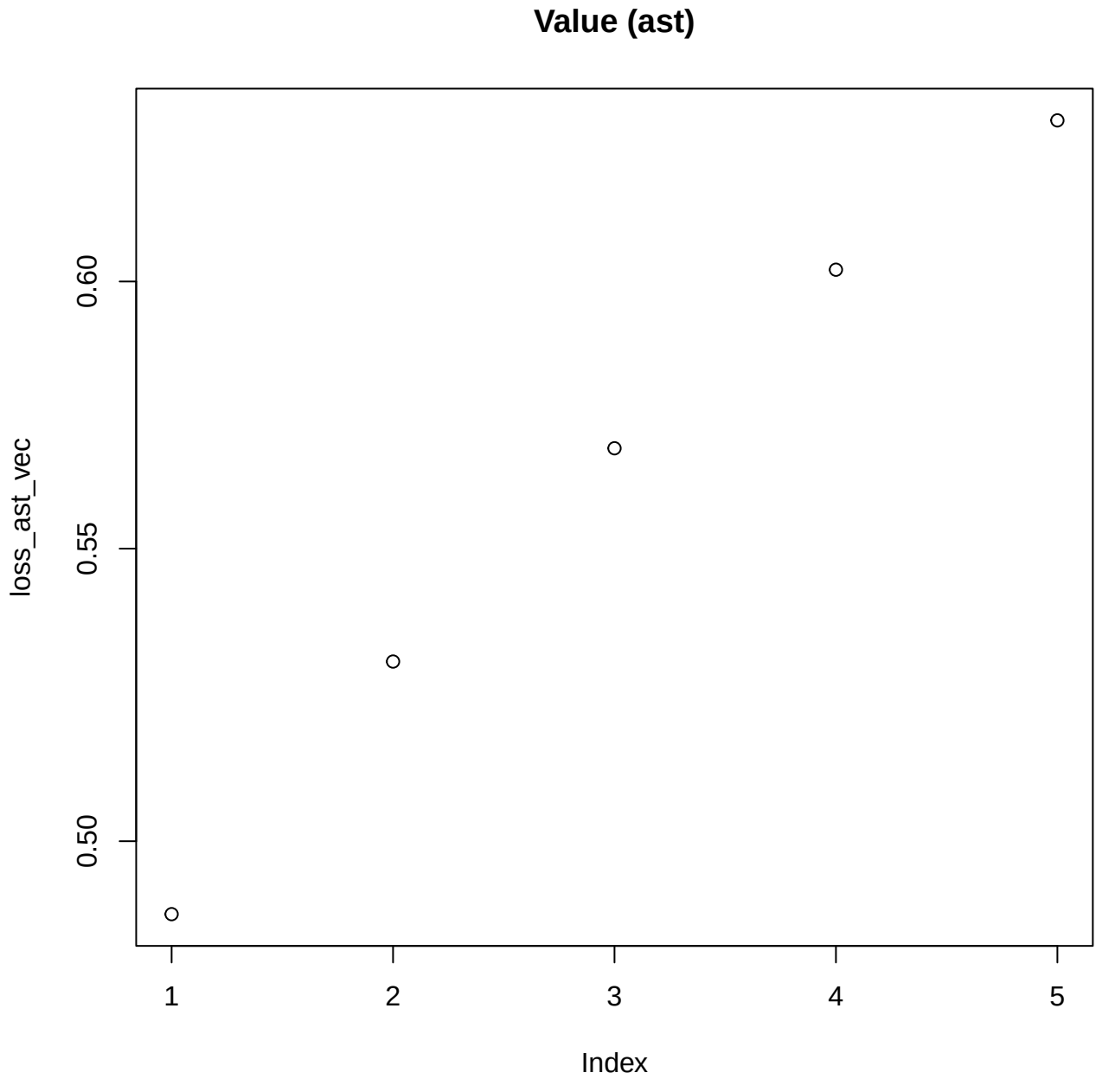

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.03
0.04
0.05
0.06
0.07
1
2
3
4
5
Index

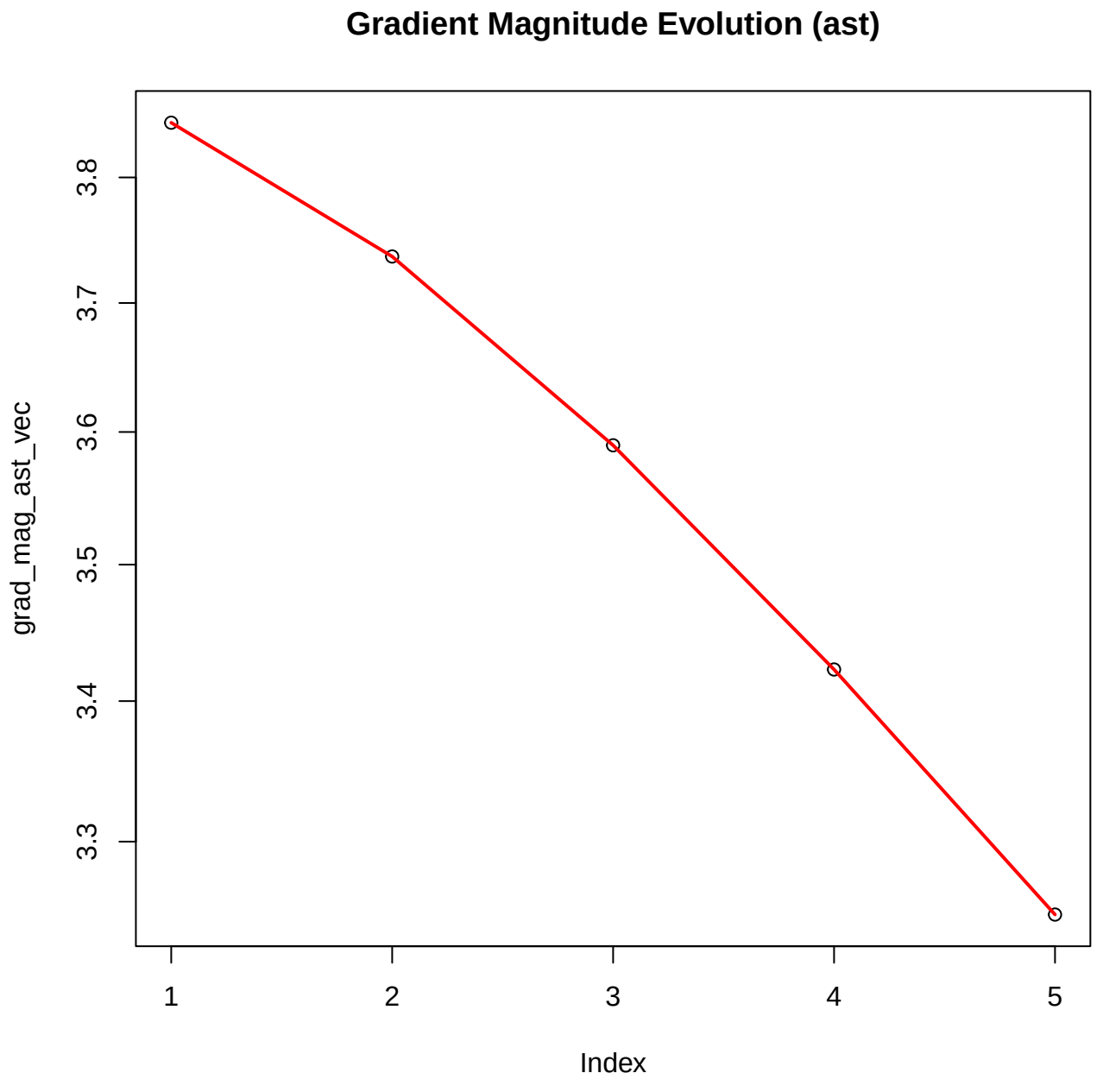

Value (ast)
loss_ast_vec
0.50
0.51
0.52
0.53
1
2
3
4
5
Index

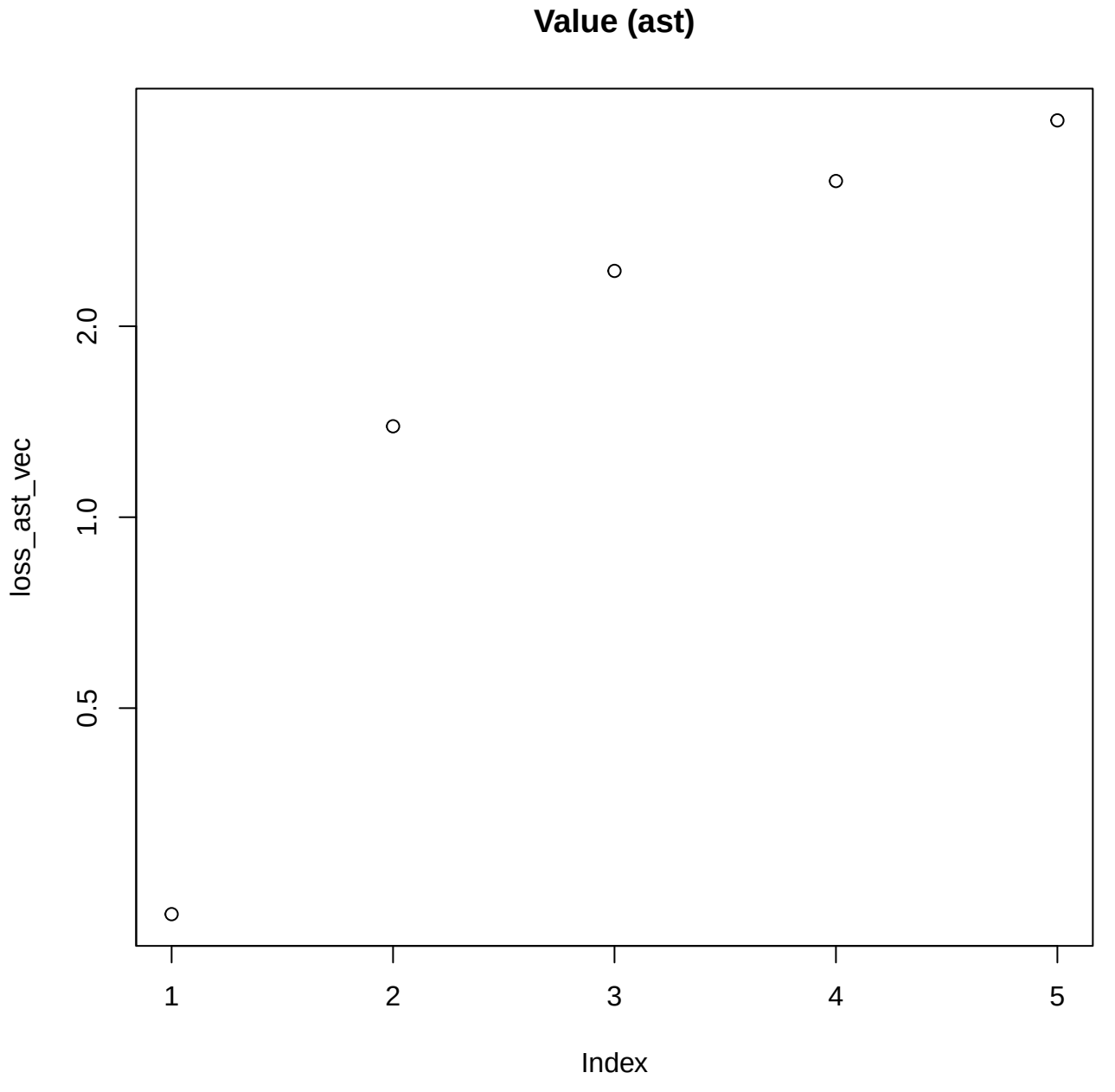

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.05
0.10
0.15
0.20
0.6
0.8
1.0
1.2
1.4
Index

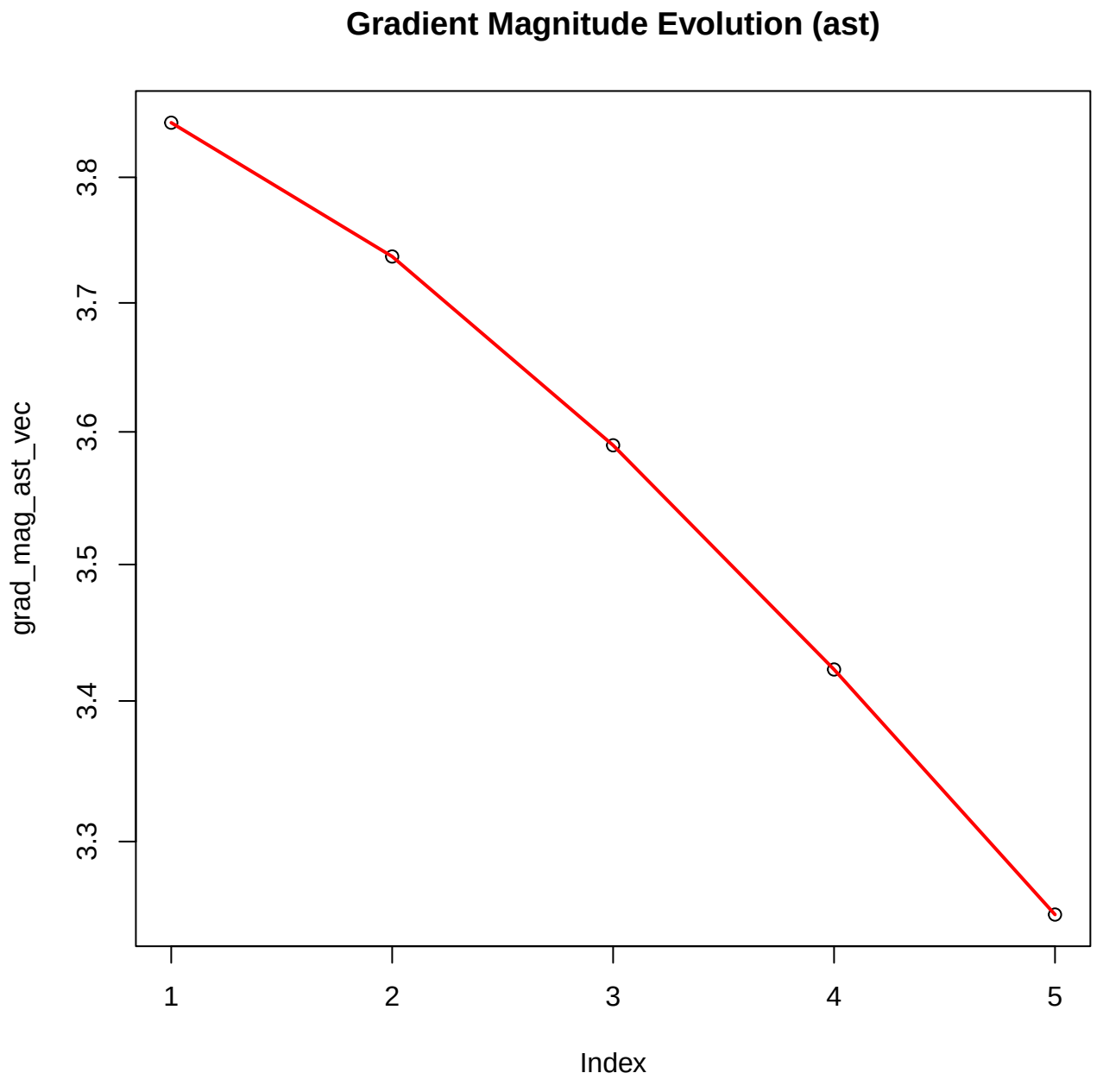

Value (ast)
loss_ast_vec
0.65
0.60
0.55
0.50
0.45
0.40
0.6
0.8
1.0
1.2
1.4
Index

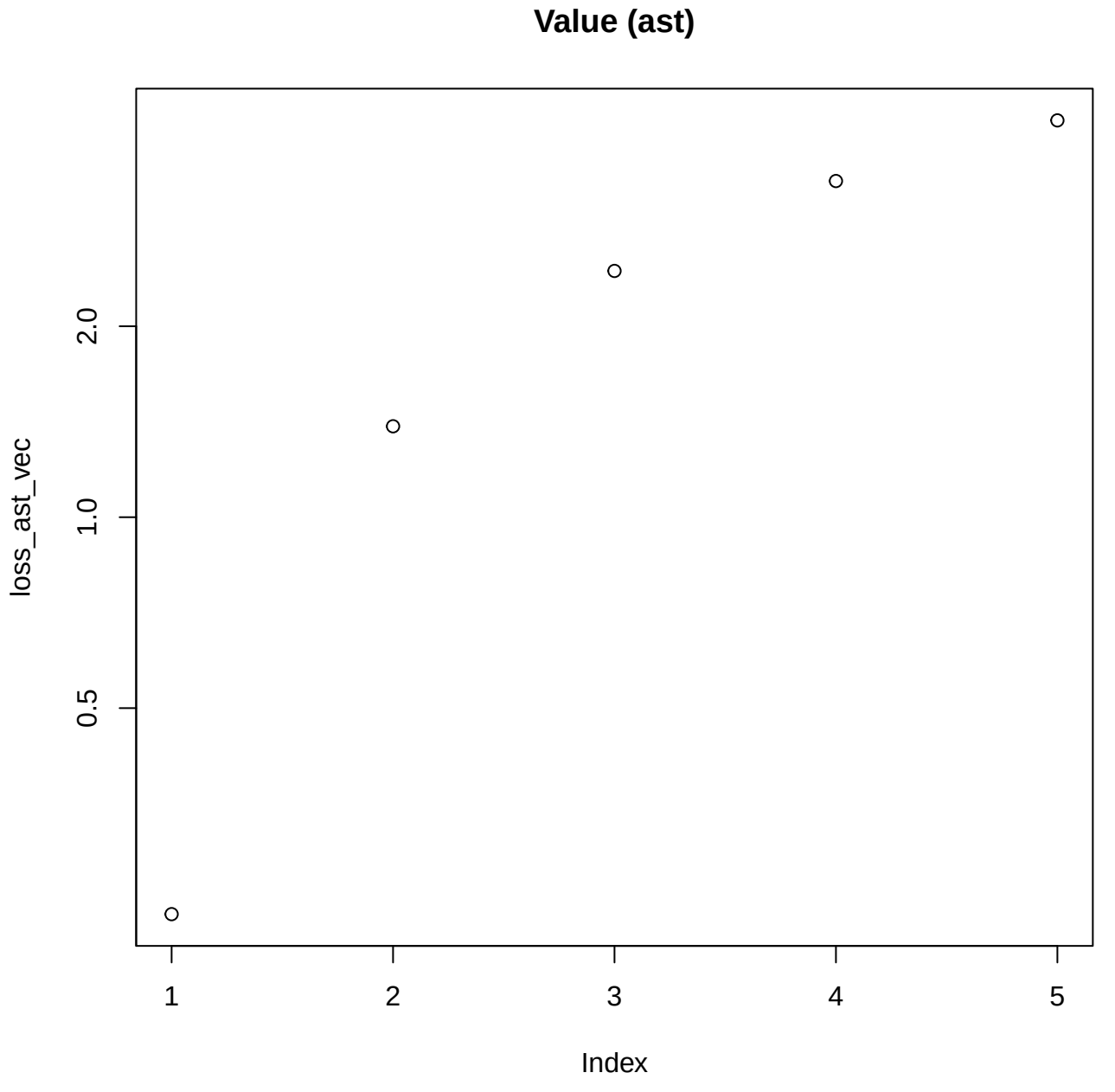

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.140
0.135
0.130
0.125
0.120
1
2
3
4
5
Index

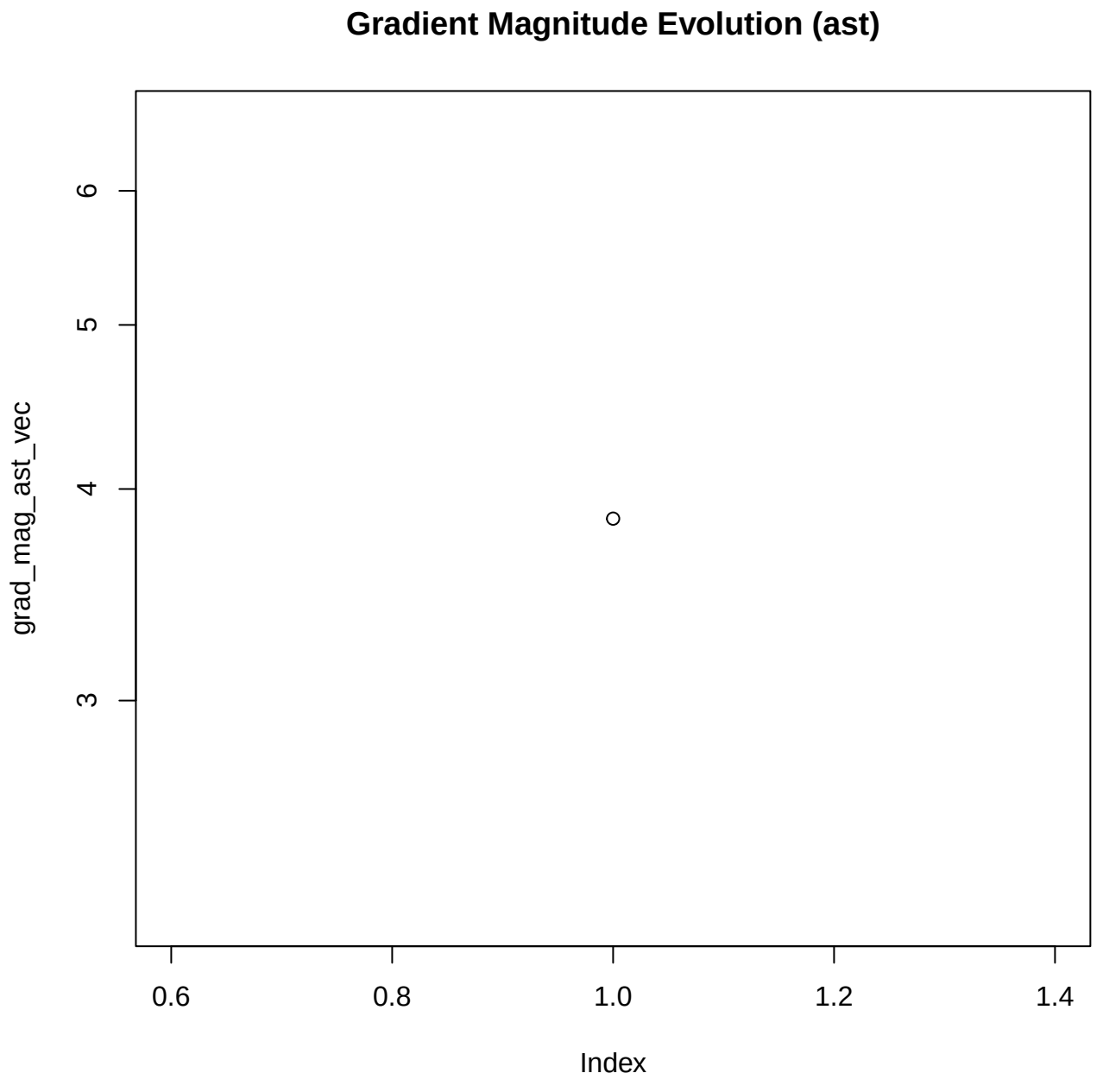

Value (ast)
loss_ast_vec
0.50
0.55
0.60
1
2
3
4
5
Index

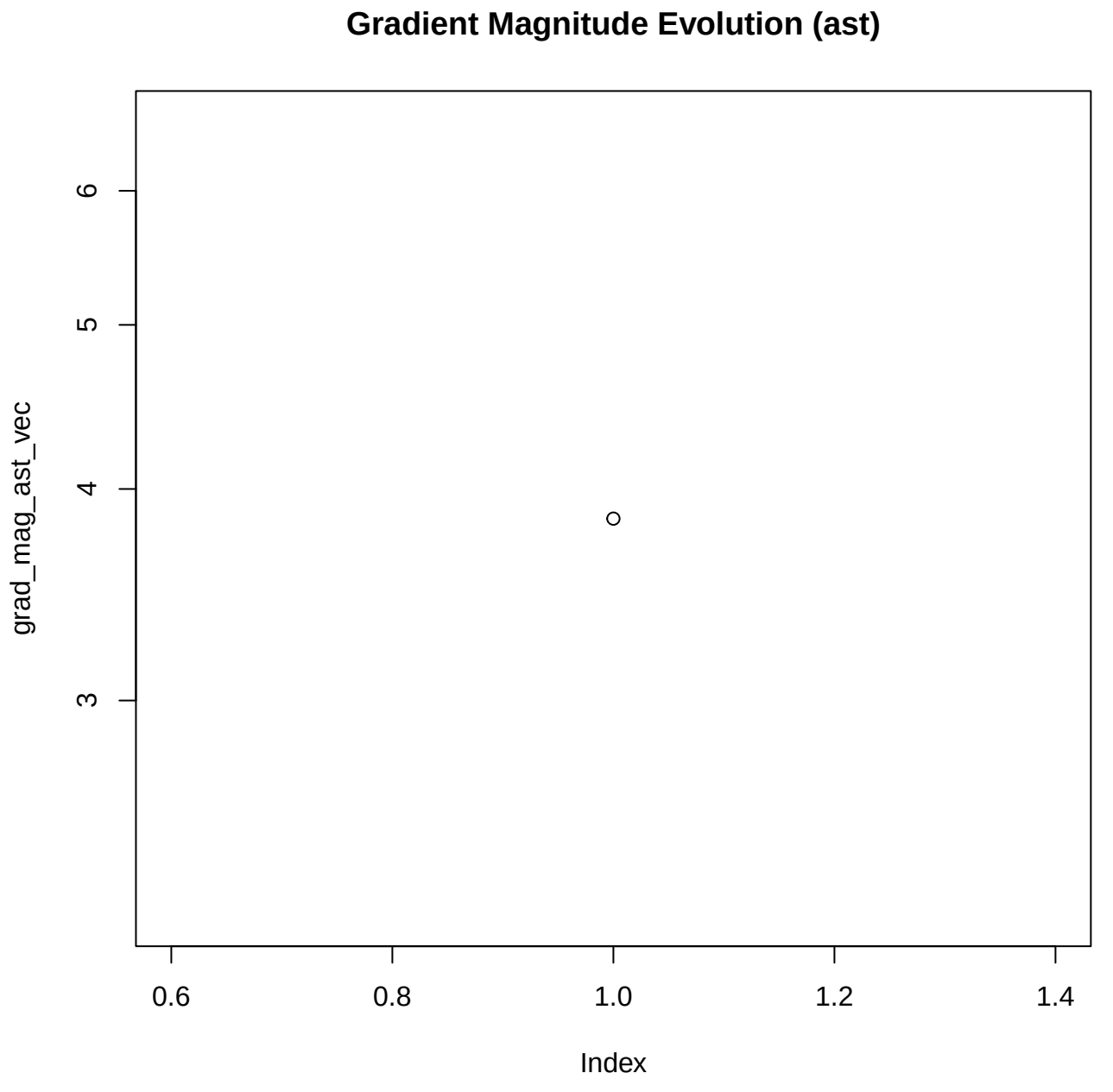

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
3.3
3.4
3.5
3.6
3.7
3.8
1
2
3
4
5
Index

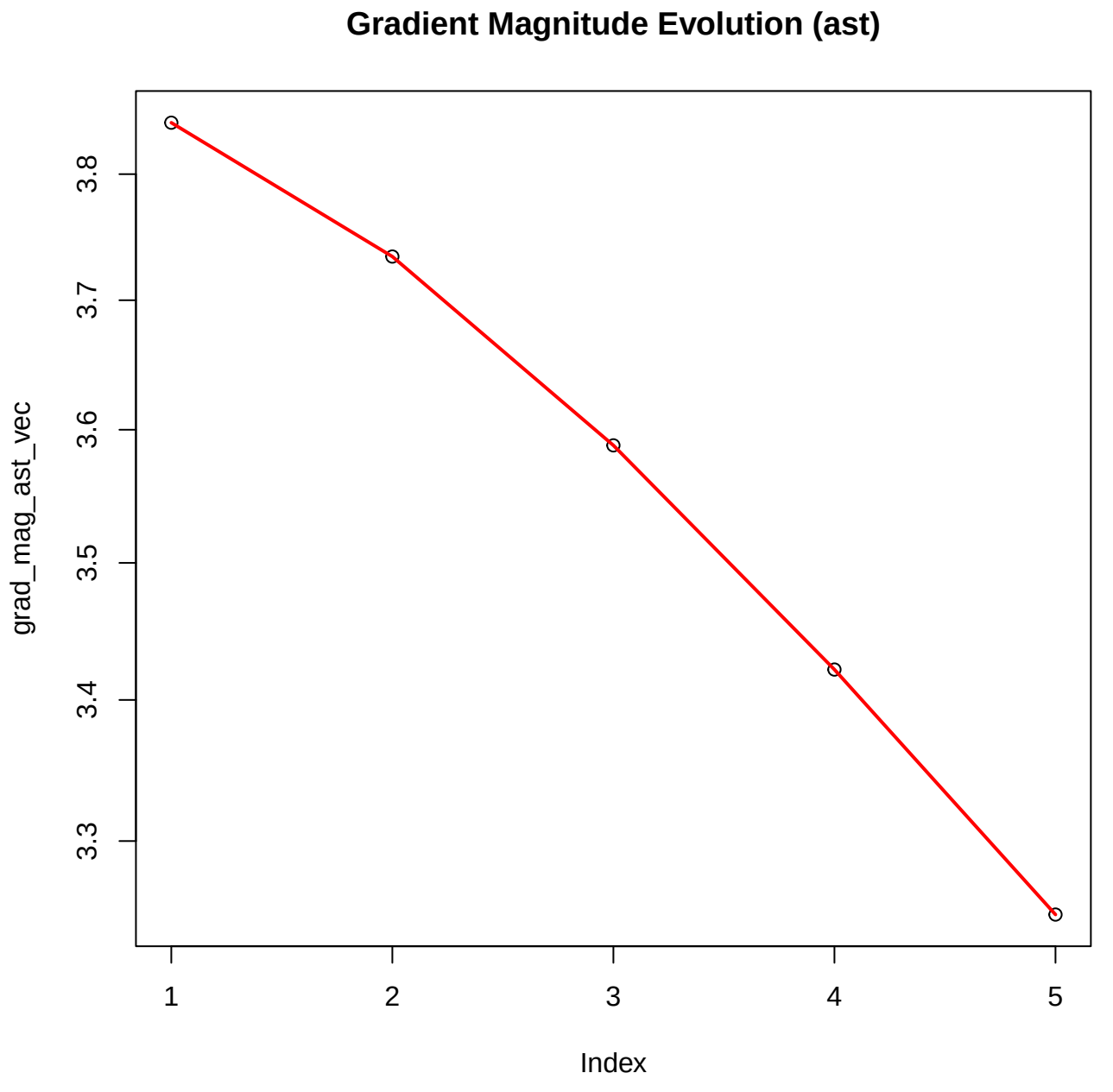

Value (ast)
loss_ast_vec
0.5
1.0
2.0
1
2
3
4
5
Index

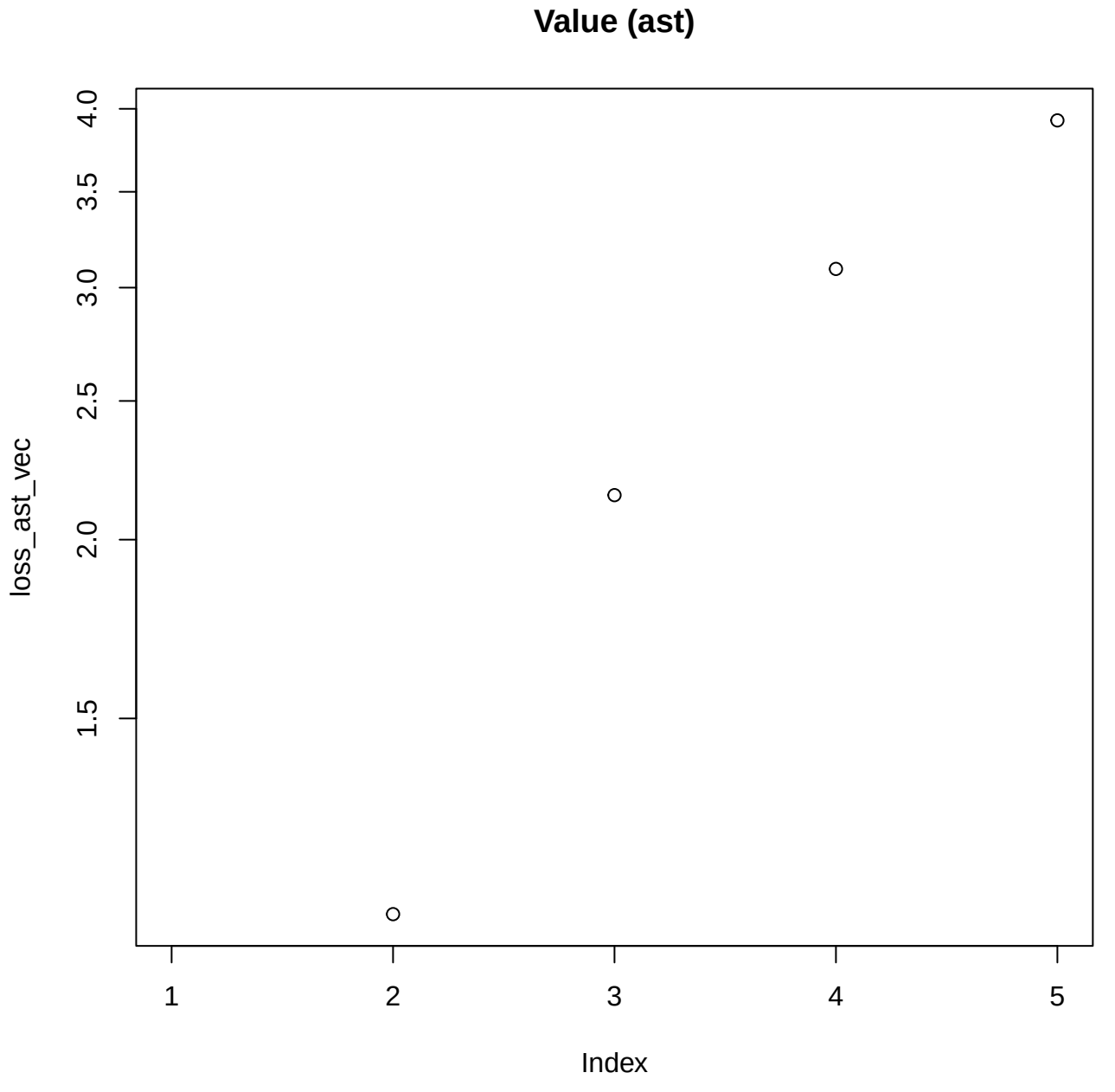

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
3.3
3.4
3.5
3.6
3.7
3.8
1
2
3
4
5
Index

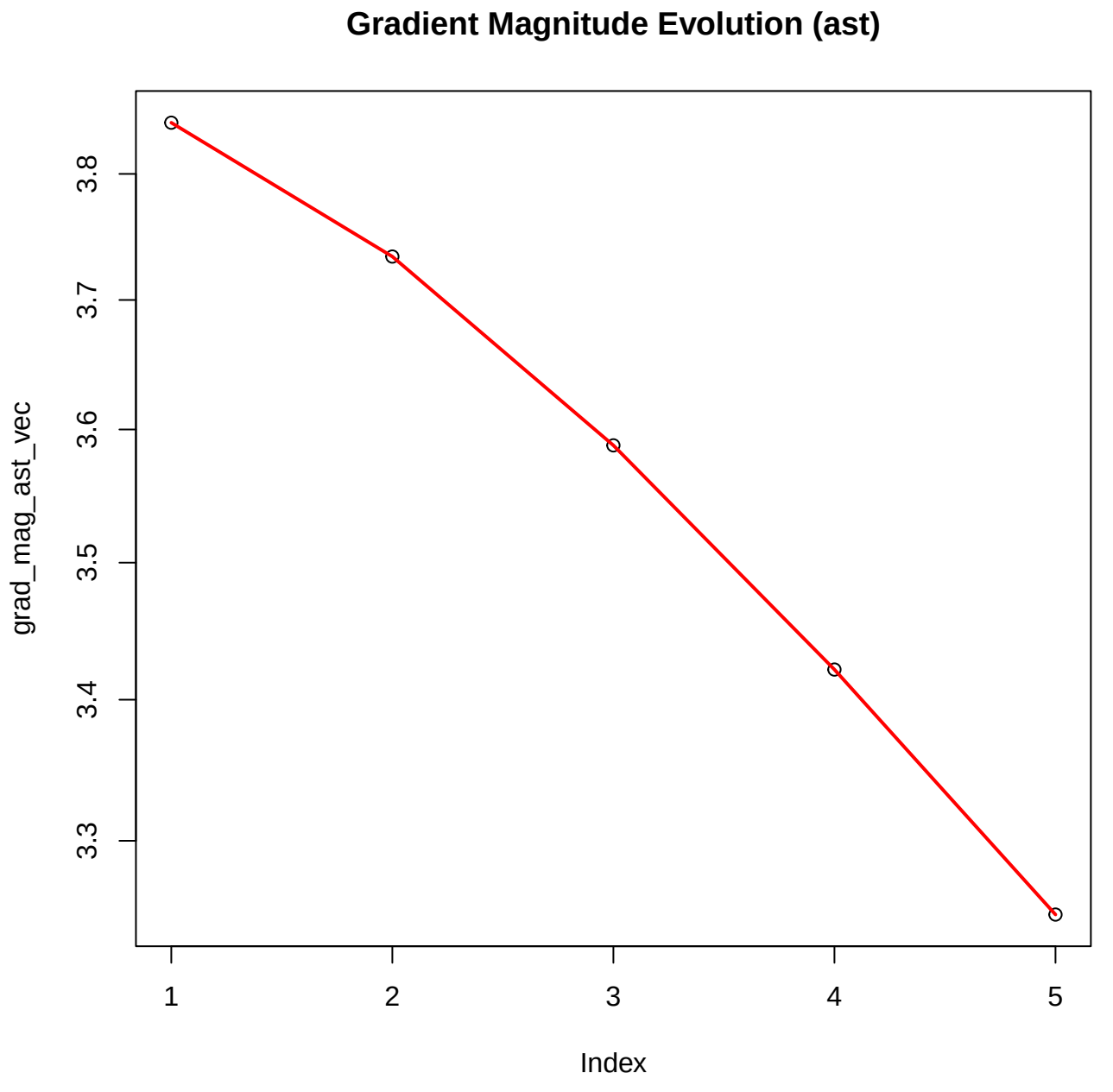

Value (ast)
loss_ast_vec
0.5
1.0
2.0
1
2
3
4
5
Index

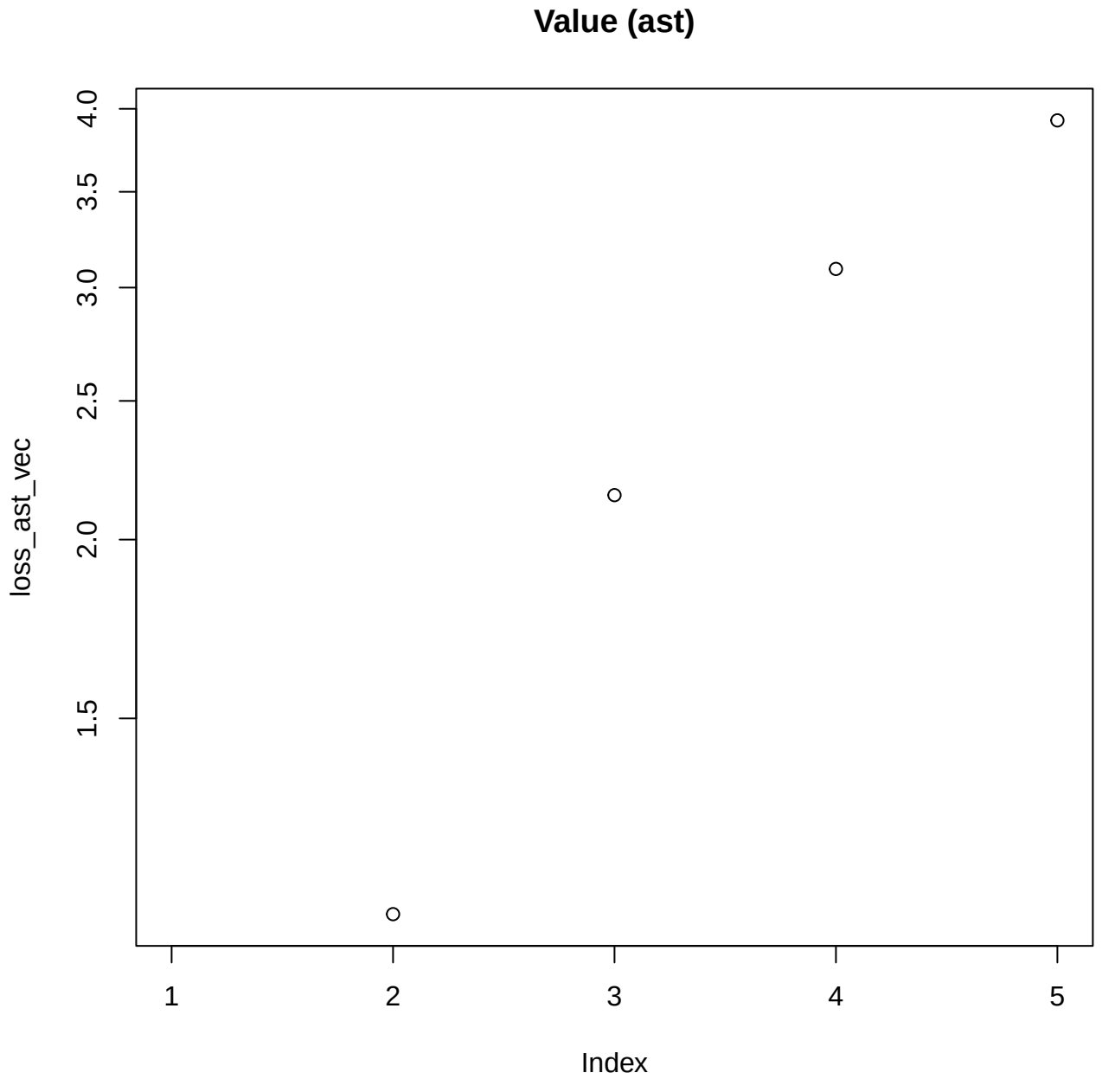

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
3
4
5
6
0.6
0.8
1.0
1.2
1.4
Index

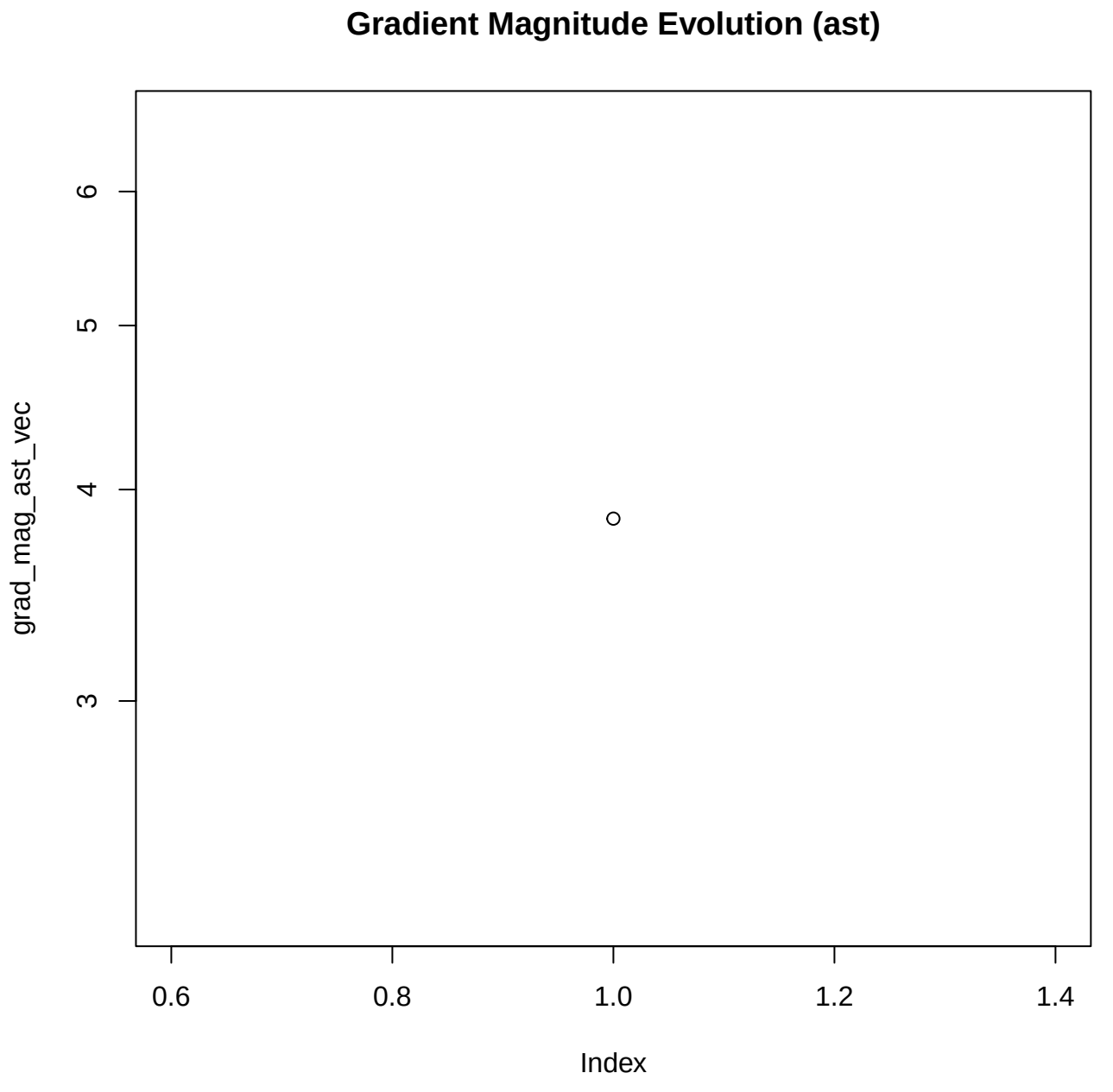

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
3
4
5
6
0.6
0.8
1.0
1.2
1.4
Index

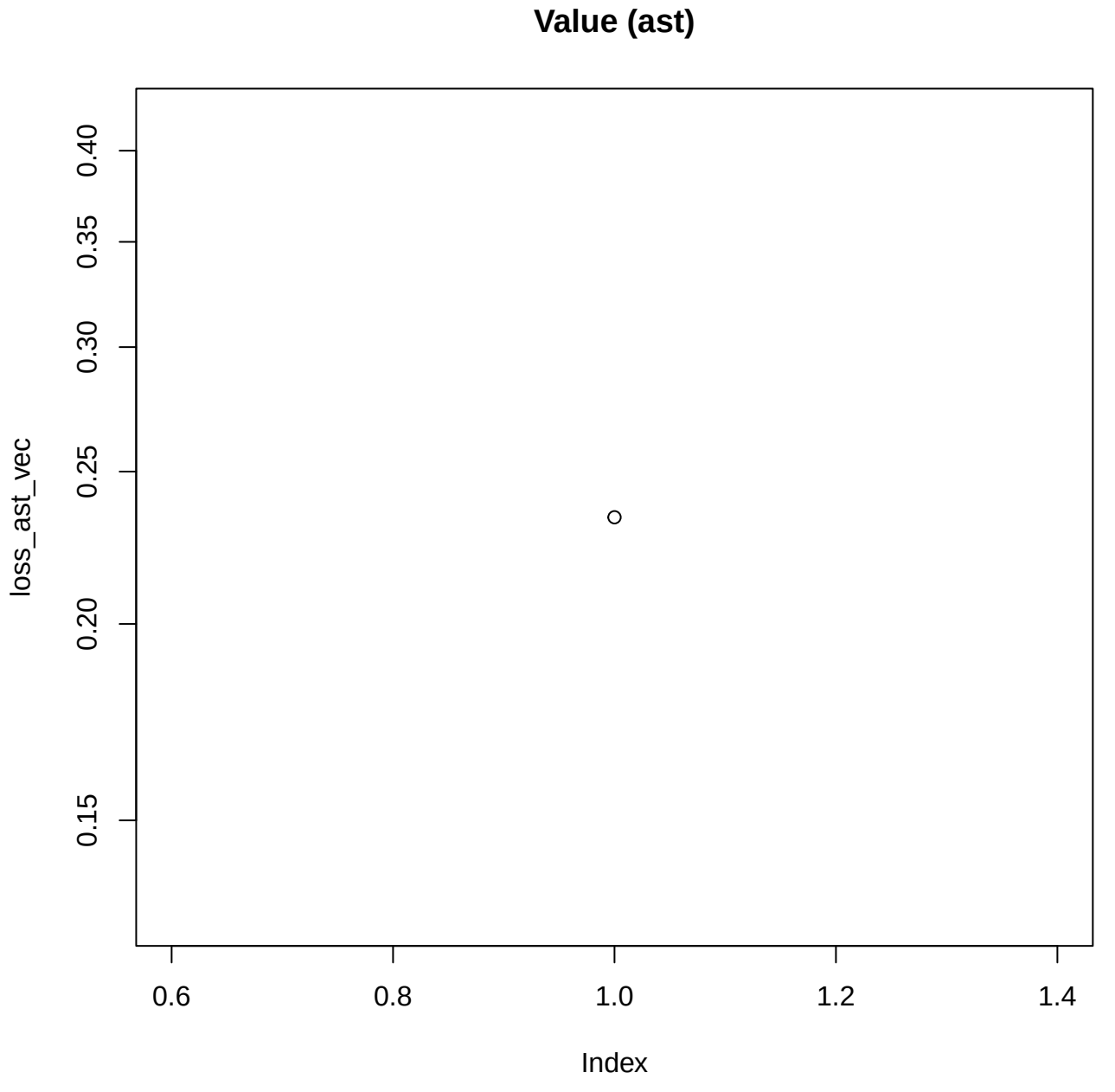

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
3.3
3.4
3.5
3.6
3.7
3.8
1
2
3
4
5
Index

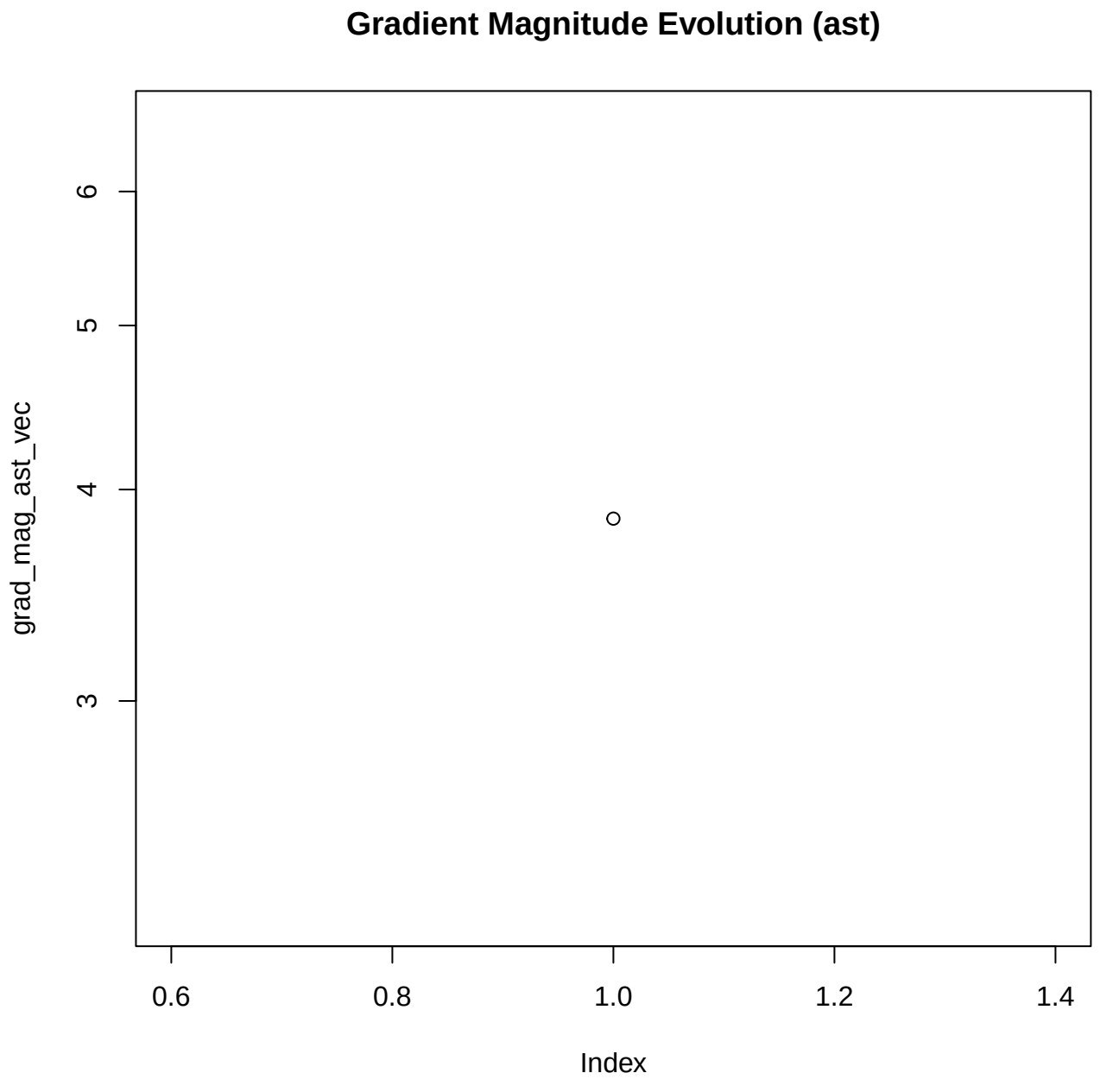

Value (ast)
loss_ast_vec
4.0
3.5
3.0
2.5
2.0
1.5
1
2
3
4
5
Index

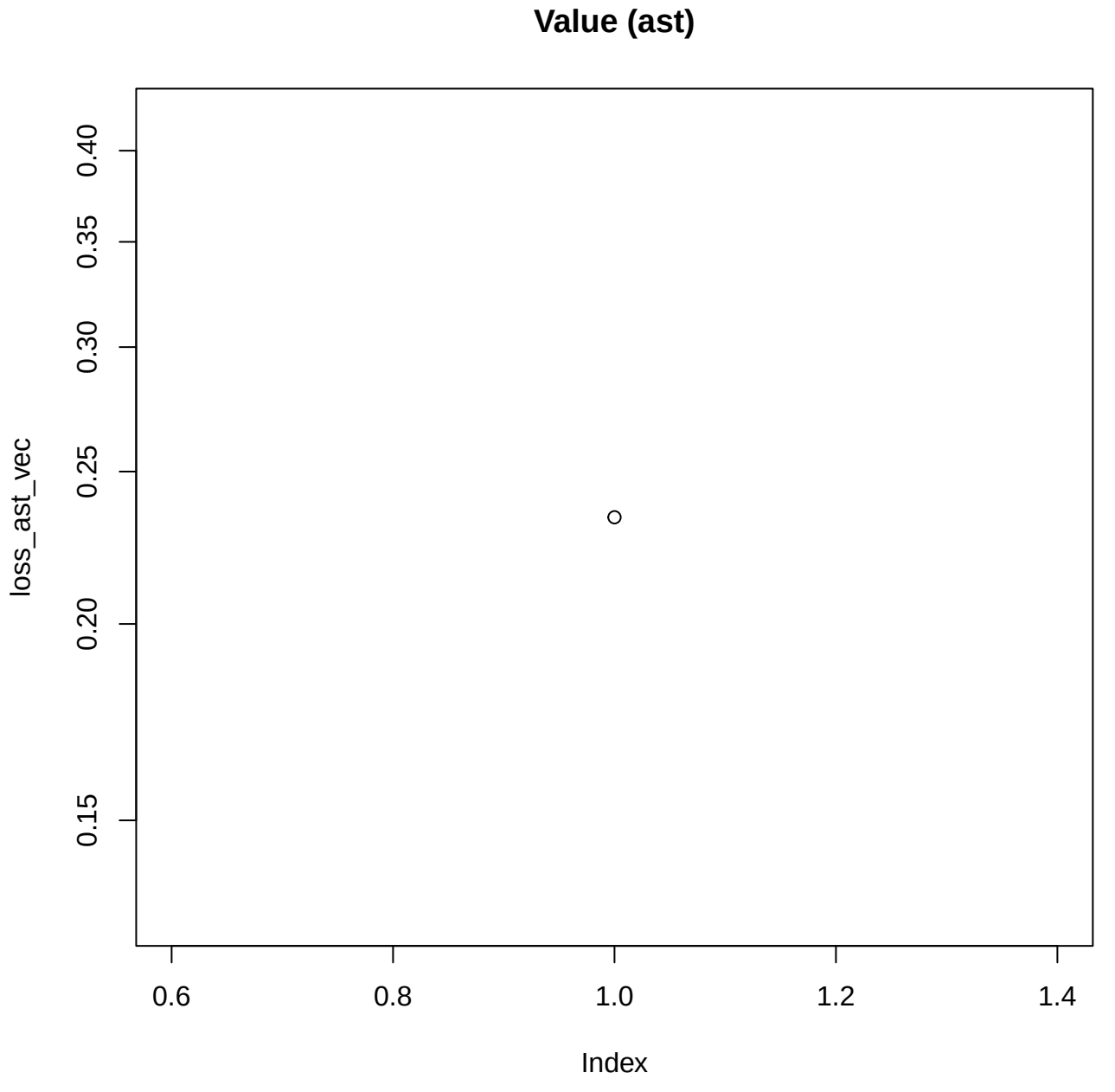

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
3.3
3.4
3.5
3.6
3.7
3.8
1
2
3
4
5
Index

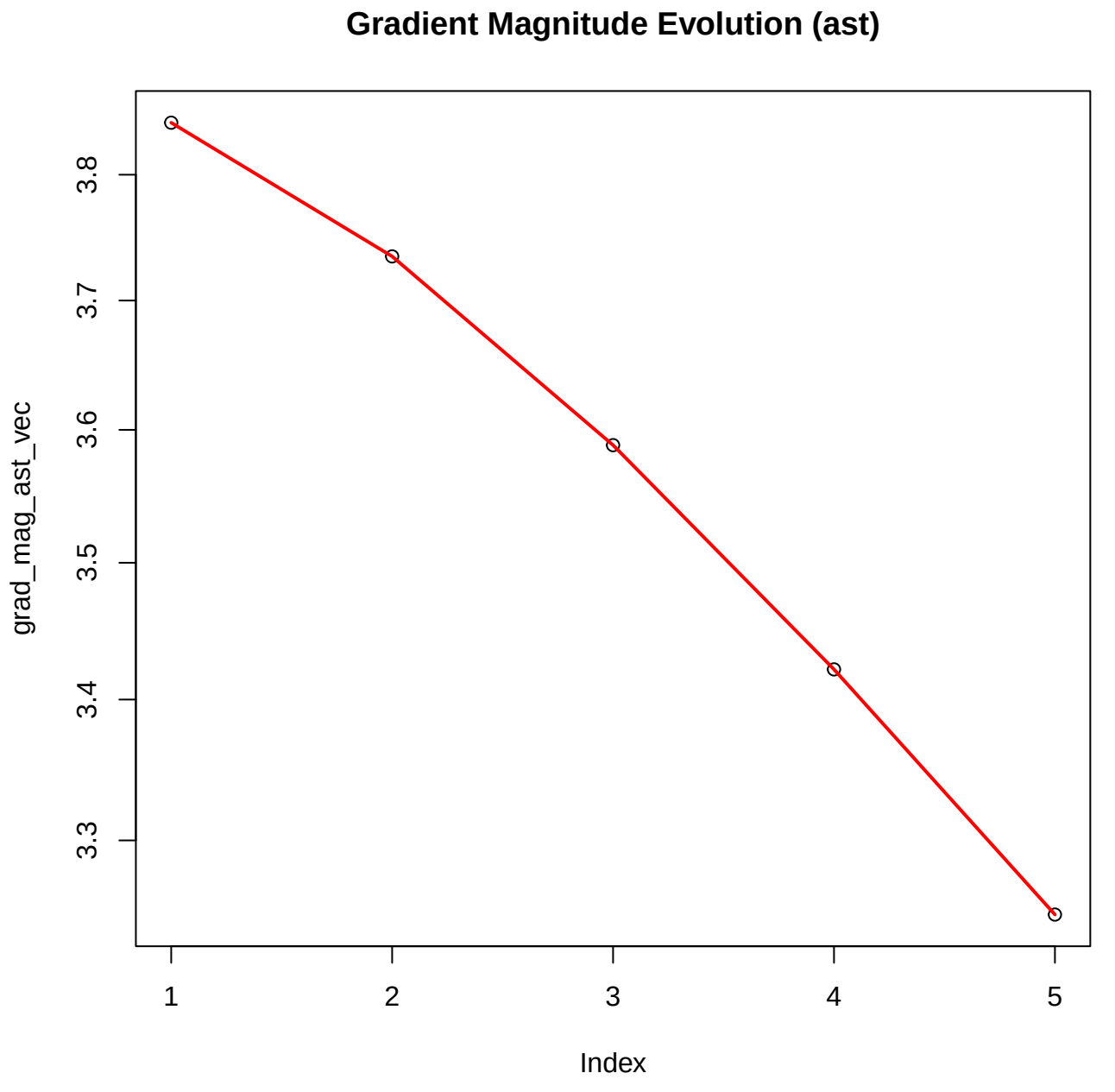

Value (ast)
loss_ast_vec
4.0
3.5
3.0
2.5
2.0
1.5
1
2
3
4
5
Index

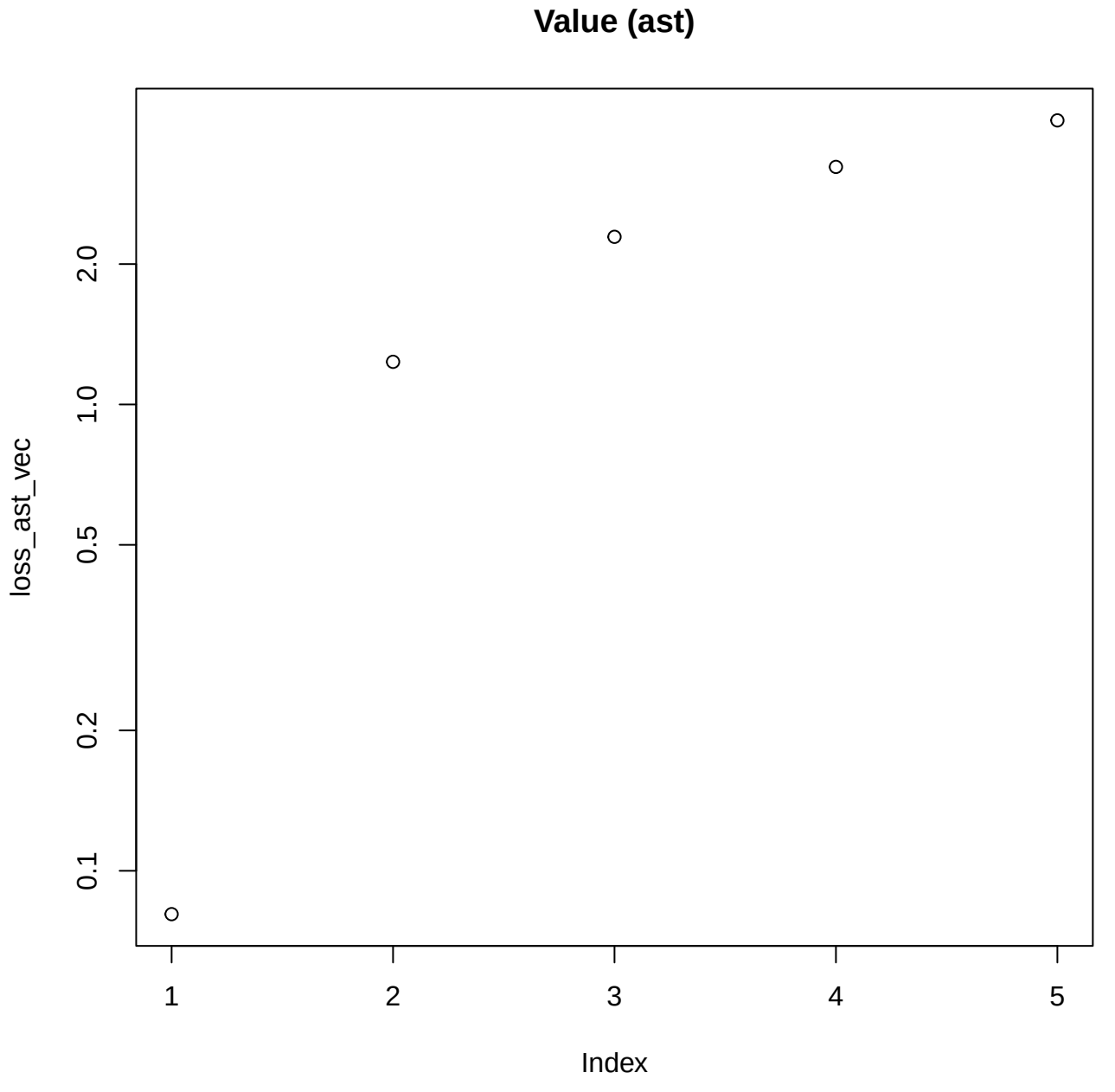

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
3
4
5
6
0.6
0.8
1.0
1.2
1.4
Index

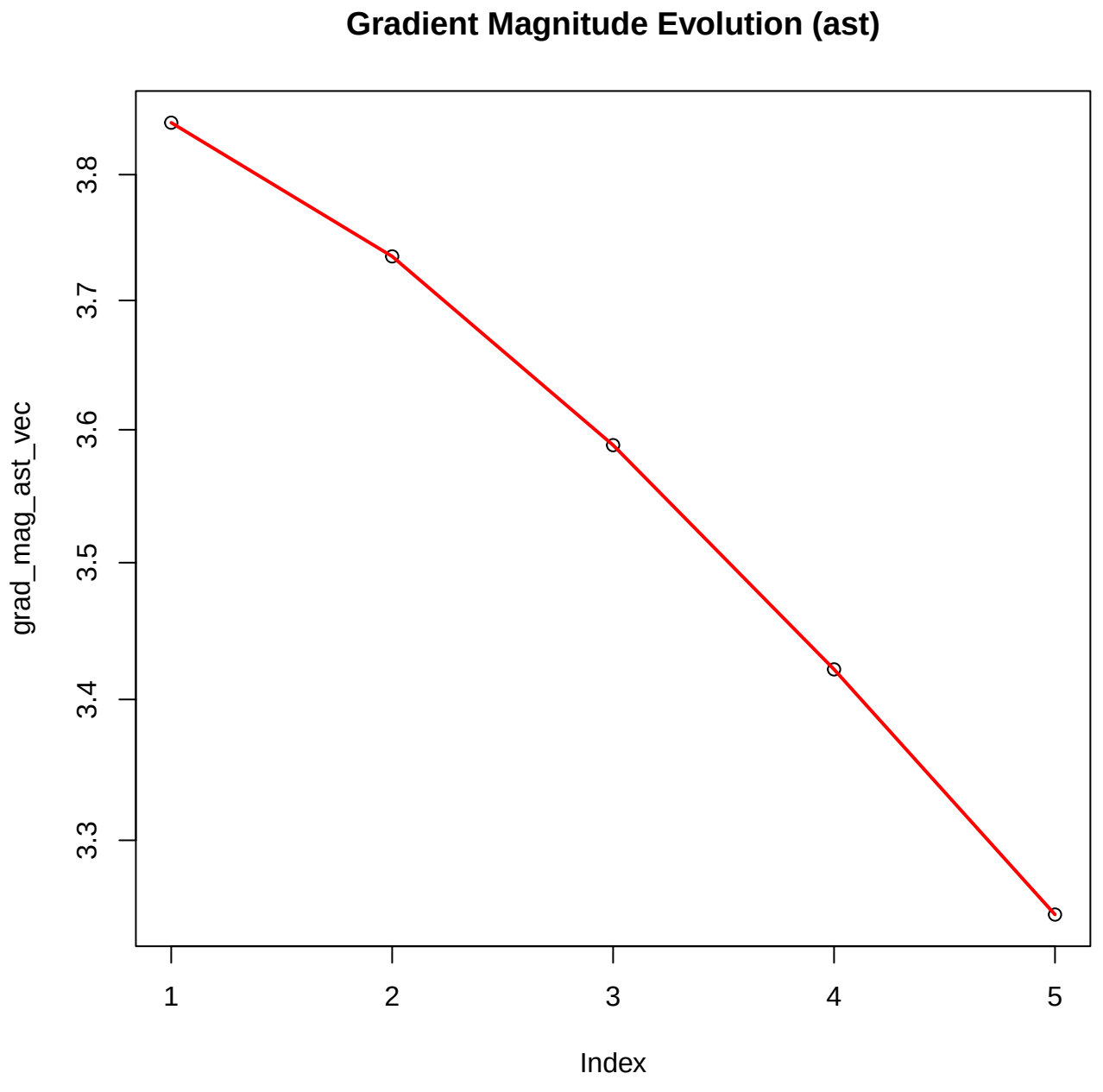

Value (ast)
loss_ast_vec
0.15
0.20
0.25
0.30
0.35
0.40
0.6
0.8
1.0
1.2
1.4
Index

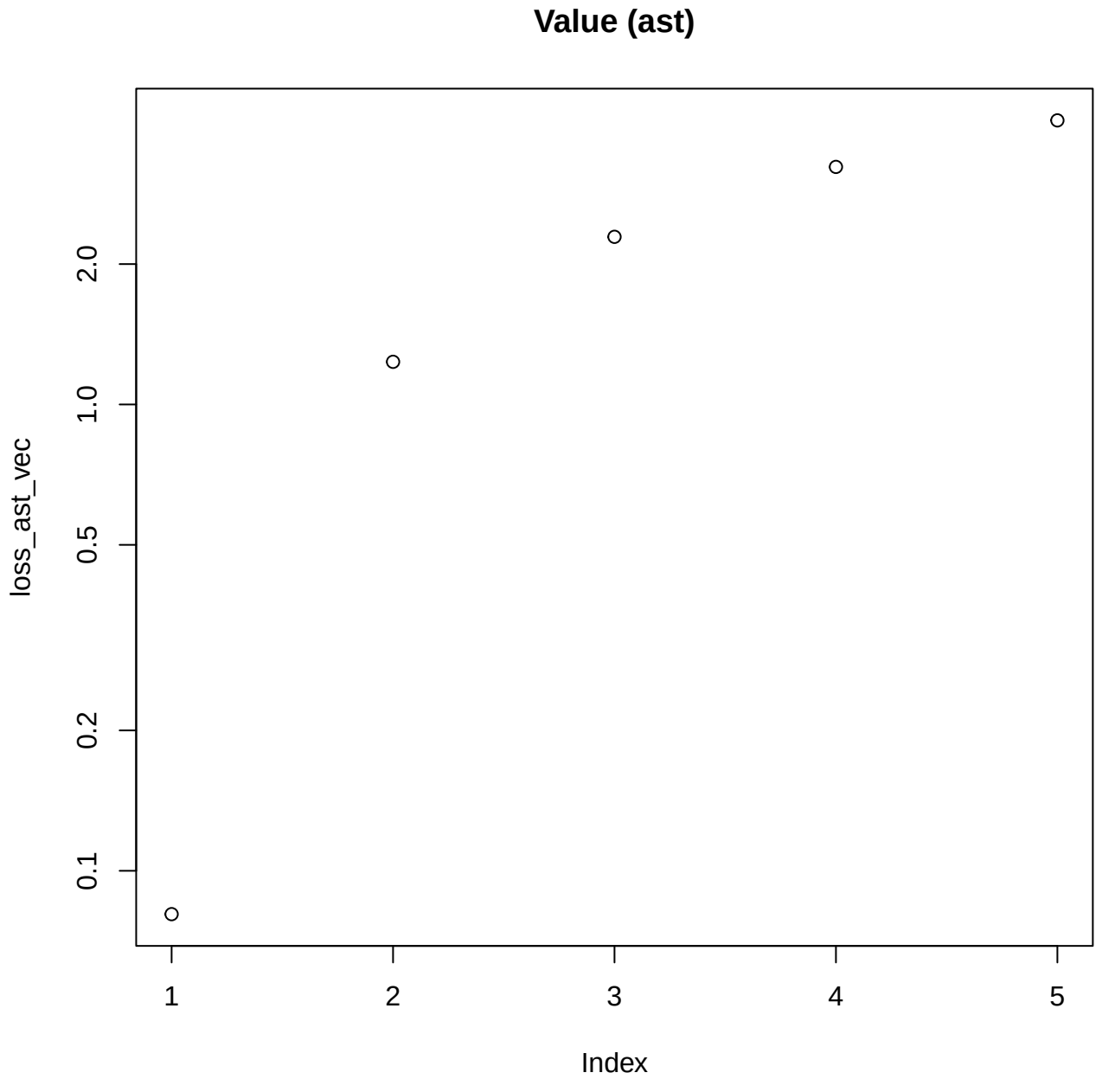

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
3
4
5
6
0.6
0.8
1.0
1.2
1.4
Index

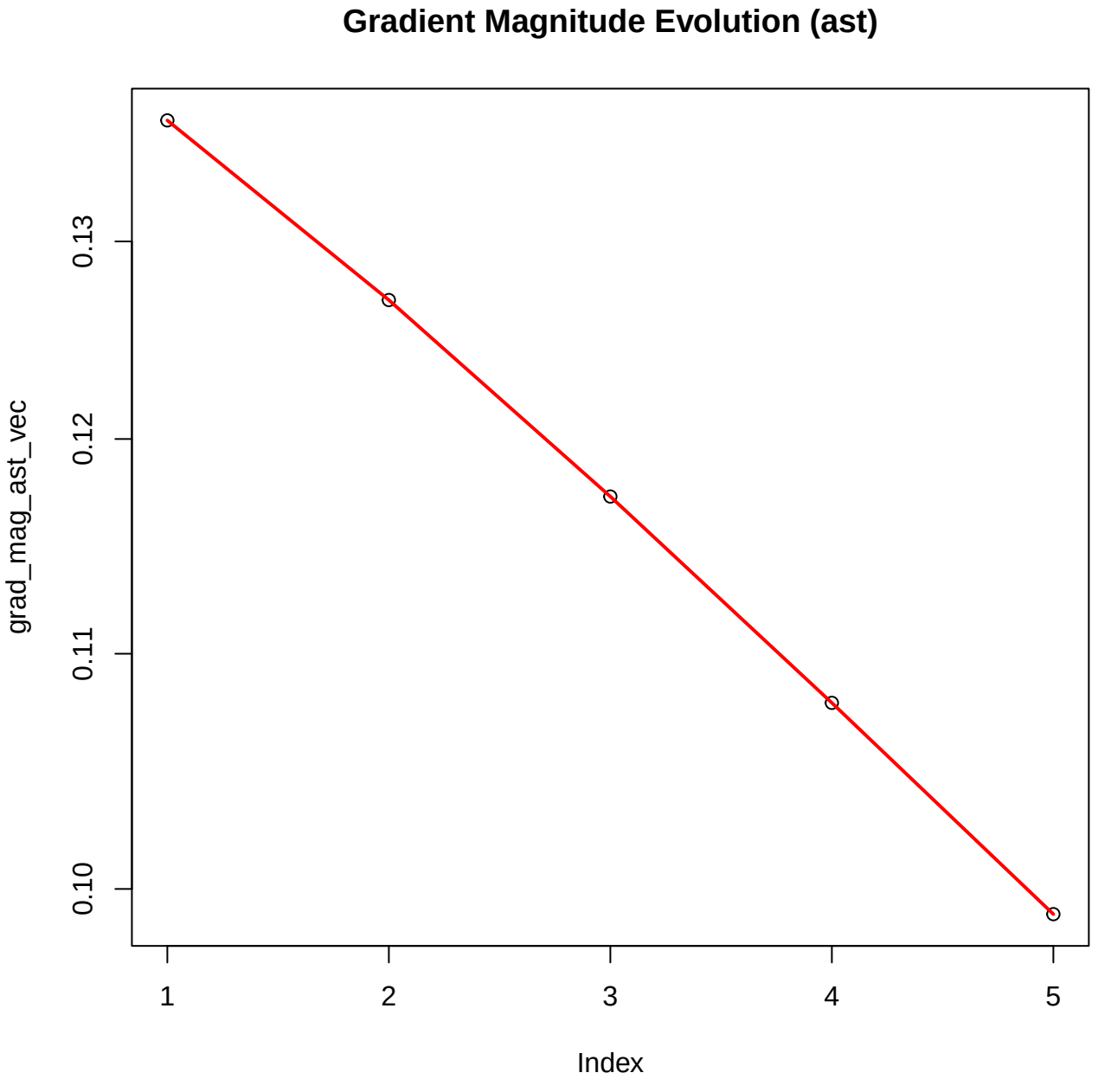

Value (ast)
loss_ast_vec
0.15
0.20
0.25
0.30
0.35
0.40
0.6
0.8
1.0
1.2
1.4
Index

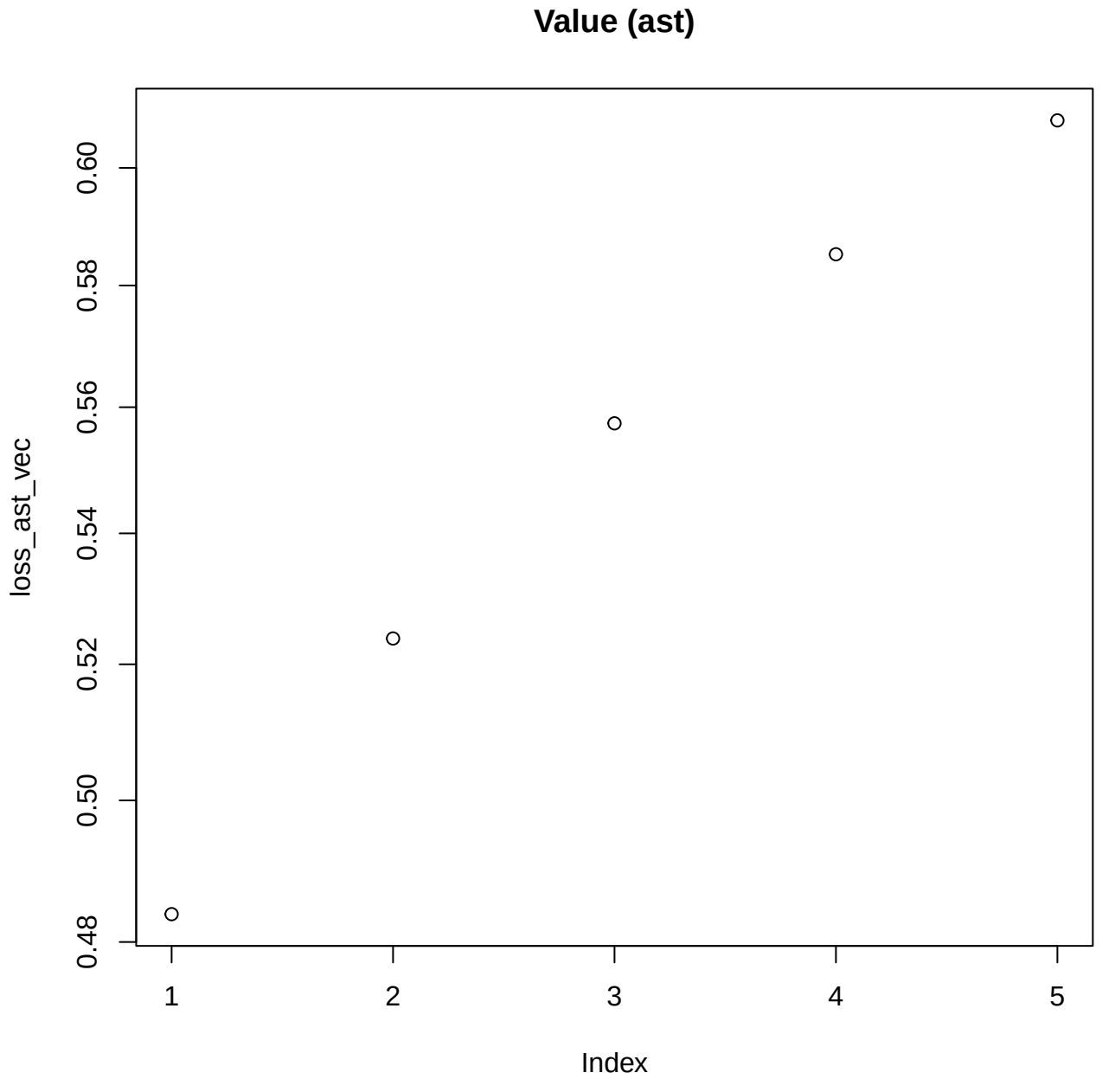

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
3.3
3.4
3.5
3.6
3.7
3.8
1
2
3
4
5
Index

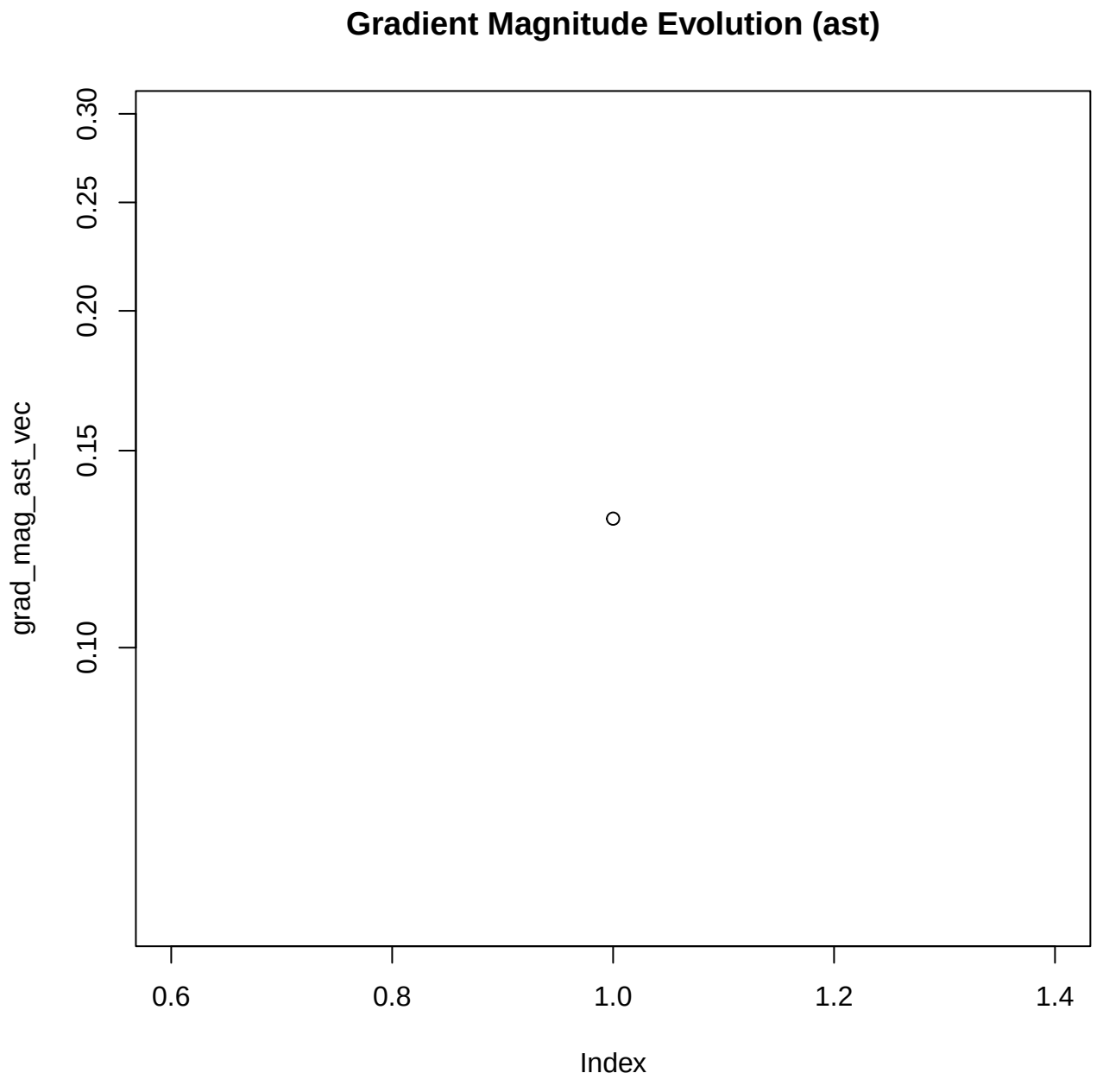

Value (ast)
loss_ast_vec
0.1
0.2
0.5
1.0
2.0
1
2
3
4
5
Index

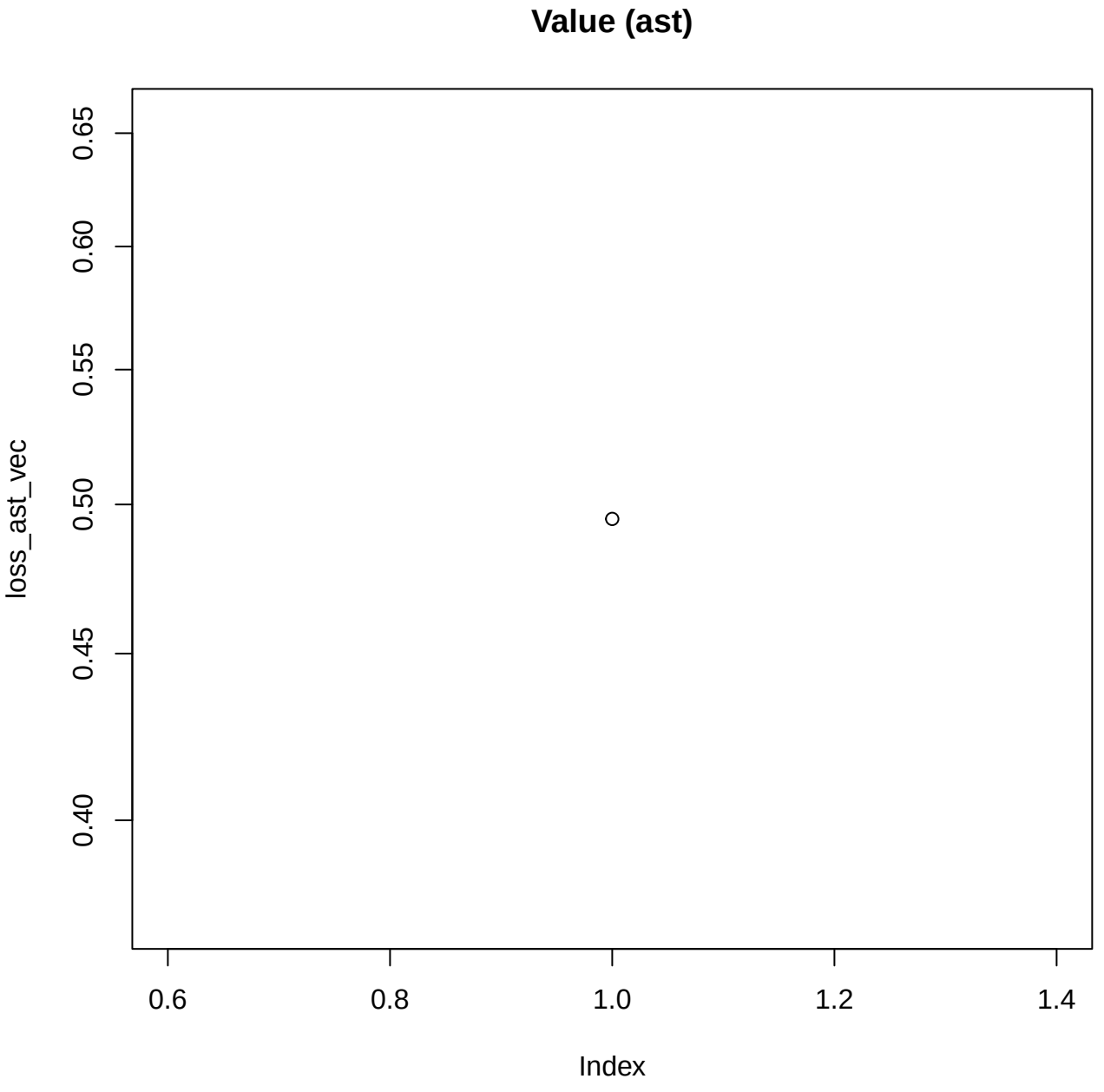

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
3.3
3.4
3.5
3.6
3.7
3.8
1
2
3
4
5
Index

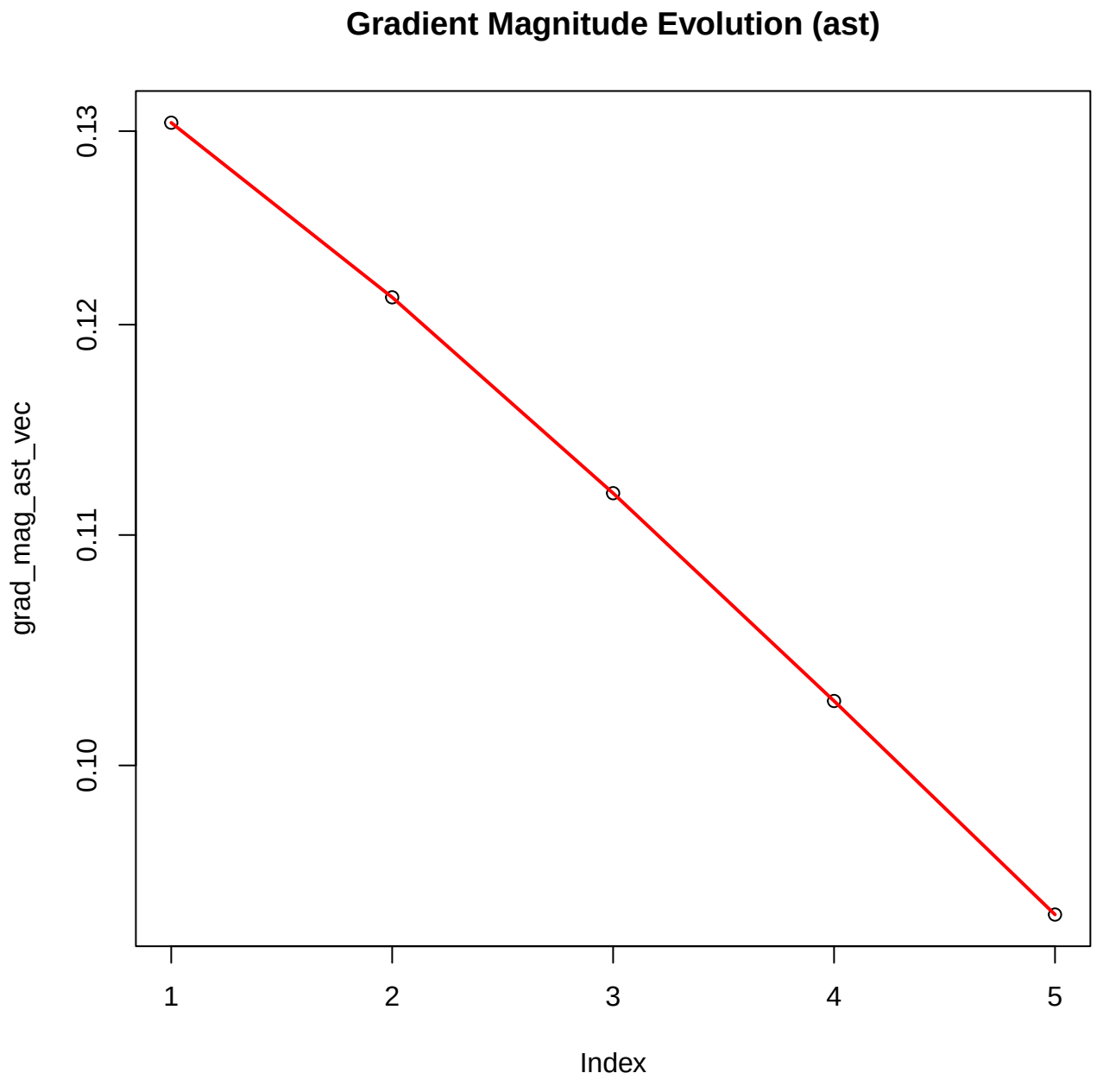

Value (ast)
loss_ast_vec
0.1
0.2
0.5
1.0
2.0
1
2
3
4
5
Index

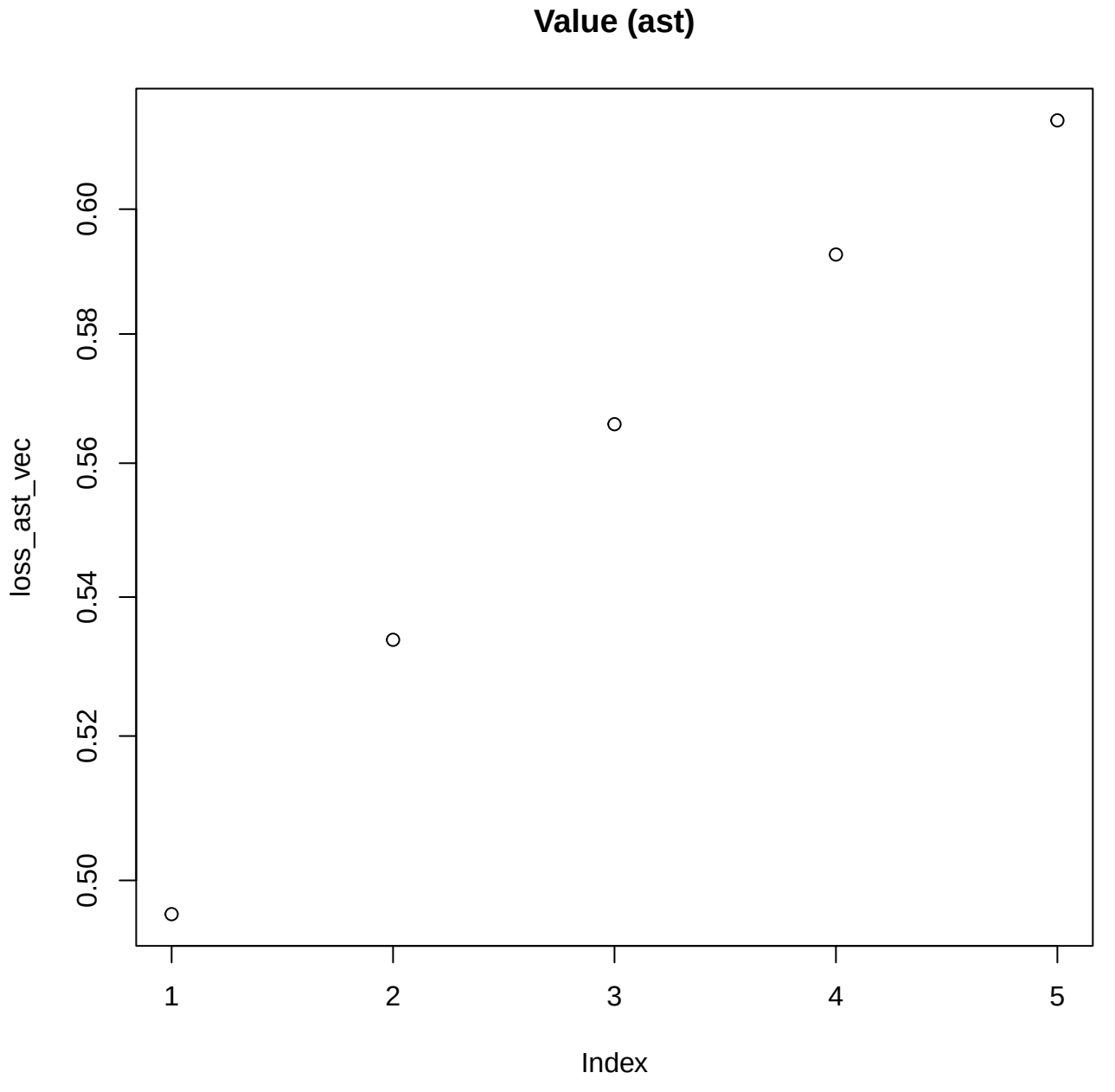

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index

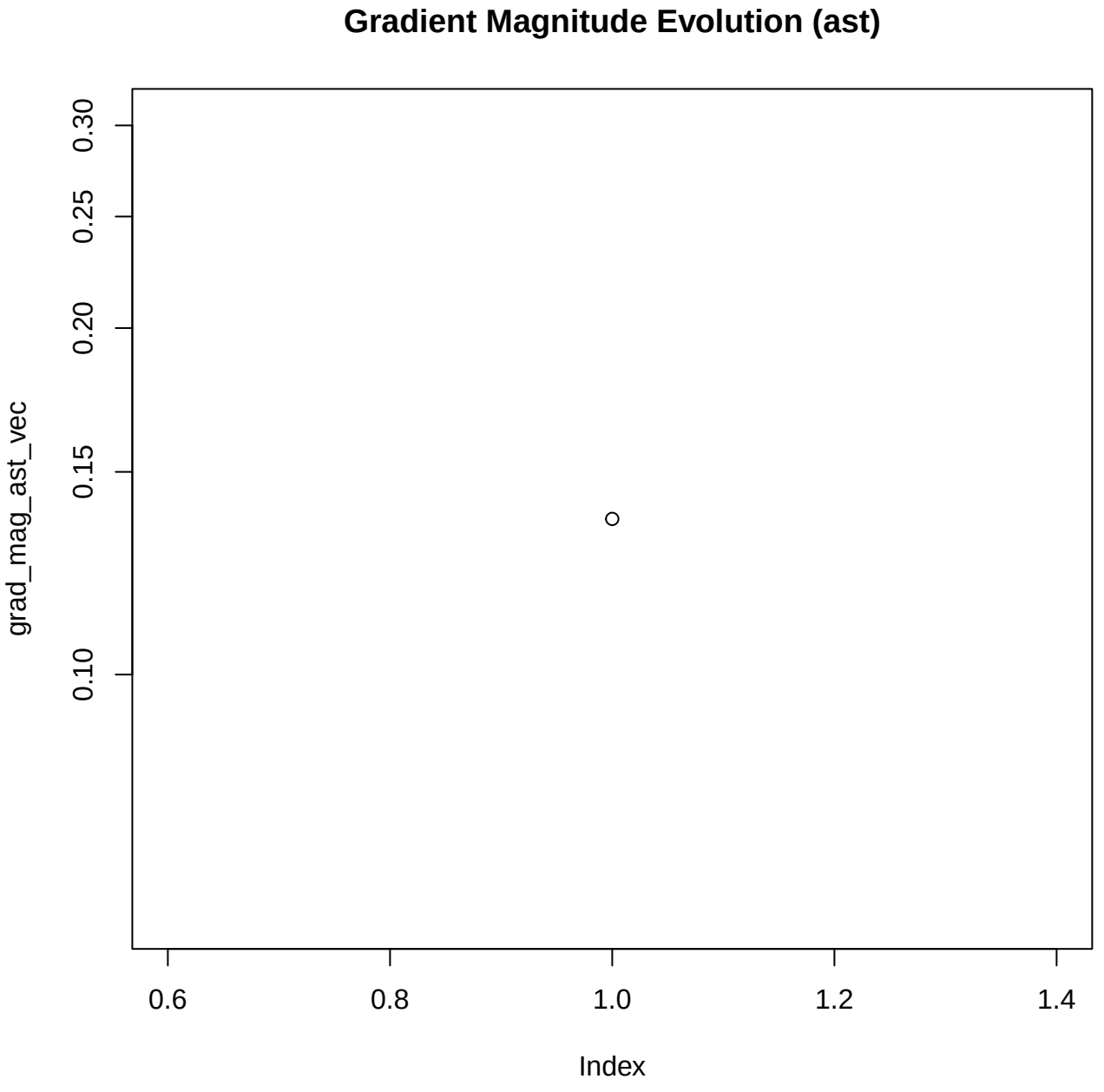

Value (ast)
loss_ast_vec
0.48
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index

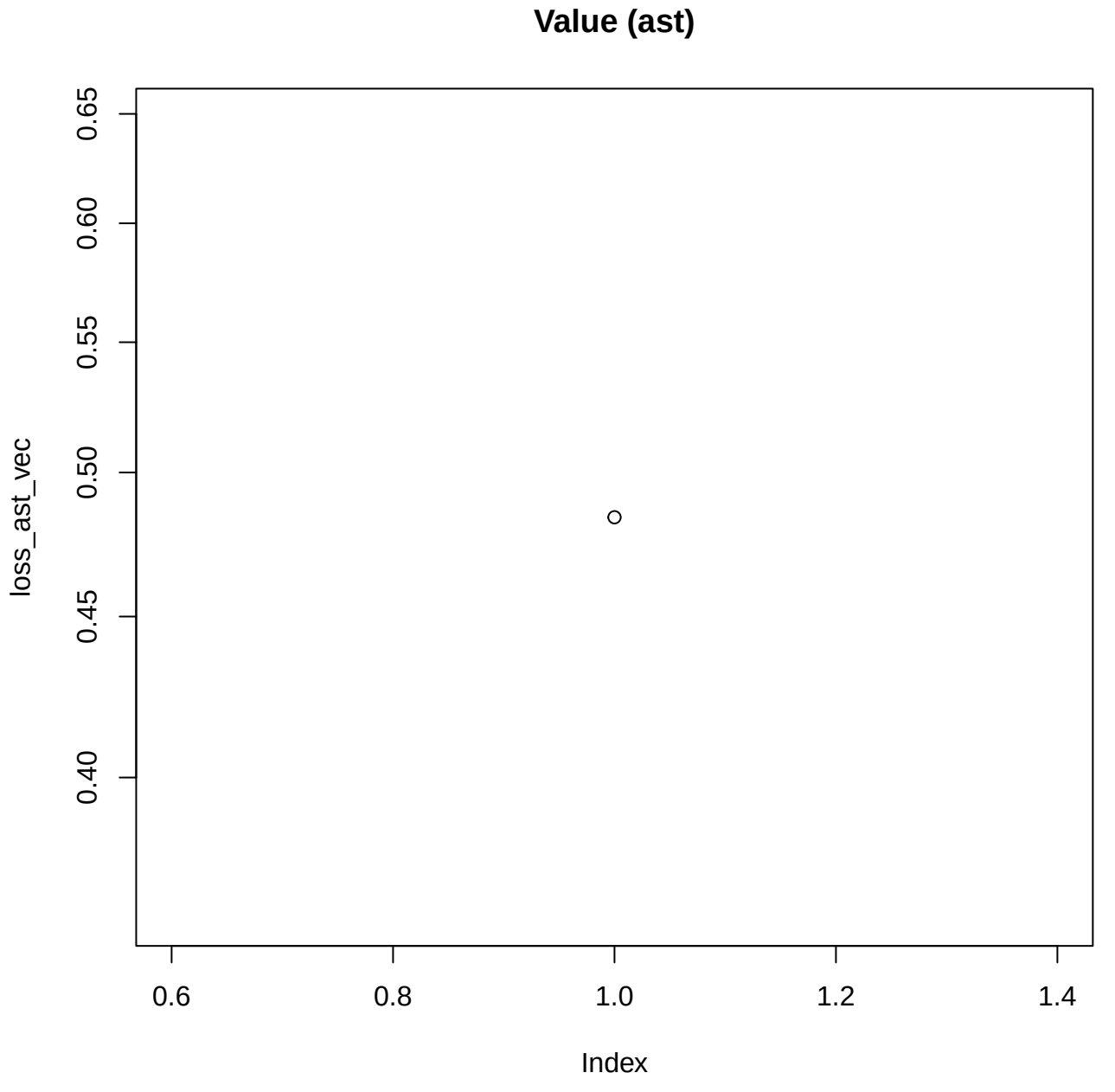

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.15
0.20
0.25
0.30
0.6
0.8
1.0
1.2
1.4
Index

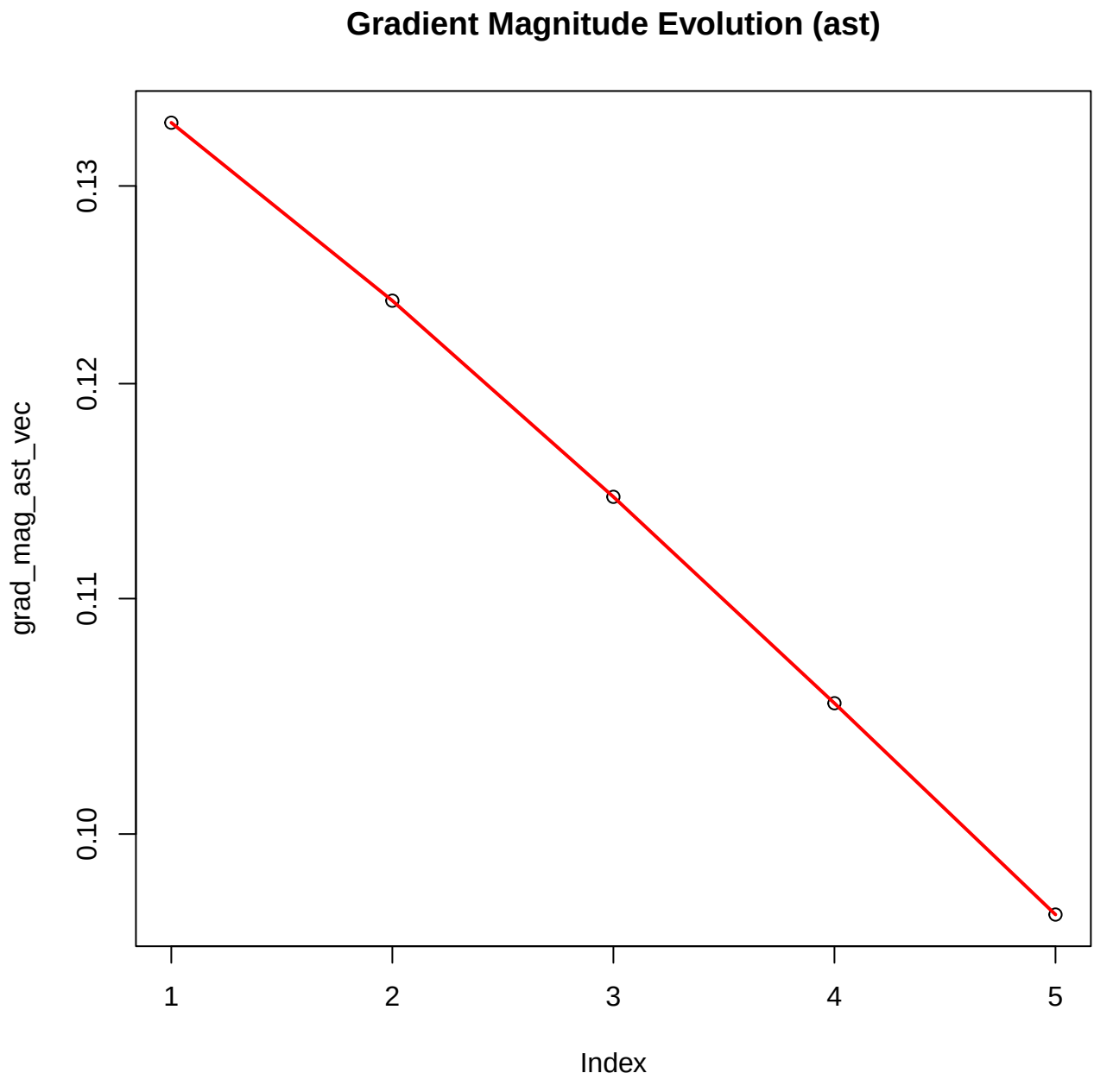

Value (ast)
loss_ast_vec
0.65
0.60
0.55
0.50
0.45
0.40
0.6
0.8
1.0
1.2
1.4
Index

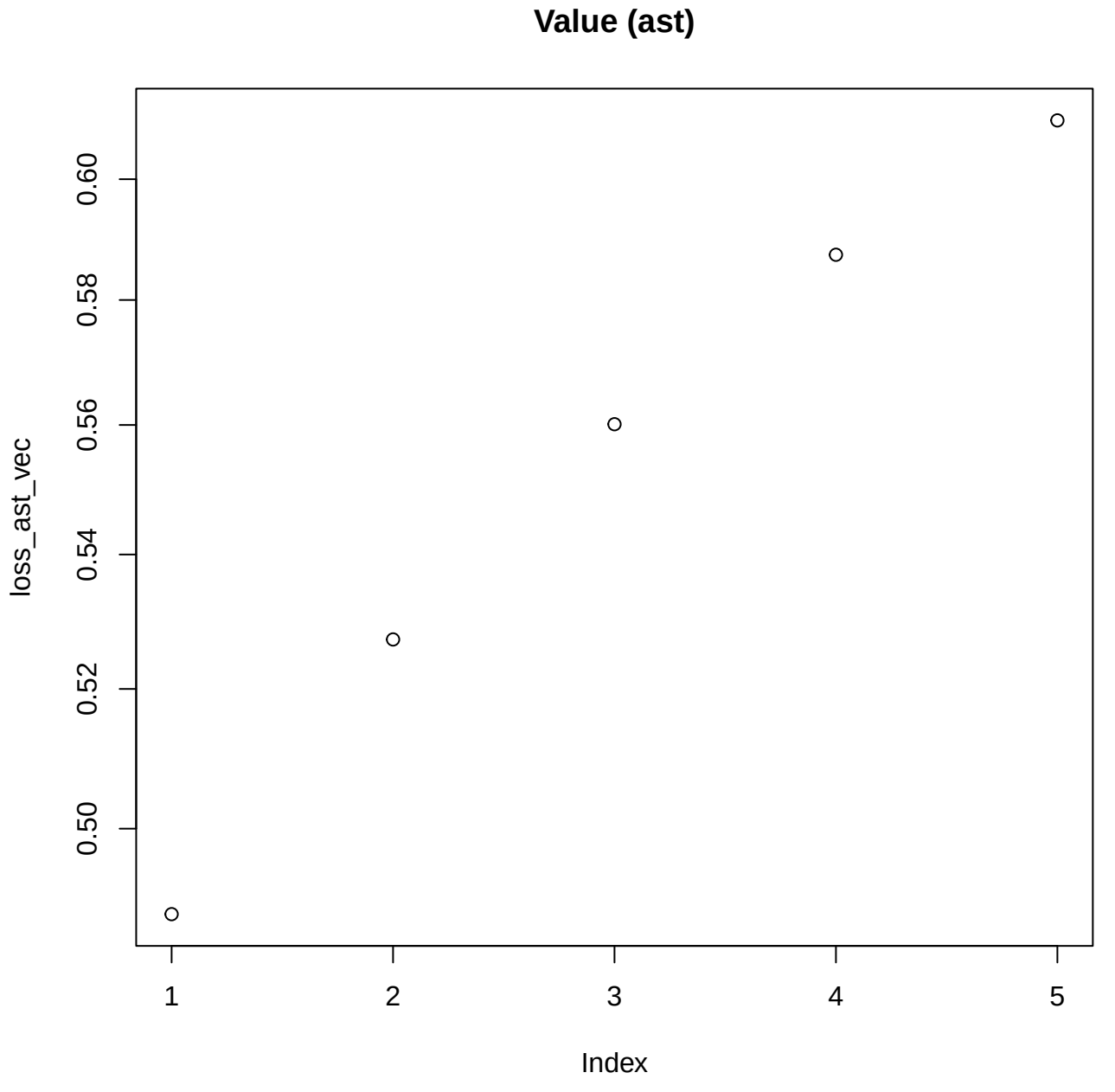

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index

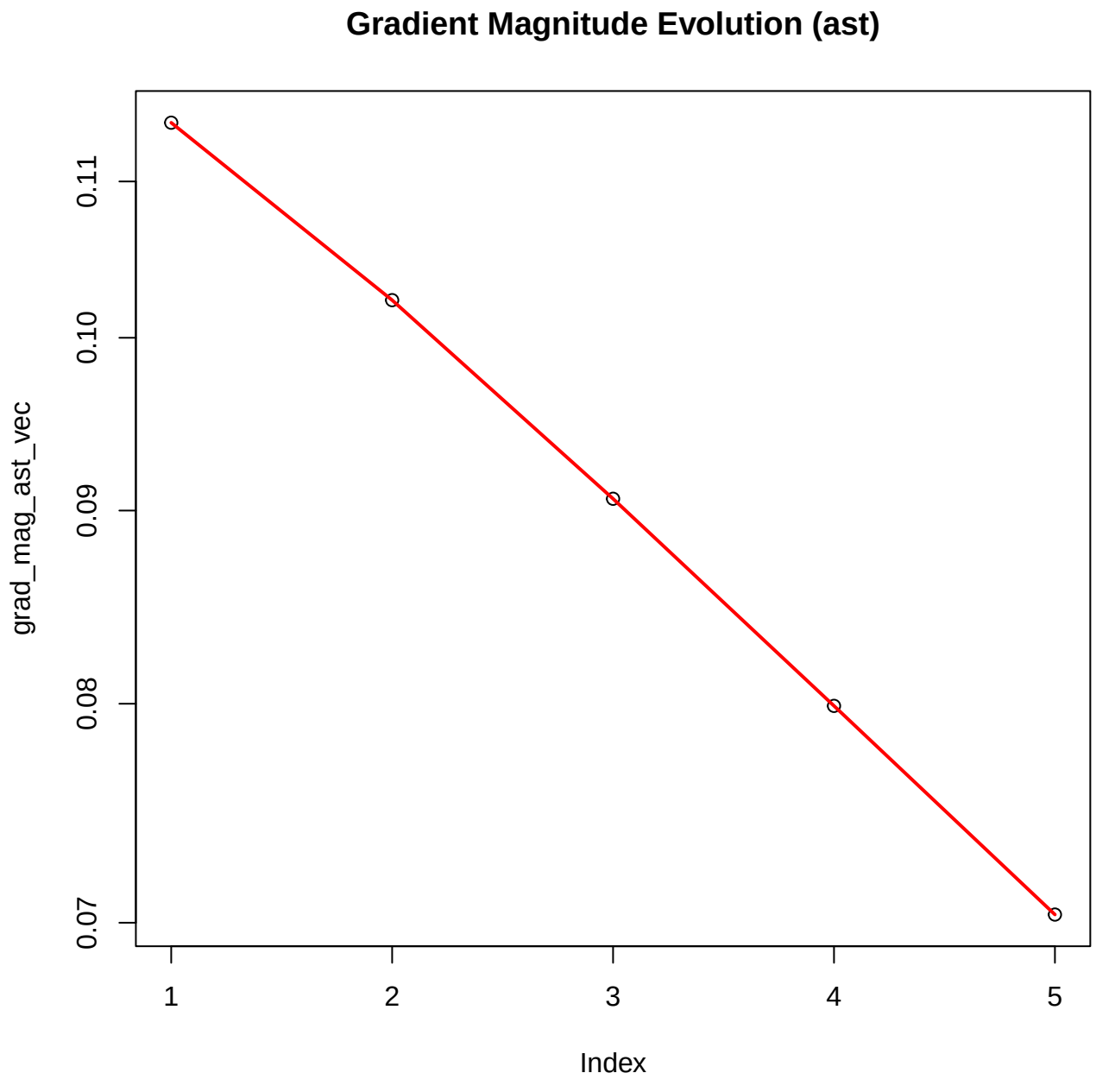

Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index

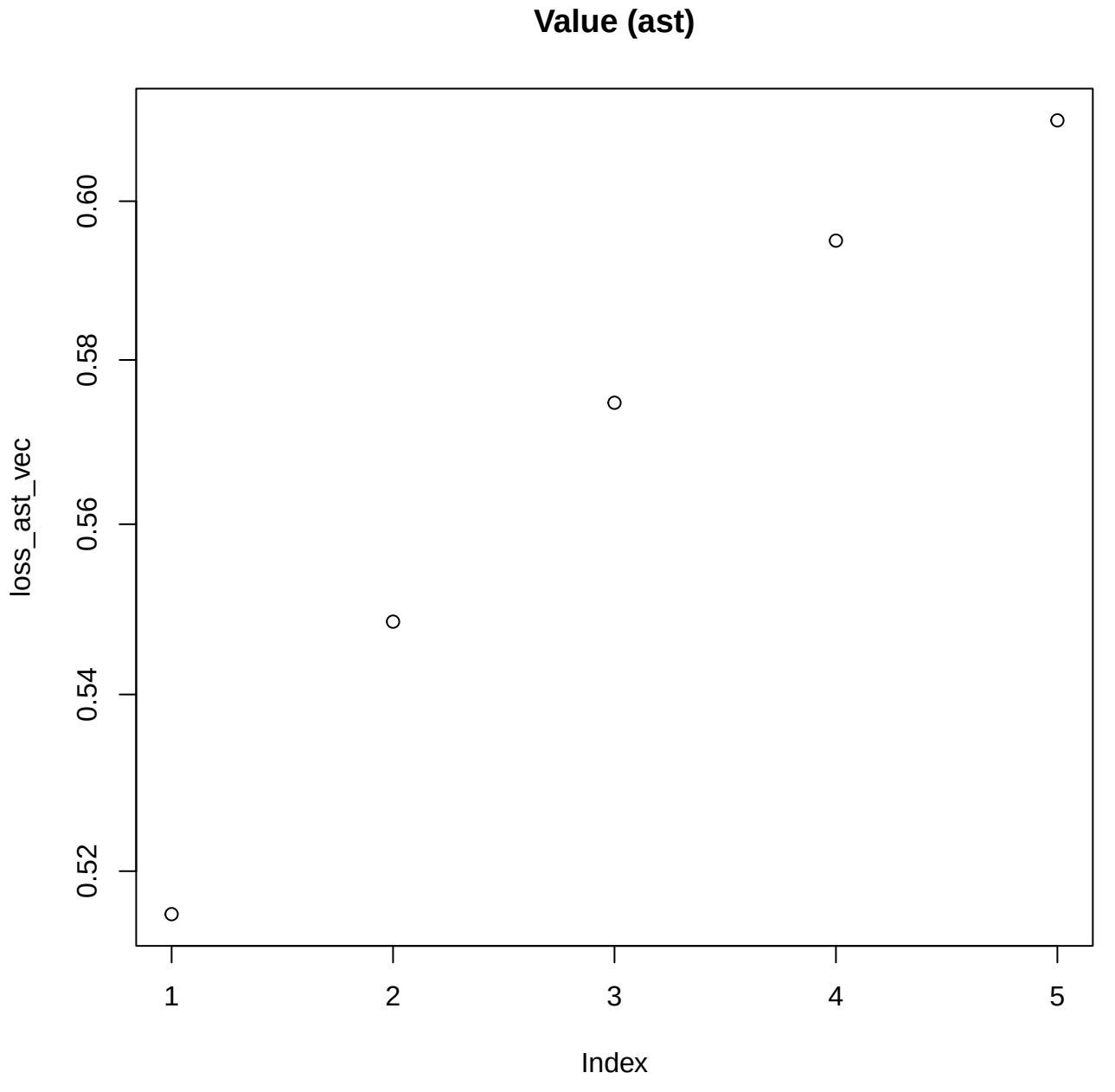

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.15
0.20
0.25
0.30
0.6
0.8
1.0
1.2
1.4
Index

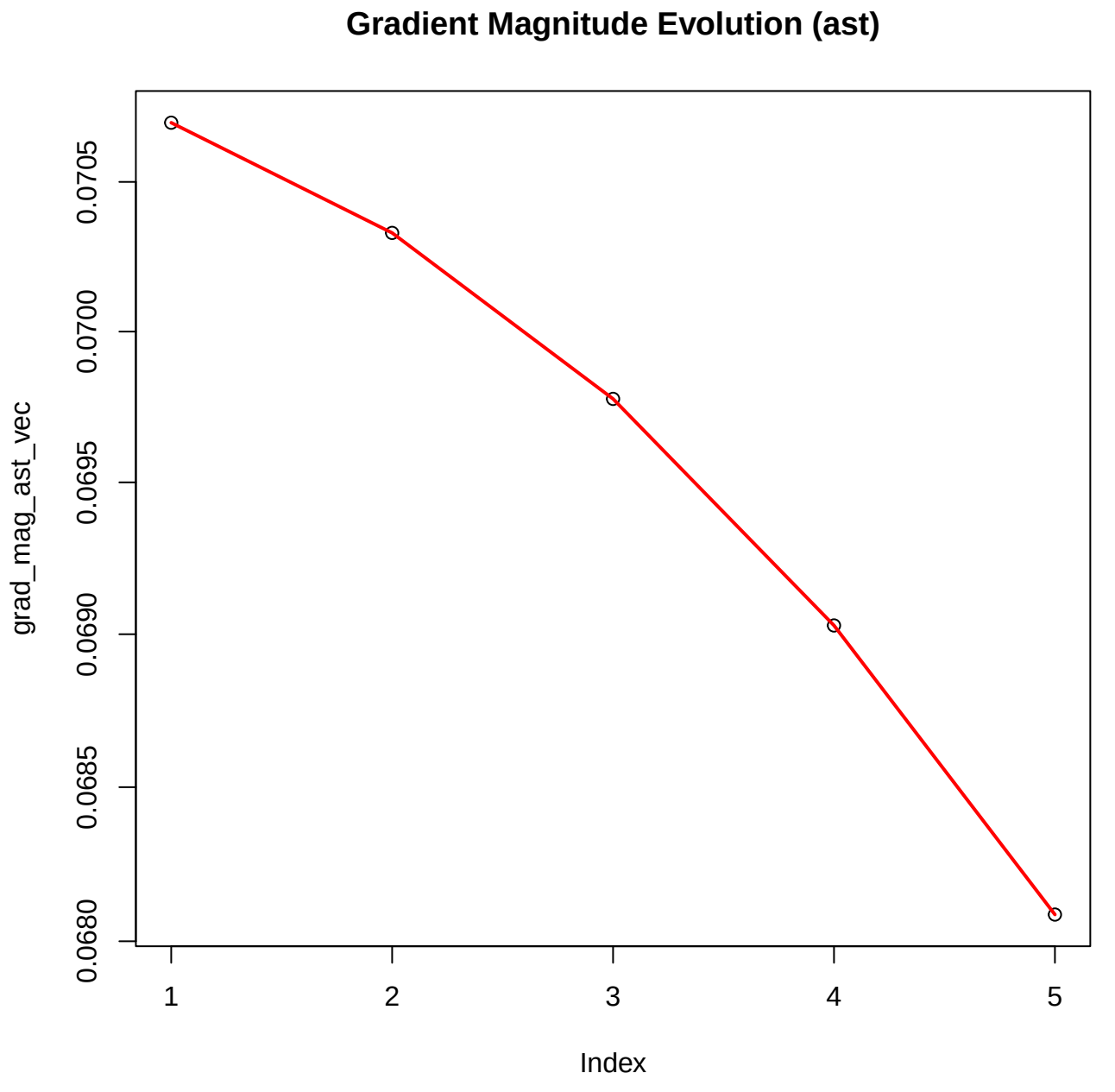

Value (ast)
loss_ast_vec
0.65
0.60
0.55
0.50
0.45
0.40
0.6
0.8
1.0
1.2
1.4
Index

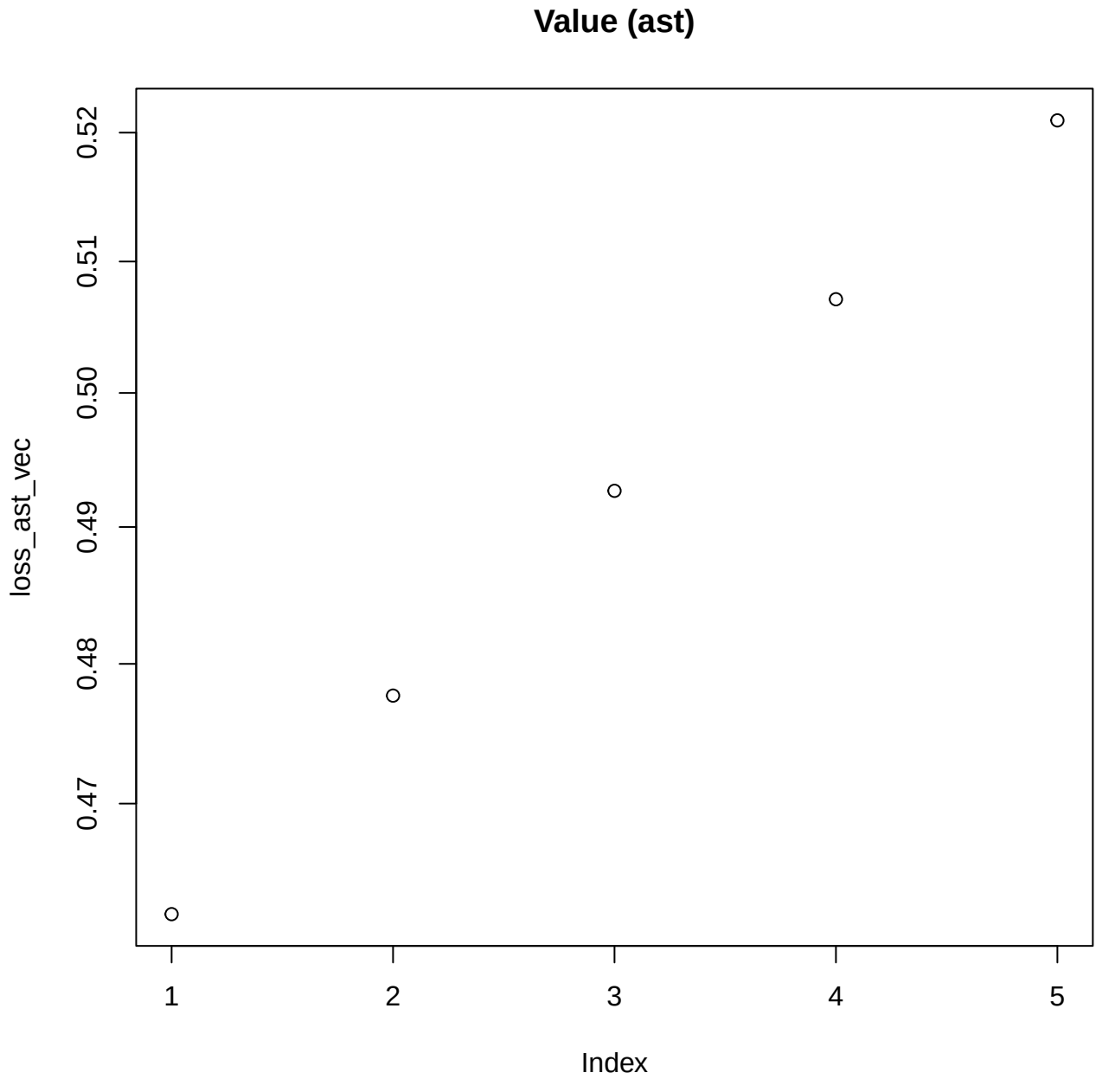

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index

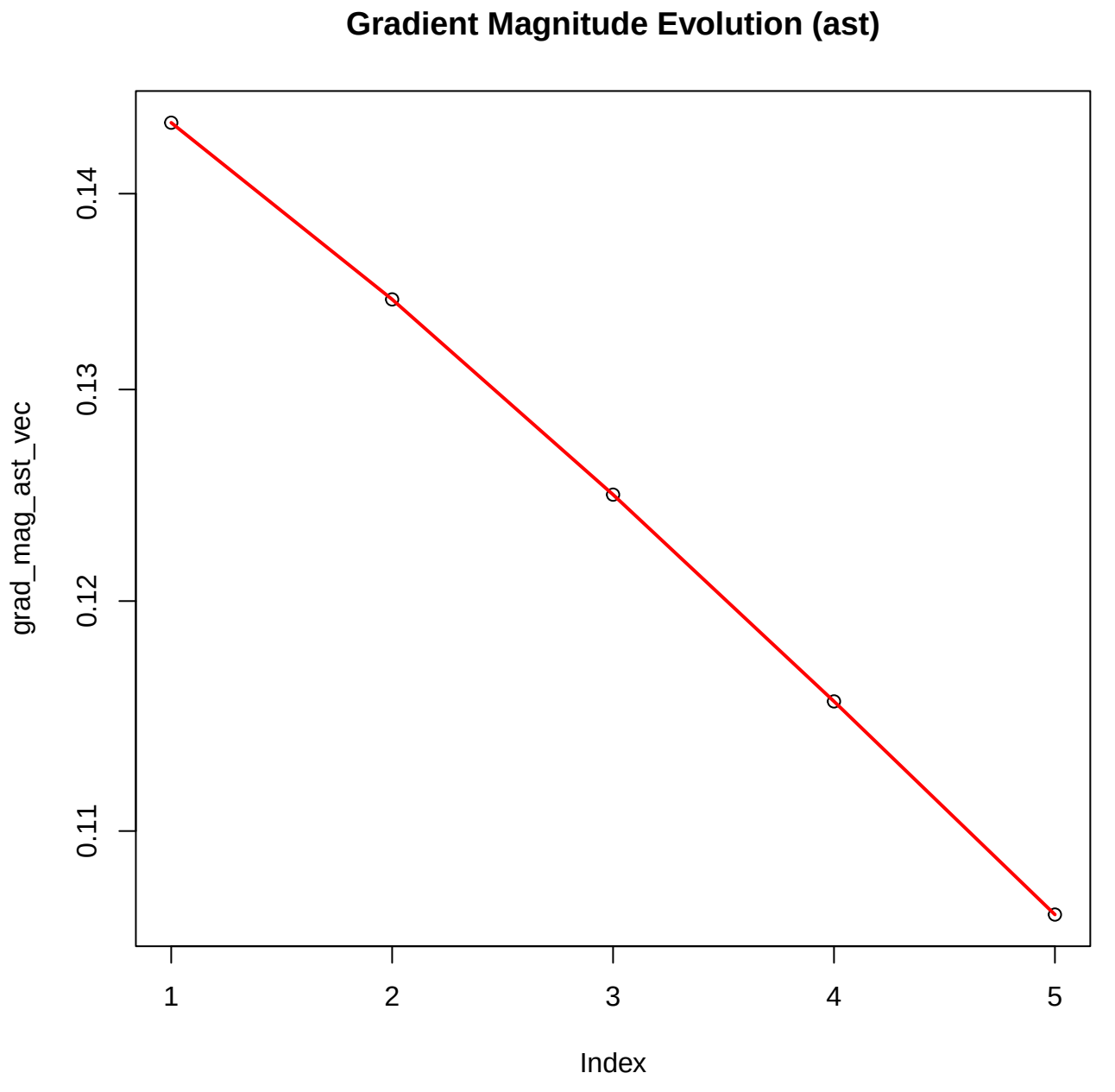

Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index

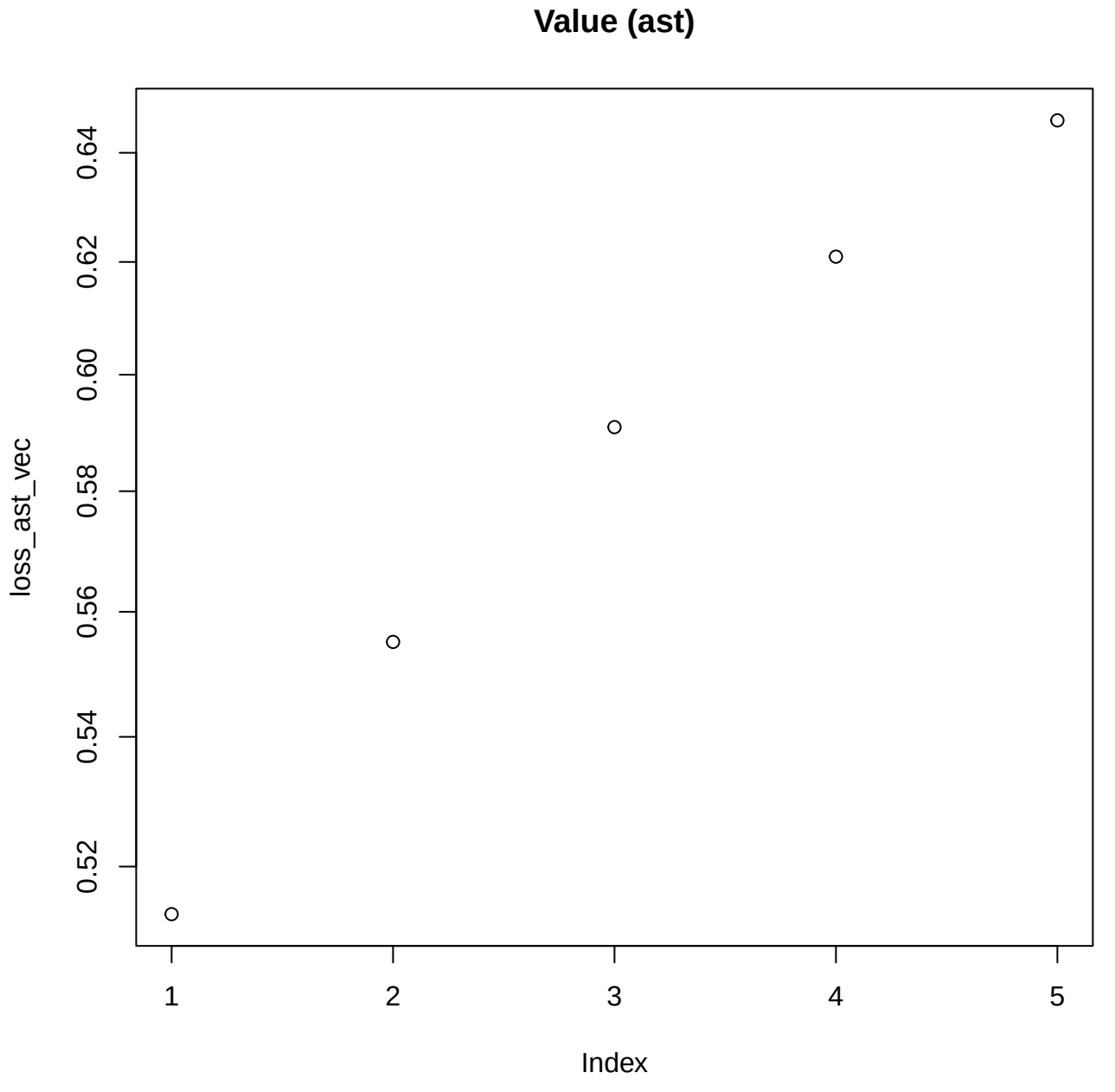

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.07
0.08
0.09
0.10
0.11
1
2
3
4
5
Index

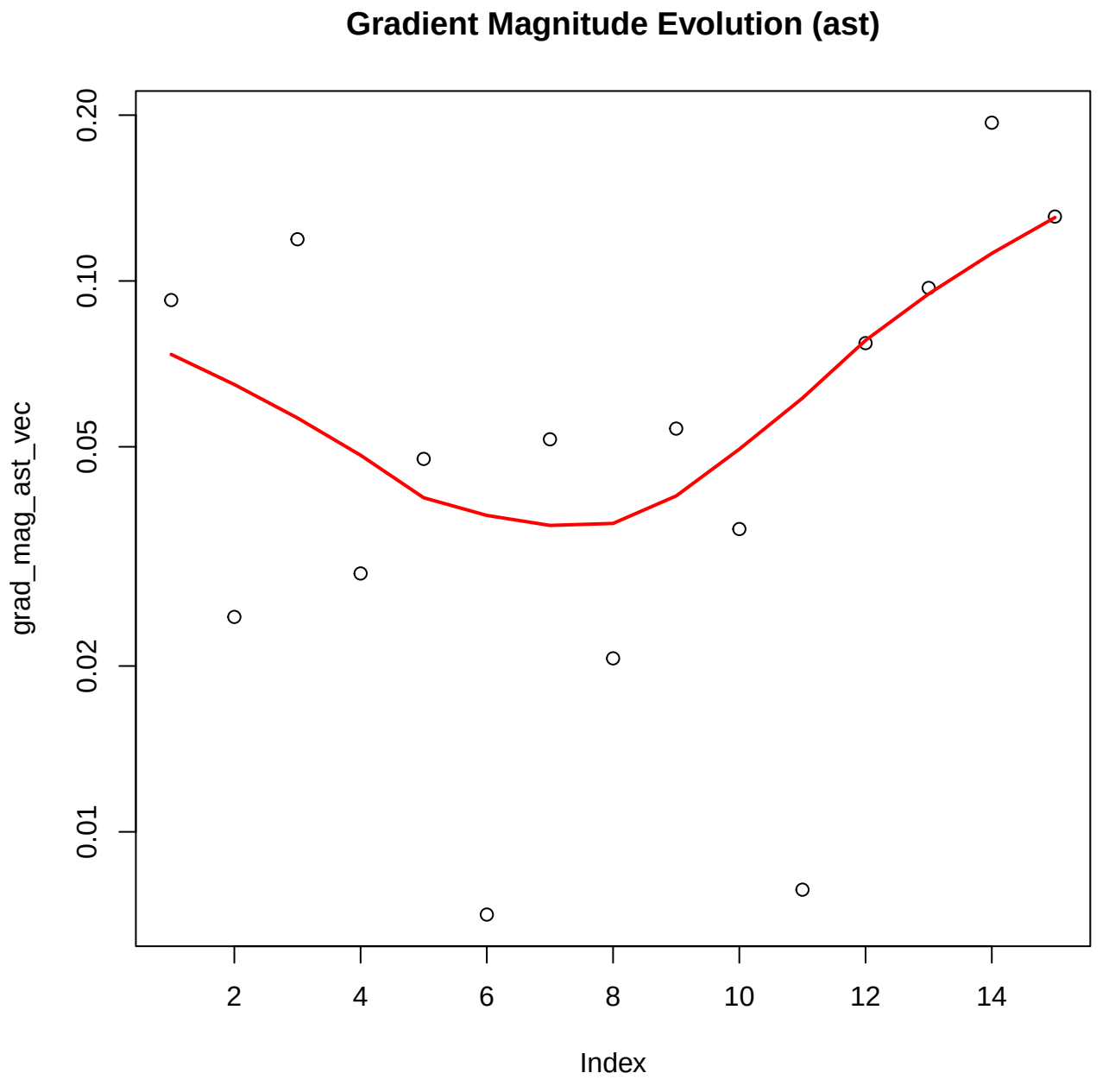

Value (ast)
loss_ast_vec
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index

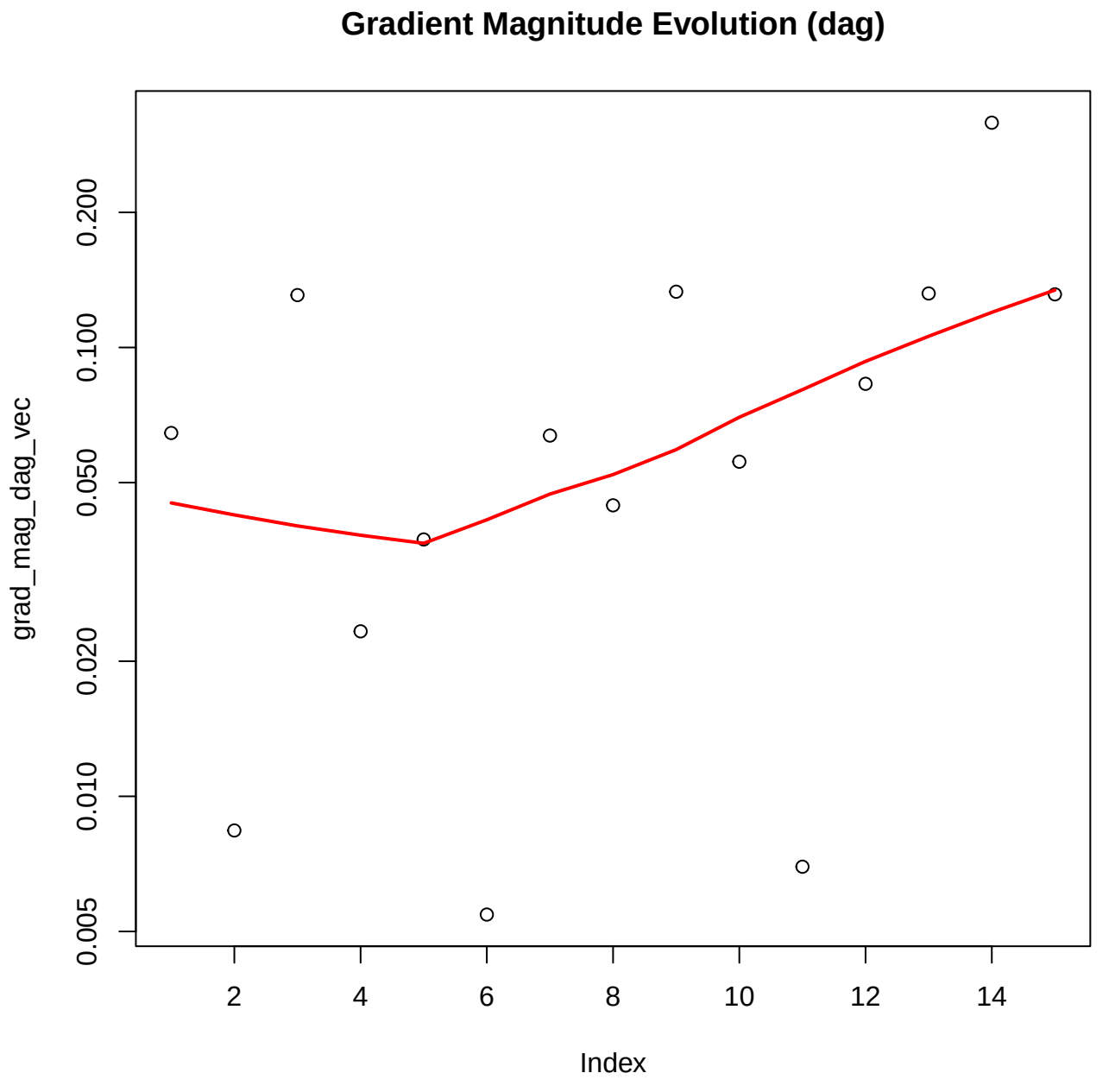

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.0680
0.0685
0.0690
0.0695
0.0700
0.0705
1
2
3
4
5
Index

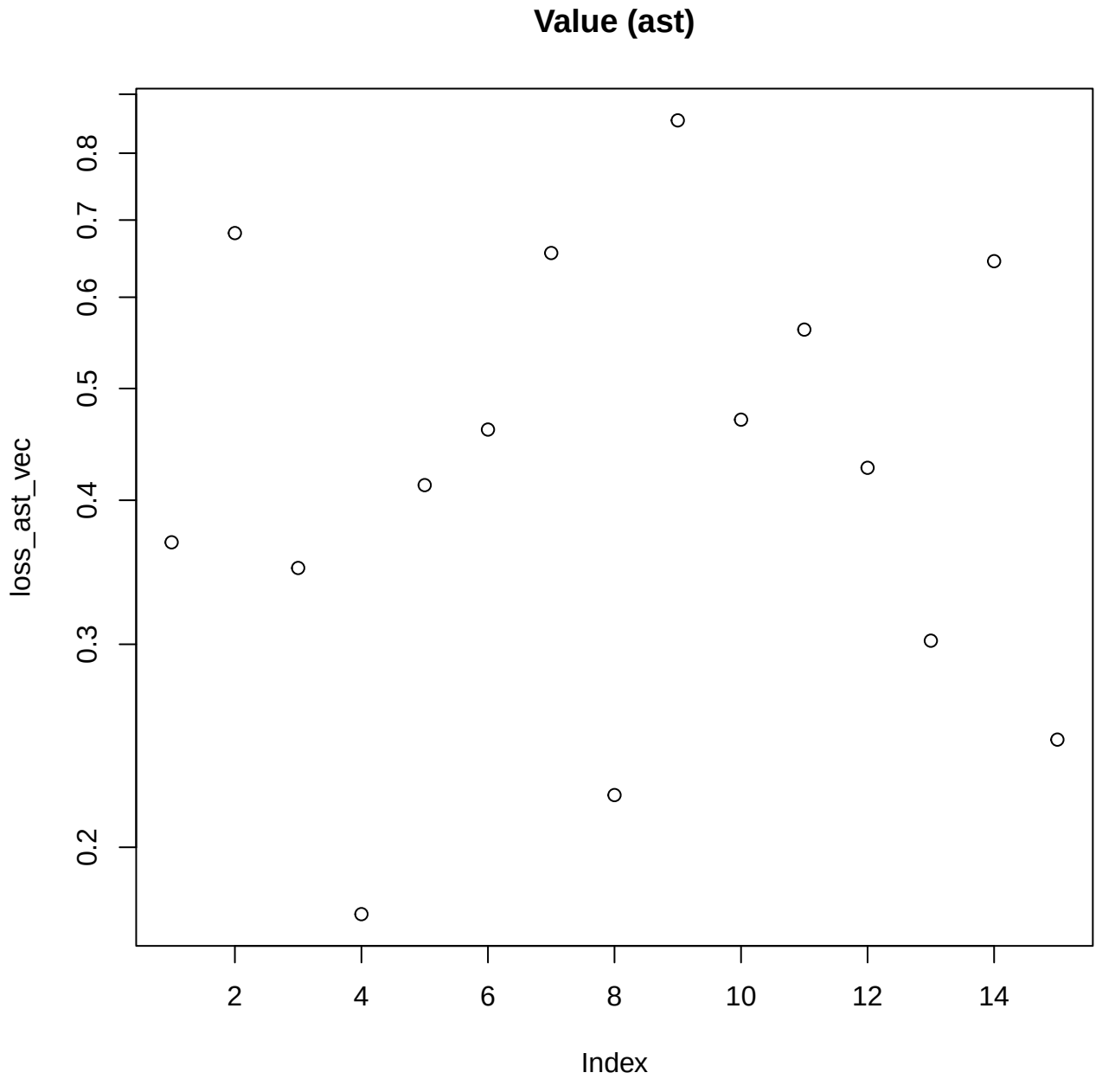

Value (ast)
loss_ast_vec
0.47
0.48
0.49
0.50
0.51
0.52
1
2
3
4
5
Index

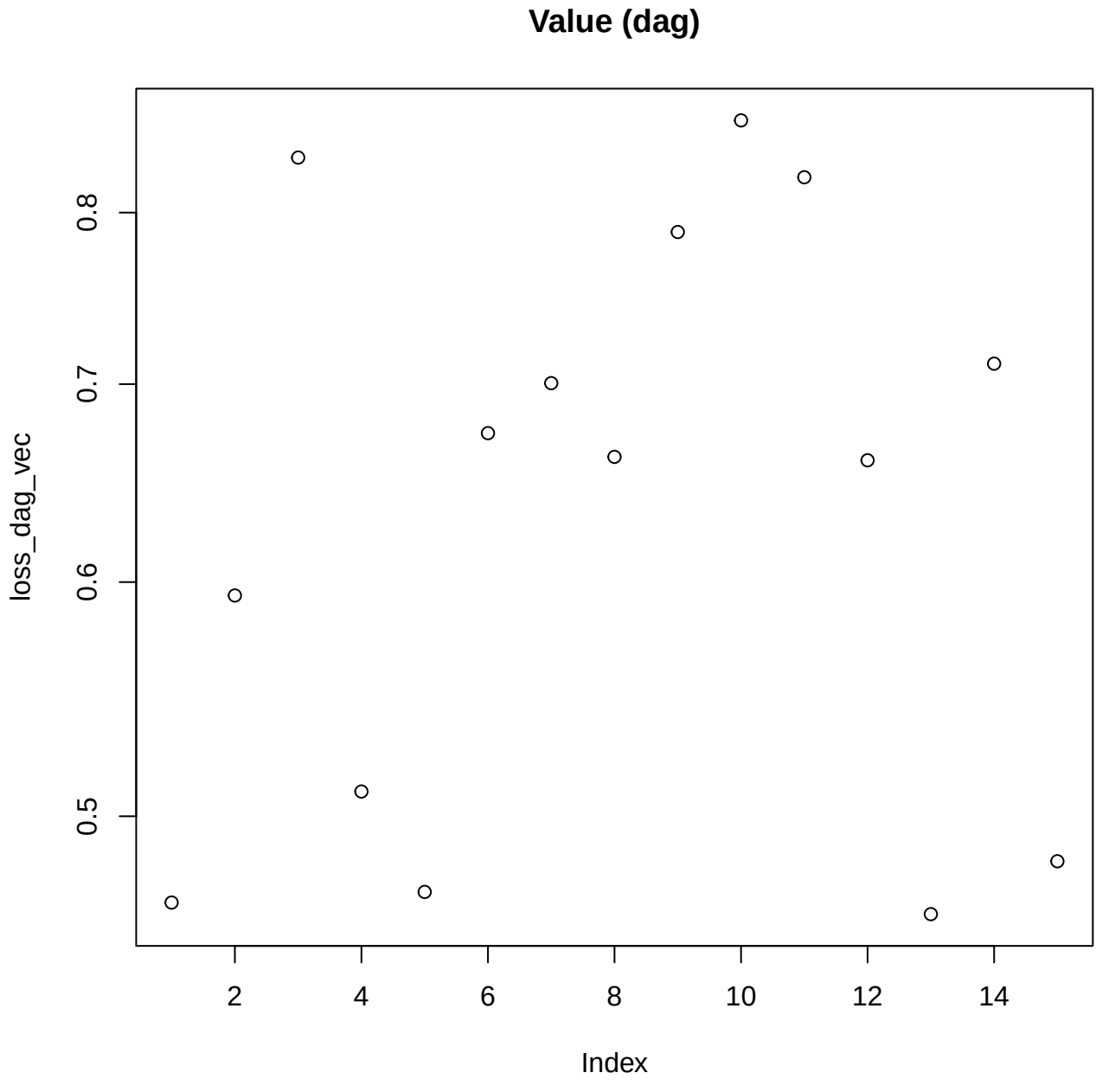

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.11
0.12
0.13
0.14
1
2
3
4
5
Index

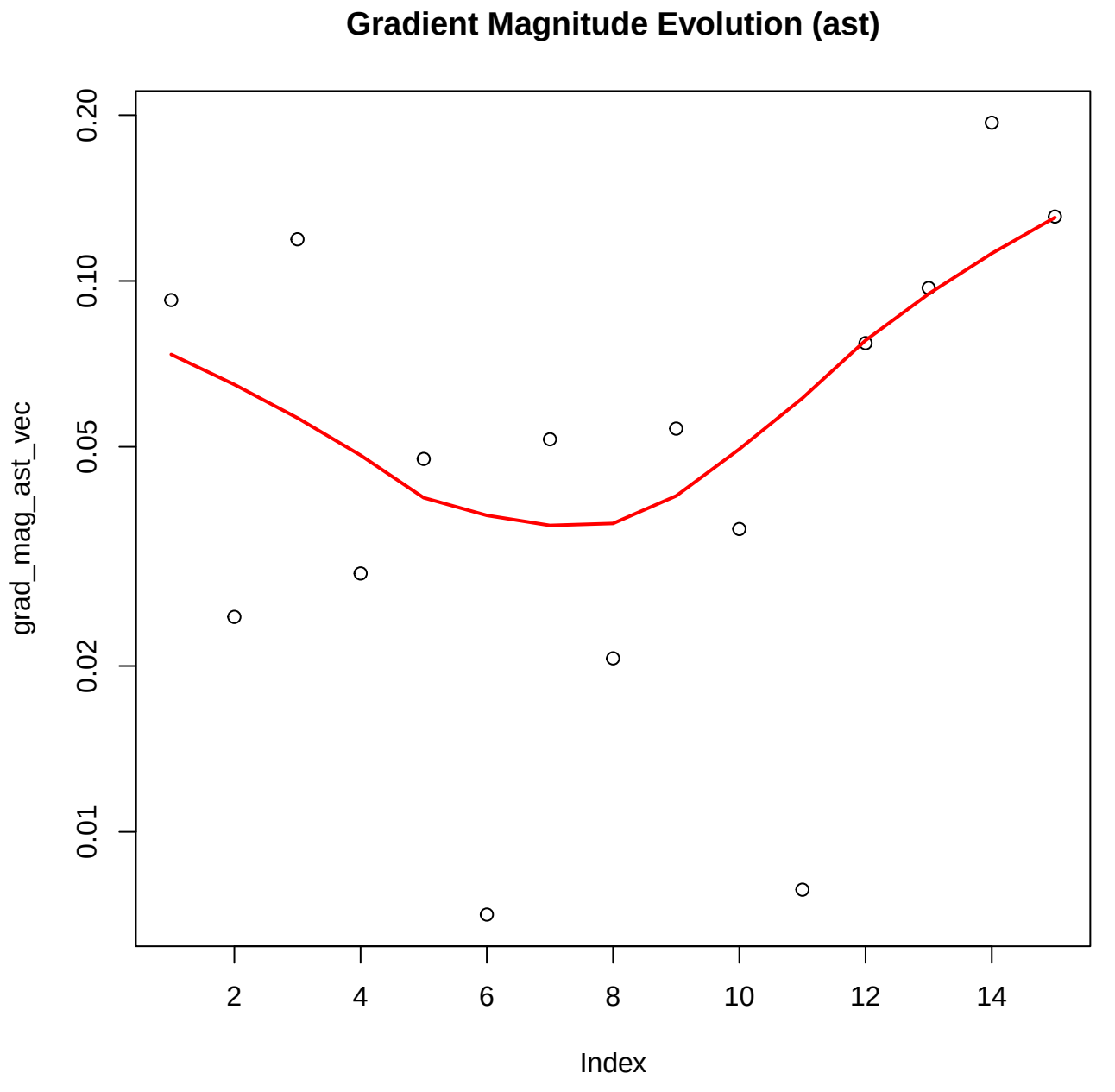

Value (ast)
loss_ast_vec
0.52
0.54
0.56
0.58
0.60
0.62
0.64
1
2
3
4
5
Index

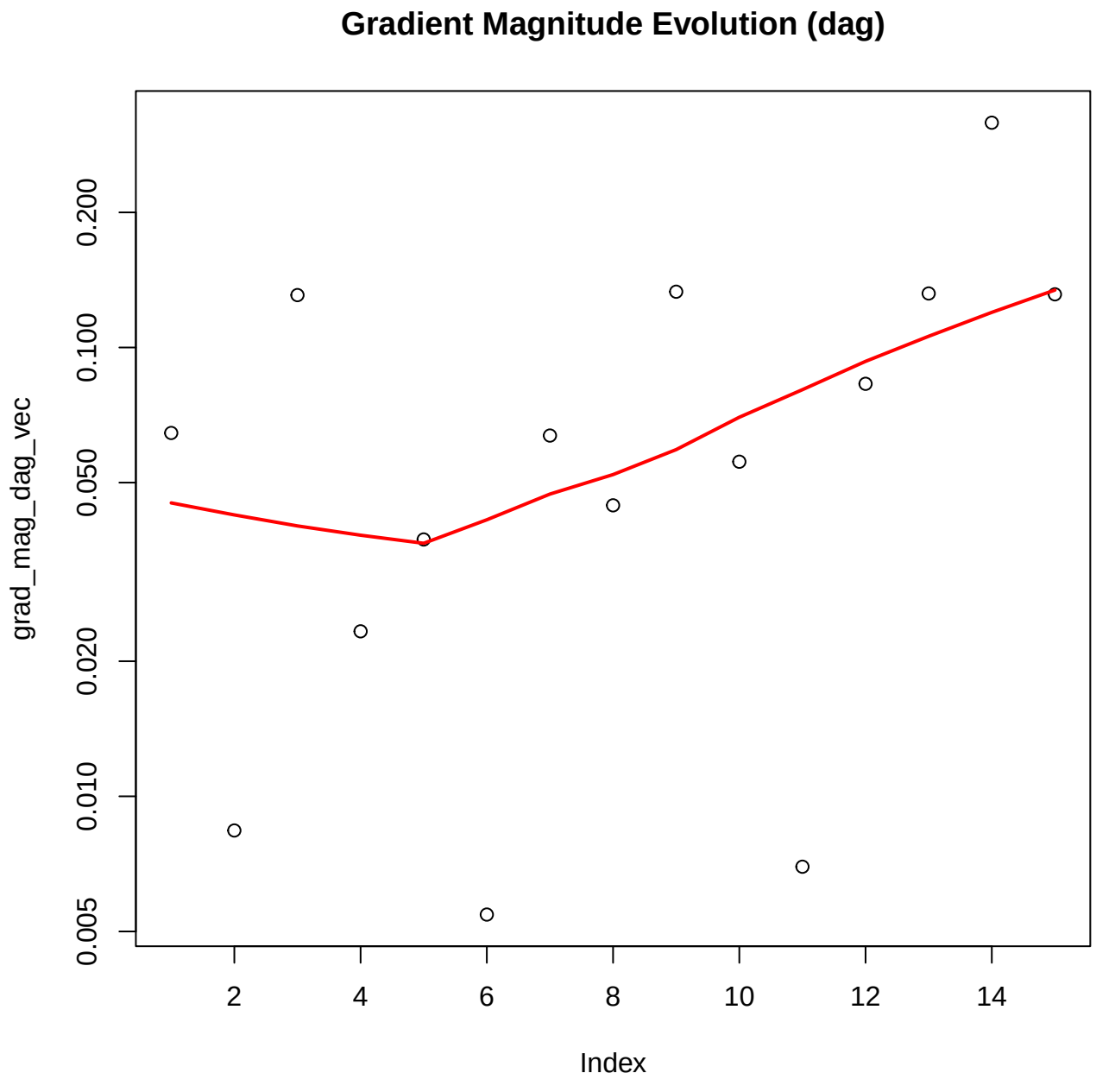

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.01
0.02
0.05
0.10
0.20
2
4
6
8
10
12
14
Index

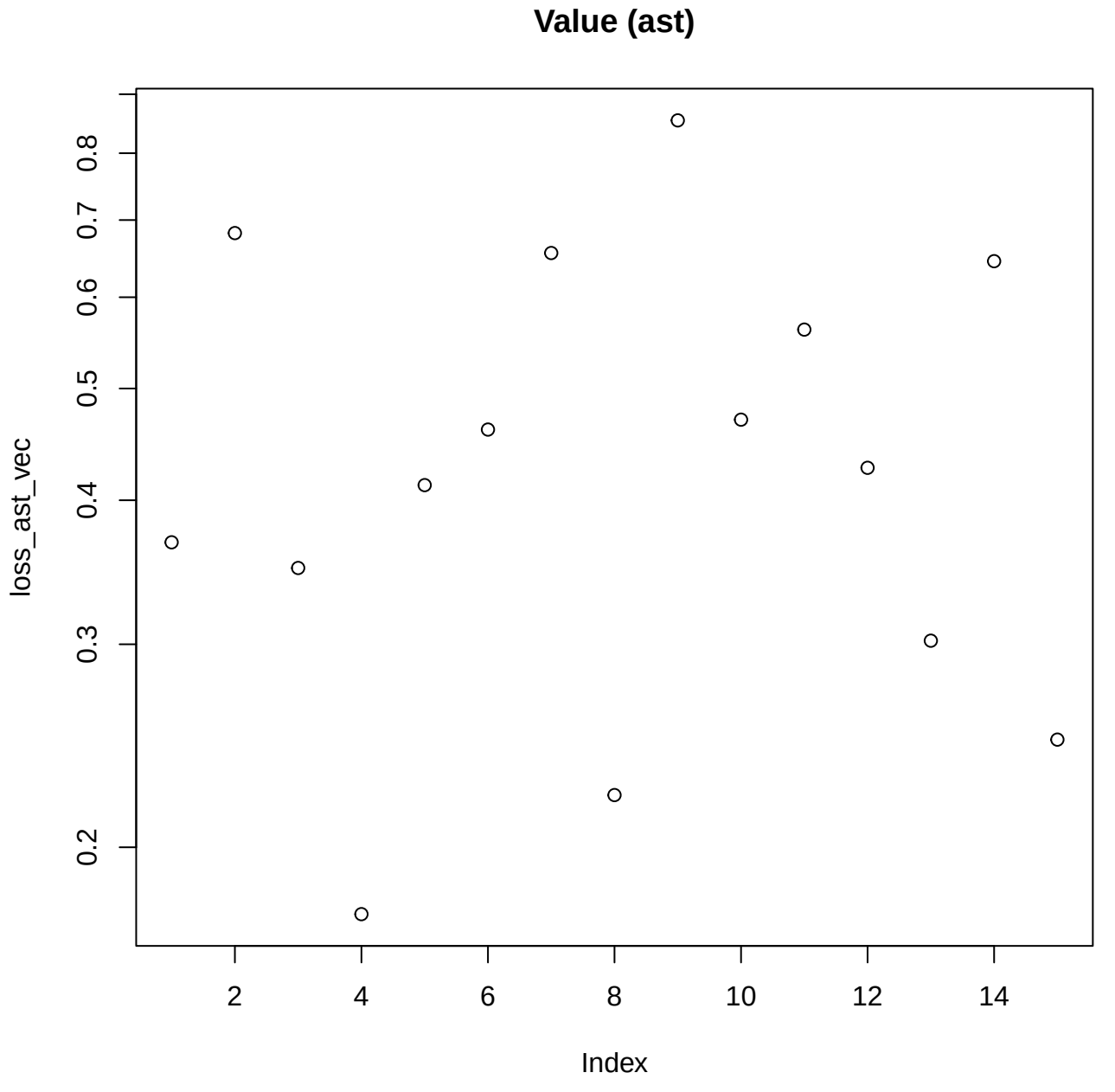

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.005
0.010
0.020
0.050
0.100
0.200
2
4
6
8
10
12
14
Index

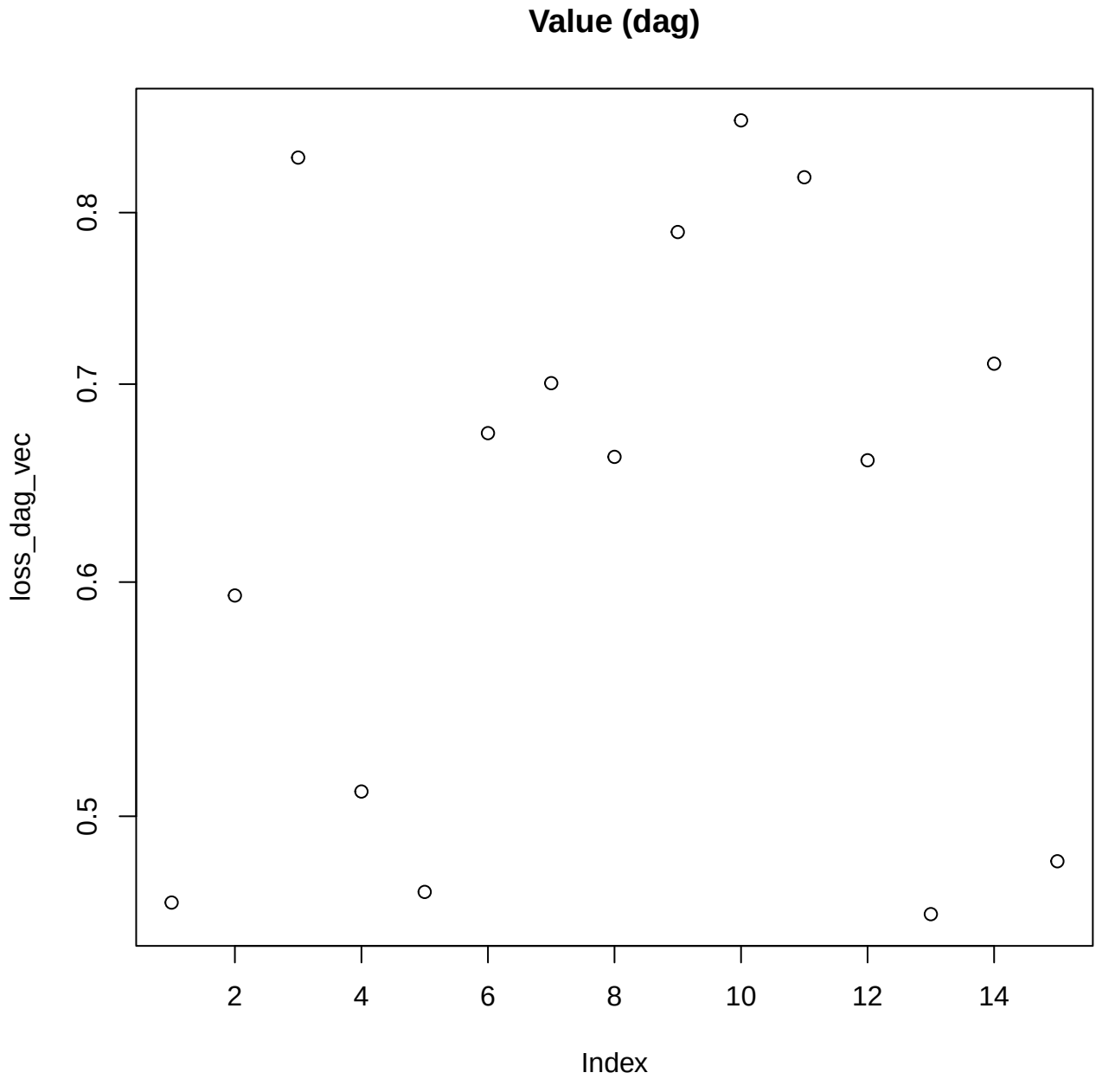

Value (ast)
loss_ast_vec
0.2
0.3
0.4
0.5
0.6
0.7
0.8
2
4
6
8
10
12
14
Index

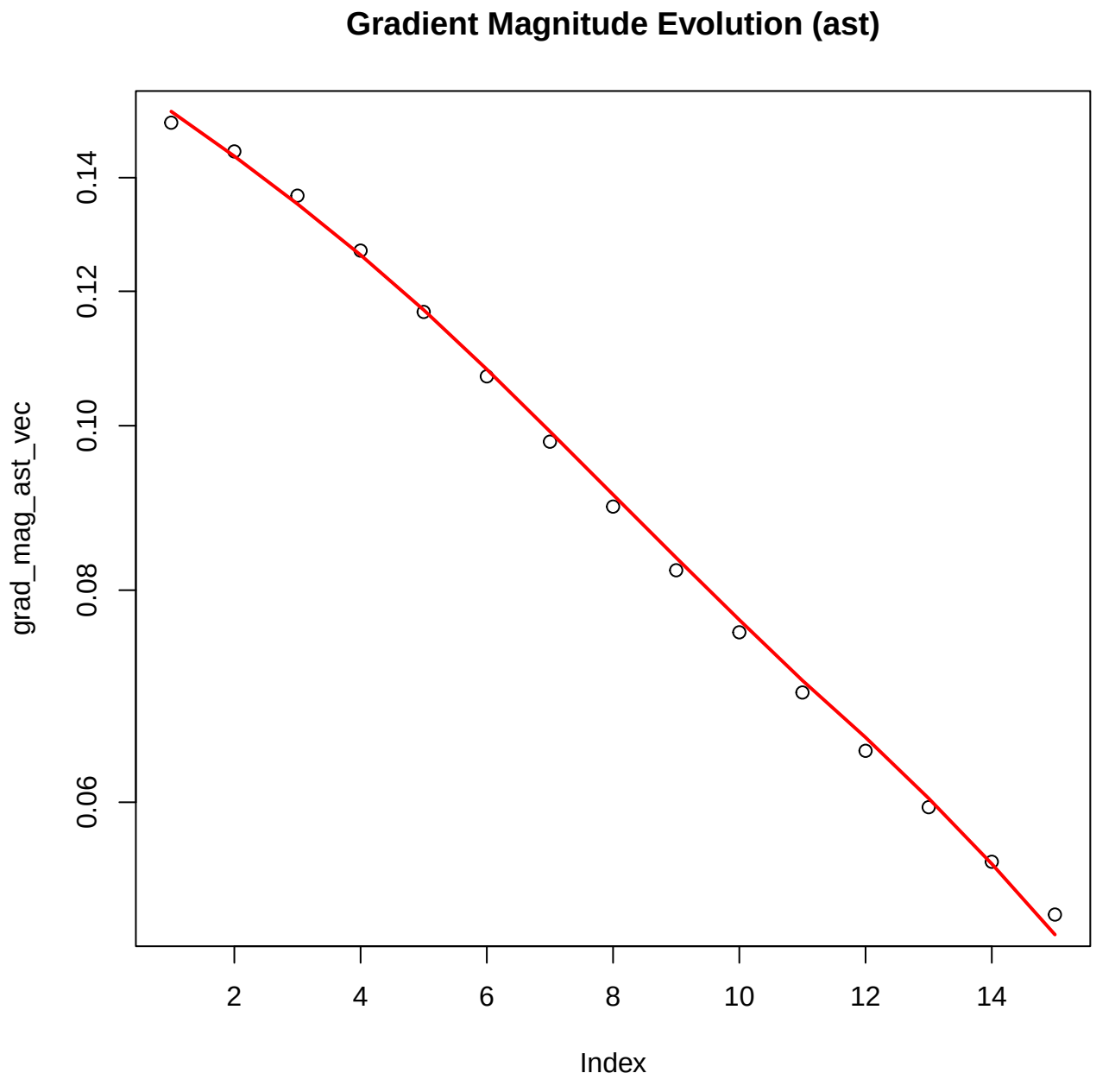

Value (dag)
loss_dag_vec
0.5
0.6
0.7
0.8
2
4
6
8
10
12
14
Index

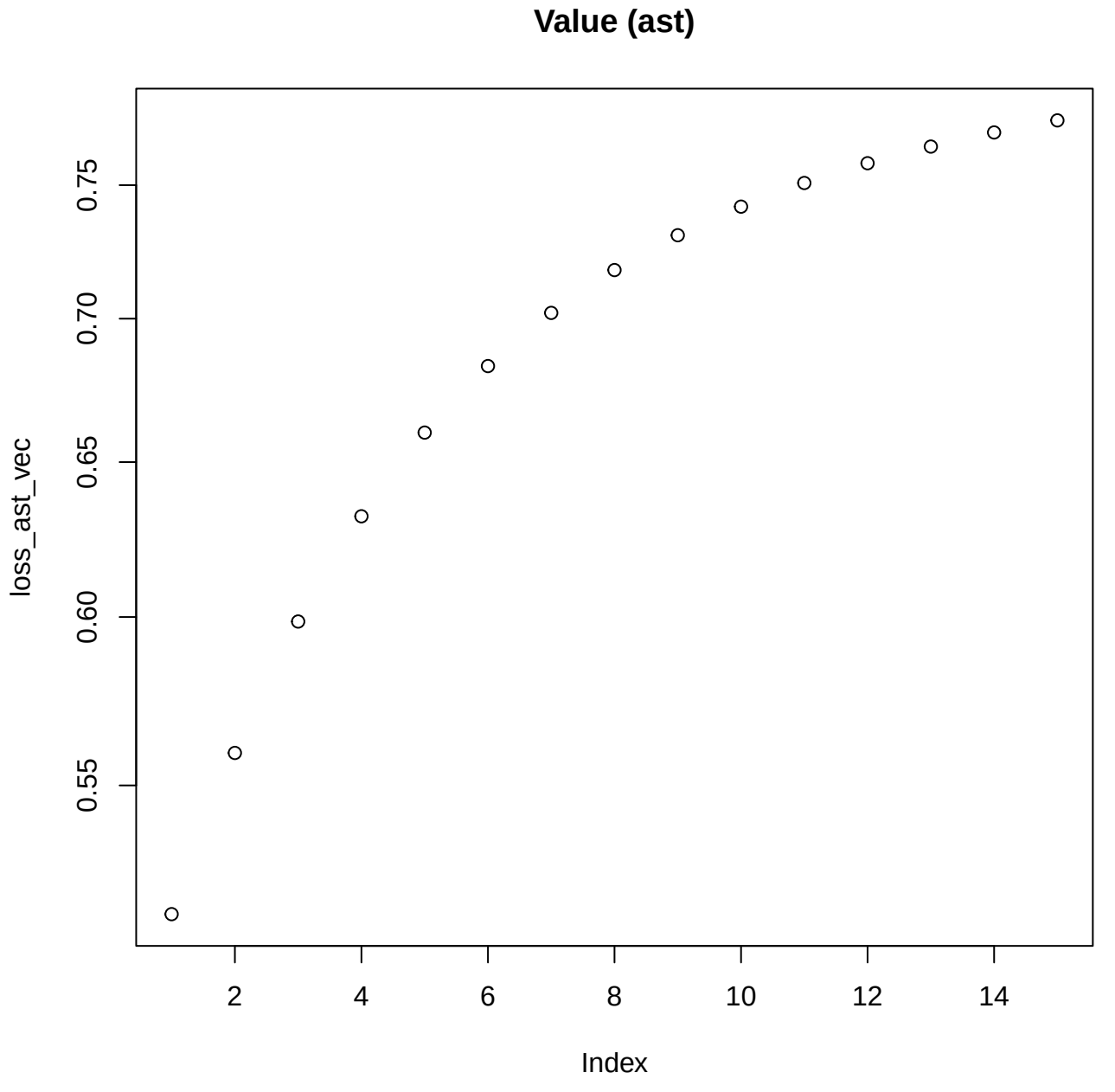

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.01
0.02
0.05
0.10
0.20
2
4
6
8
10
12
14
Index

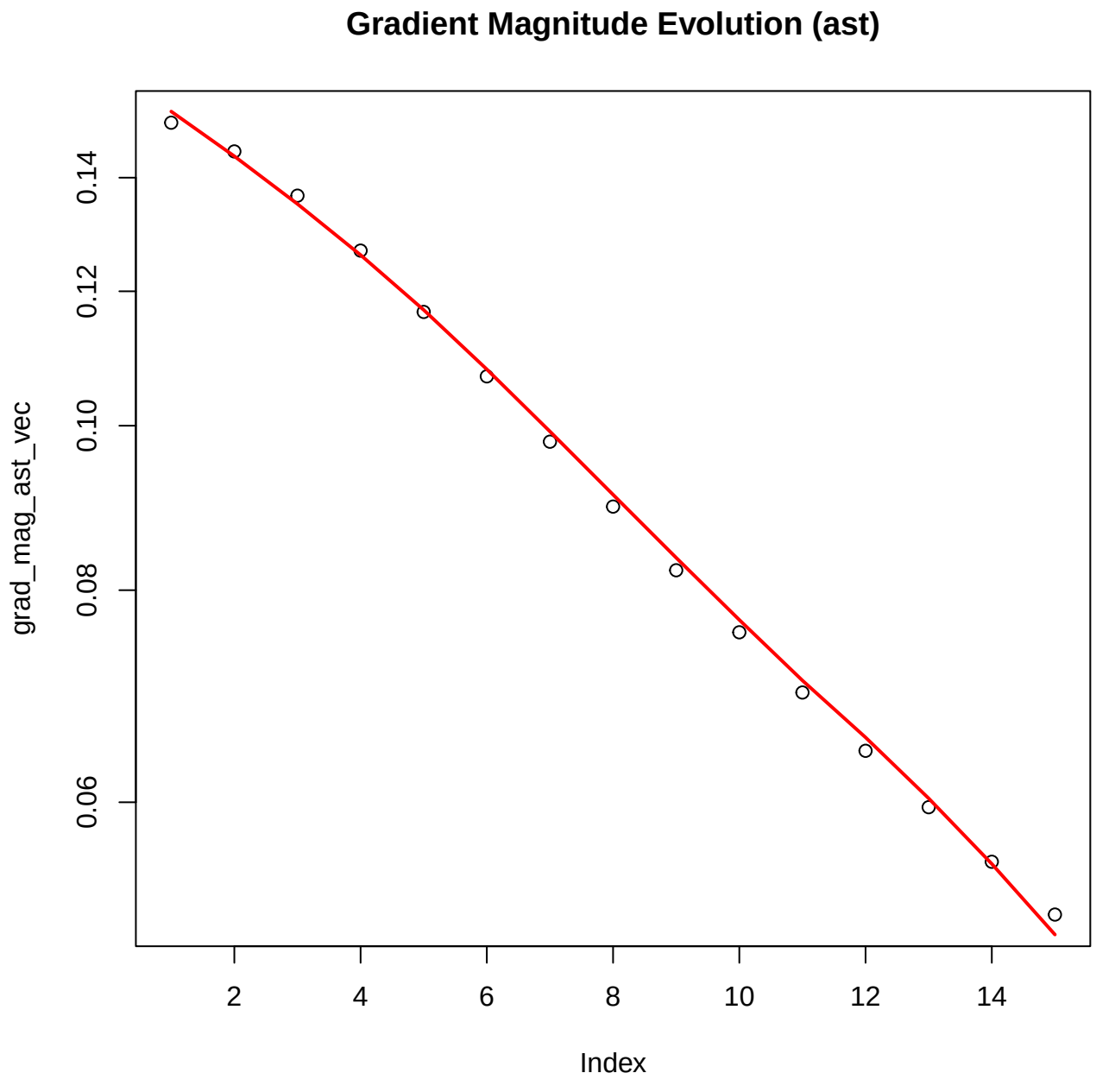

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.005
0.010
0.020
0.050
0.100
0.200
2
4
6
8
10
12
14
Index

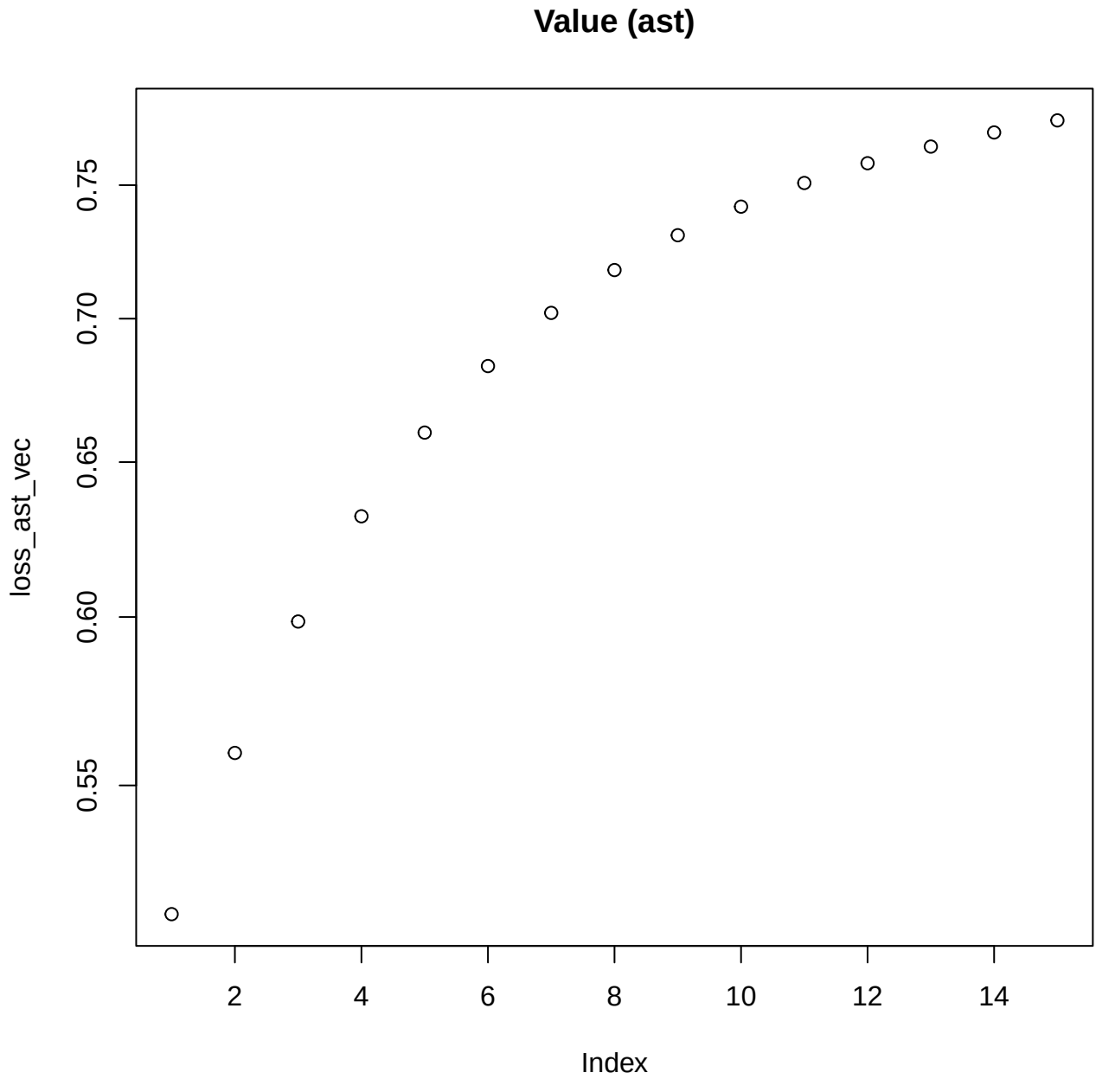

Value (ast)
loss_ast_vec
0.2
0.3
0.4
0.5
0.6
0.7
0.8
2
4
6
8
10
12
14
Index

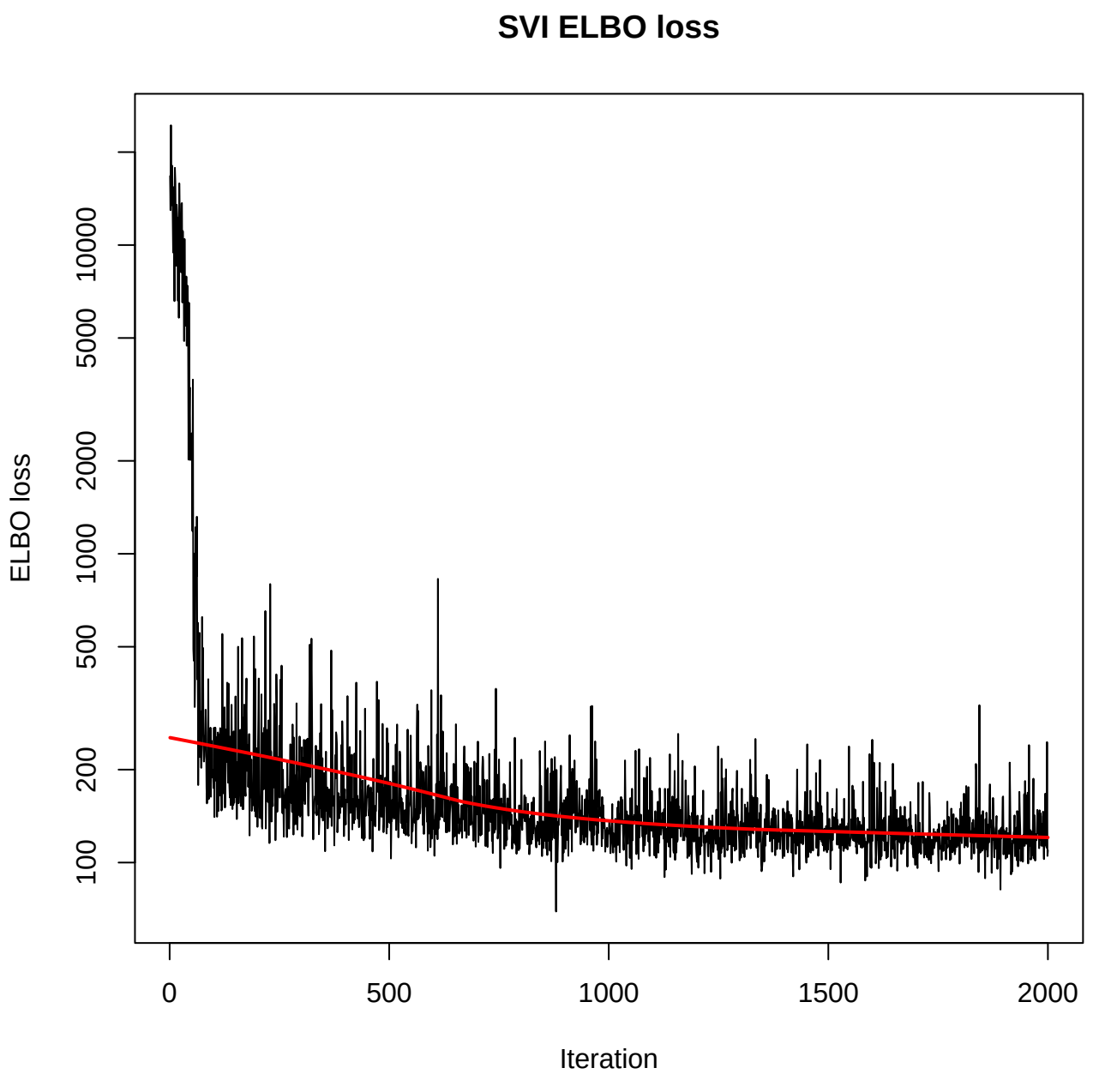

Value (dag)
loss_dag_vec
0.5
0.6
0.7
0.8
2
4
6
8
10
12
14
Index

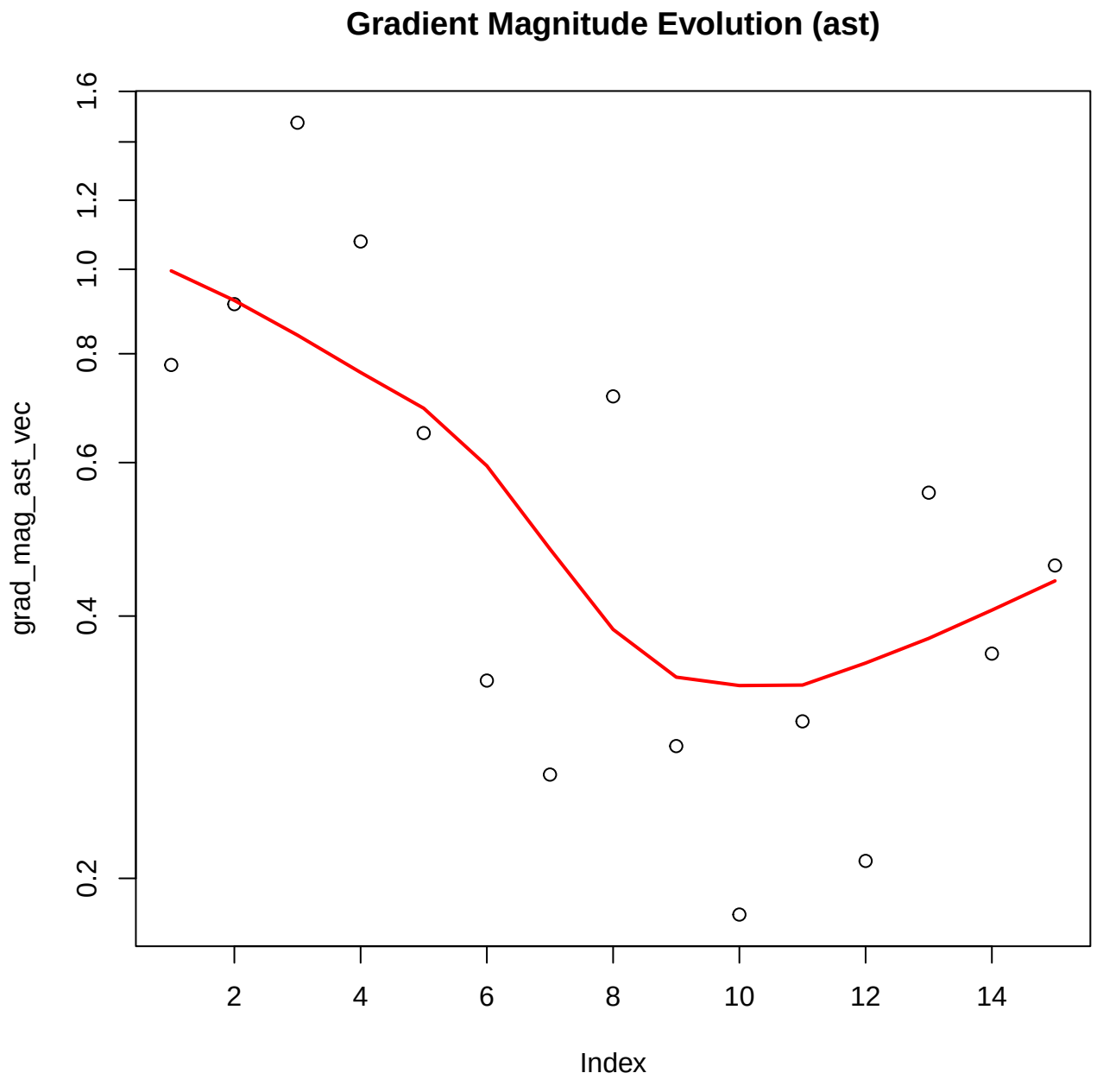

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.06
0.08
0.10
0.12
0.14
2
4
6
8
10
12
14
Index

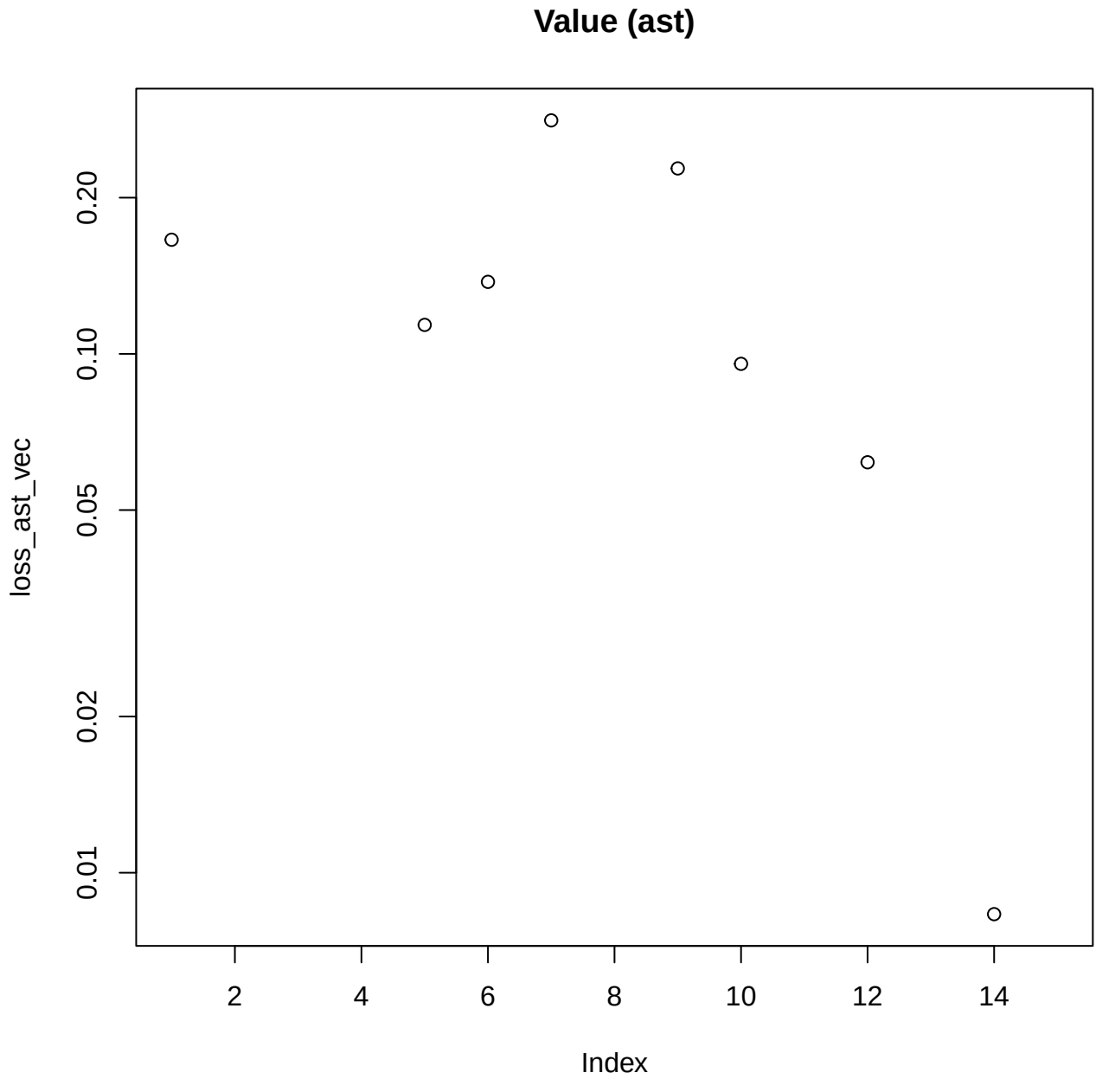

Value (ast)
loss_ast_vec
0.55
0.60
0.65
0.70
0.75
2
4
6
8
10
12
14
Index

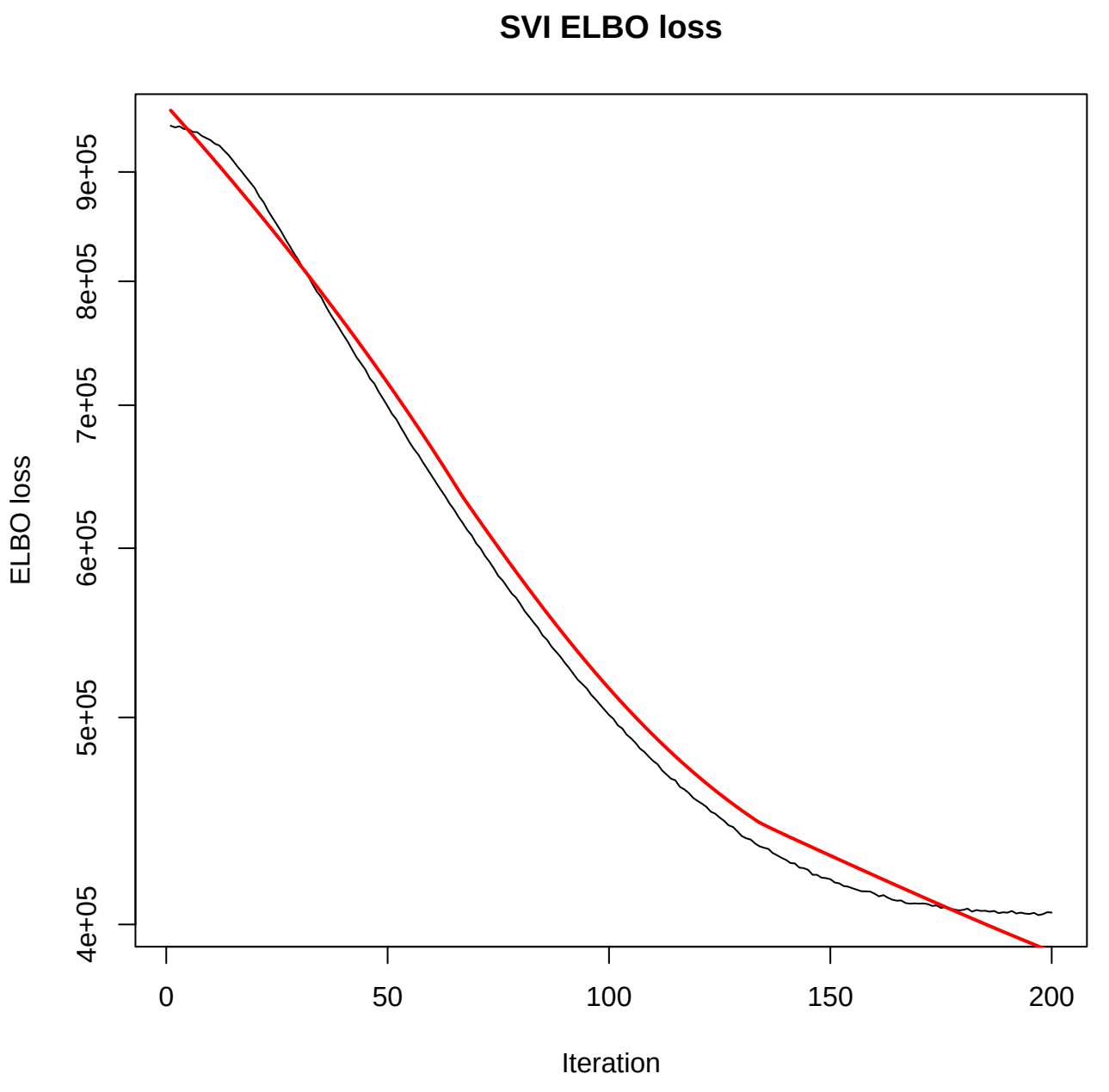

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.06
0.08
0.10
0.12
0.14
2
4
6
8
10
12
14
Index

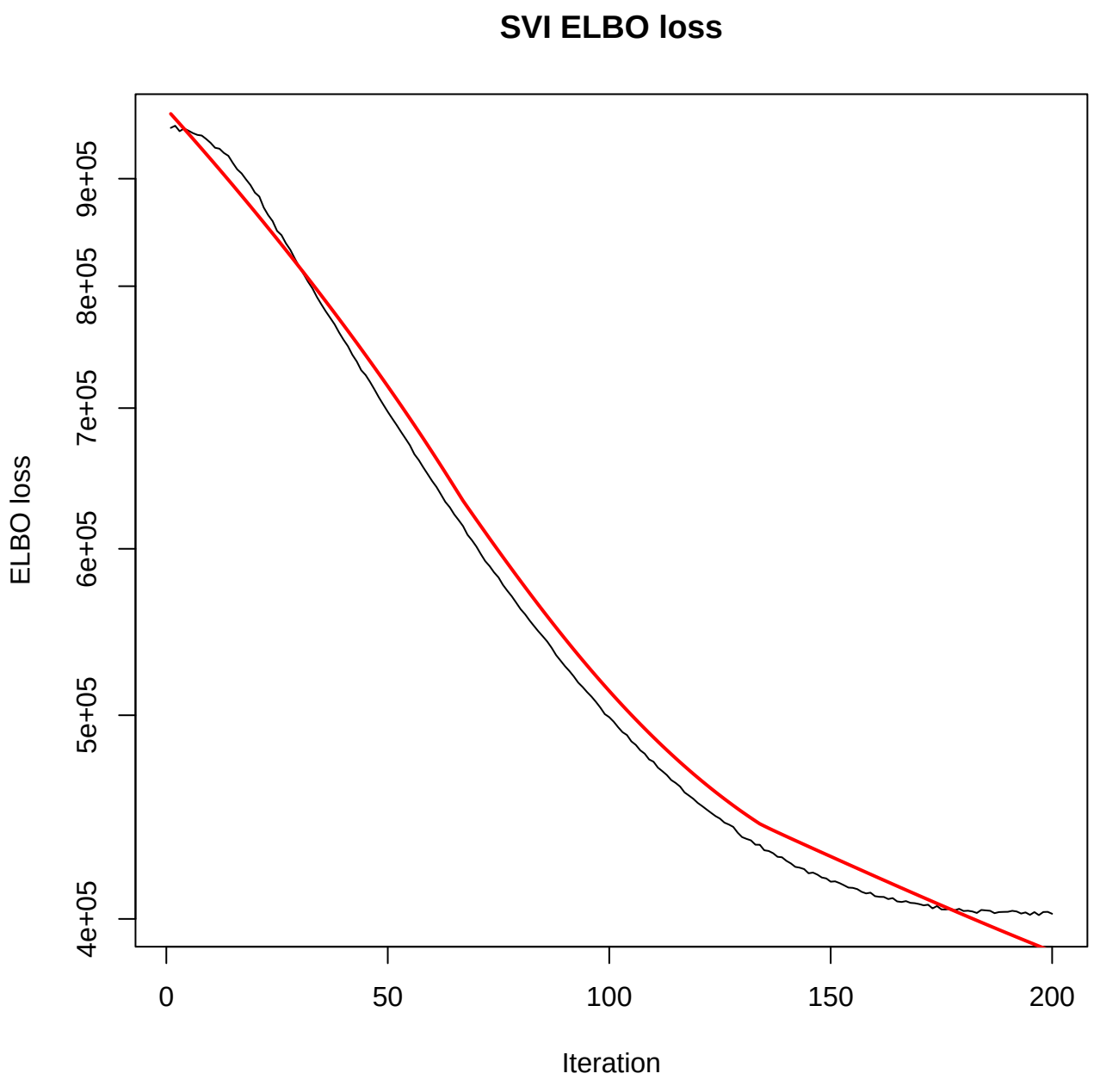

Value (ast)
loss_ast_vec
0.55
0.60
0.65
0.70
0.75
2
4
6
8
10
12
14
Index

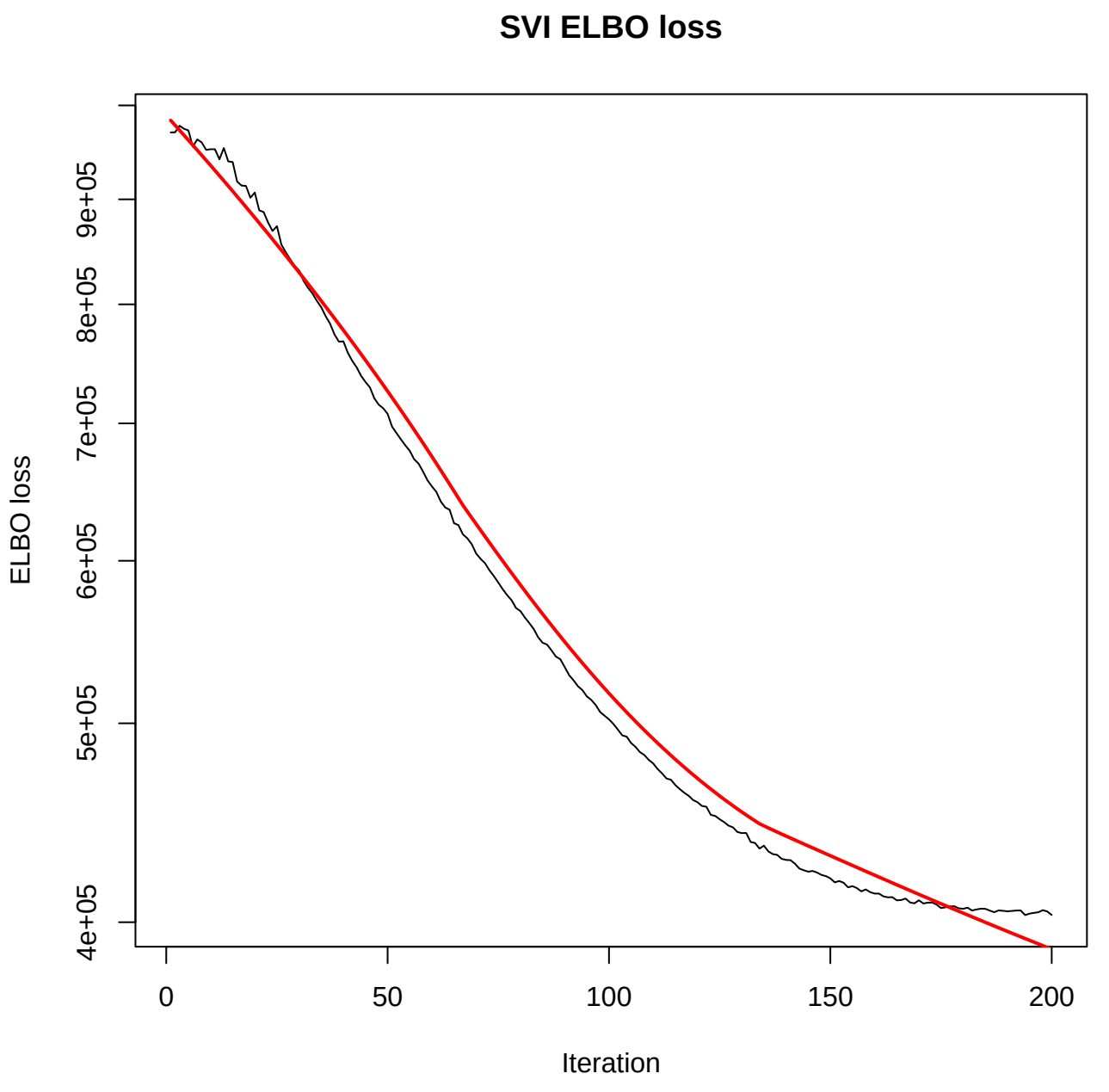

SVI ELBO loss
ELBO loss
10000
5000
2000
1000
500
200
100
0
500
1000
1500
2000
Iteration

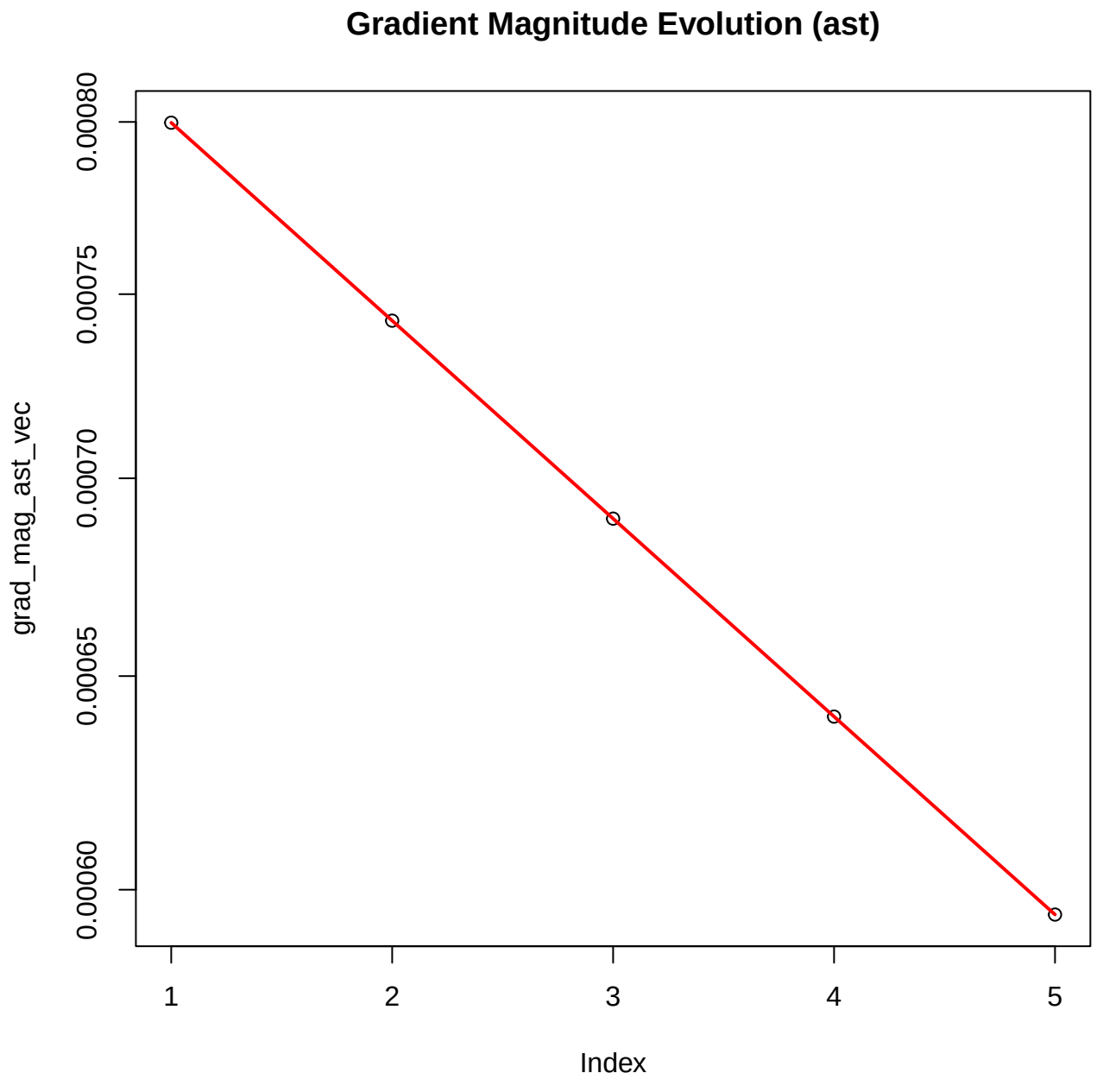

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.2
0.4
0.6
0.8
1.0
1.2
1.6
2
4
6
8
10
12
14
Index

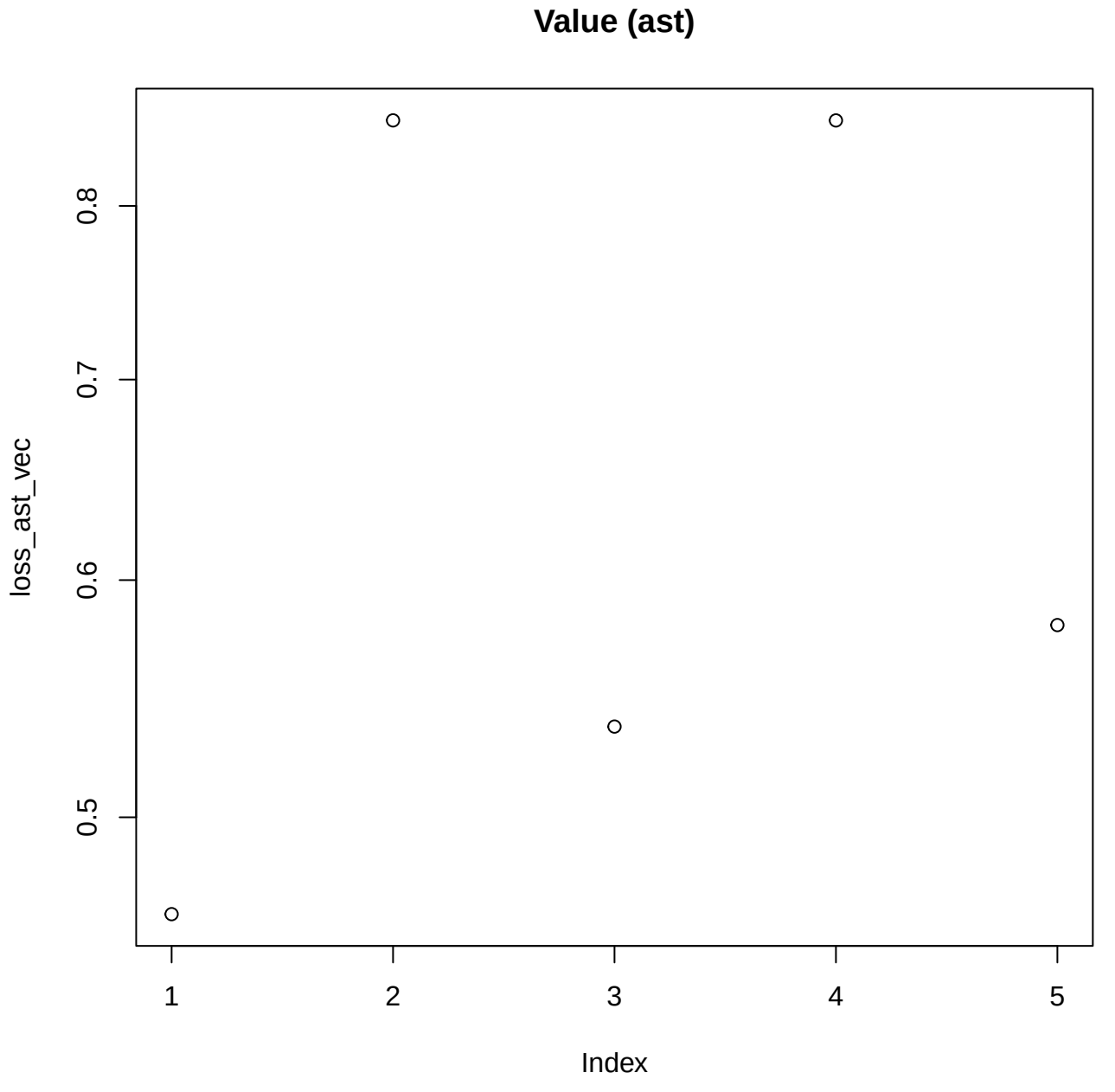

Value (ast)
loss_ast_vec
0.01
0.02
0.05
0.10
0.20
2
4
6
8
10
12
14
Index

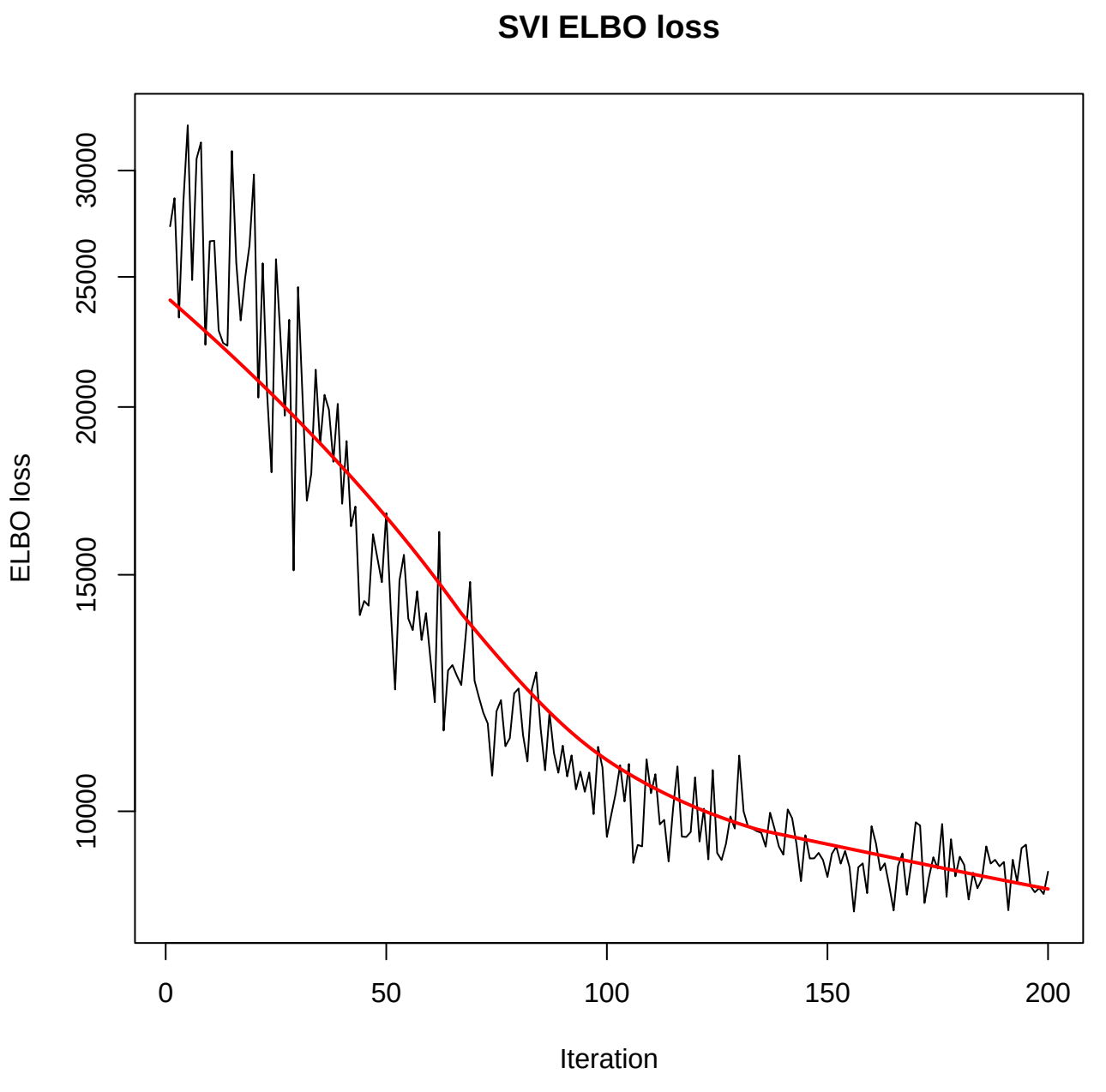

SVI ELBO loss
ELBO loss
9e+05
8e+05
7e+05
6e+05
5e+05
4e+05
0
50
100
150
200
Iteration

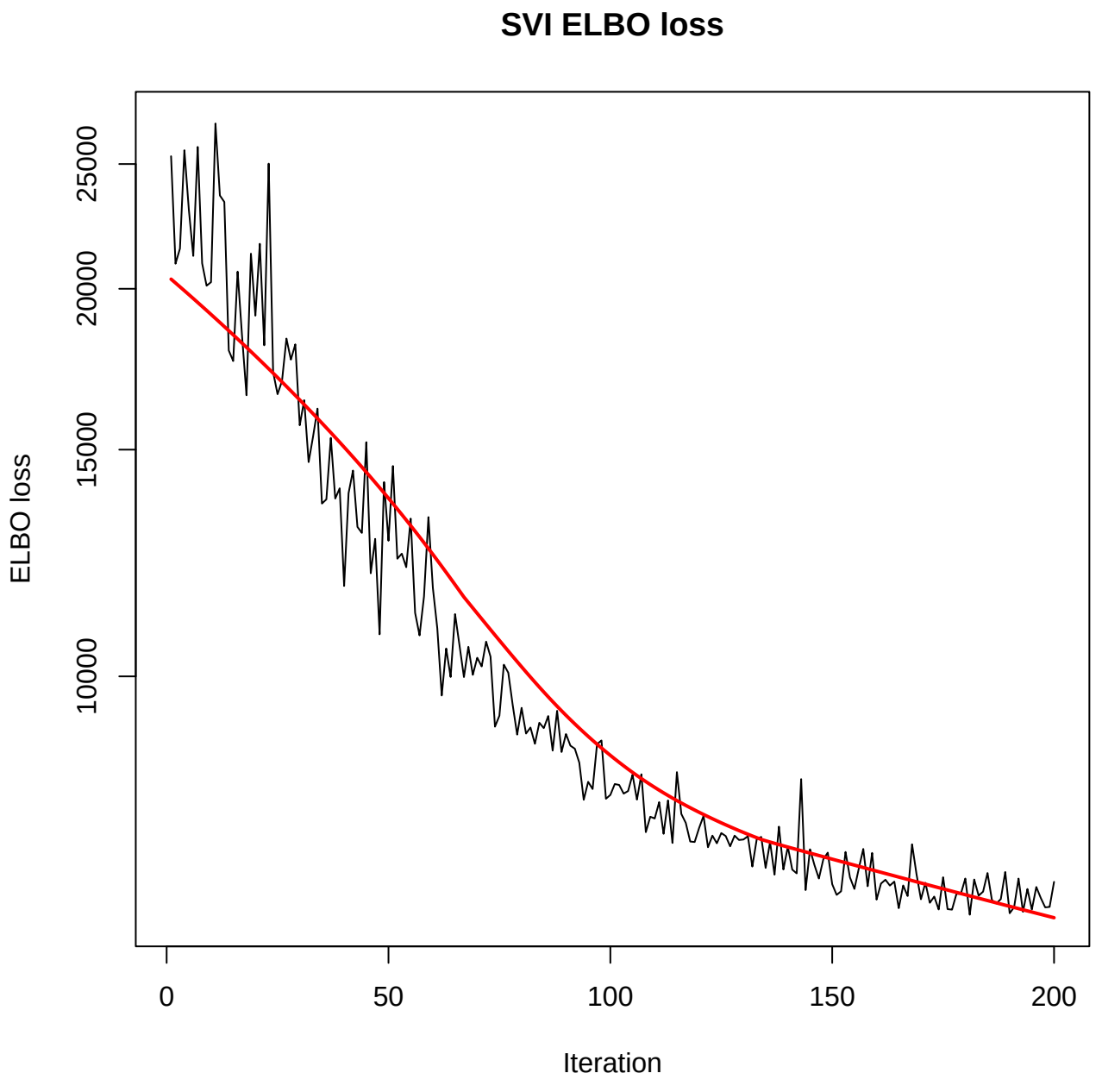

SVI ELBO loss
ELBO loss
9e+05
8e+05
7e+05
6e+05
5e+05
4e+05
0
50
100
150
200
Iteration

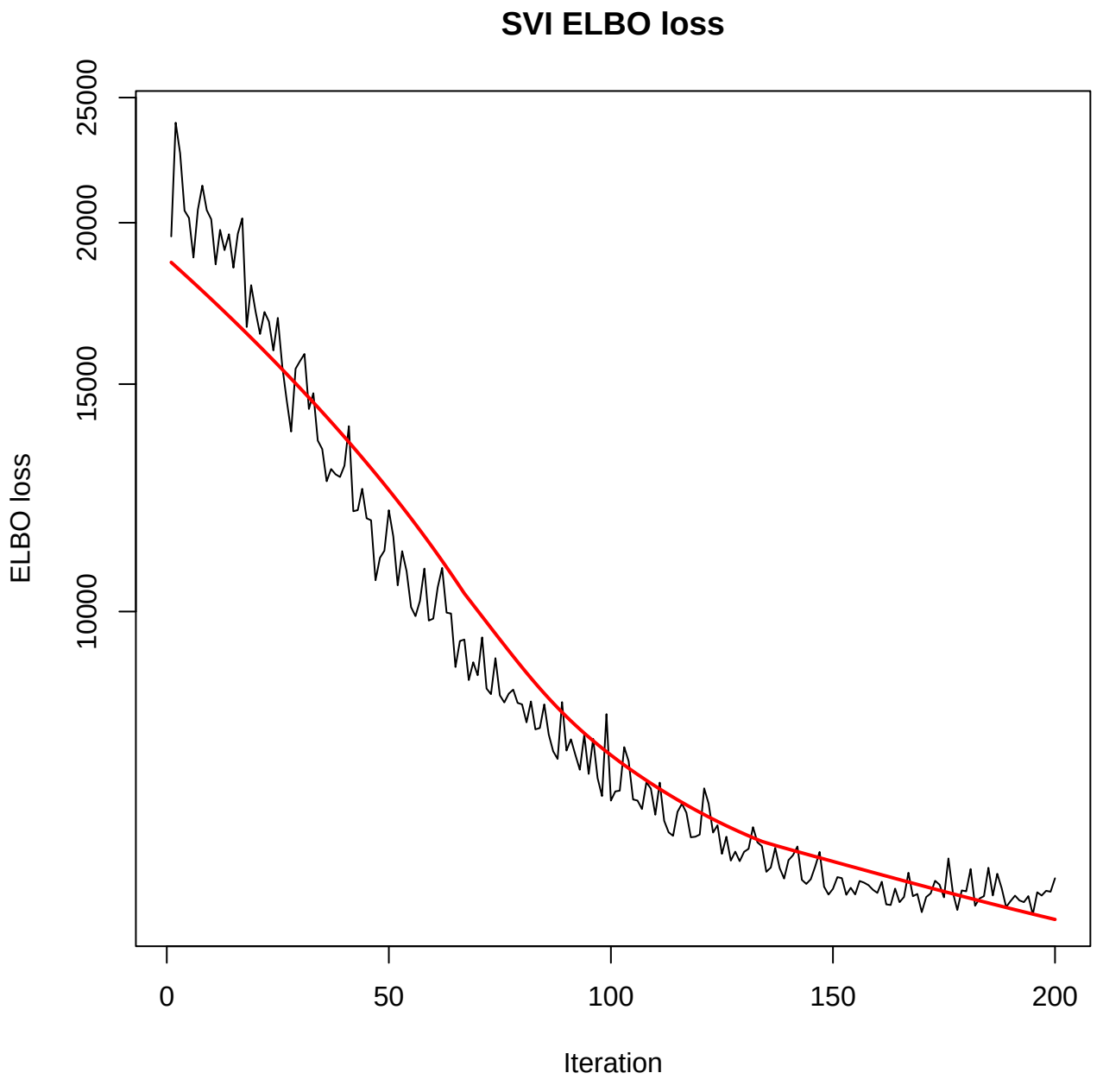

SVI ELBO loss
ELBO loss
9e+05
8e+05
7e+05
6e+05
5e+05
4e+05
0
50
100
150
200
Iteration

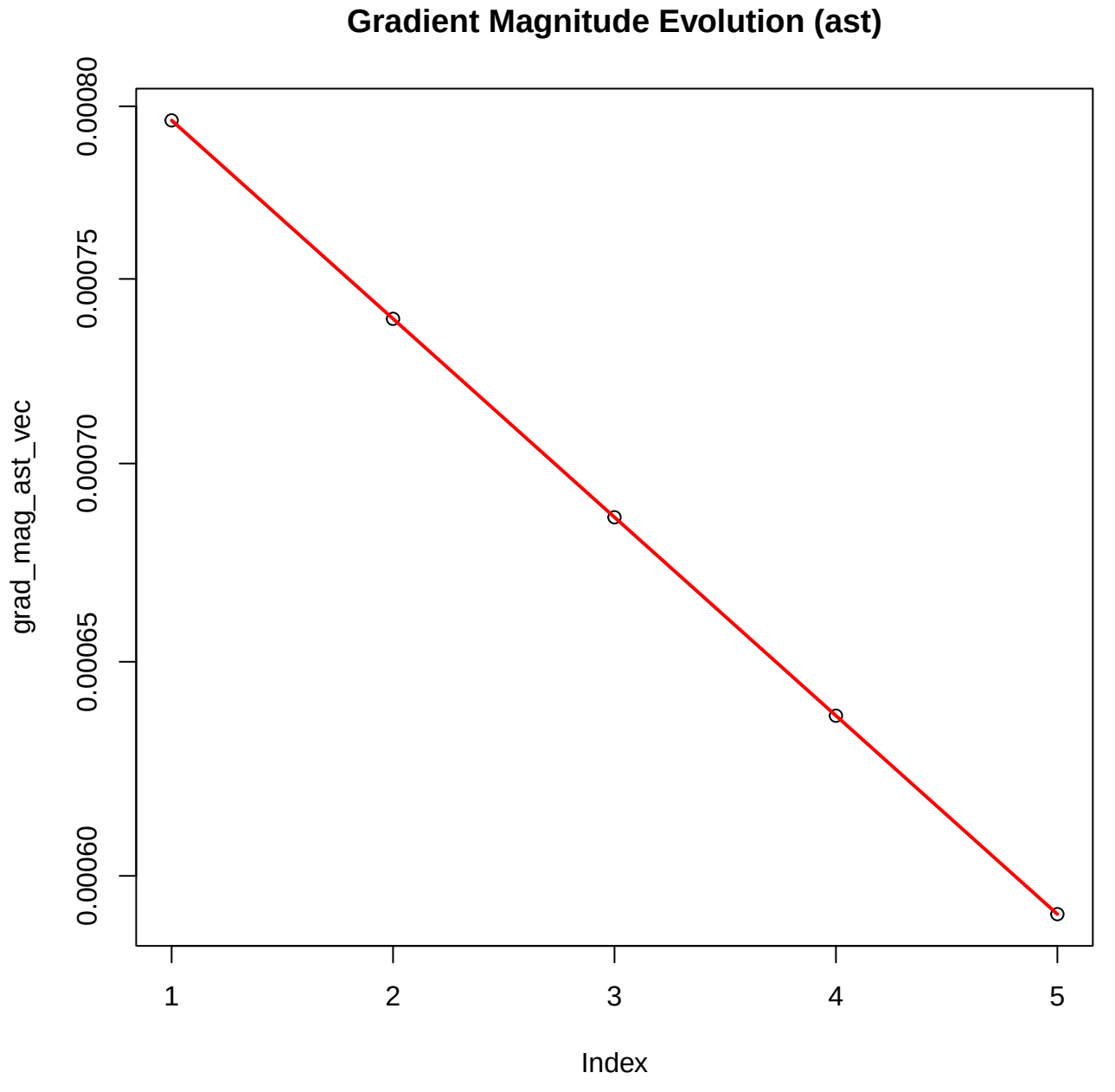

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
0.00080
1
2
3
4
5
Index

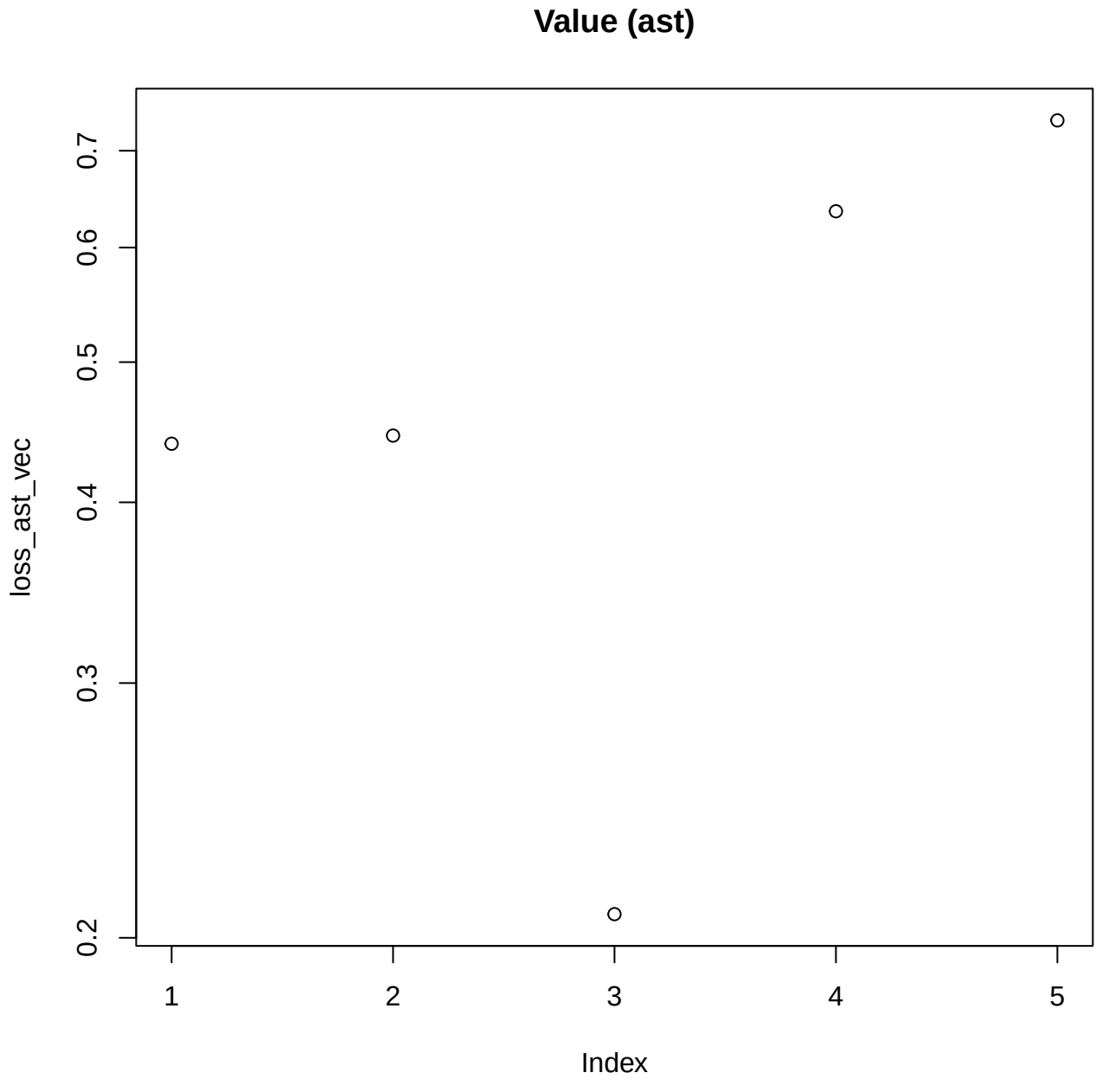

Value (ast)
loss_ast_vec
0.5
0.6
0.7
0.8
1
2
3
4
5
Index

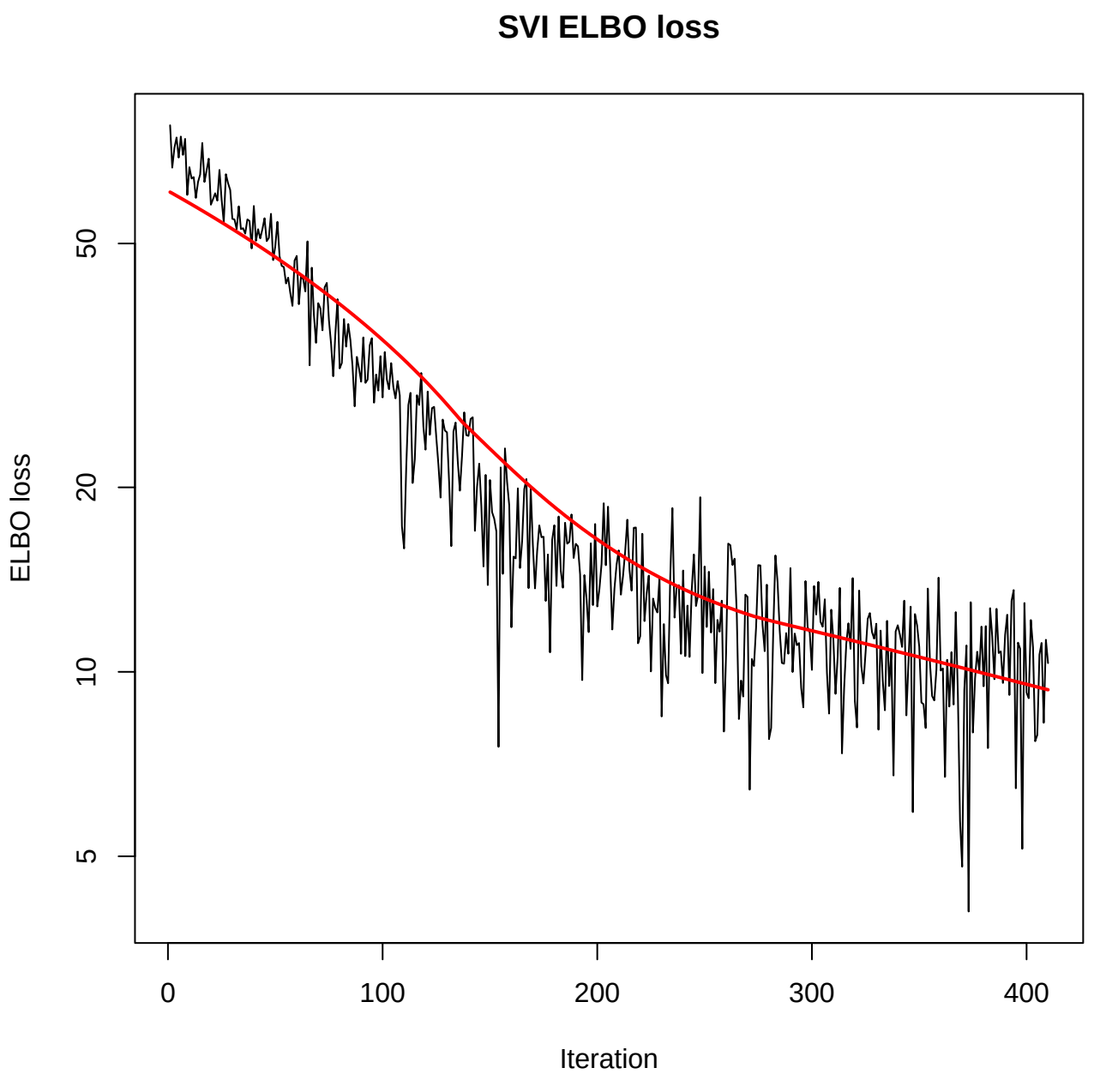

SVI ELBO loss
ELBO loss
30000
25000
20000
15000
10000
0
50
100
150
200
Iteration

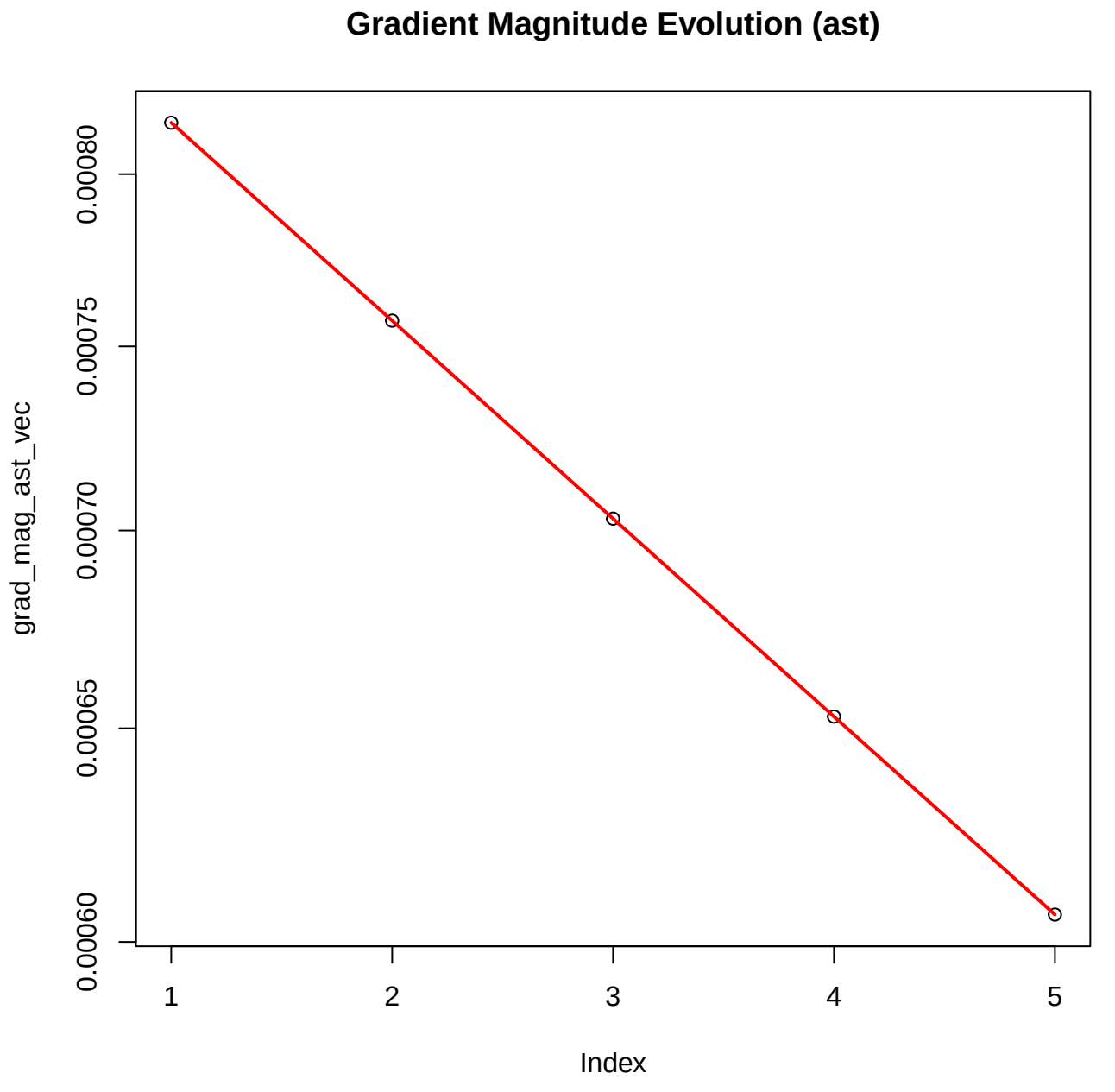

SVI ELBO loss
ELBO loss
25000
20000
15000
10000
0
50
100
150
200
Iteration

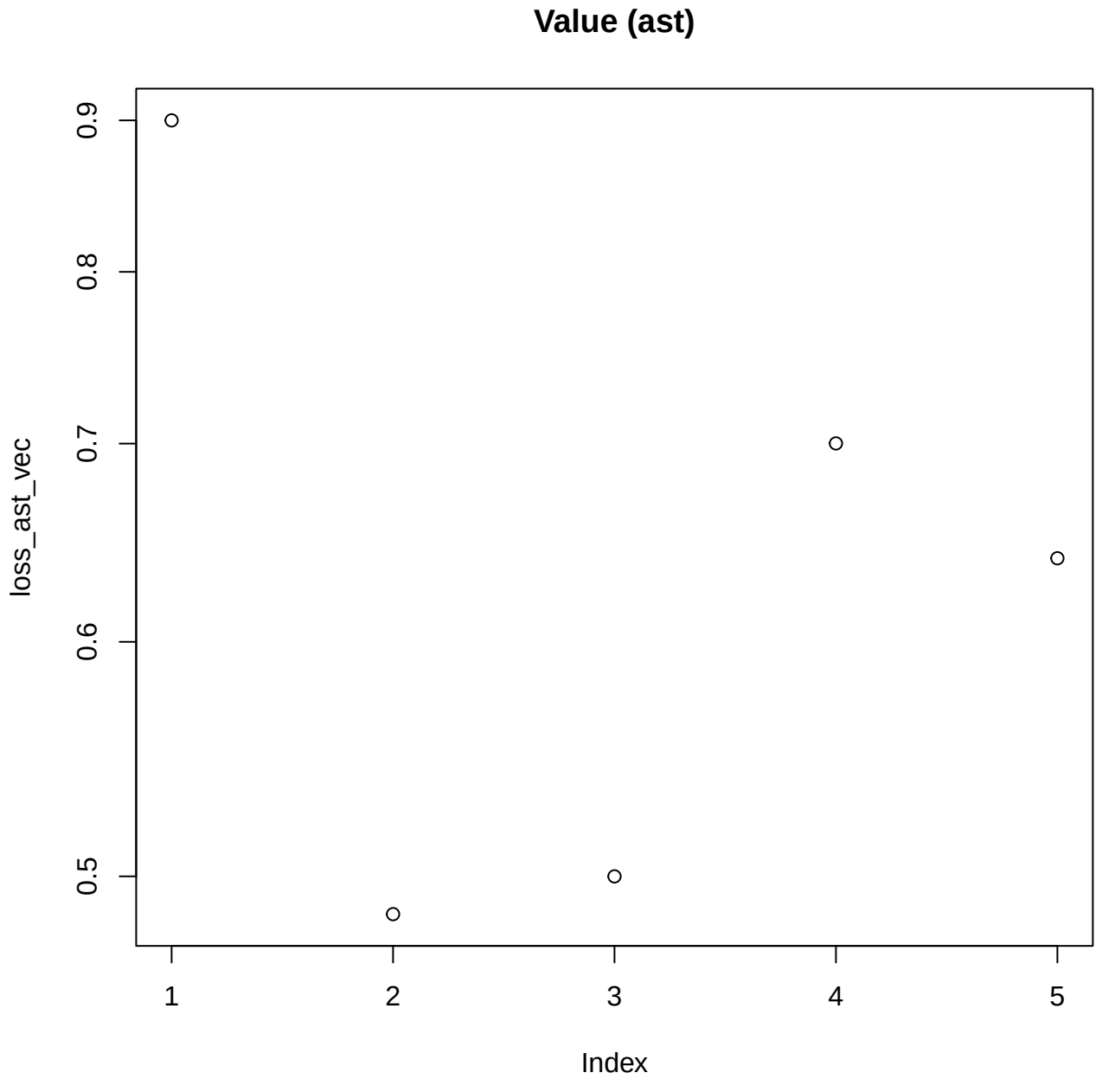

SVI ELBO loss
ELBO loss
25000
20000
15000
10000
0
50
100
150
200
Iteration

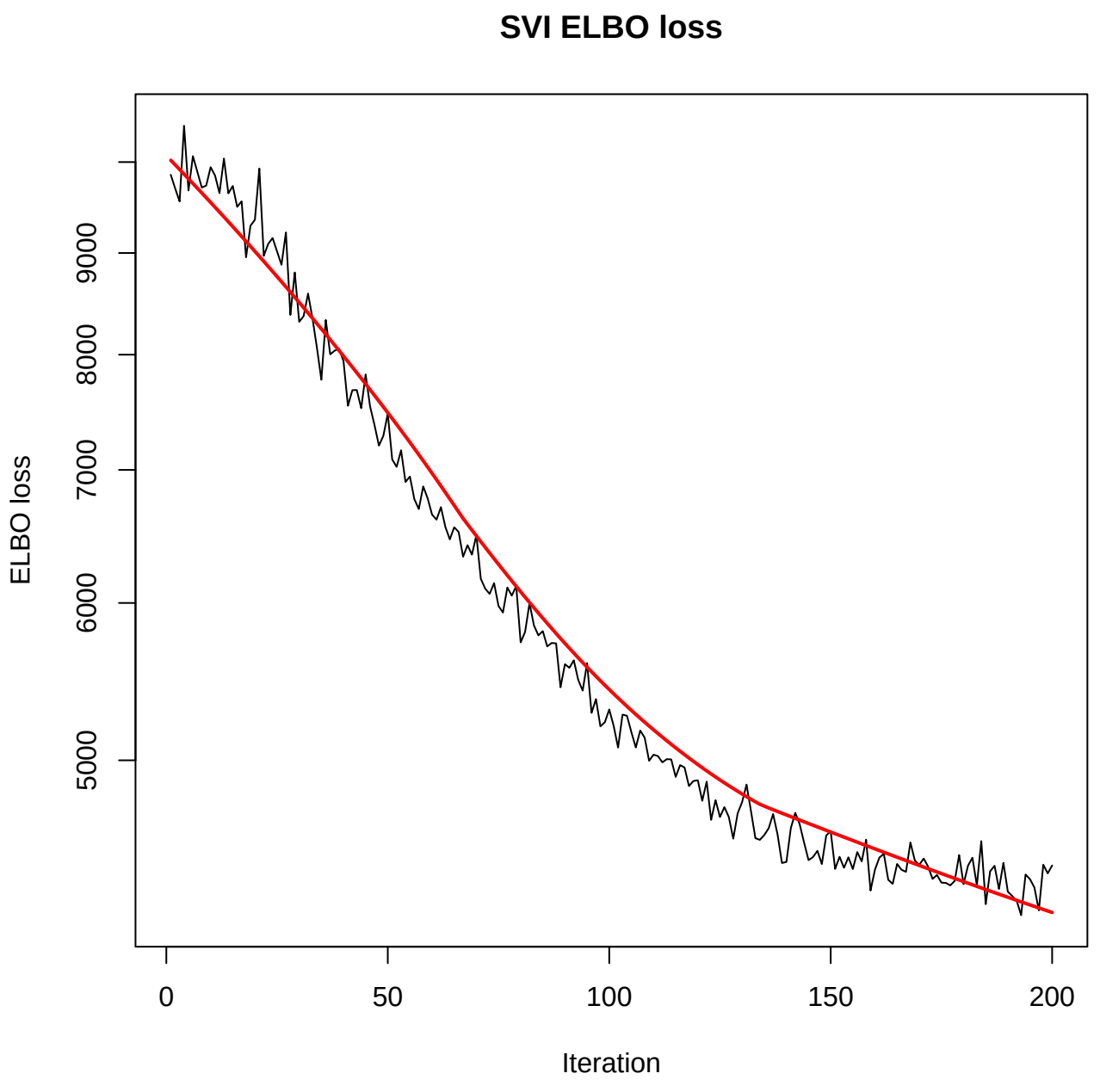

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
0.00080
1
2
3
4
5
Index

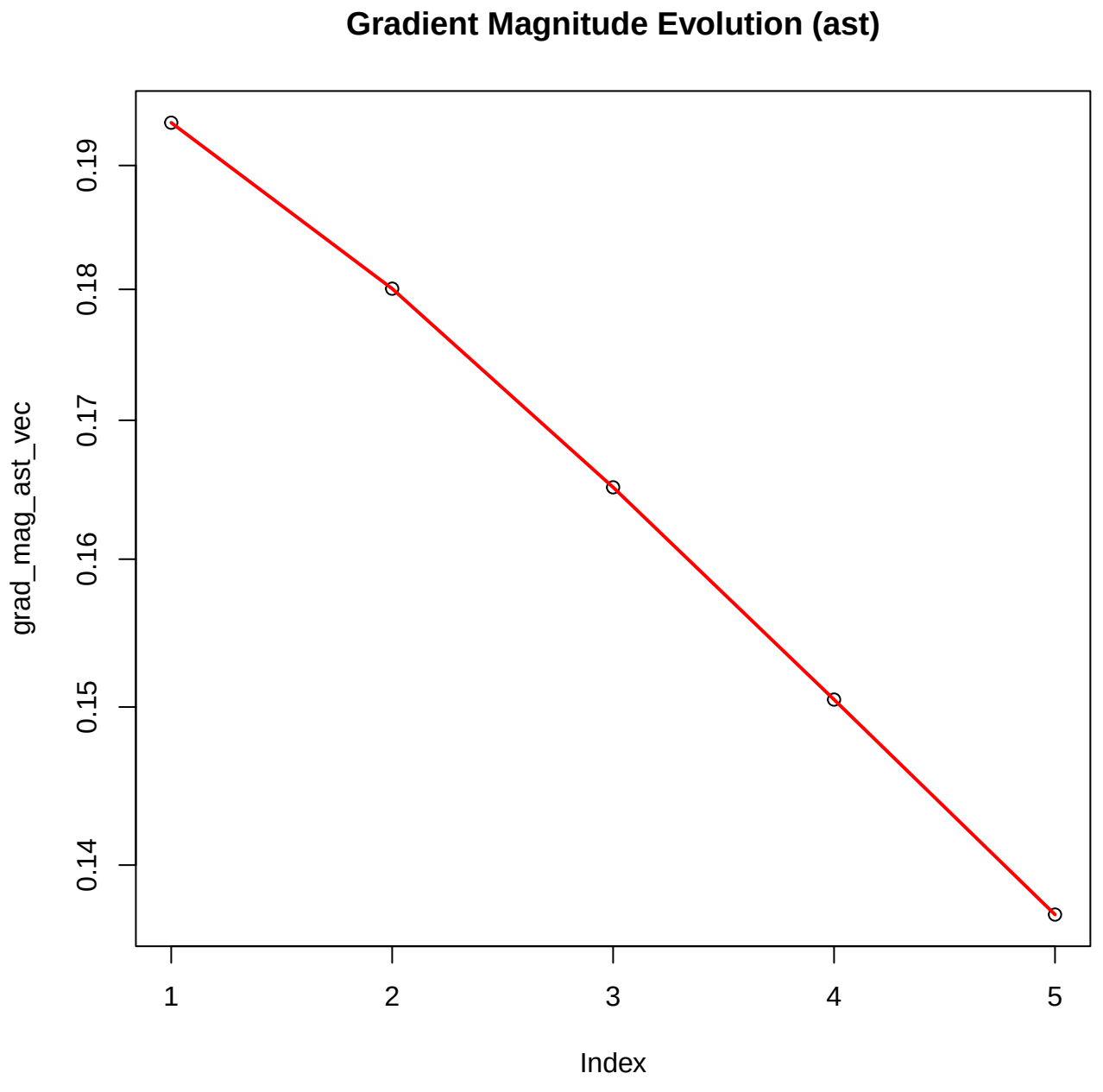

Value (ast)
loss_ast_vec
0.2
0.3
0.4
0.5
0.6
0.7
1
2
3
4
5
Index

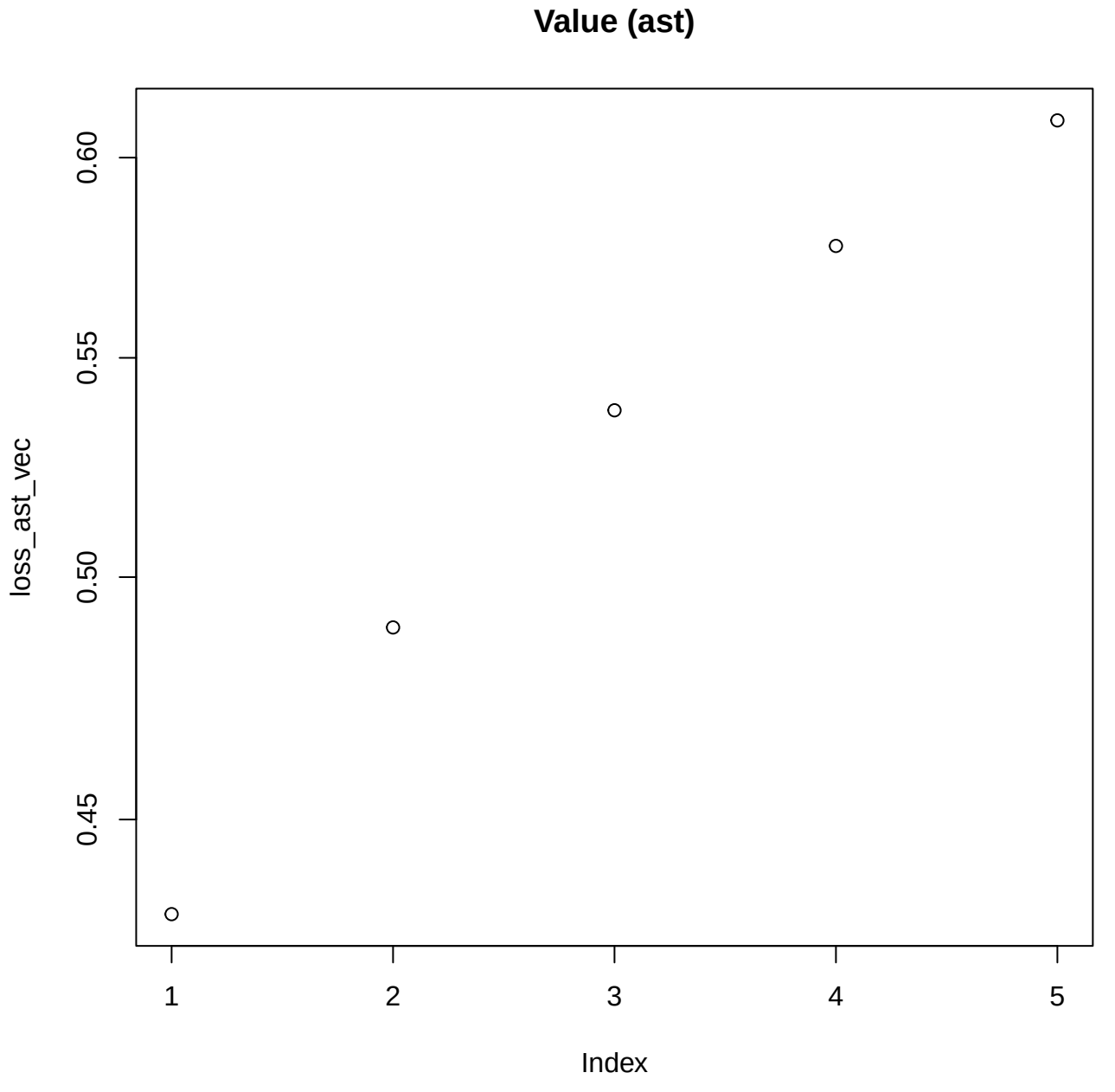

SVI ELBO loss
ELBO loss
50
20
10
5
0
100
200
300
400
Iteration

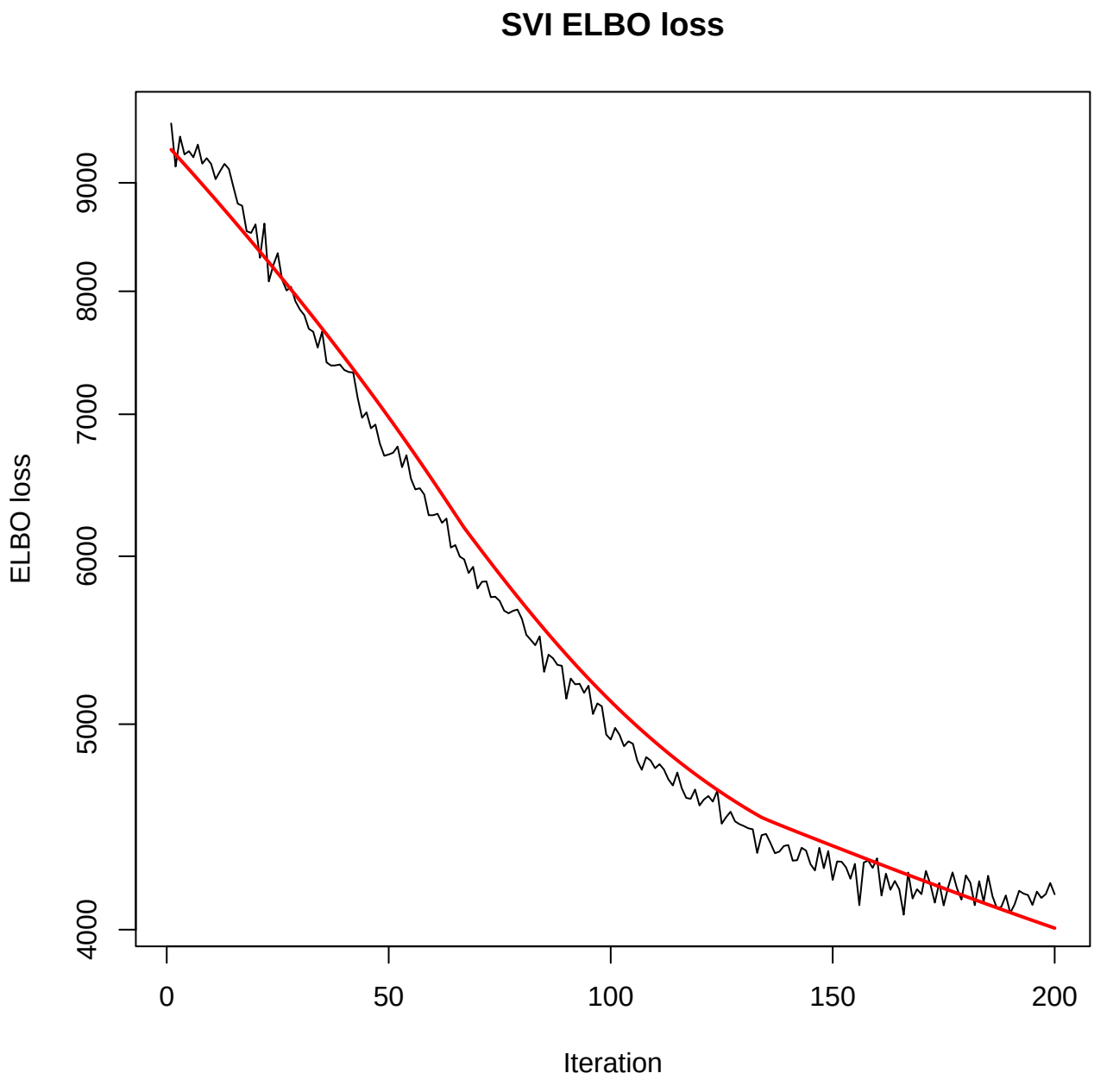

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
0.00080
1
2
3
4
5
Index

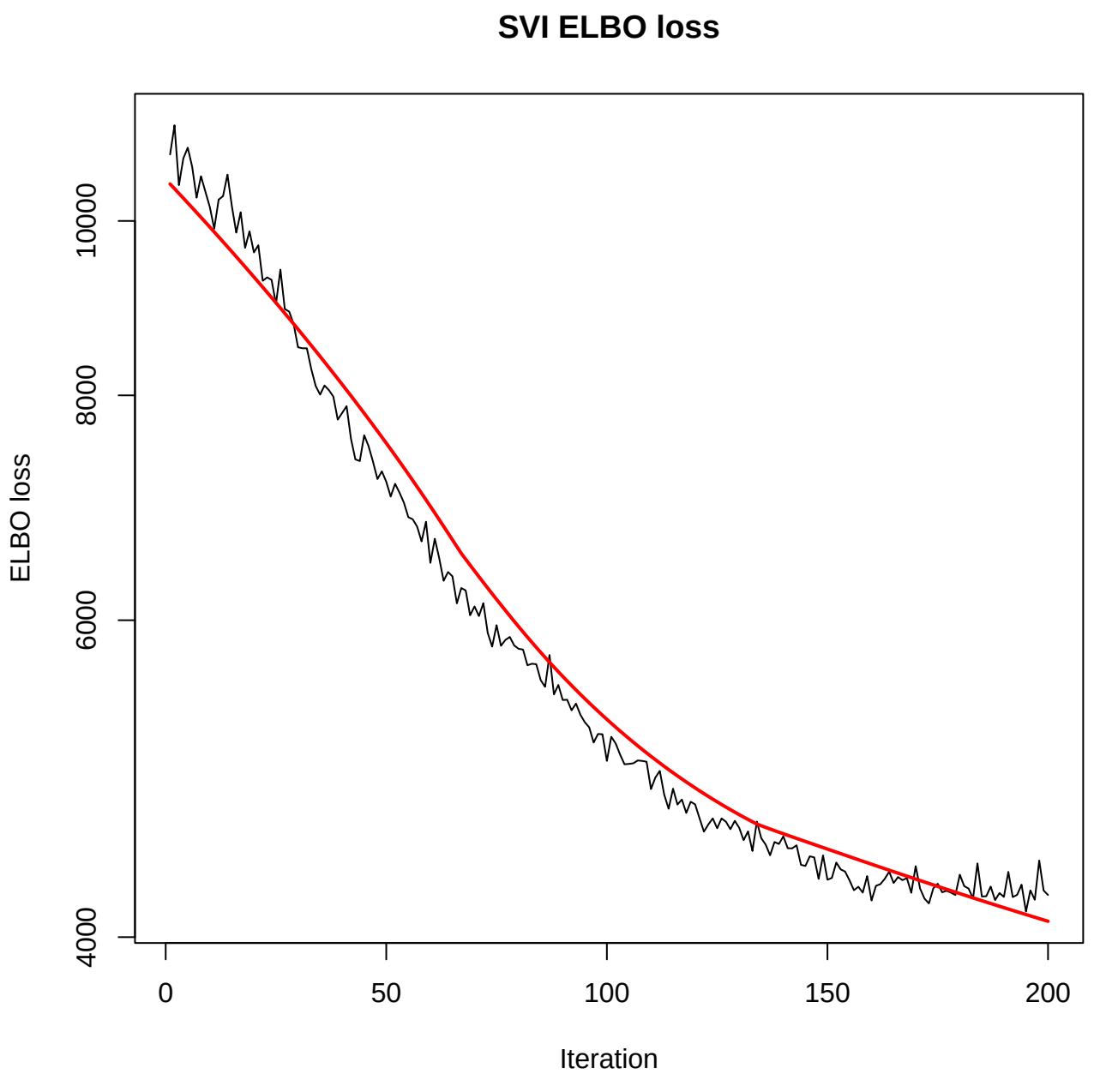

Value (ast)
loss_ast_vec
0.5
0.6
0.7
0.8
0.9
1
2
3
4
5
Index

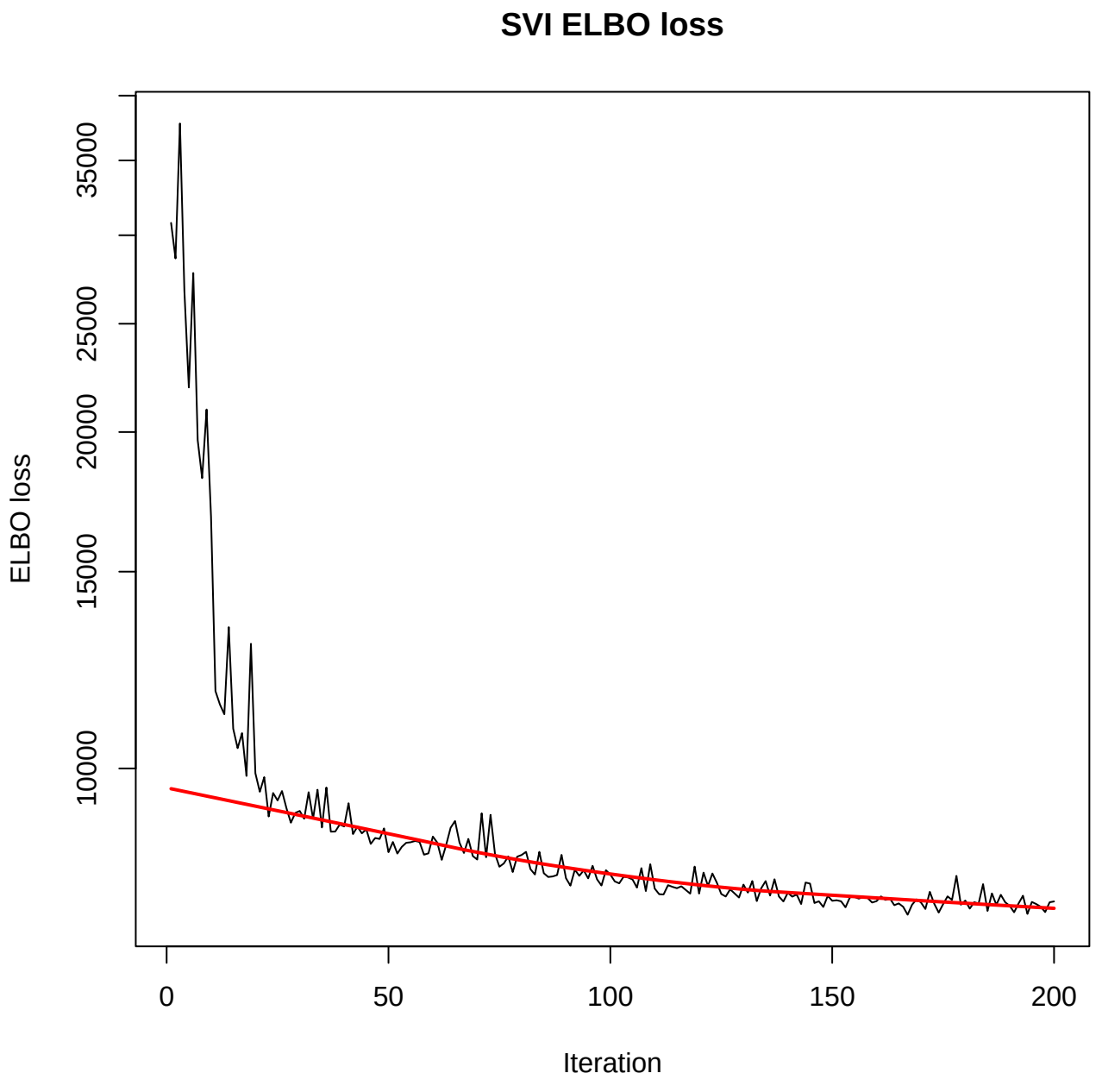

SVI ELBO loss
ELBO loss
5000
6000
7000
8000
9000
0
50
100
150
200
Iteration

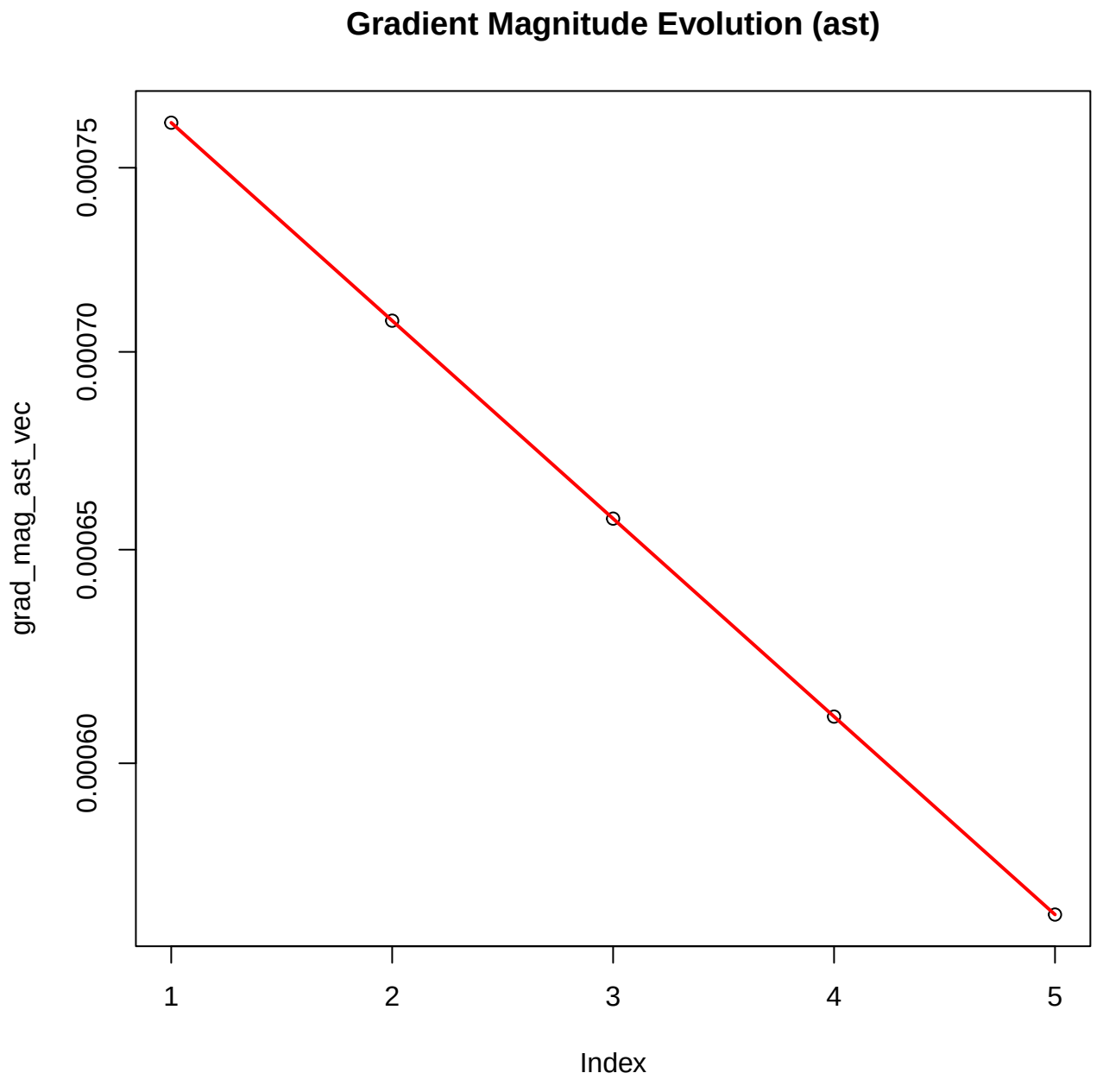

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.14
0.15
0.16
0.17
0.18
0.19
1
2
3
4
5
Index

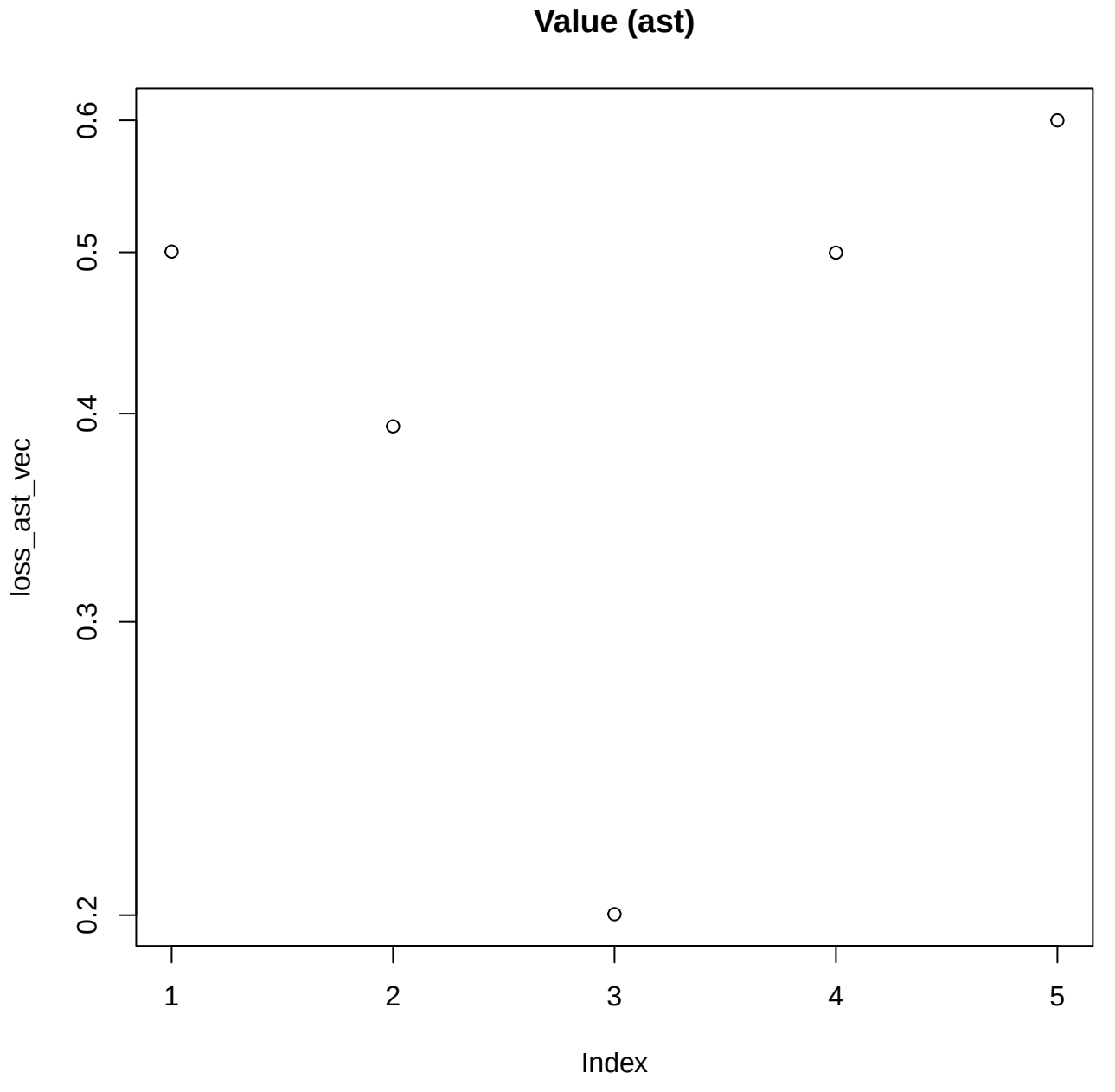

Value (ast)
loss_ast_vec
0.45
0.50
0.55
0.60
1
2
3
4
5
Index

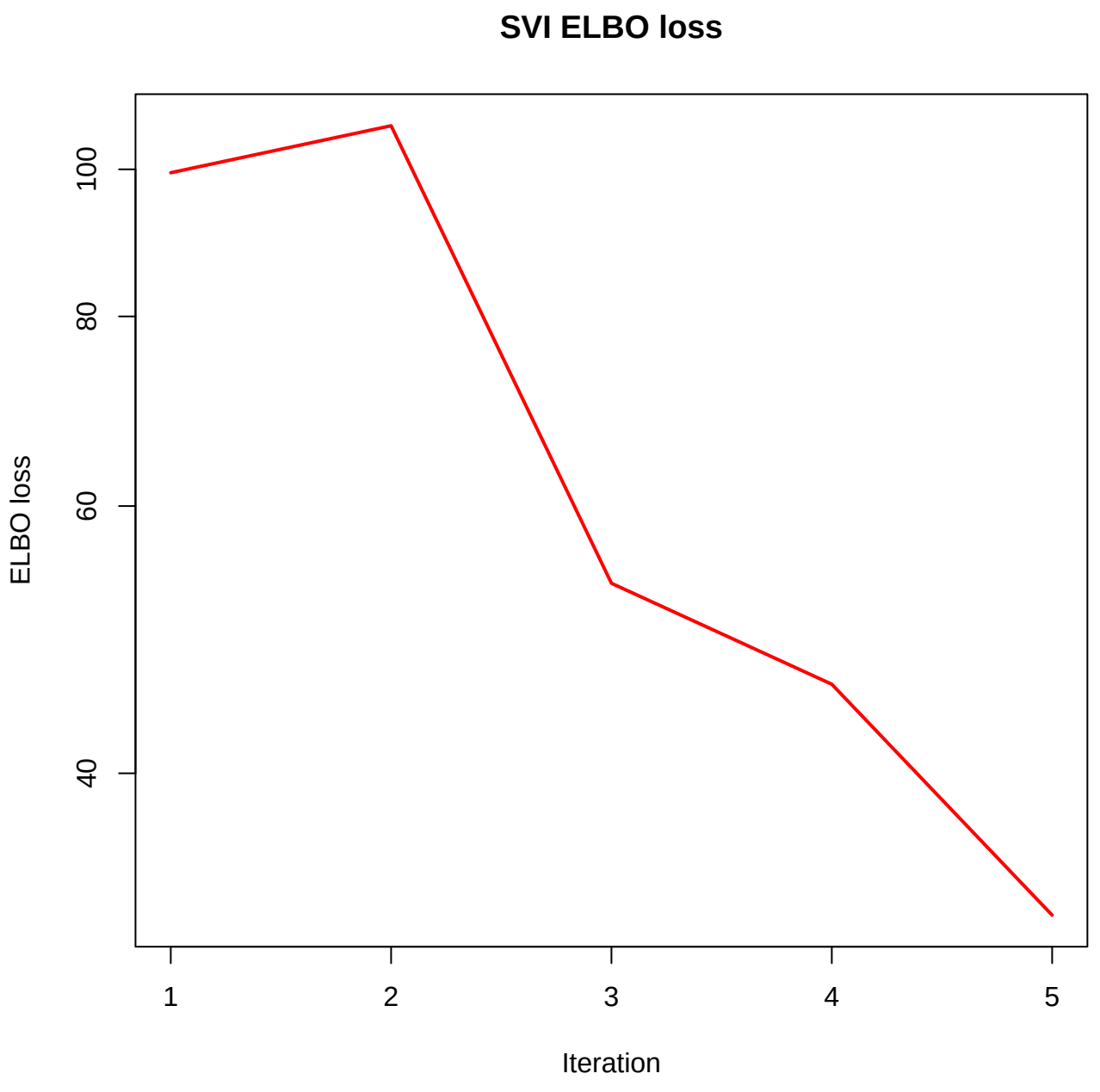

SVI ELBO loss
ELBO loss
9000
8000
7000
6000
5000
4000
0
50
100
150
200
Iteration

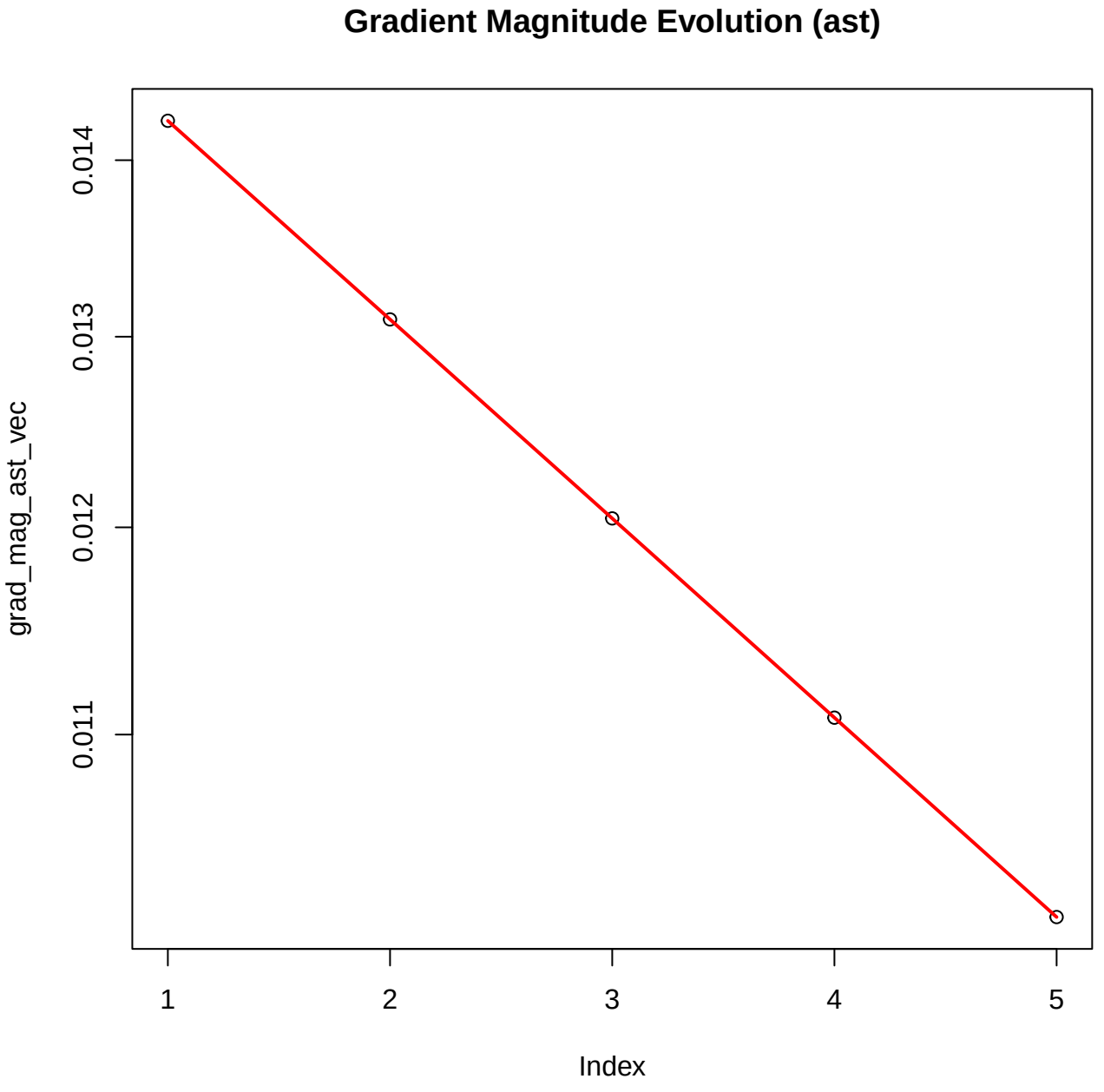

SVI ELBO loss
ELBO loss
10000
8000
6000
4000
0
50
100
150
200
Iteration

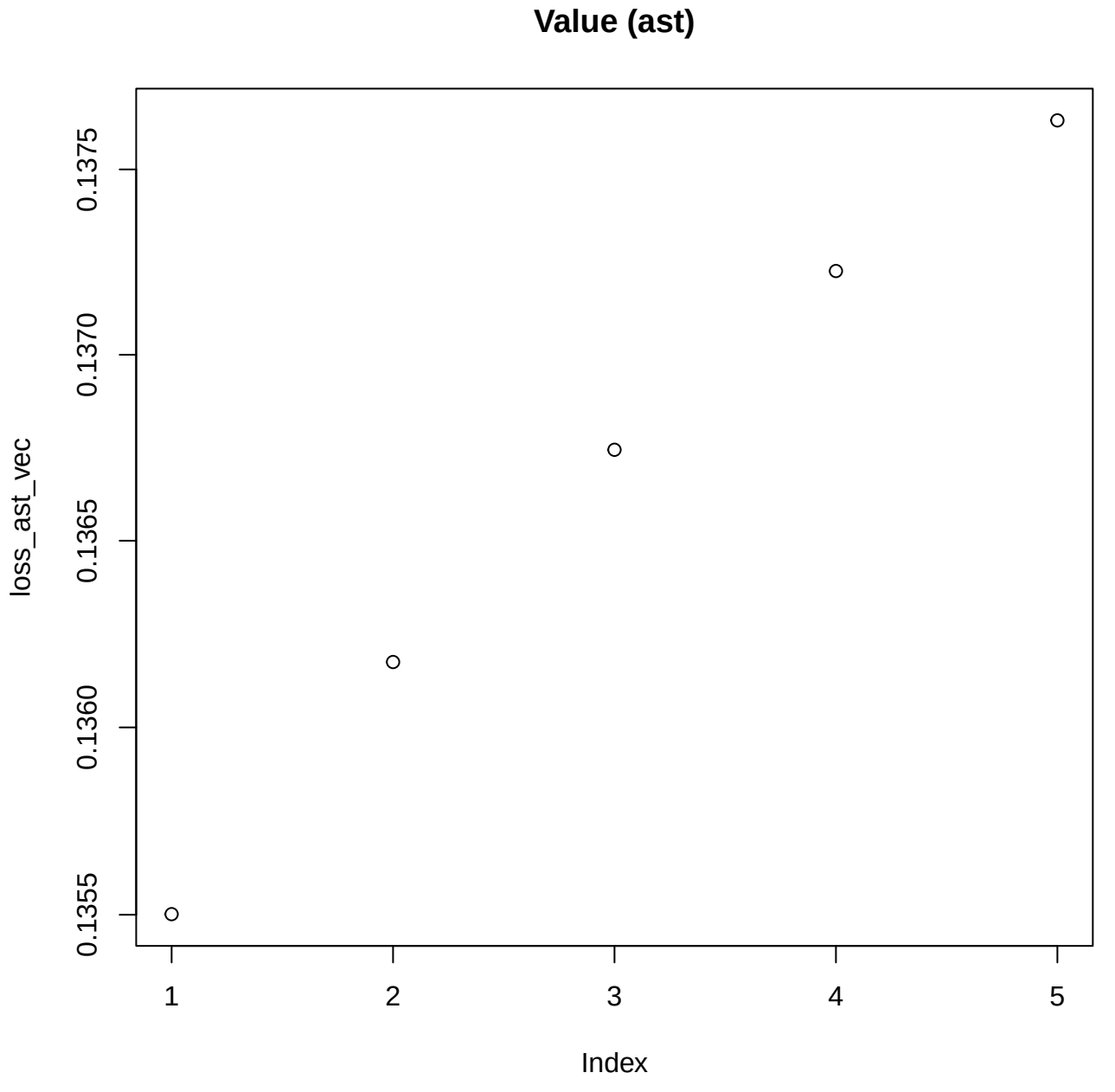

SVI ELBO loss
ELBO loss
35000
25000
20000
15000
10000
0
50
100
150
200
Iteration

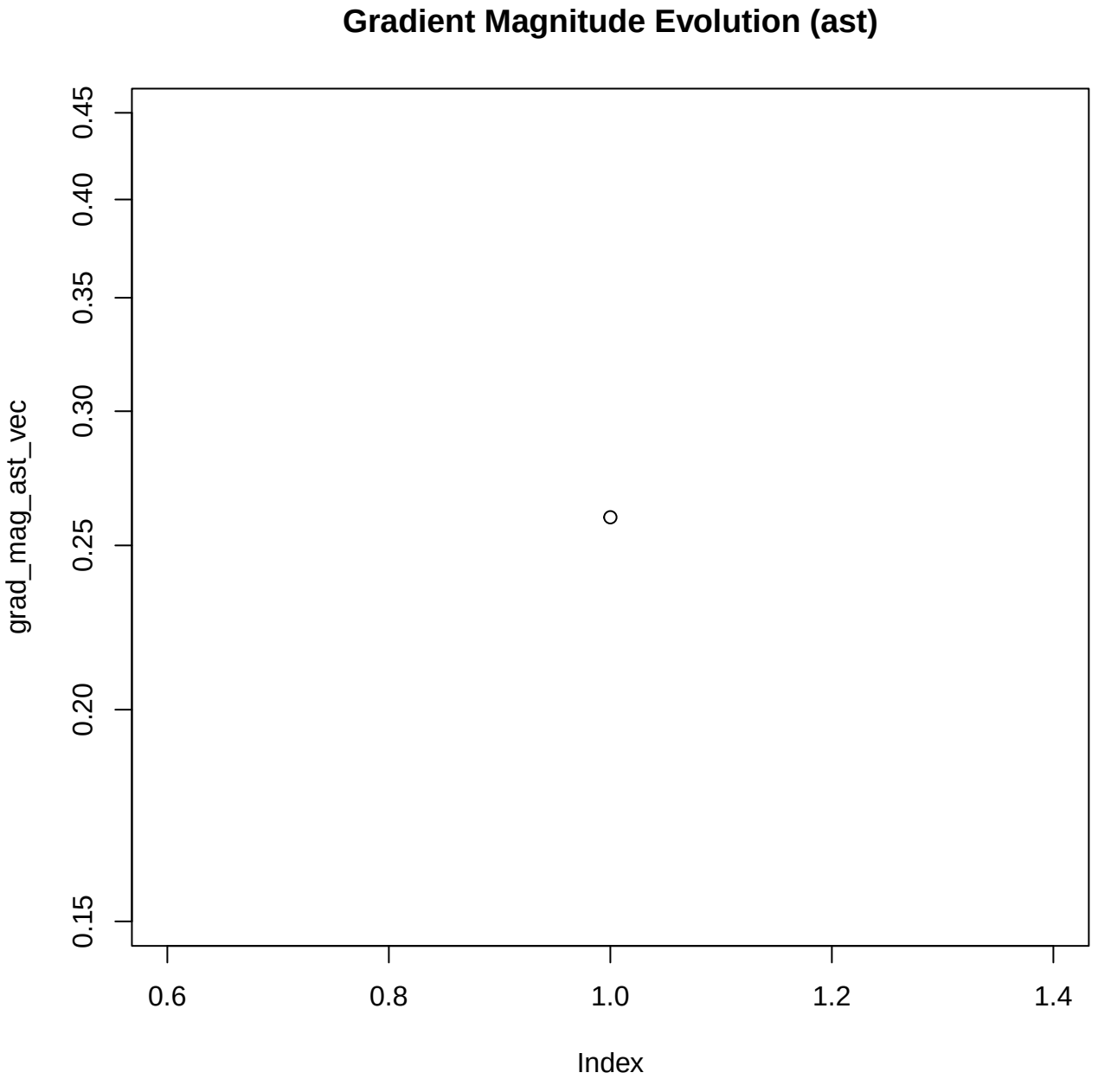

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
1
2
3
4
5
Index

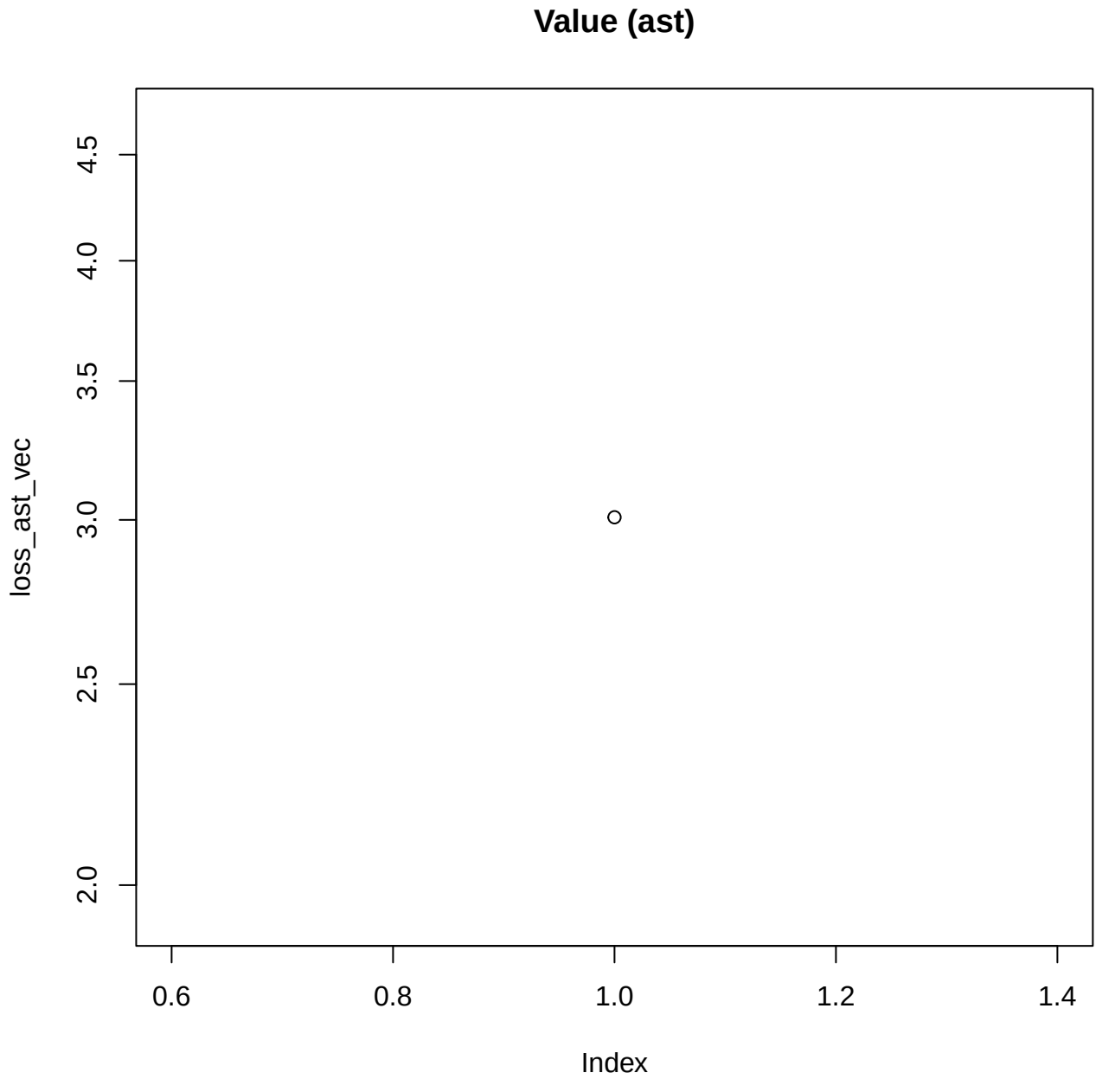

Value (ast)
loss_ast_vec
0.2
0.3
0.4
0.5
0.6
1
2
3
4
5
Index

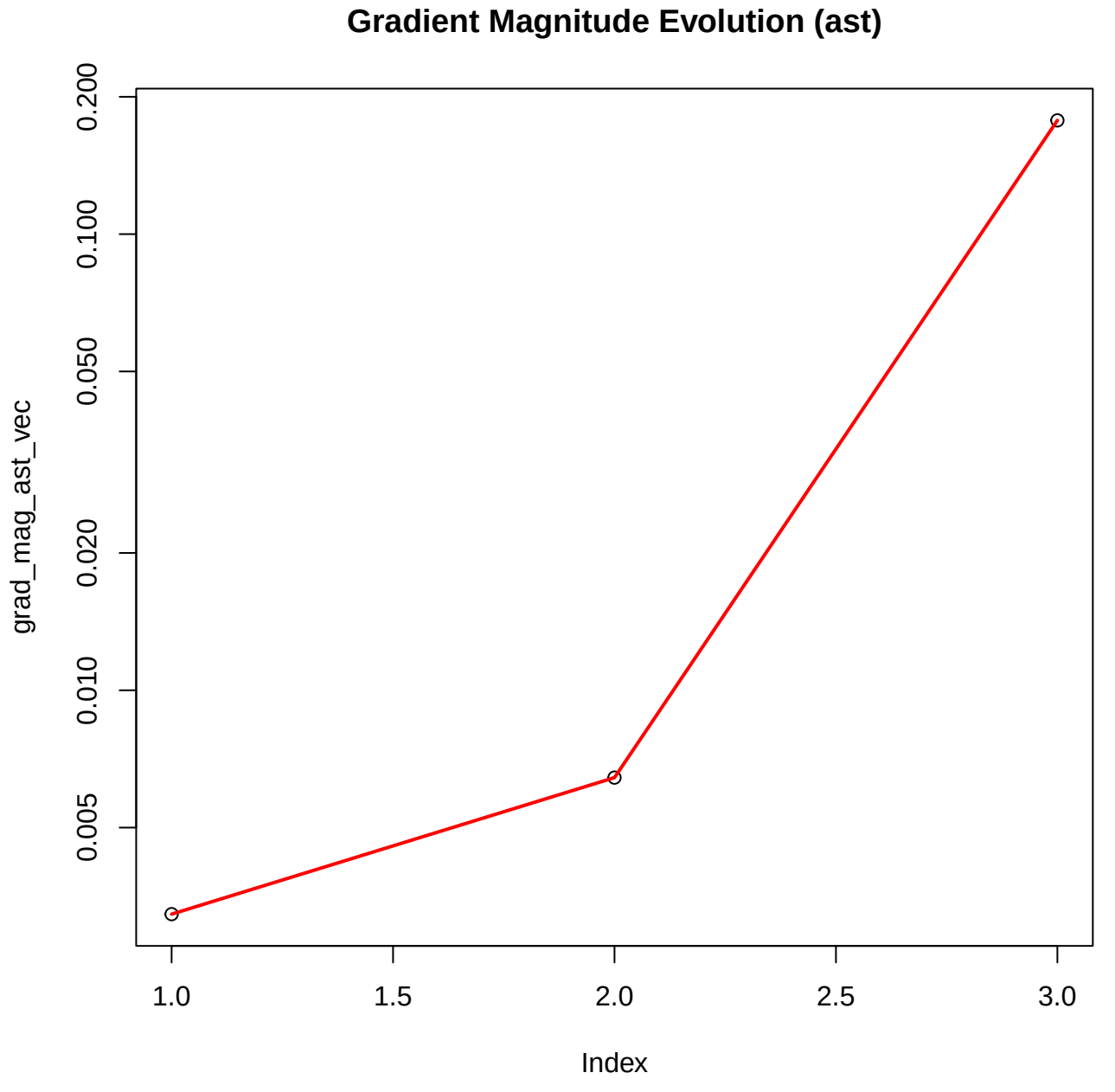

SVI ELBO loss
ELBO loss
100
80
60
40
1
2
3
4
5
Iteration

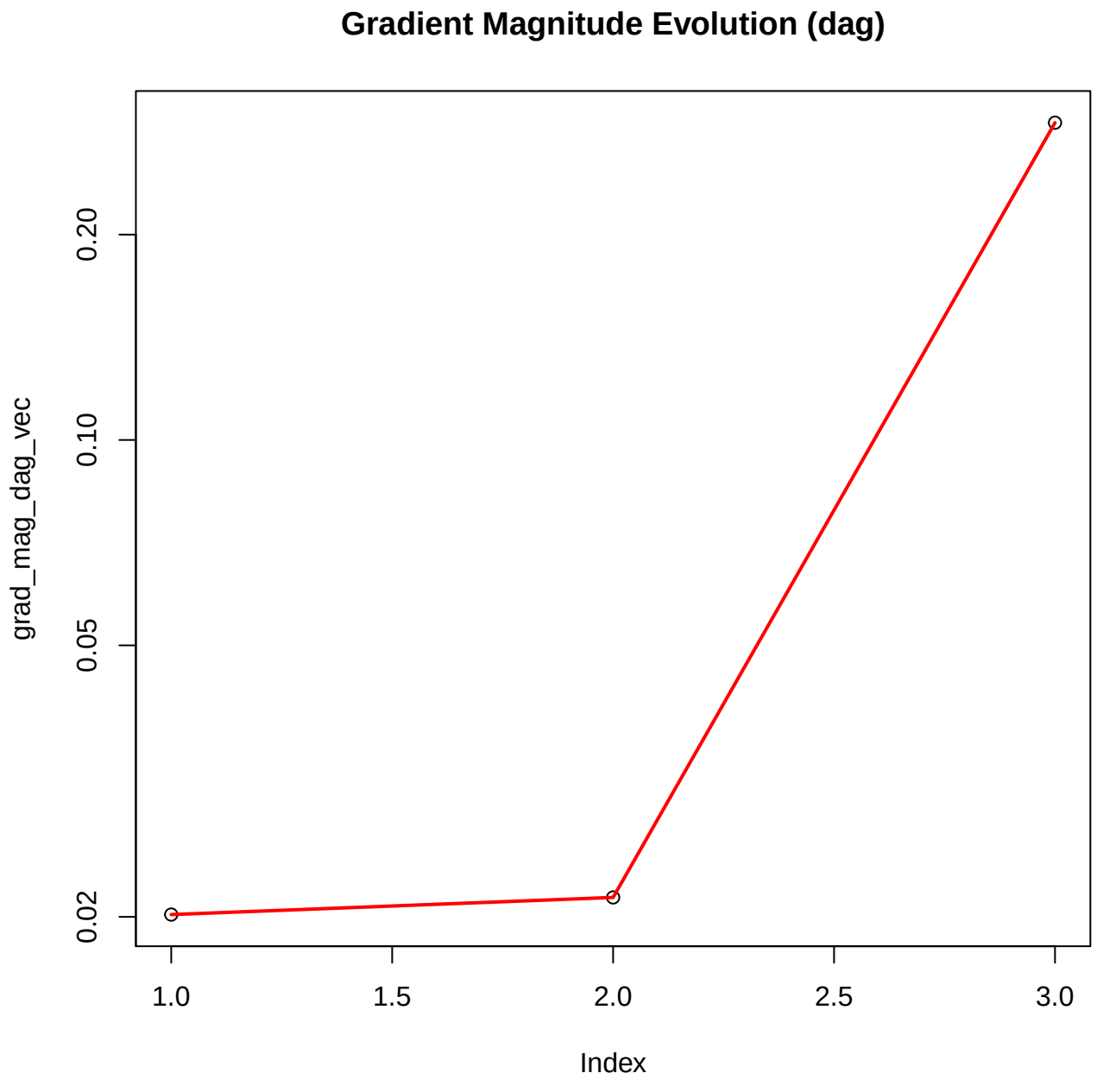

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.011
0.012
0.013
0.014
1
2
3
4
5
Index

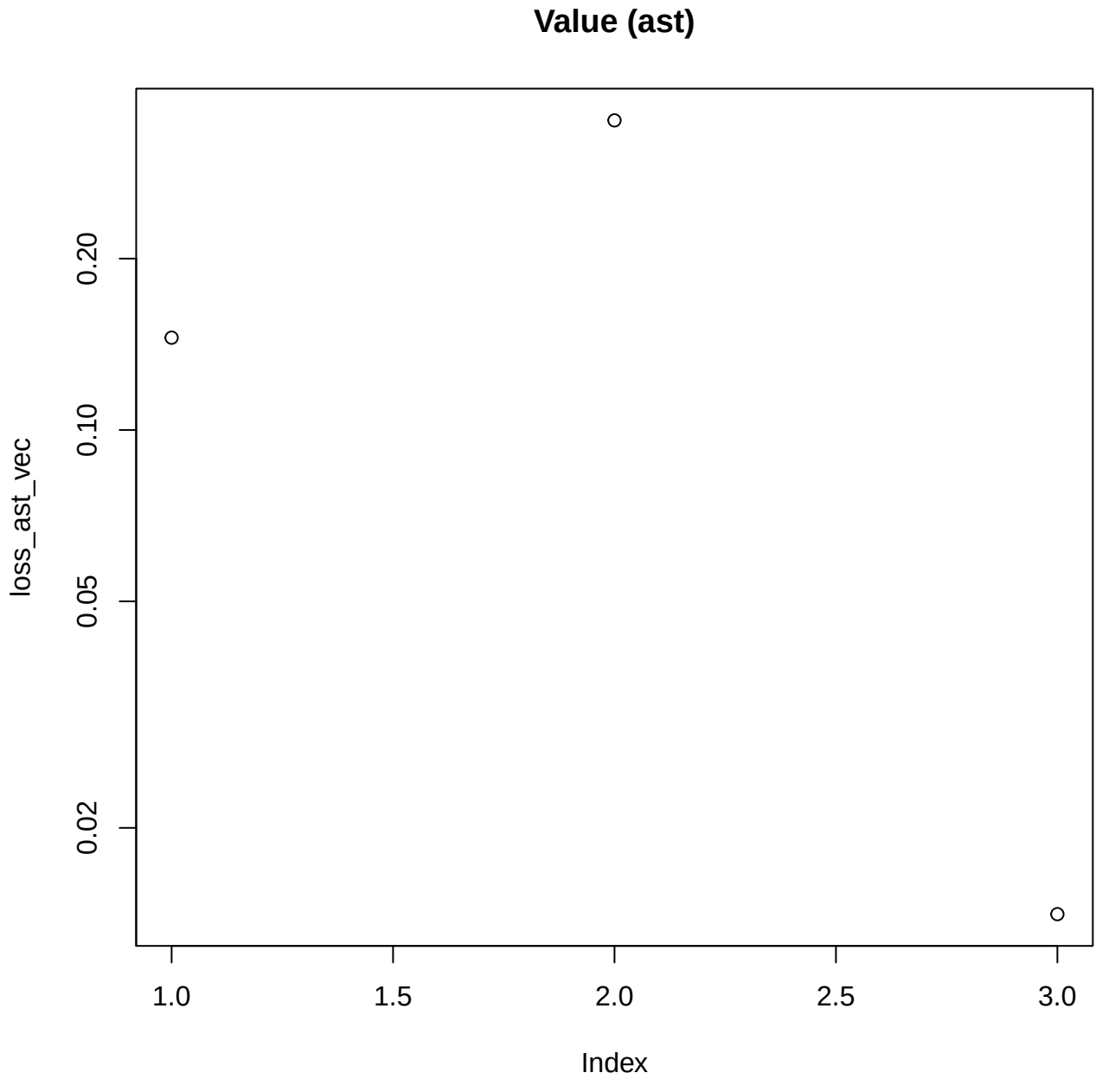

Value (ast)
loss_ast_vec
0.1355
0.1360
0.1365
0.1370
0.1375
1
2
3
4
5
Index

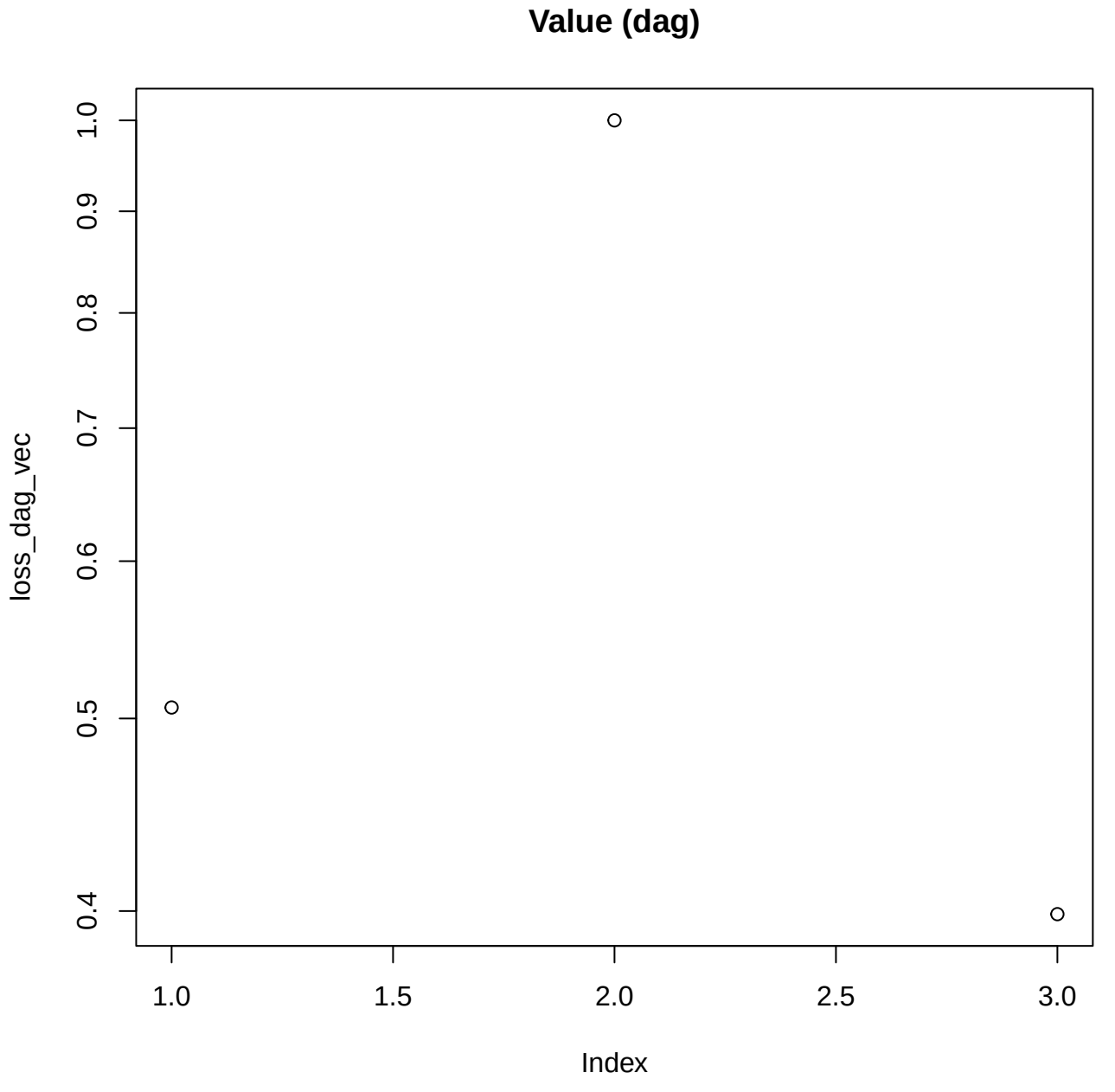

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.15
0.20
0.25
0.30
0.35
0.40
0.45
0.6
0.8
1.0
1.2
1.4
Index

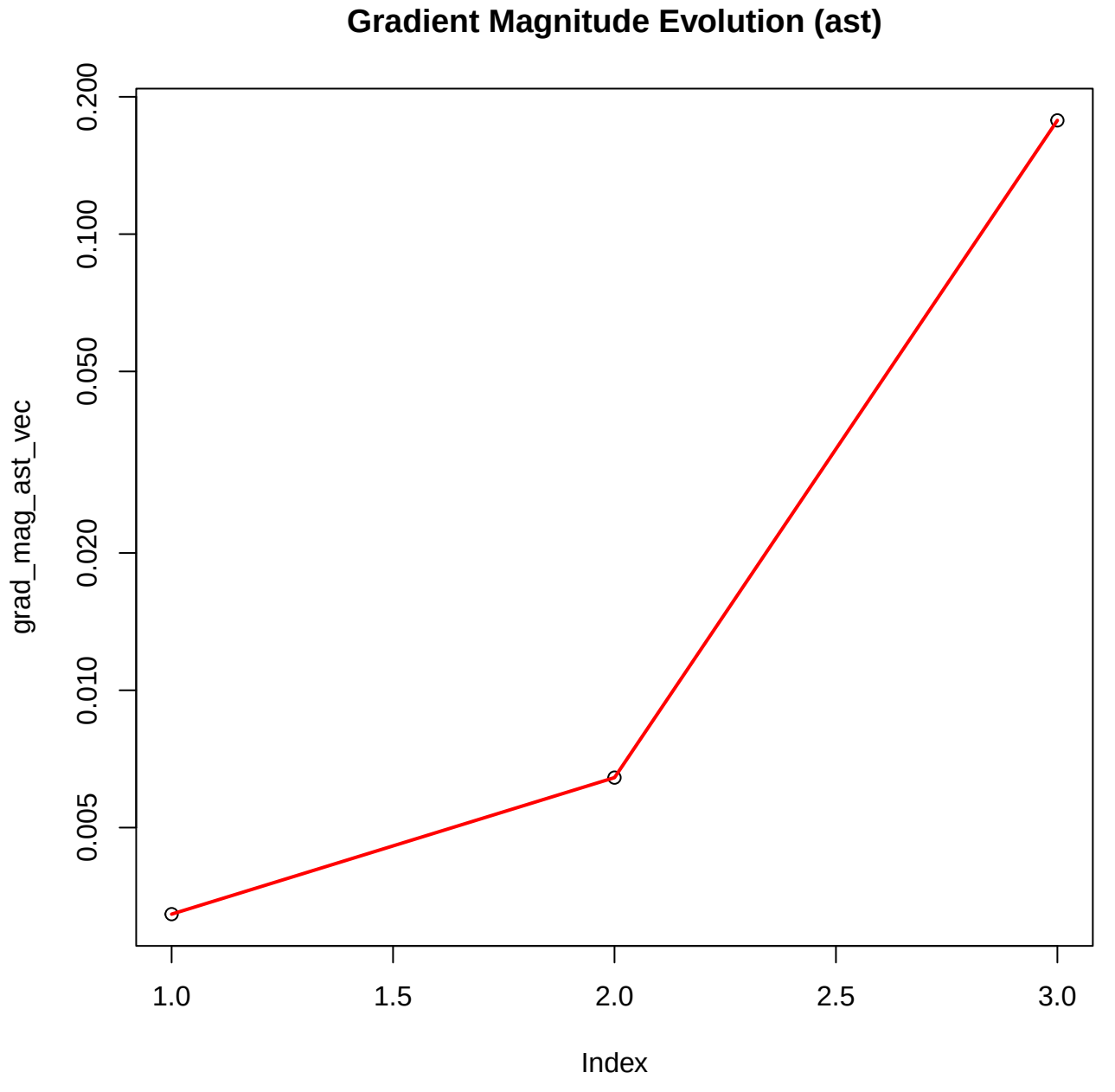

Value (ast)
loss_ast_vec
4.5
4.0
3.5
3.0
2.5
2.0
0.6
0.8
1.0
1.2
1.4
Index

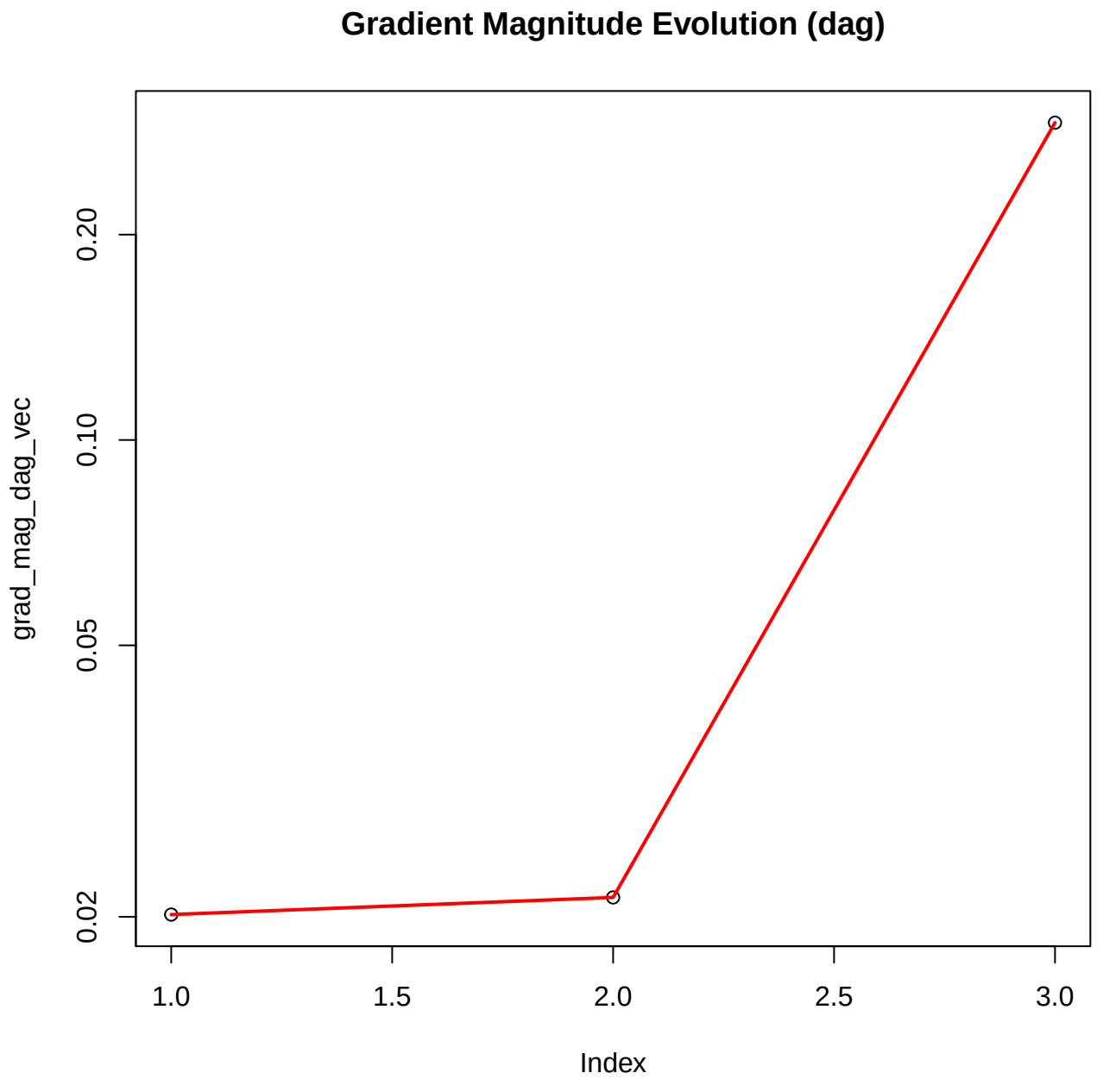

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.005
0.010
0.020
0.050
0.100
0.200
1.0
1.5
2.0
2.5
3.0
Index

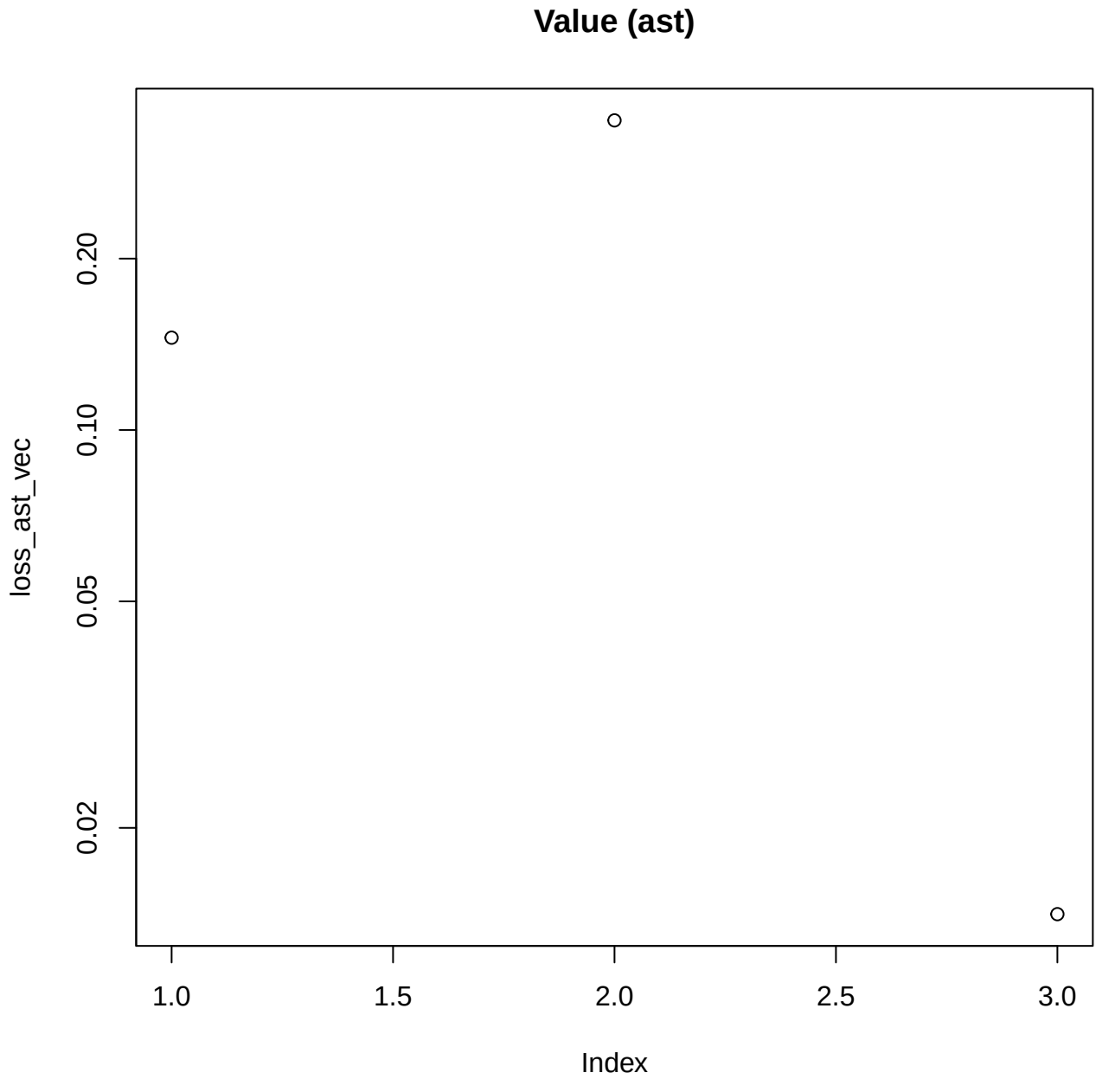

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.02
0.05
0.10
0.20
1.0
1.5
2.0
2.5
3.0
Index

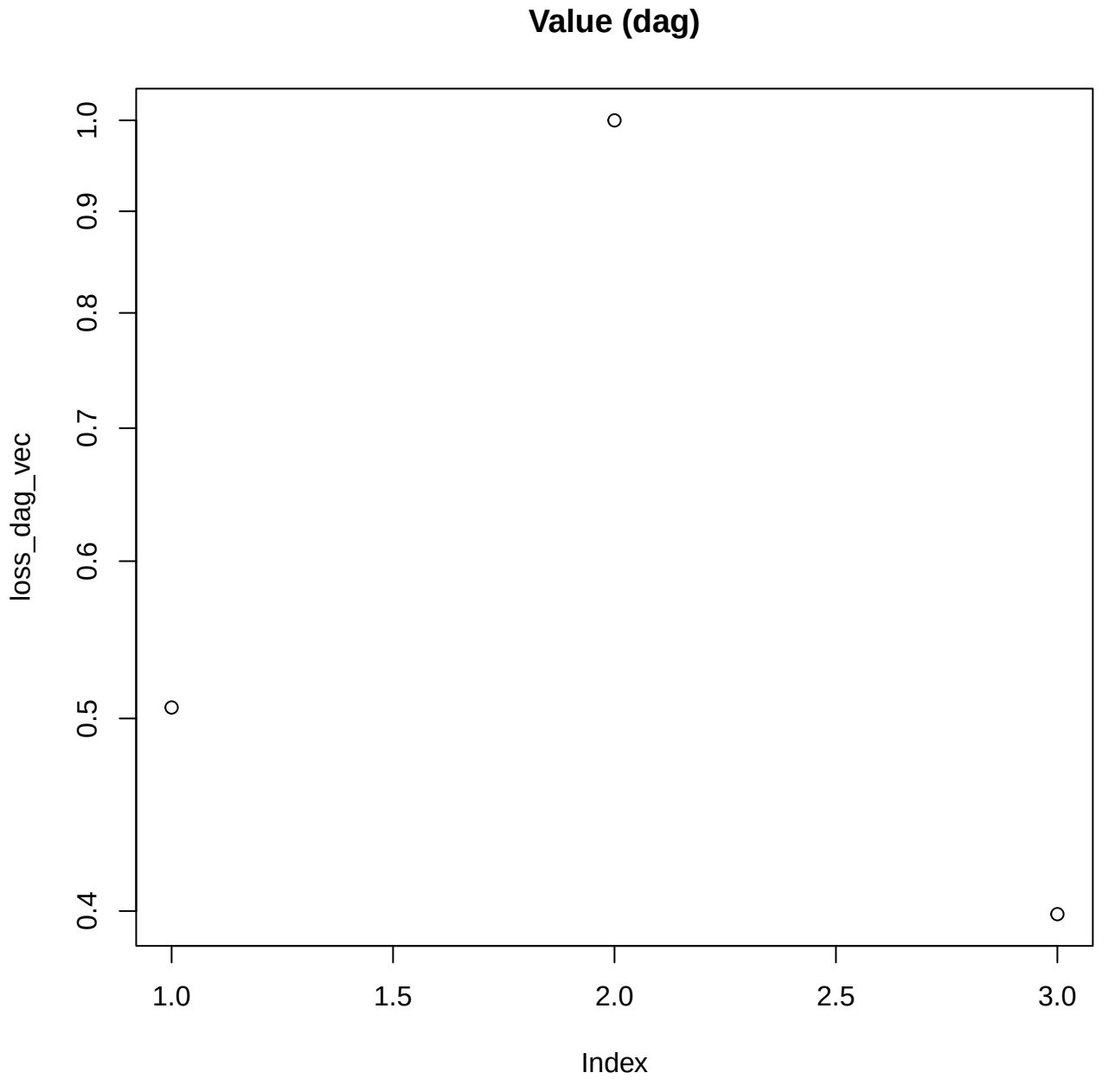

Value (ast)
loss_ast_vec
0.02
0.05
0.10
0.20
1.0
1.5
2.0
2.5
3.0
Index

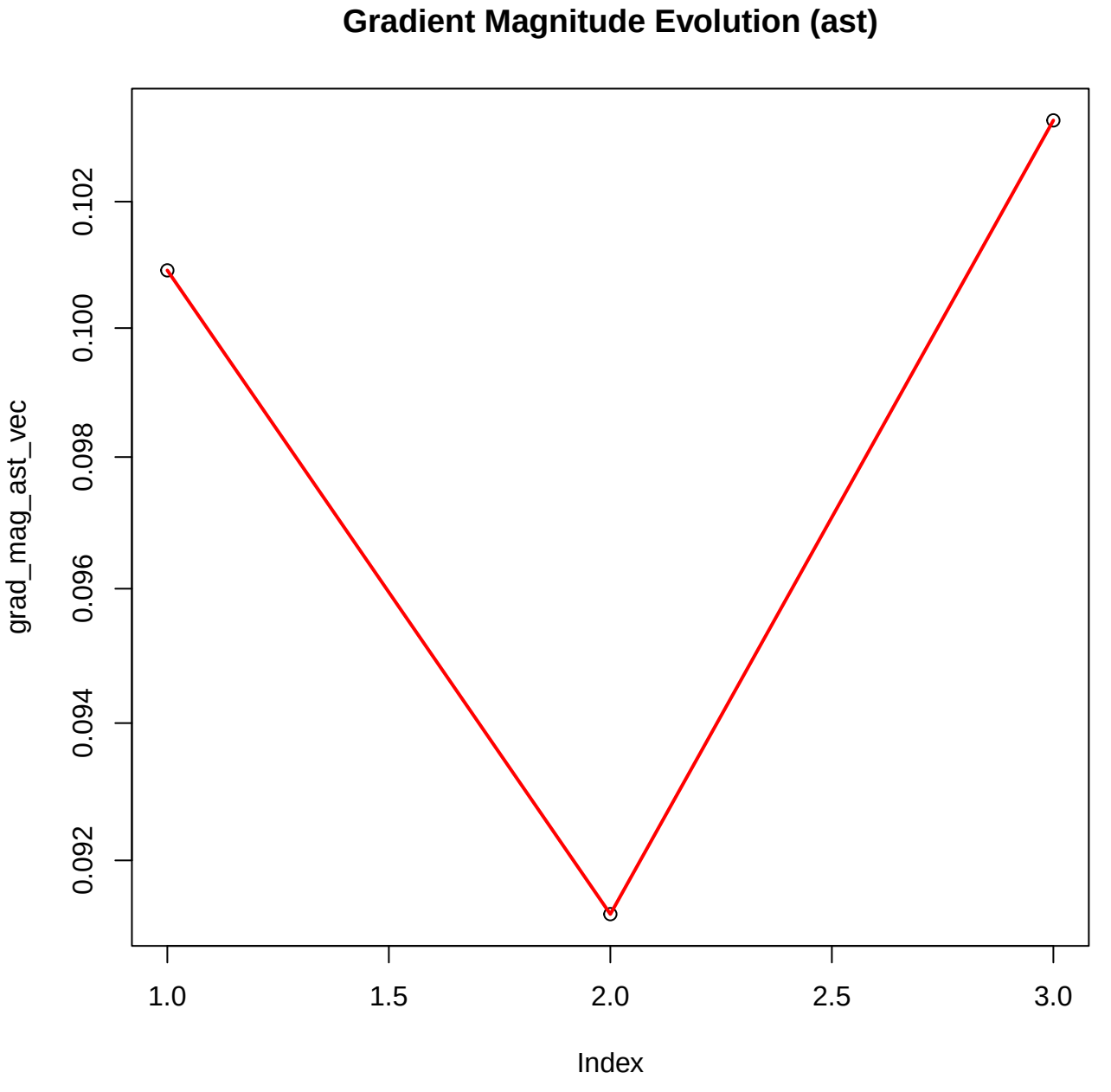

Value (dag)
loss_dag_vec
1.0
0.9
0.8
0.7
0.6
0.5
0.4
1.0
1.5
2.0
2.5
3.0
Index

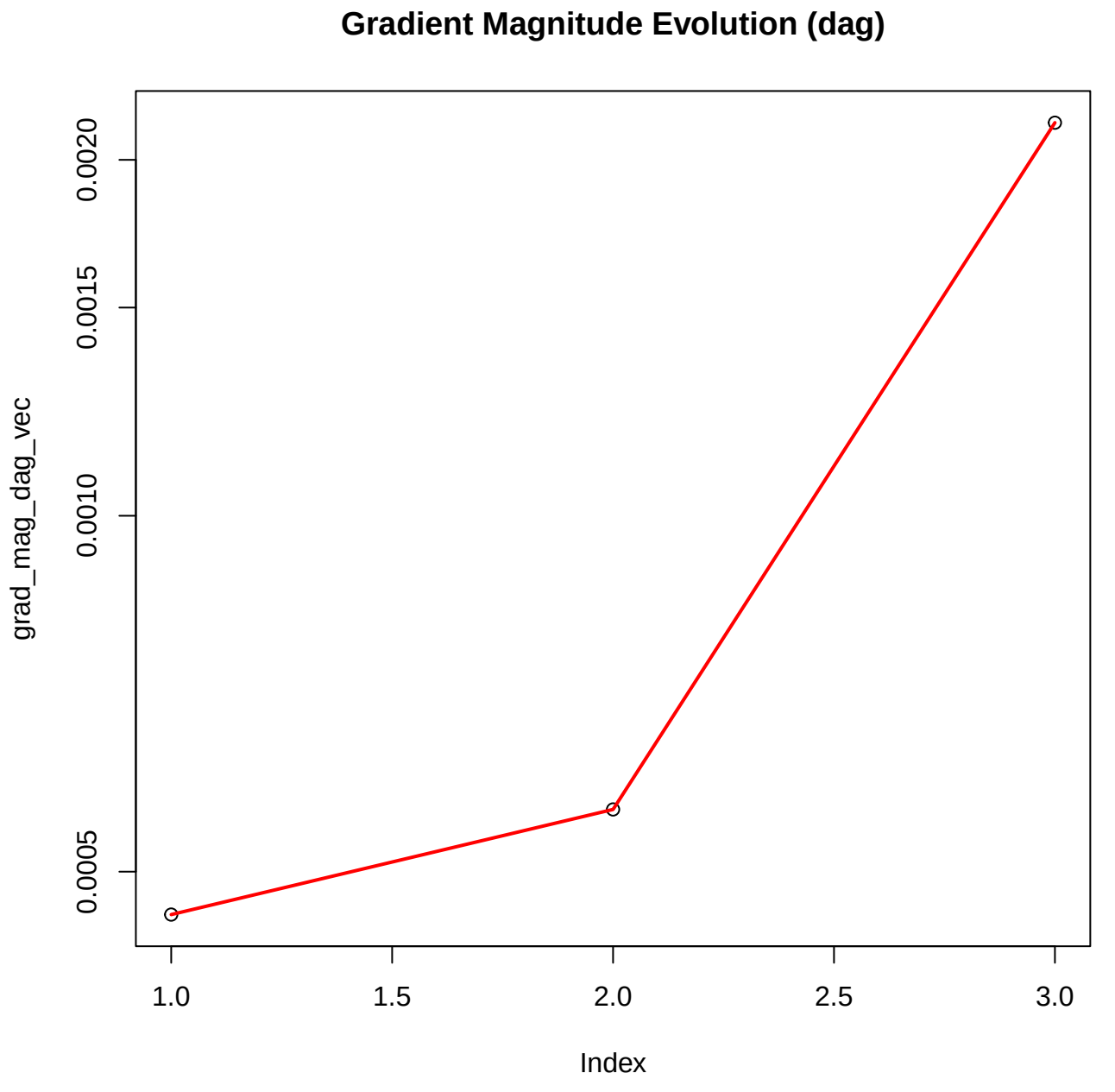

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.005
0.010
0.020
0.050
0.100
0.200
1.0
1.5
2.0
2.5
3.0
Index

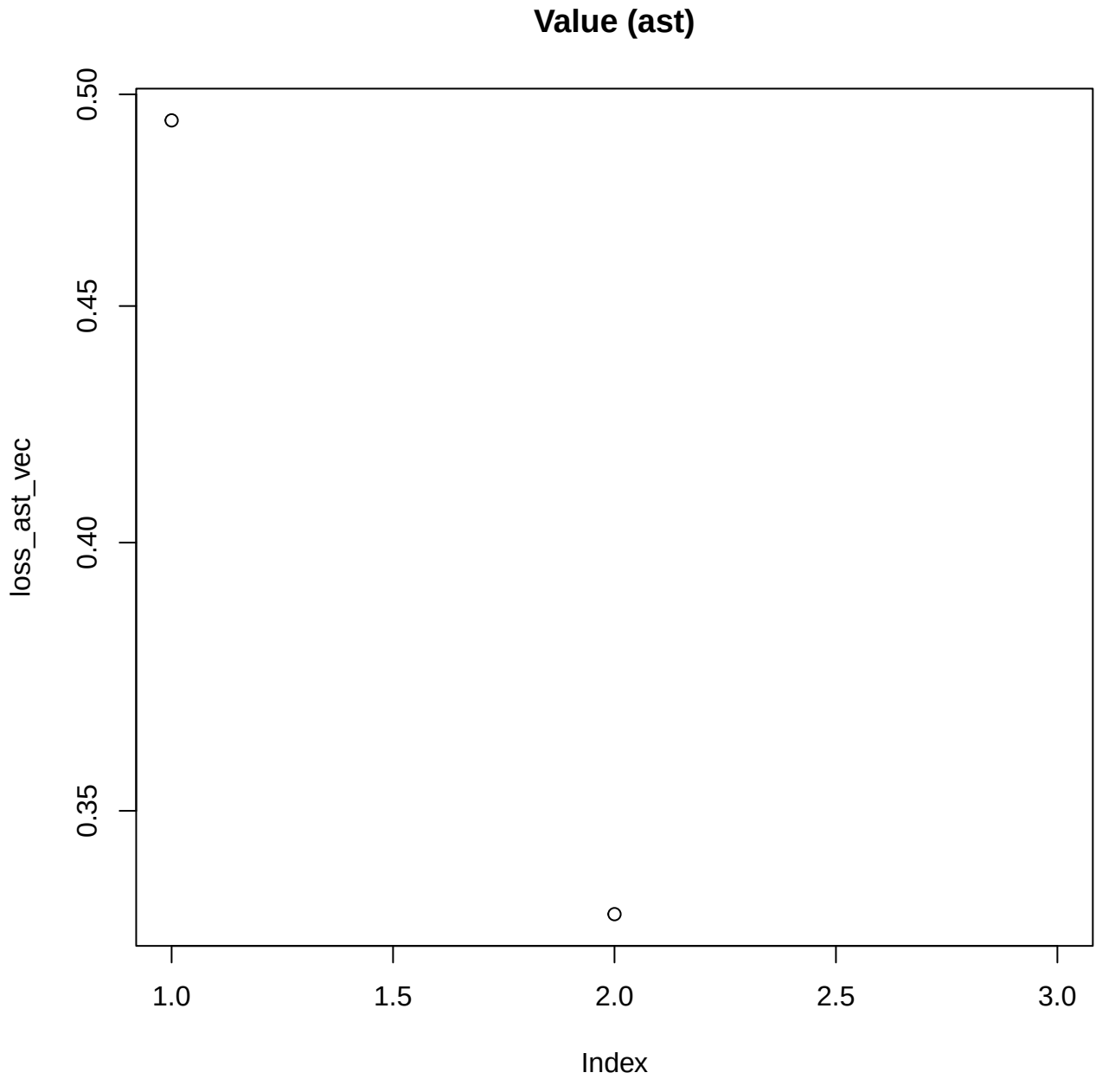

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.02
0.05
0.10
0.20
1.0
1.5
2.0
2.5
3.0
Index

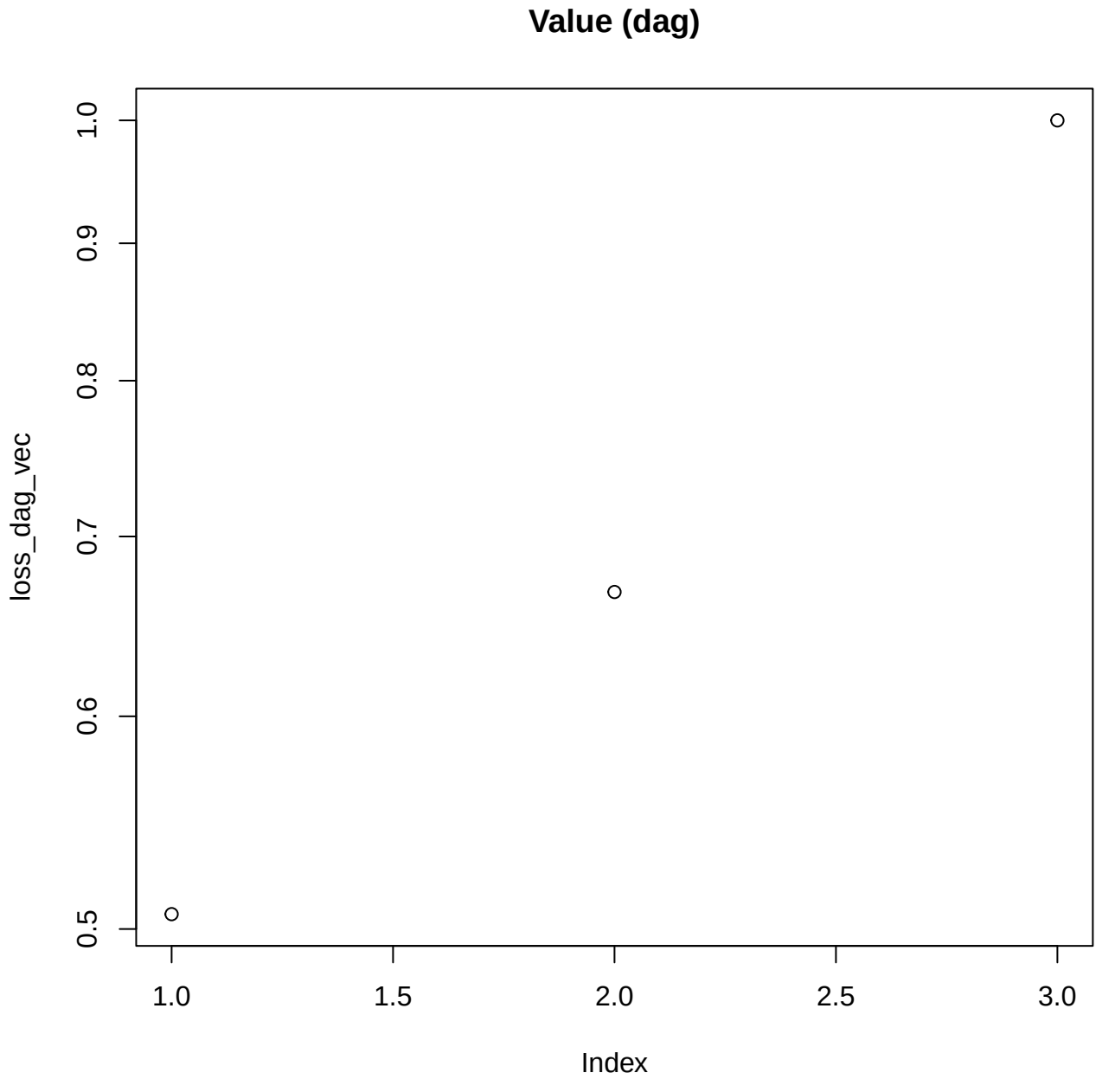

Value (ast)
loss_ast_vec
0.02
0.05
0.10
0.20
1.0
1.5
2.0
2.5
3.0
Index

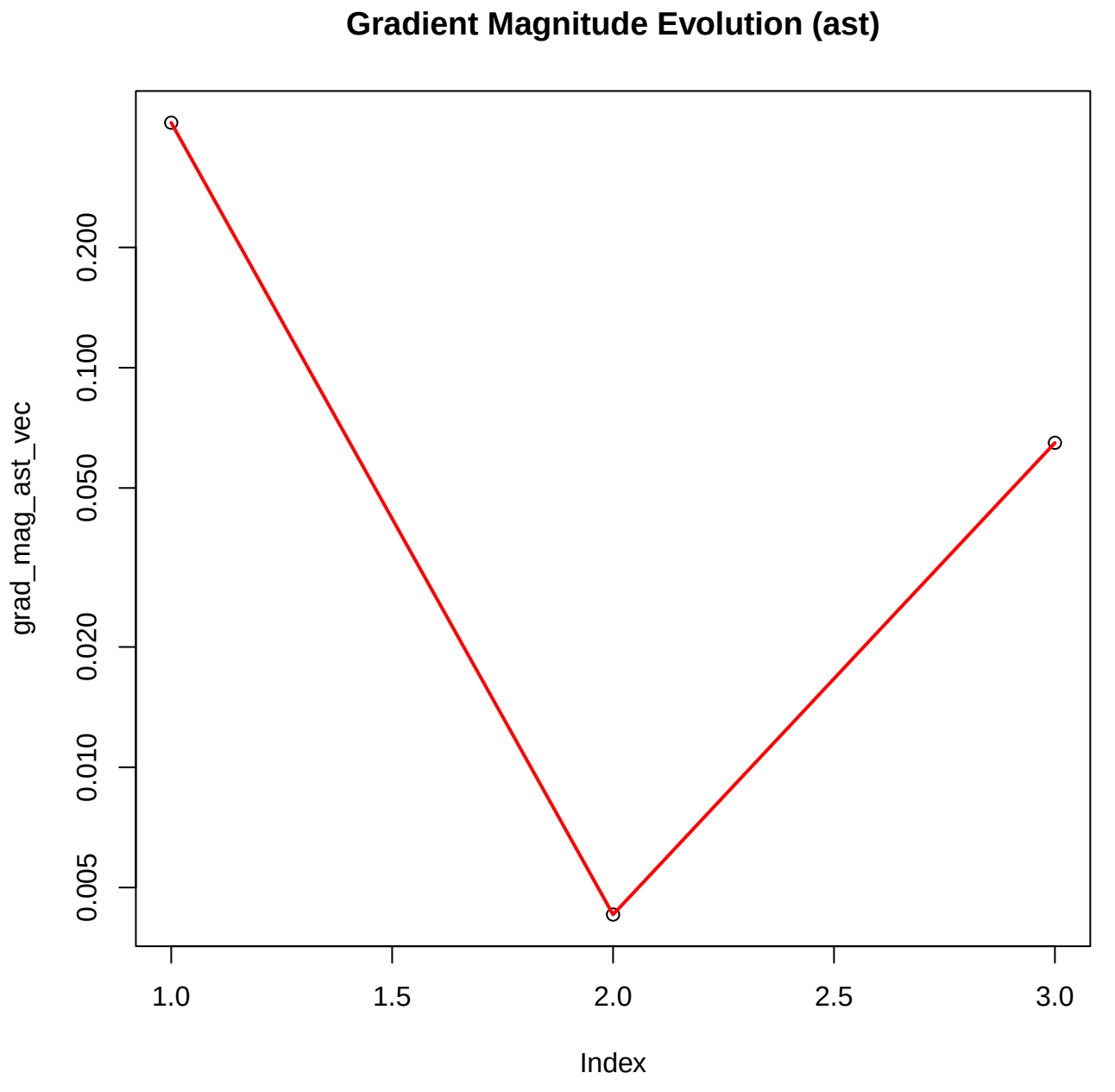

Value (dag)
loss_dag_vec
1.0
0.9
0.8
0.7
0.6
0.5
0.4
1.0
1.5
2.0
2.5
3.0
Index

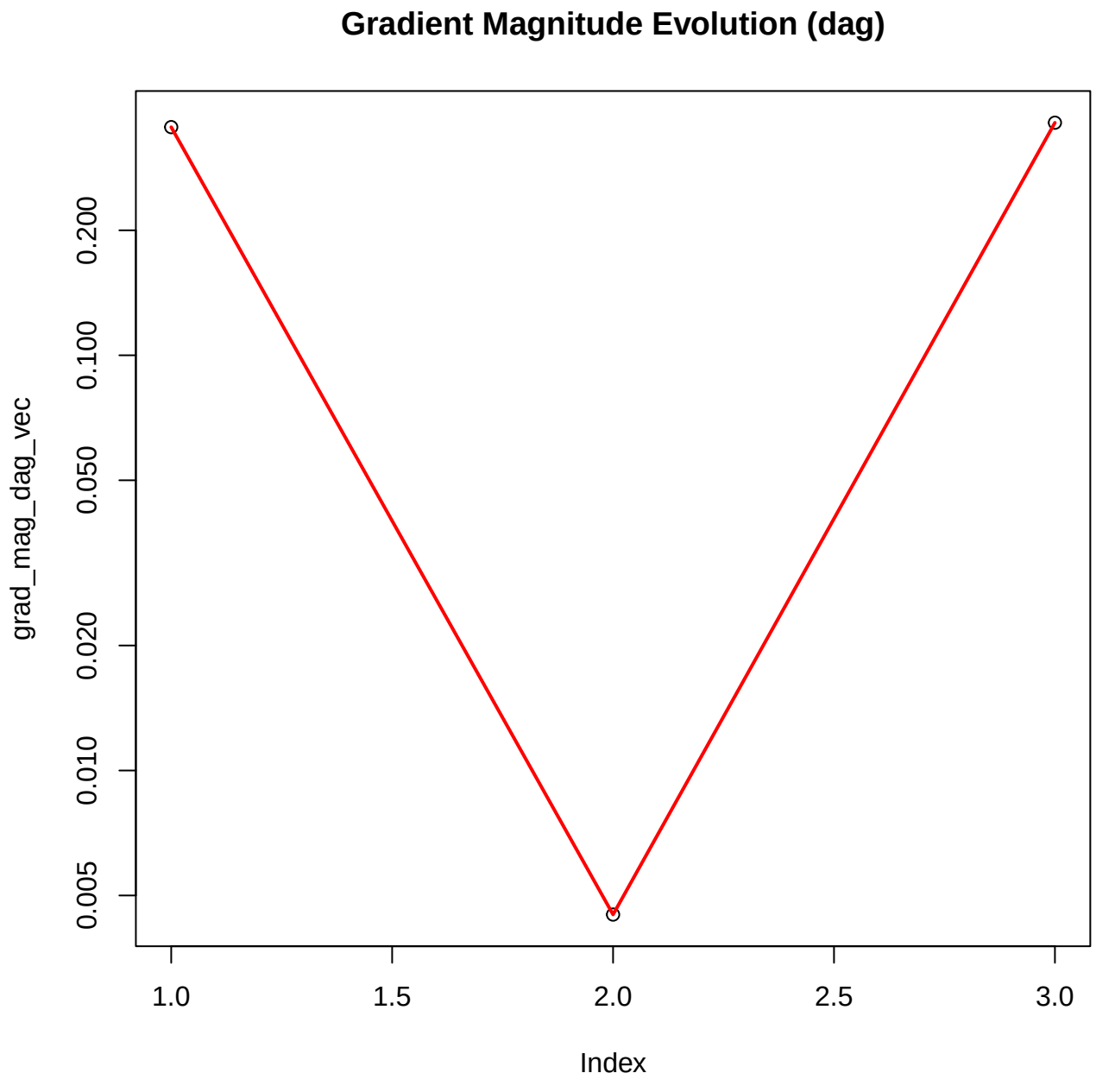

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.092
0.094
0.096
0.098
0.100
0.102
1.0
1.5
2.0
2.5
3.0
Index

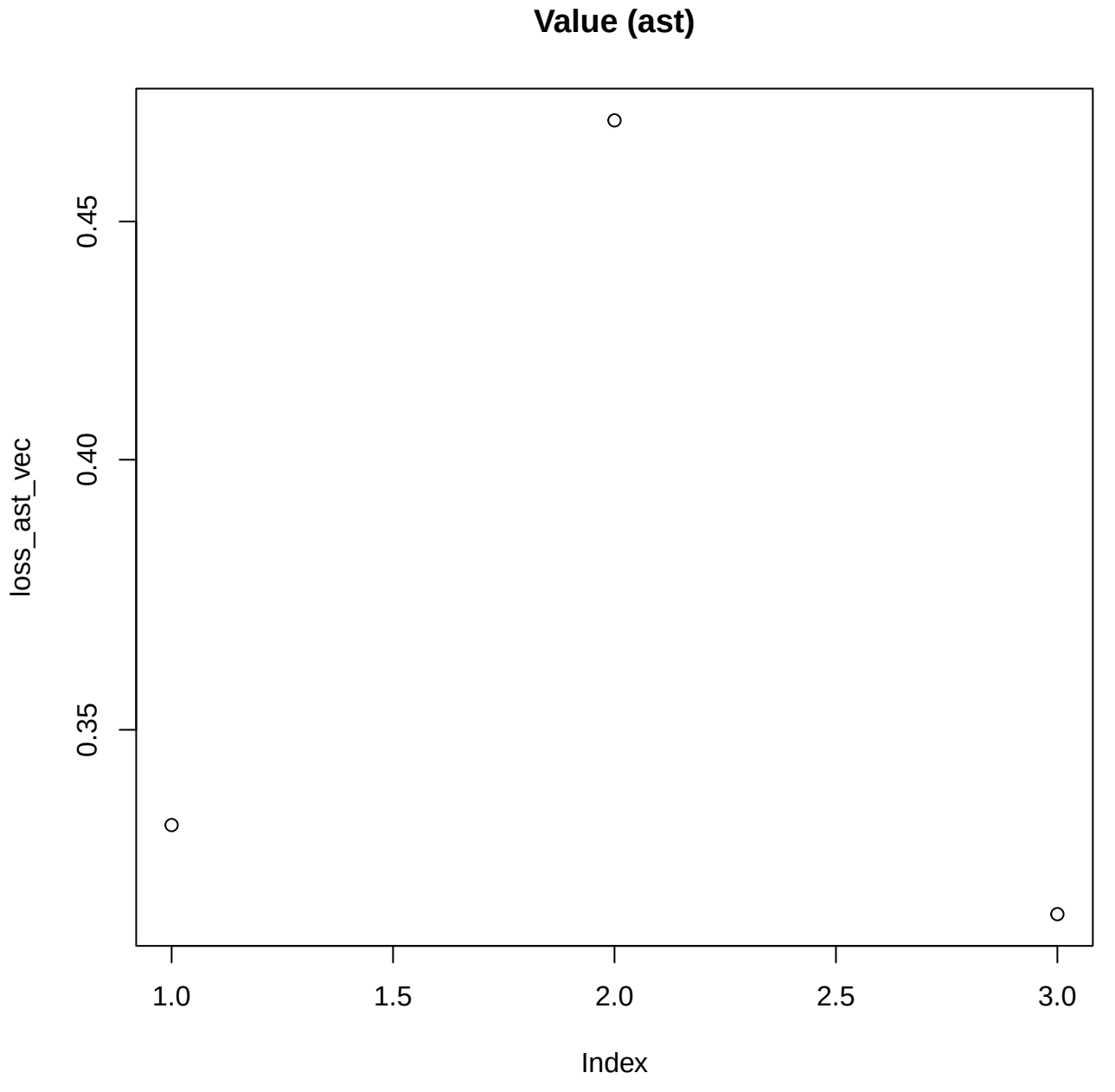

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.0005
0.0010
0.0015
0.0020
1.0
1.5
2.0
2.5
3.0
Index

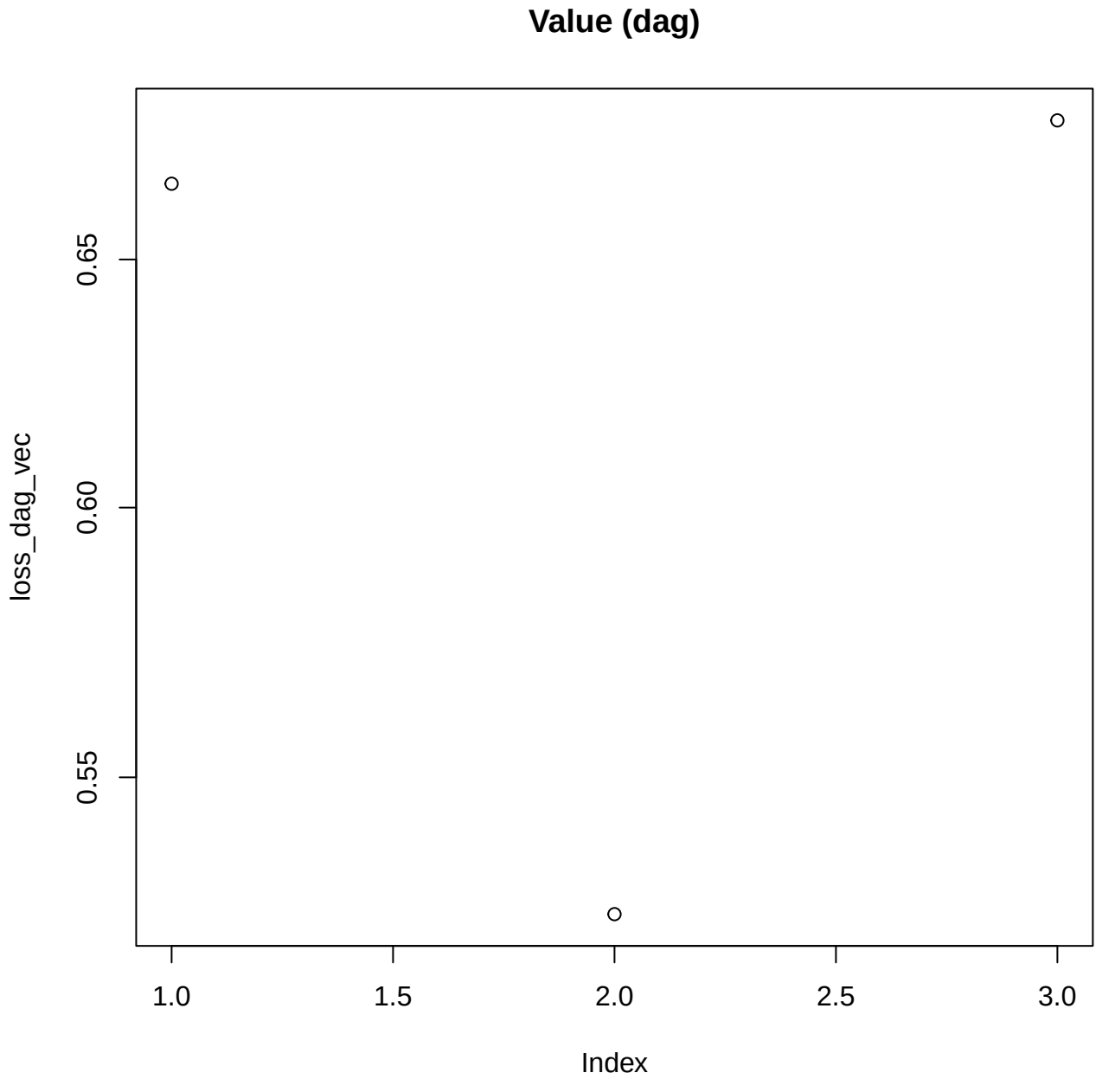

Value (ast)
loss_ast_vec
0.50
0.45
0.40
0.35
1.0
1.5
2.0
2.5
3.0
Index

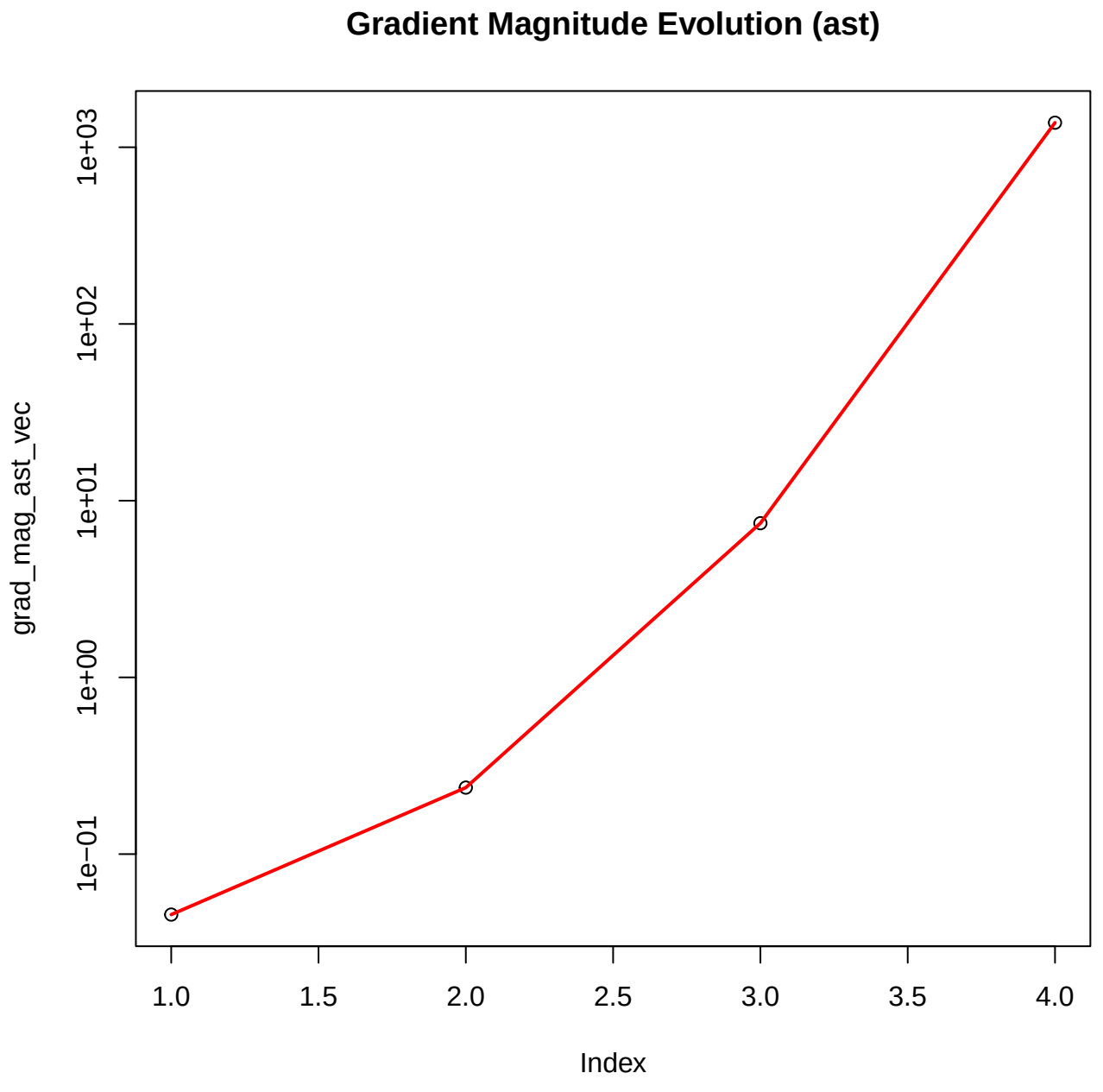

Value (dag)
loss_dag_vec
0.5
0.6
0.7
0.8
0.9
1.0
1.0
1.5
2.0
2.5
3.0
Index

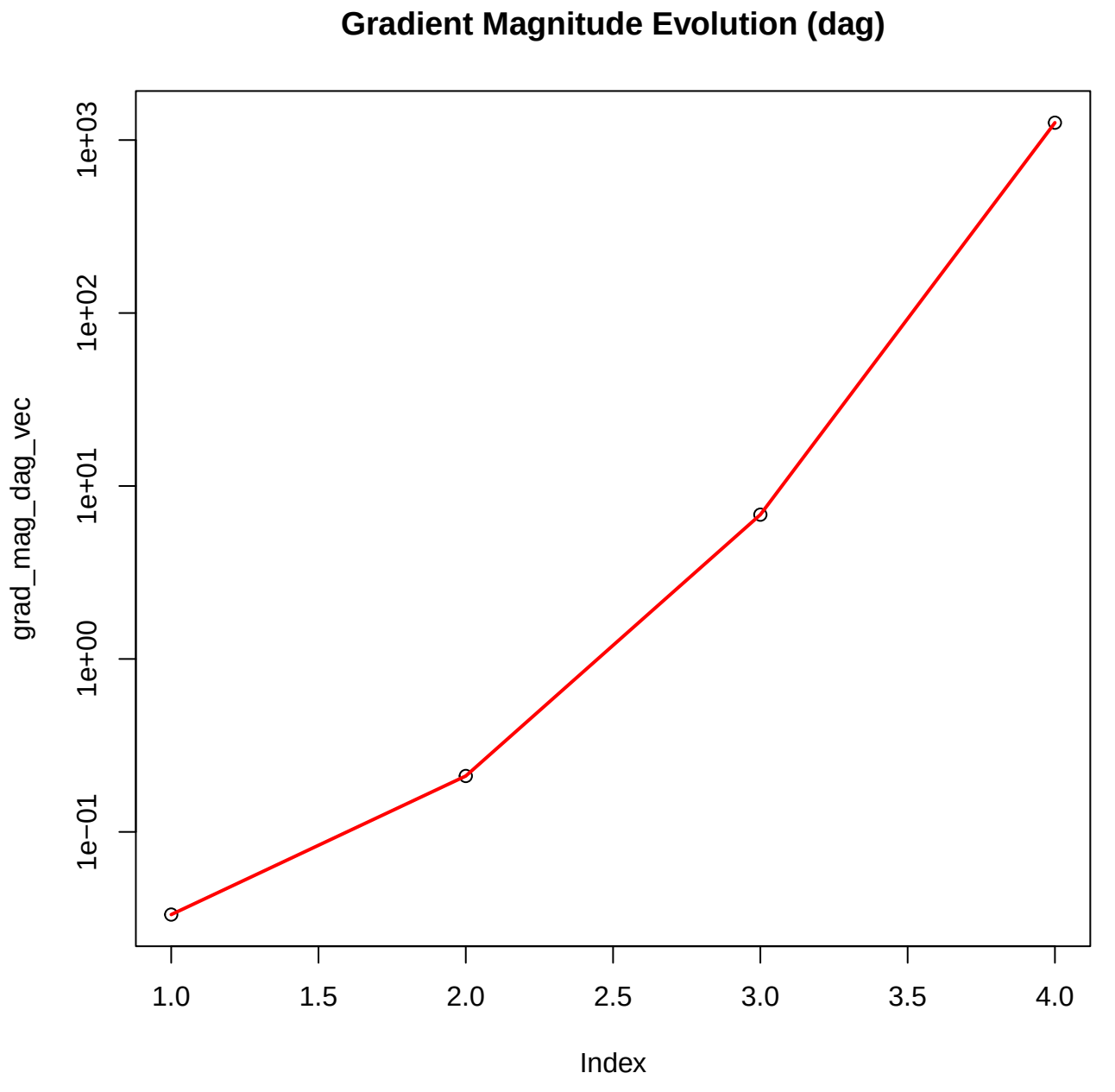

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.005
0.010
0.020
0.050
0.100
0.200
1.0
1.5
2.0
2.5
3.0
Index

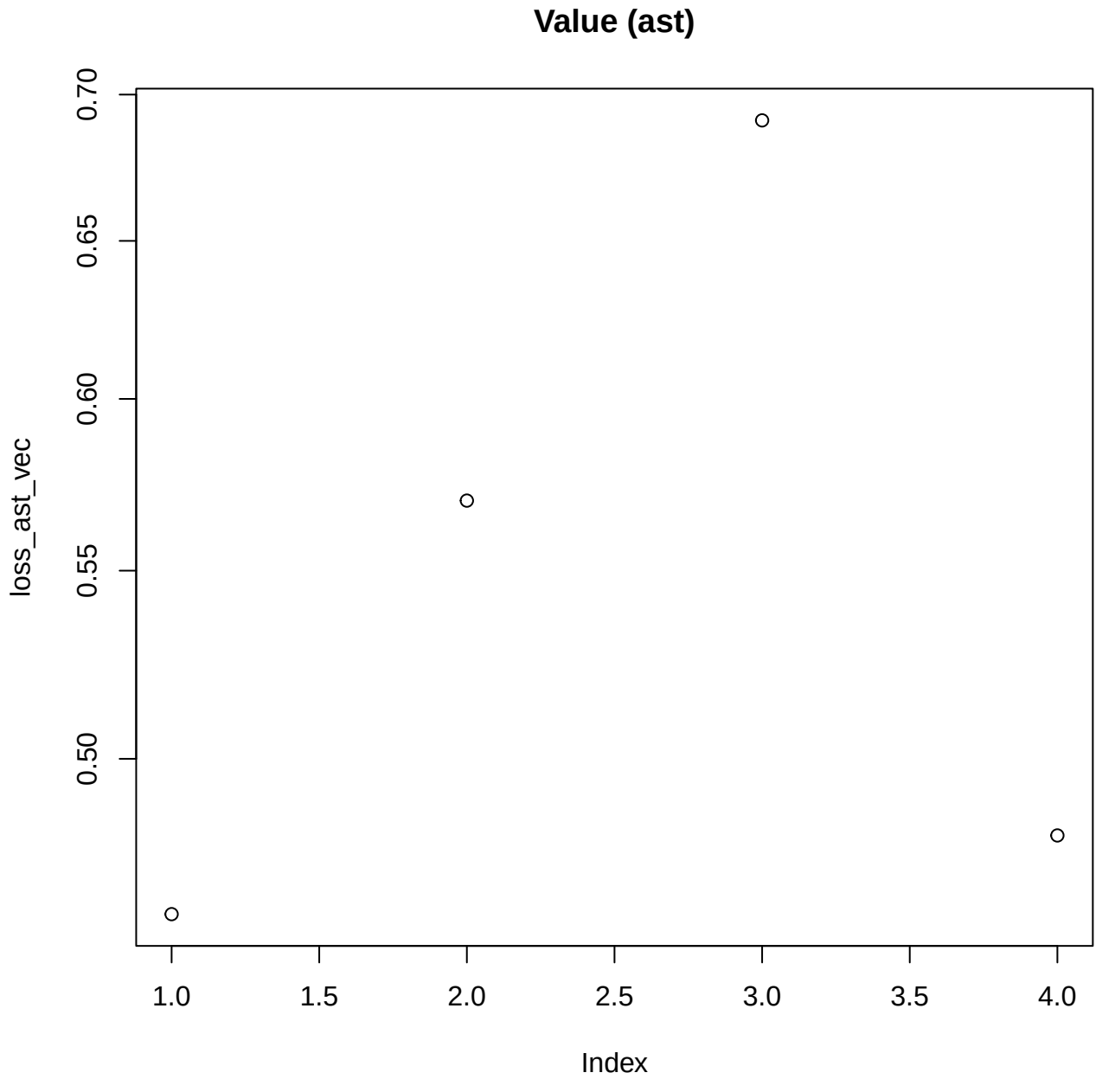

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.005
0.010
0.020
0.050
0.100
0.200
1.0
1.5
2.0
2.5
3.0
Index

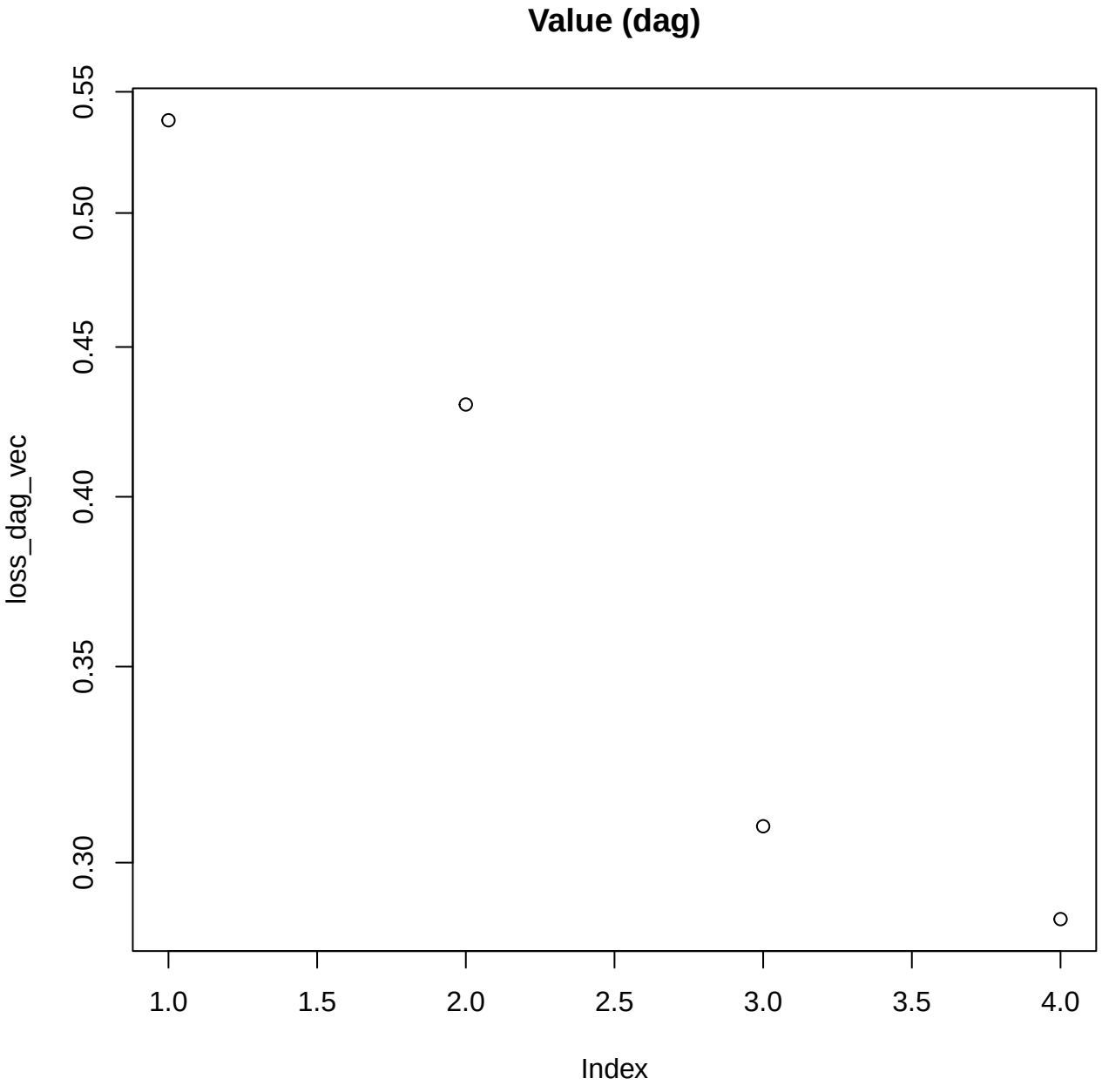

Value (ast)
loss_ast_vec
0.35
0.40
0.45
1.0
1.5
2.0
2.5
3.0
Index

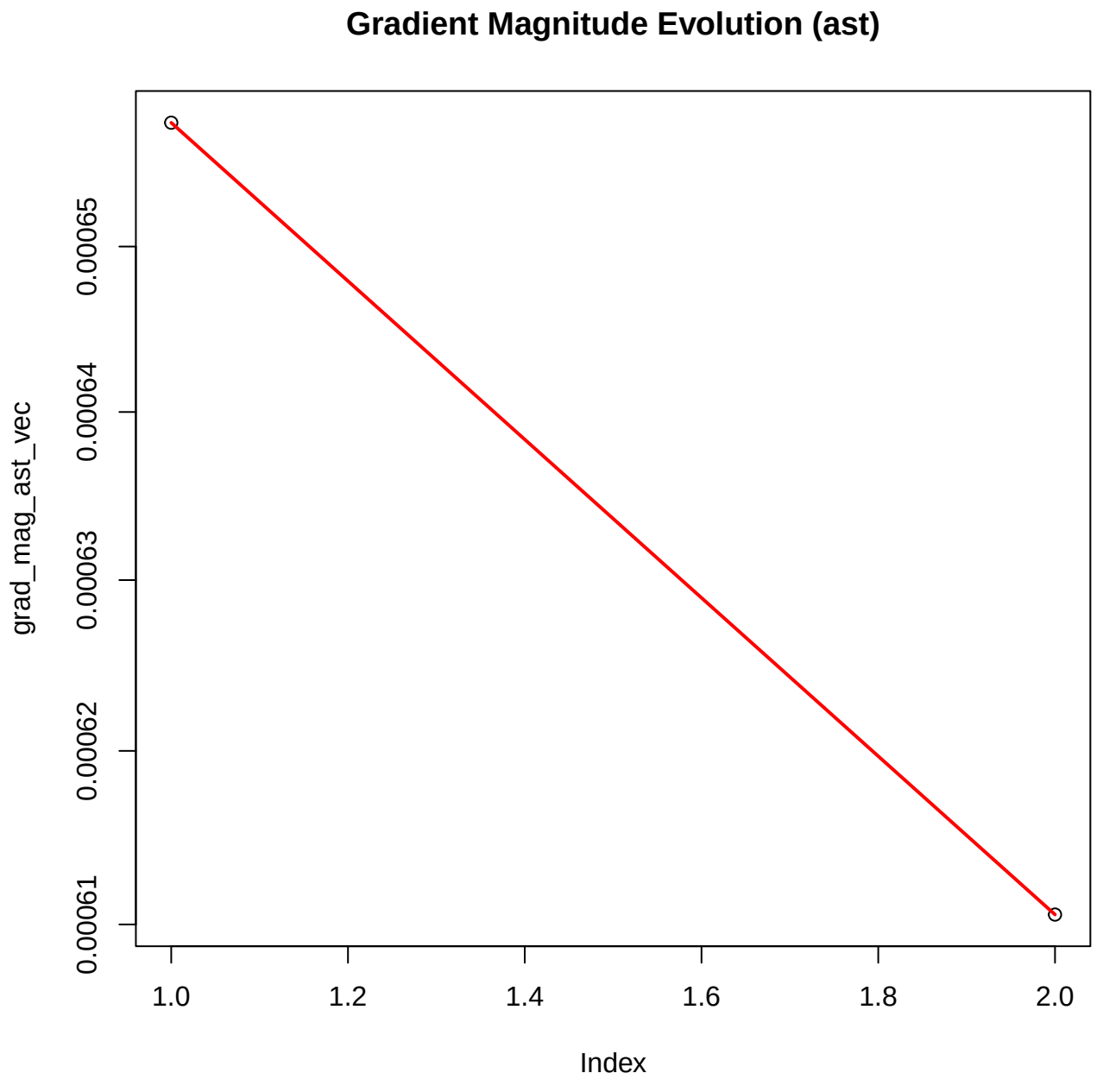

Value (dag)
loss_dag_vec
0.55
0.60
0.65
1.0
1.5
2.0
2.5
3.0
Index

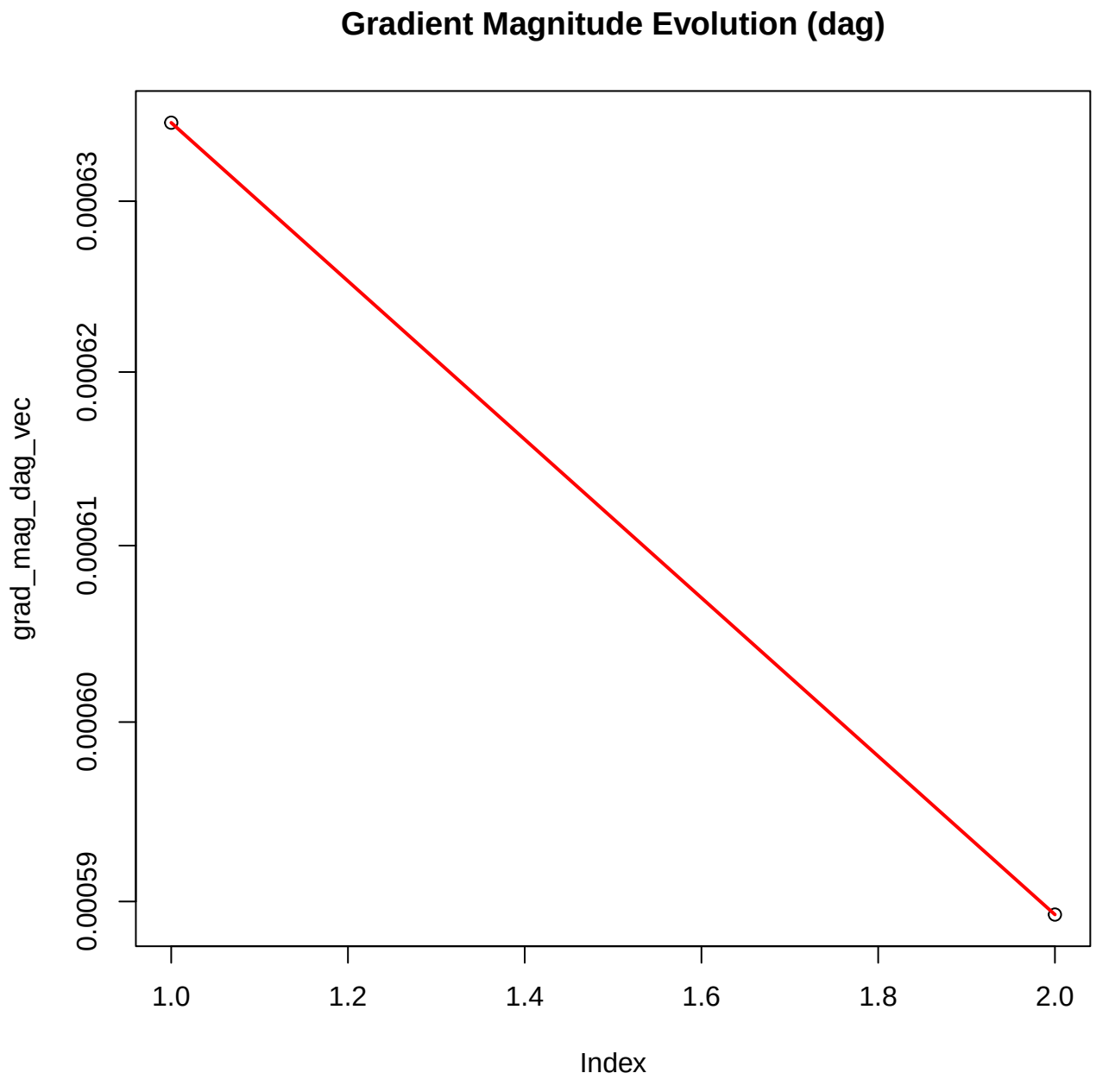

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e−01
1e+00
1e+01
1e+02
1e+03
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Index

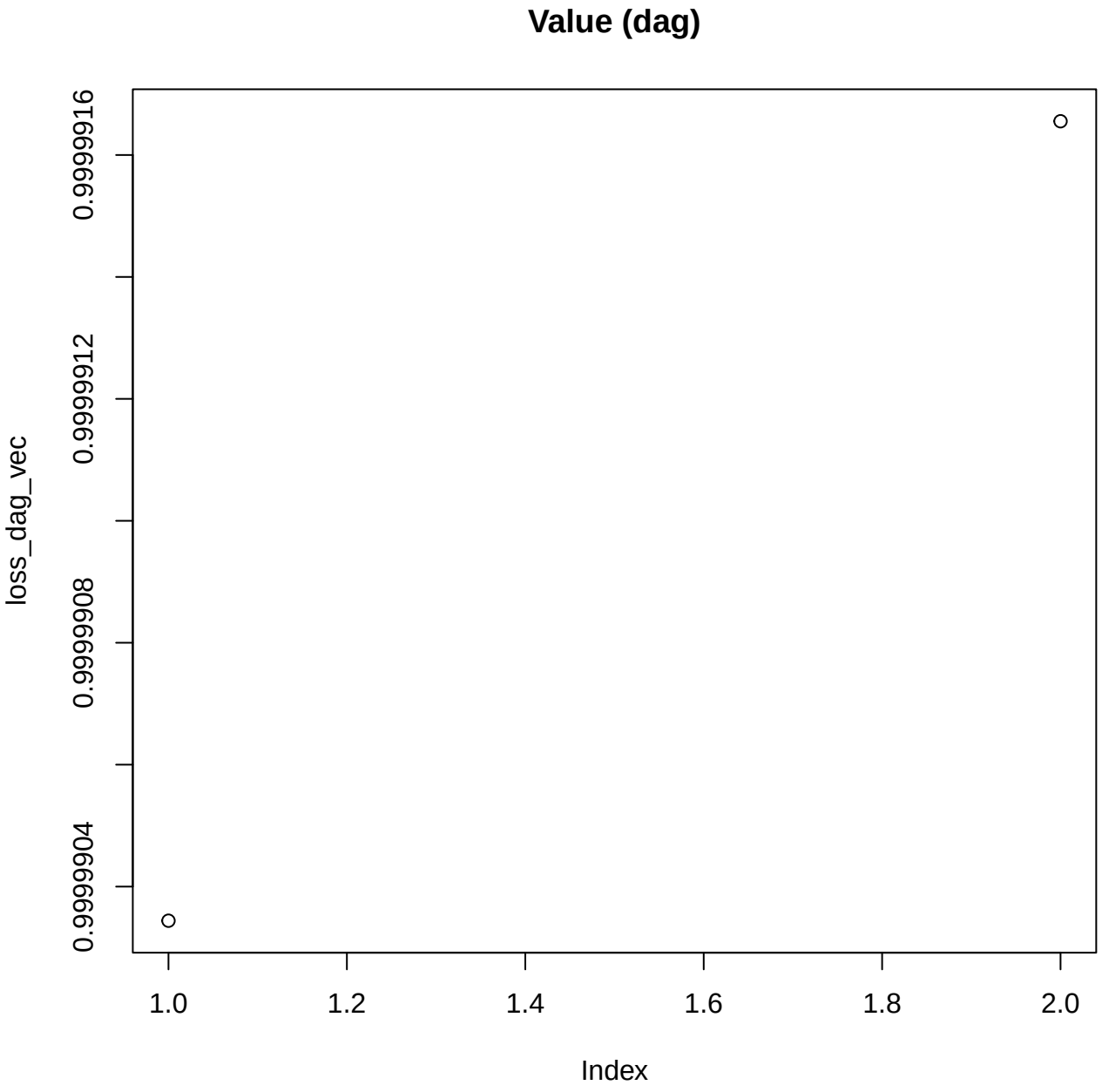

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
1e−01
1e+00
1e+01
1e+02
1e+03
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Index

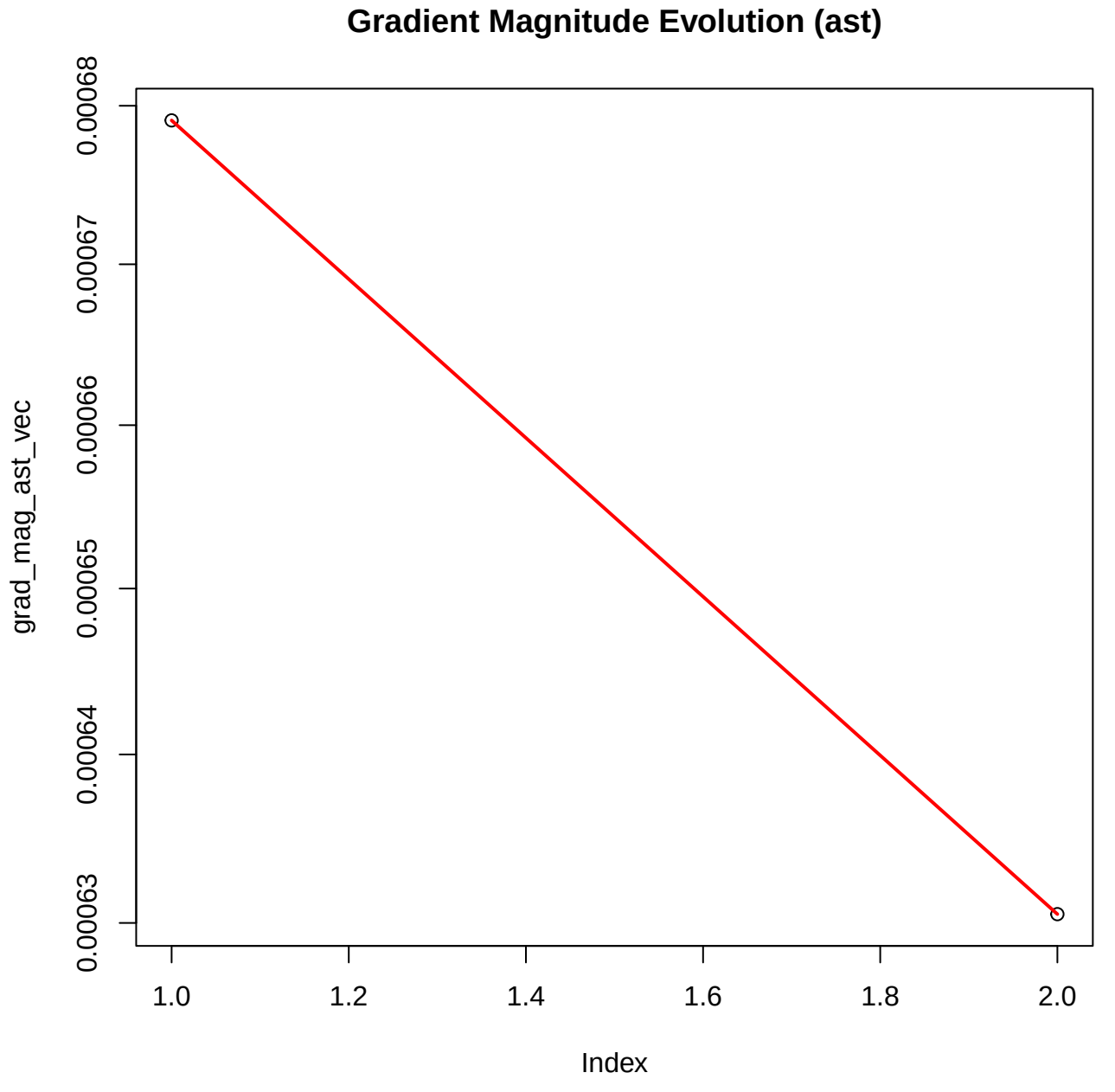

Value (ast)
loss_ast_vec
0.50
0.55
0.60
0.65
0.70
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Index

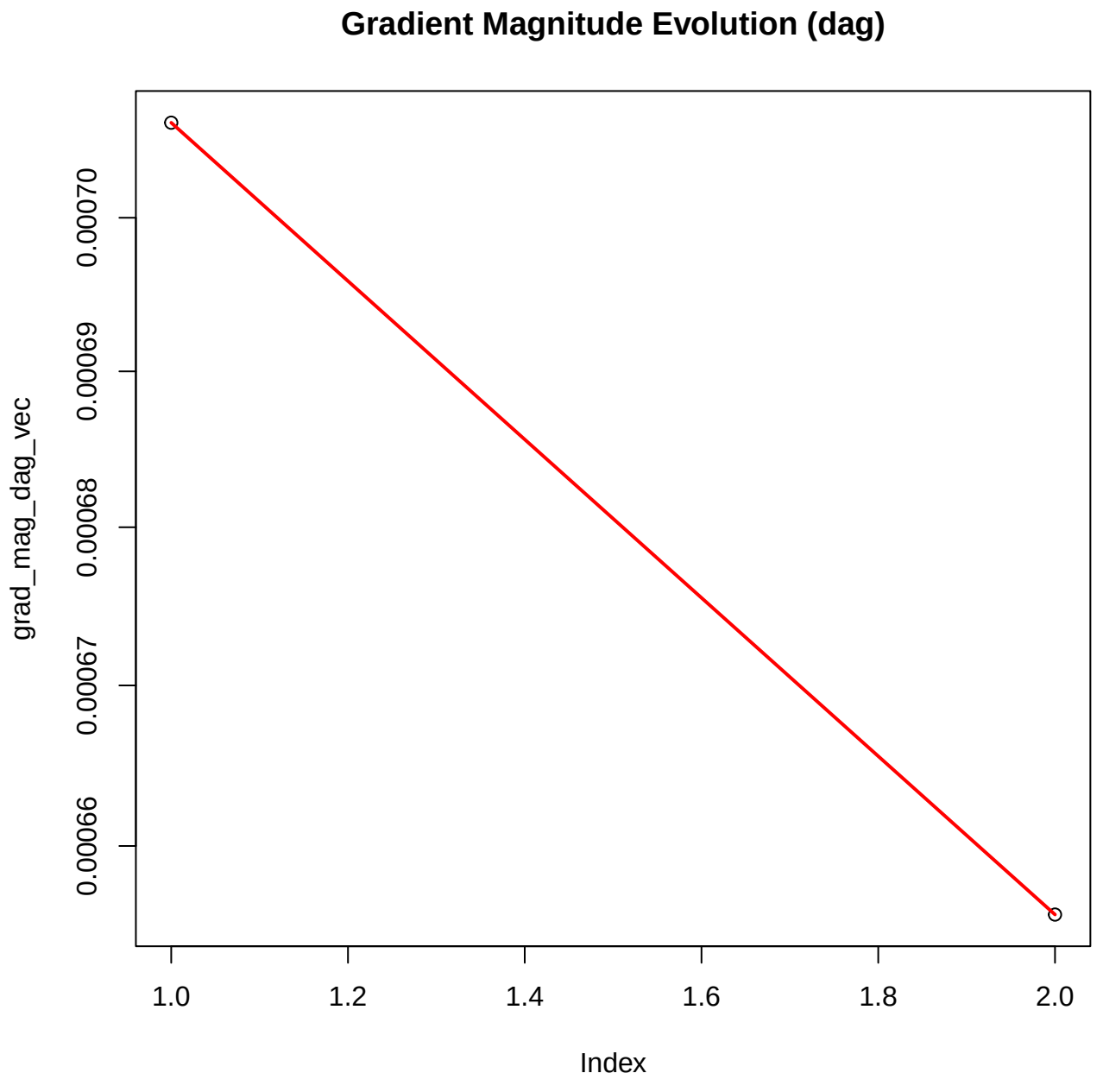

Value (dag)
loss_dag_vec
0.55
0.50
0.45
0.40
0.35
0.30
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Index

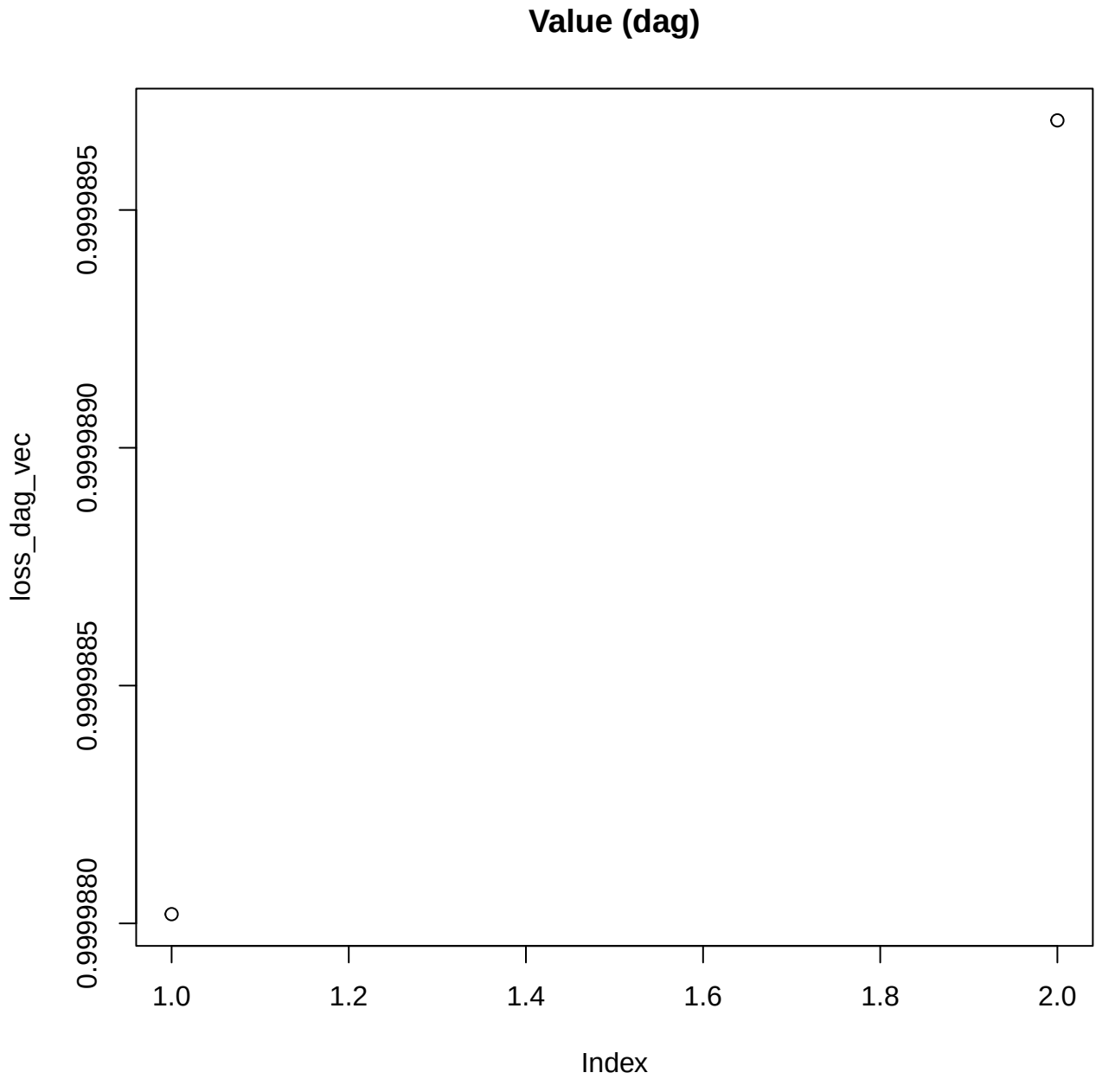

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00061
0.00062
0.00063
0.00064
0.00065
1.0
1.2
1.4
1.6
1.8
2.0
Index

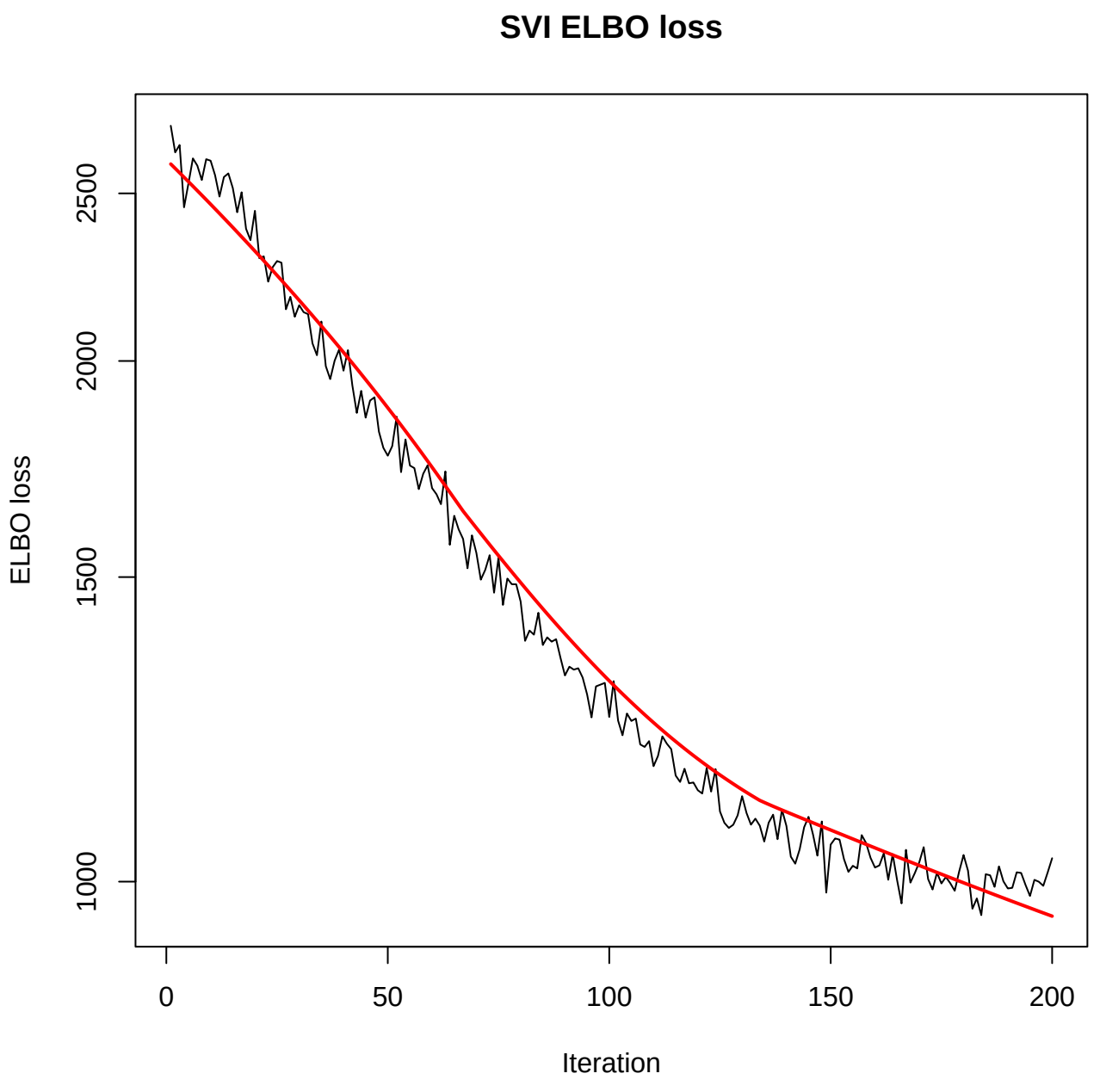

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.00059
0.00060
0.00061
0.00062
0.00063
1.0
1.2
1.4
1.6
1.8
2.0
Index

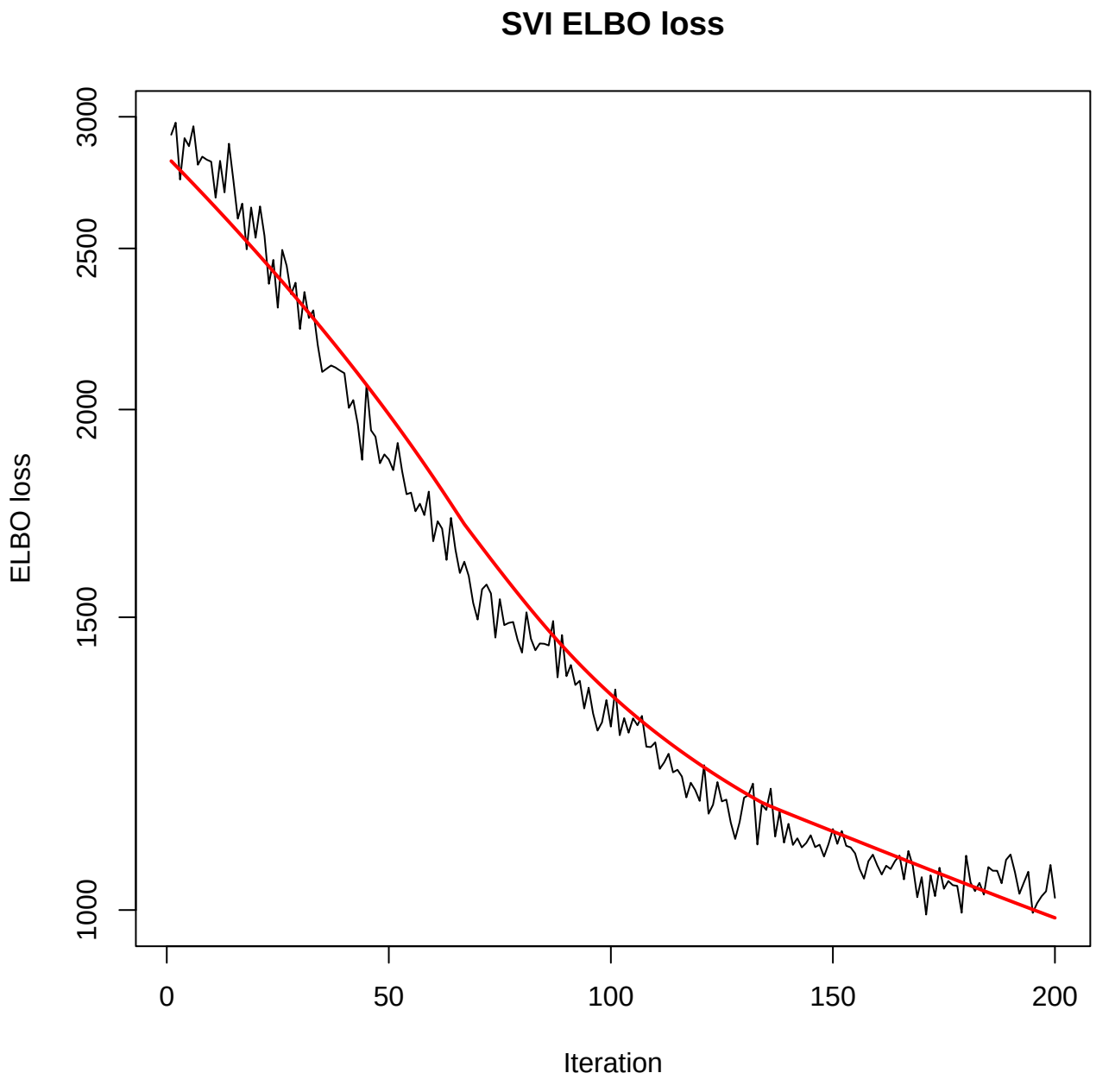

Value (dag)
loss_dag_vec
0.9999904
0.9999908
0.9999912
0.9999916
1.0
1.2
1.4
1.6
1.8
2.0
Index

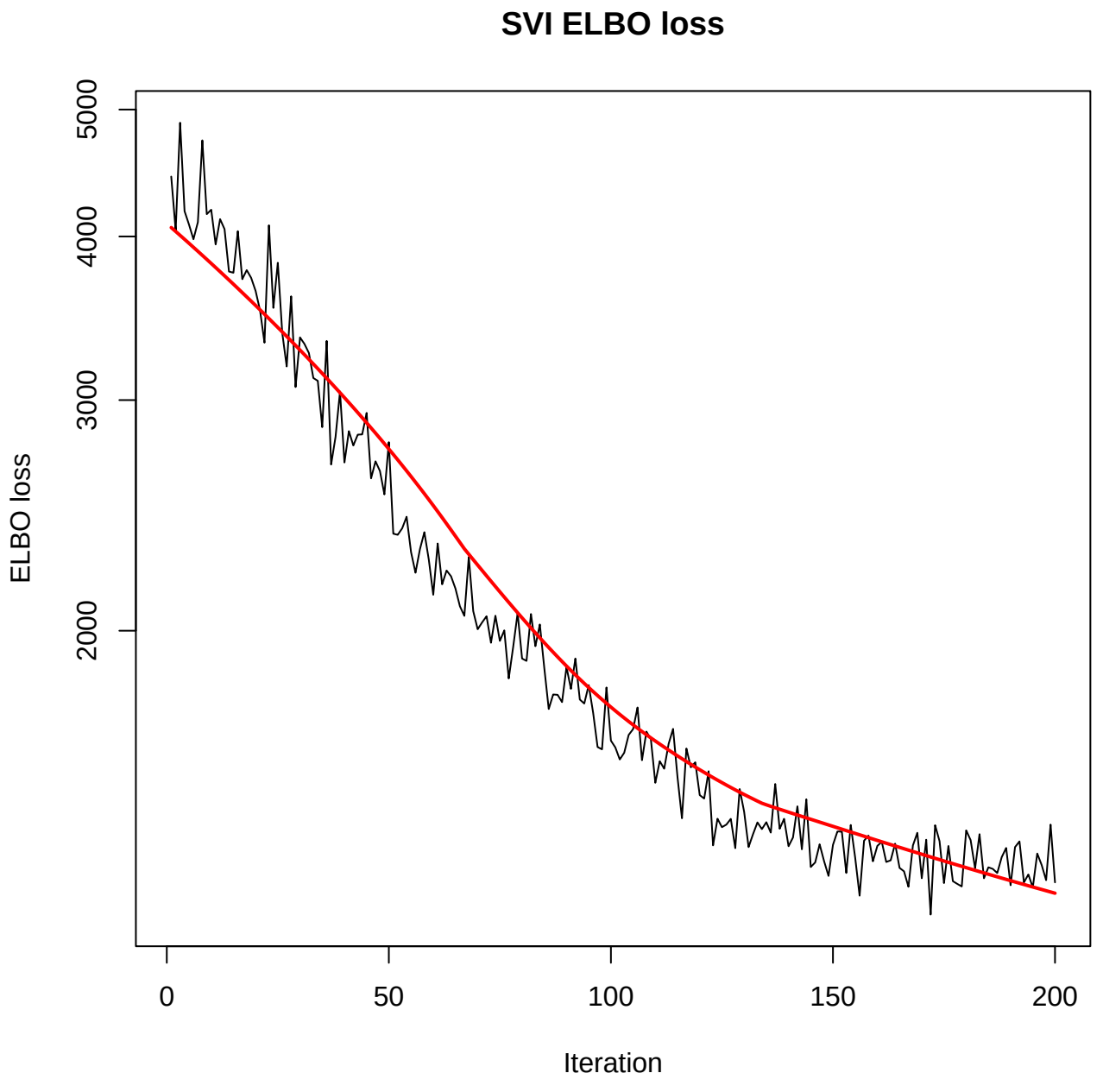

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00063
0.00064
0.00065
0.00066
0.00067
0.00068
1.0
1.2
1.4
1.6
1.8
2.0
Index

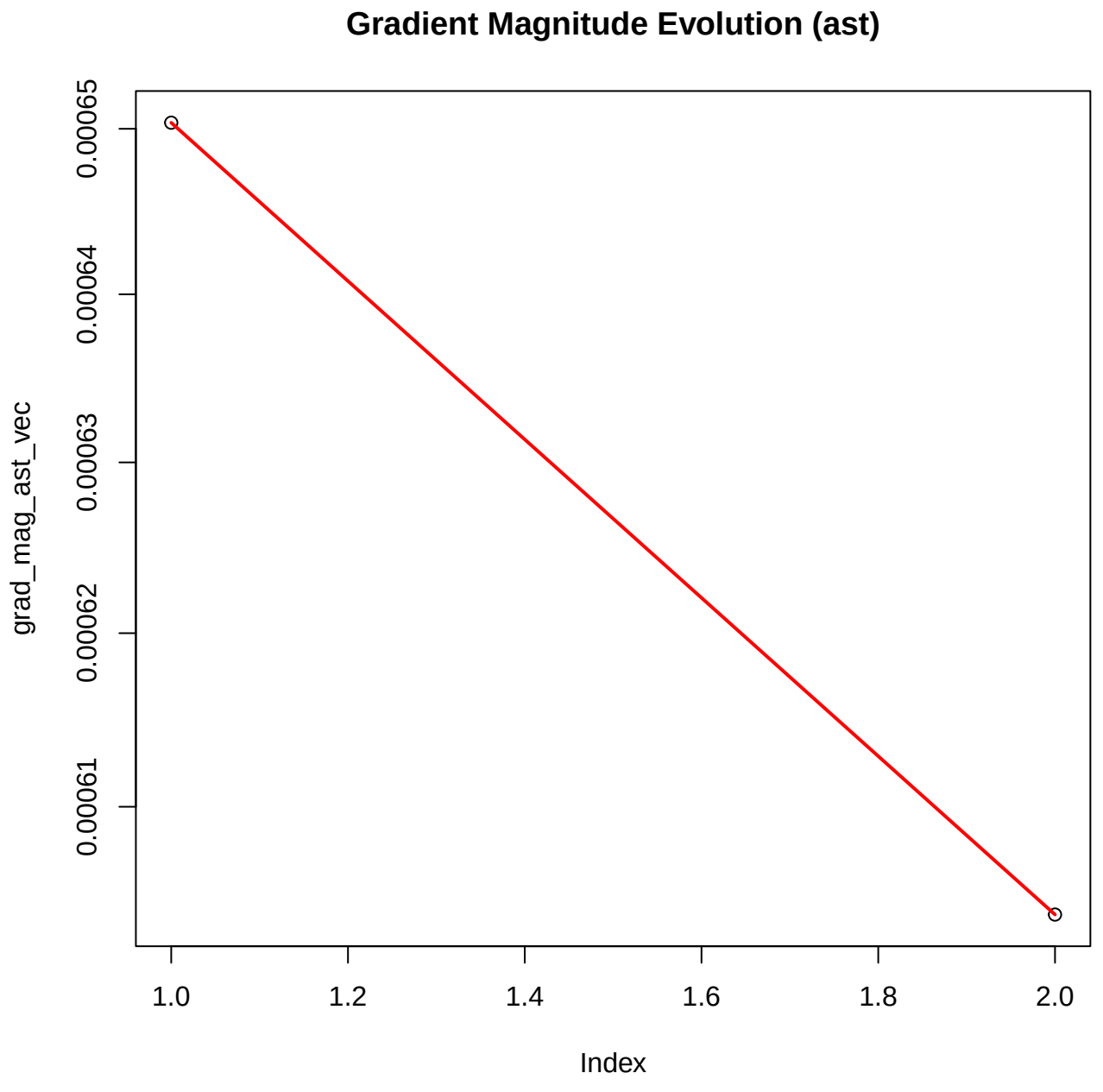

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.00066
0.00067
0.00068
0.00069
0.00070
1.0
1.2
1.4
1.6
1.8
2.0
Index

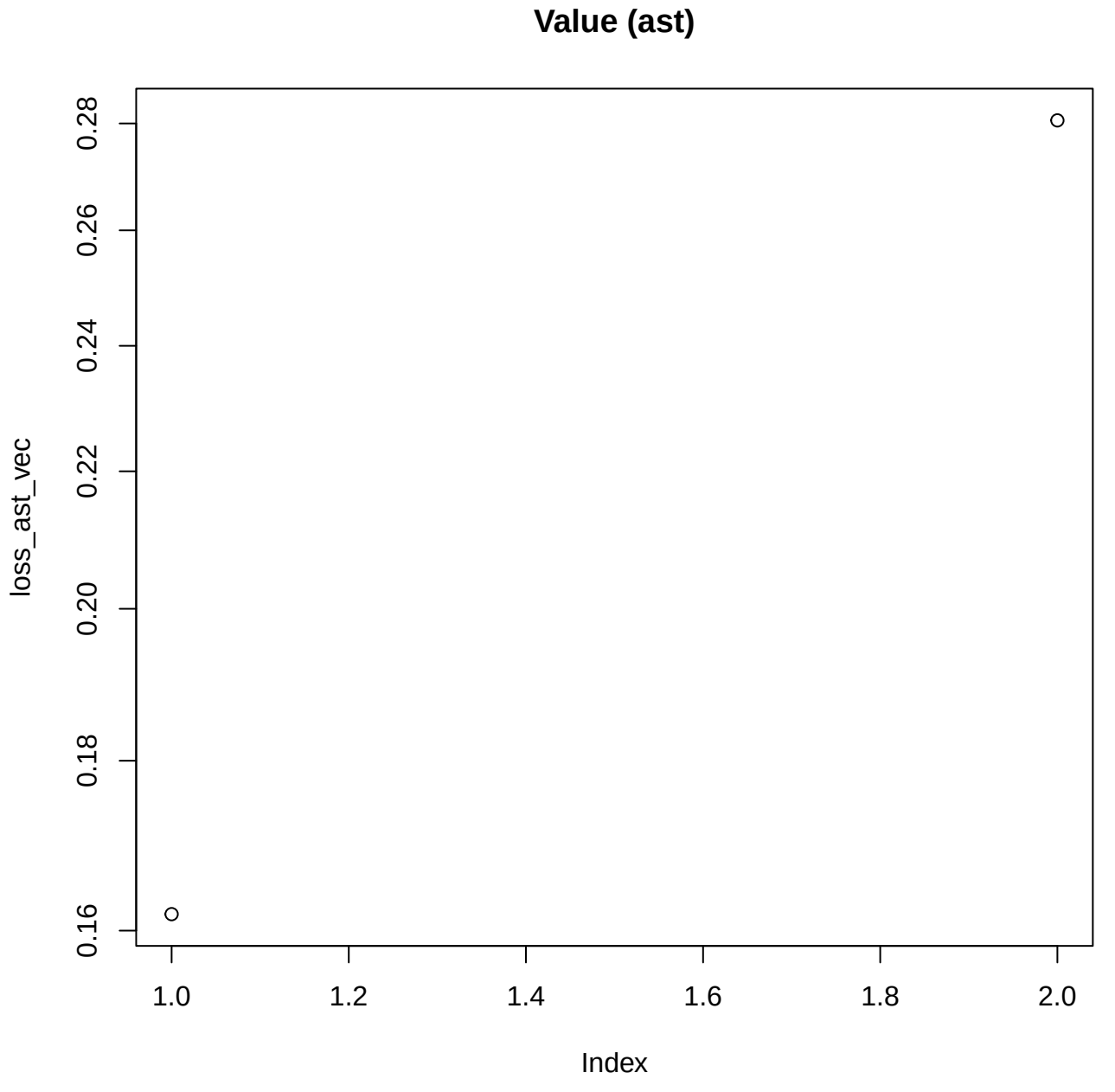

Value (dag)
loss_dag_vec
0.9999880
0.9999885
0.9999890
0.9999895
1.0
1.2
1.4
1.6
1.8
2.0
Index

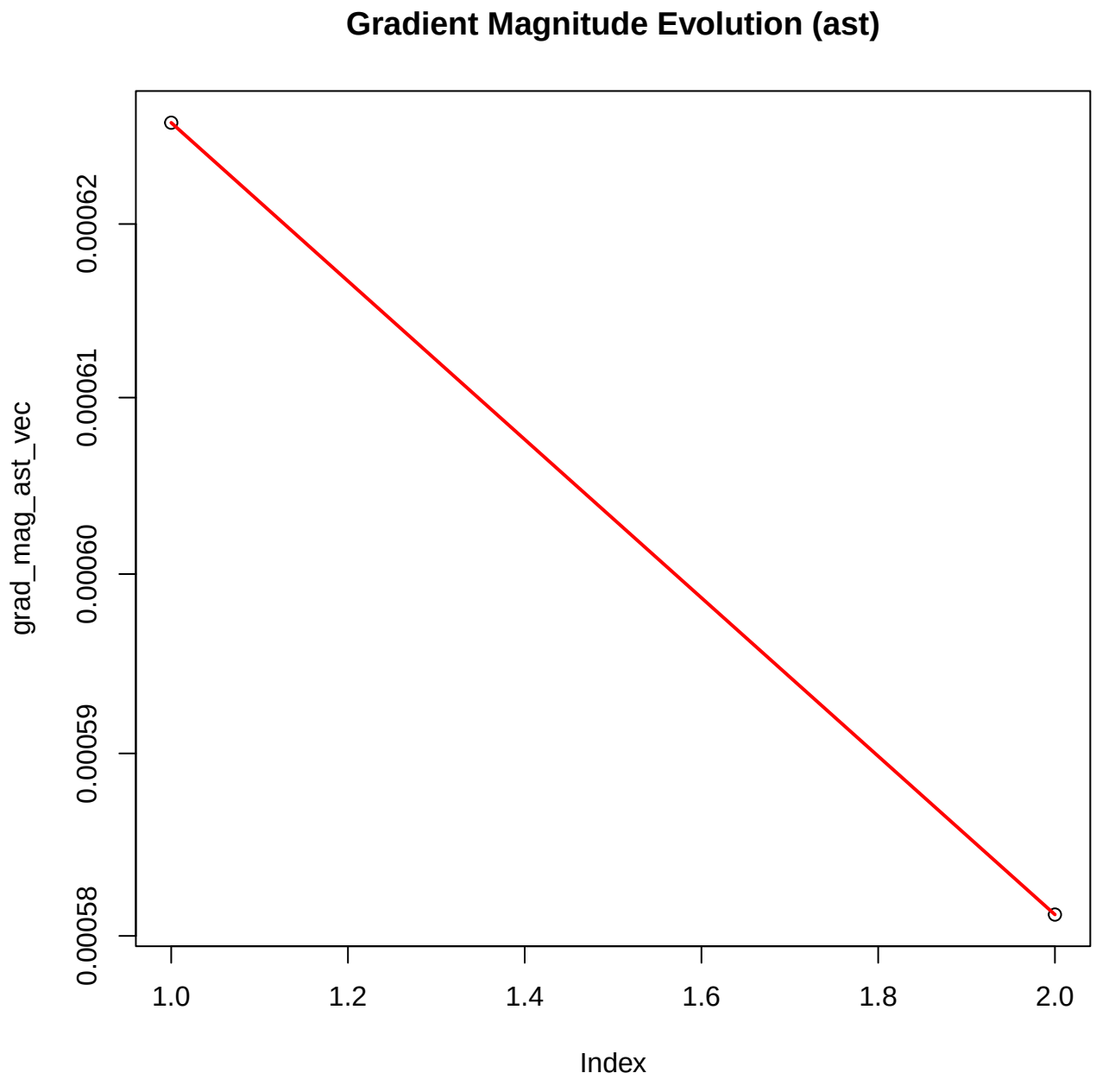

SVI ELBO loss
ELBO loss
2500
2000
1500
1000
0
50
100
150
200
Iteration

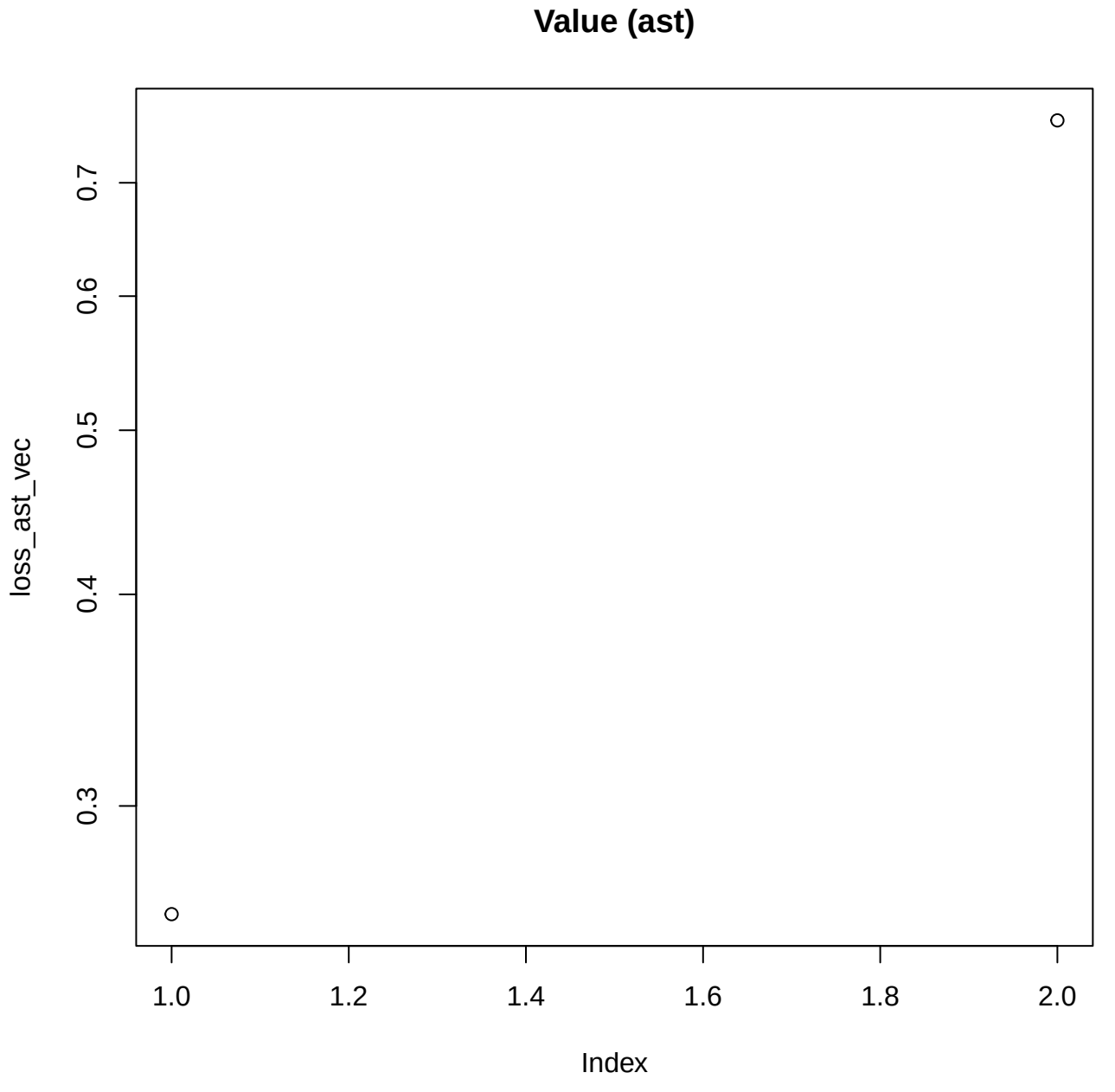

SVI ELBO loss
ELBO loss
3000
2500
2000
1500
1000
0
50
100
150
200
Iteration

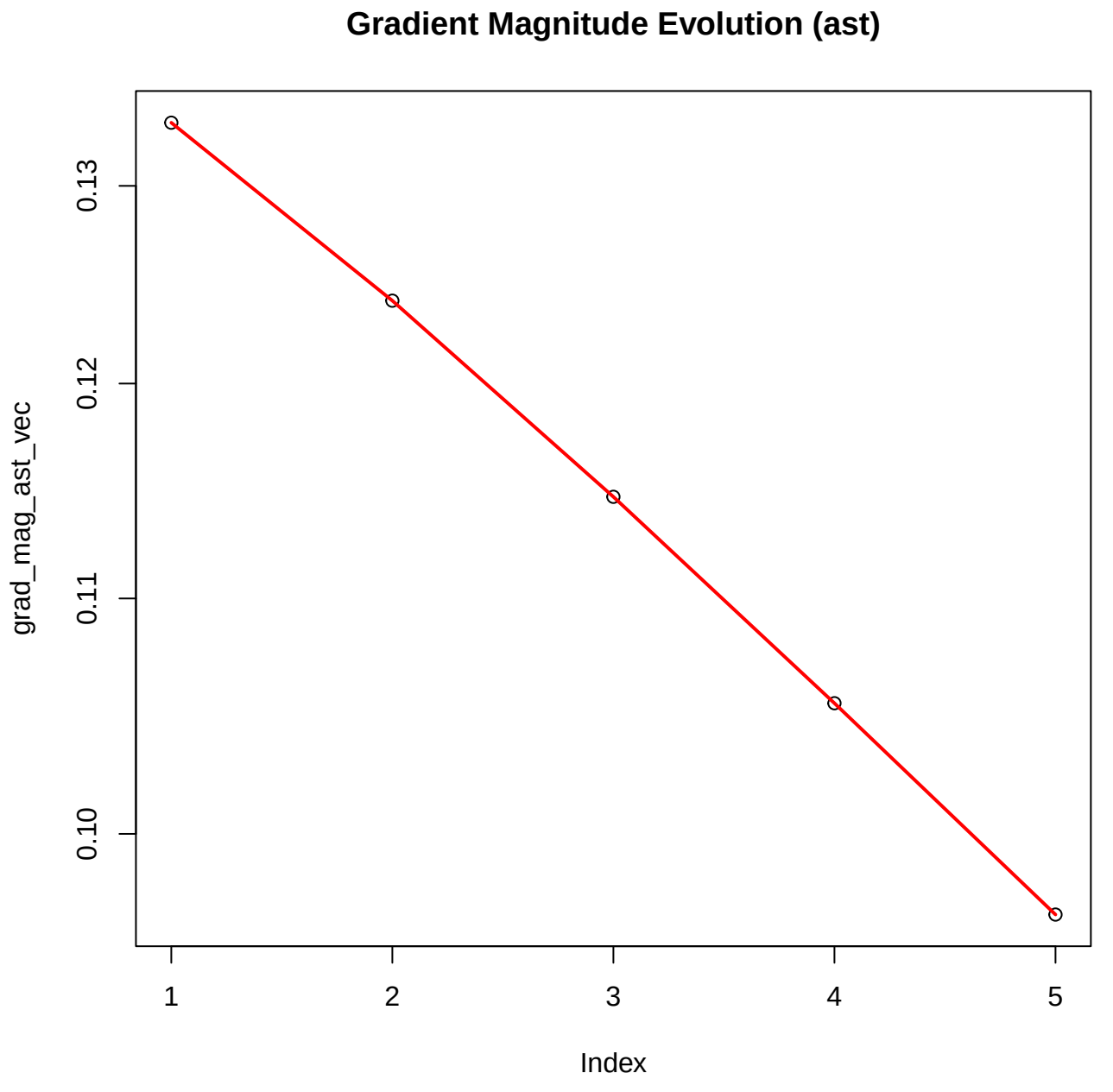

SVI ELBO loss
ELBO loss
5000
4000
3000
2000
0
50
100
150
200
Iteration

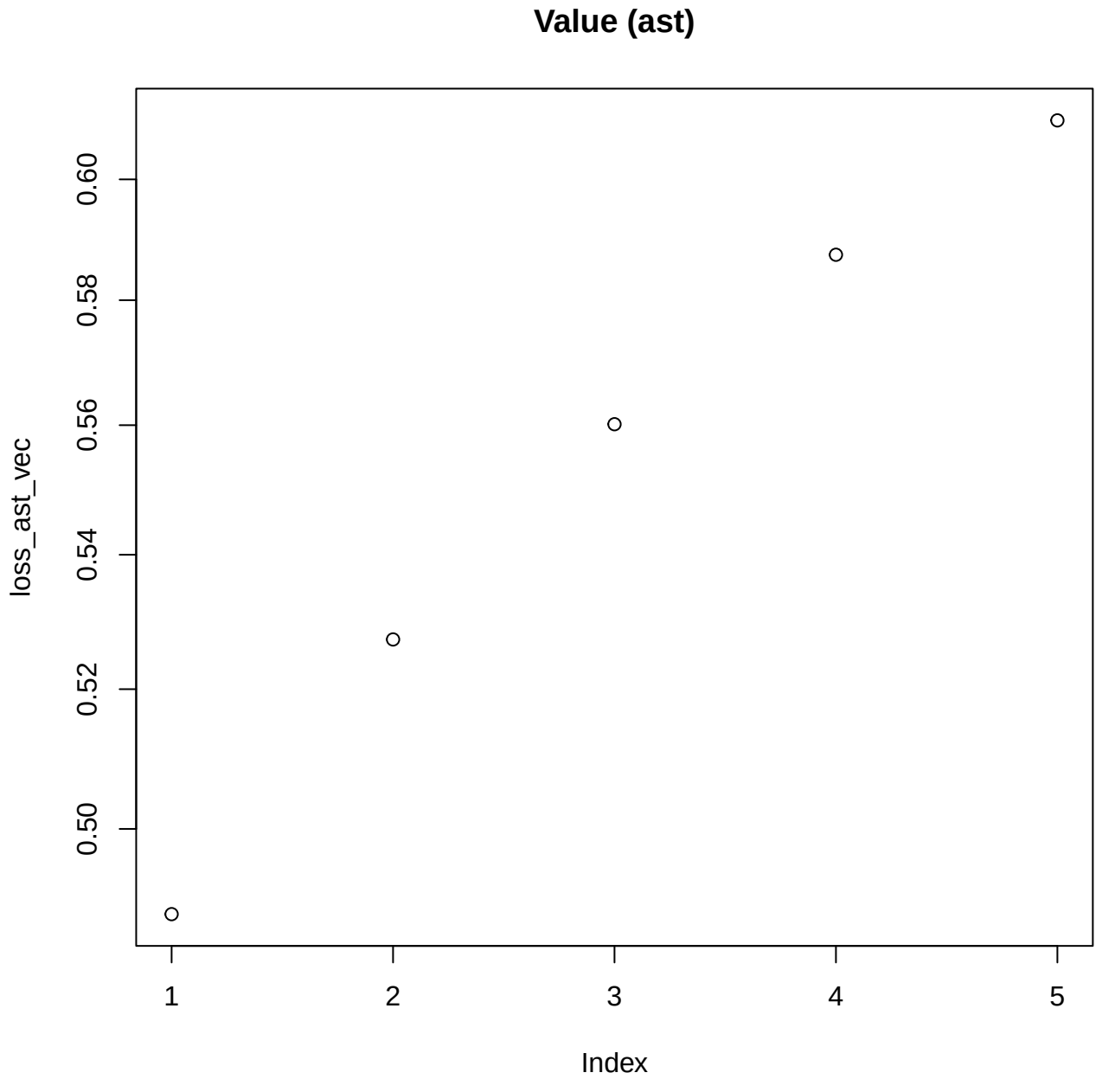

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00061
0.00062
0.00063
0.00064
0.00065
1.0
1.2
1.4
1.6
1.8
2.0
Index

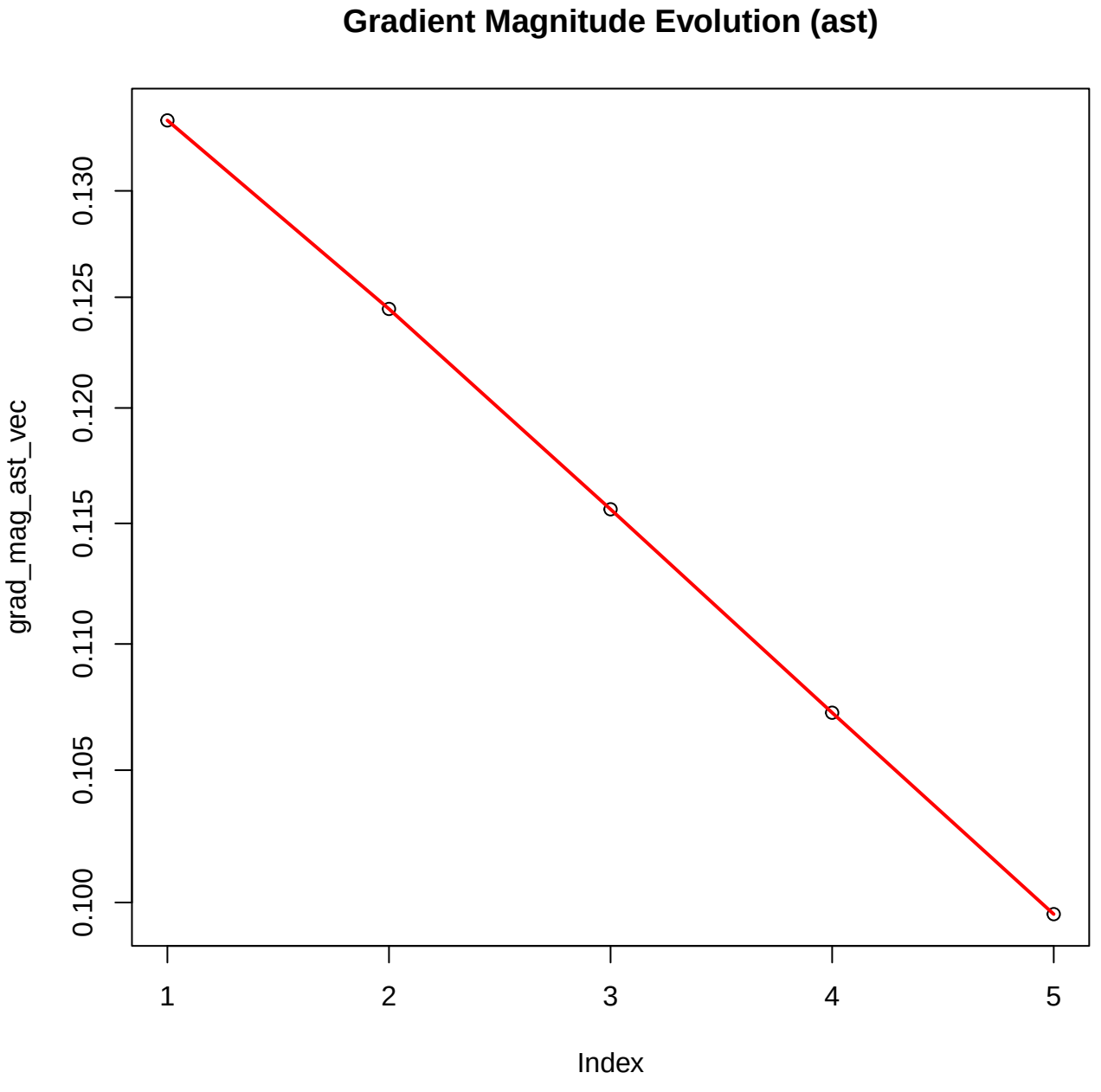

Value (ast)
loss_ast_vec
0.16
0.18
0.20
0.22
0.24
0.26
0.28
1.0
1.2
1.4
1.6
1.8
2.0
Index

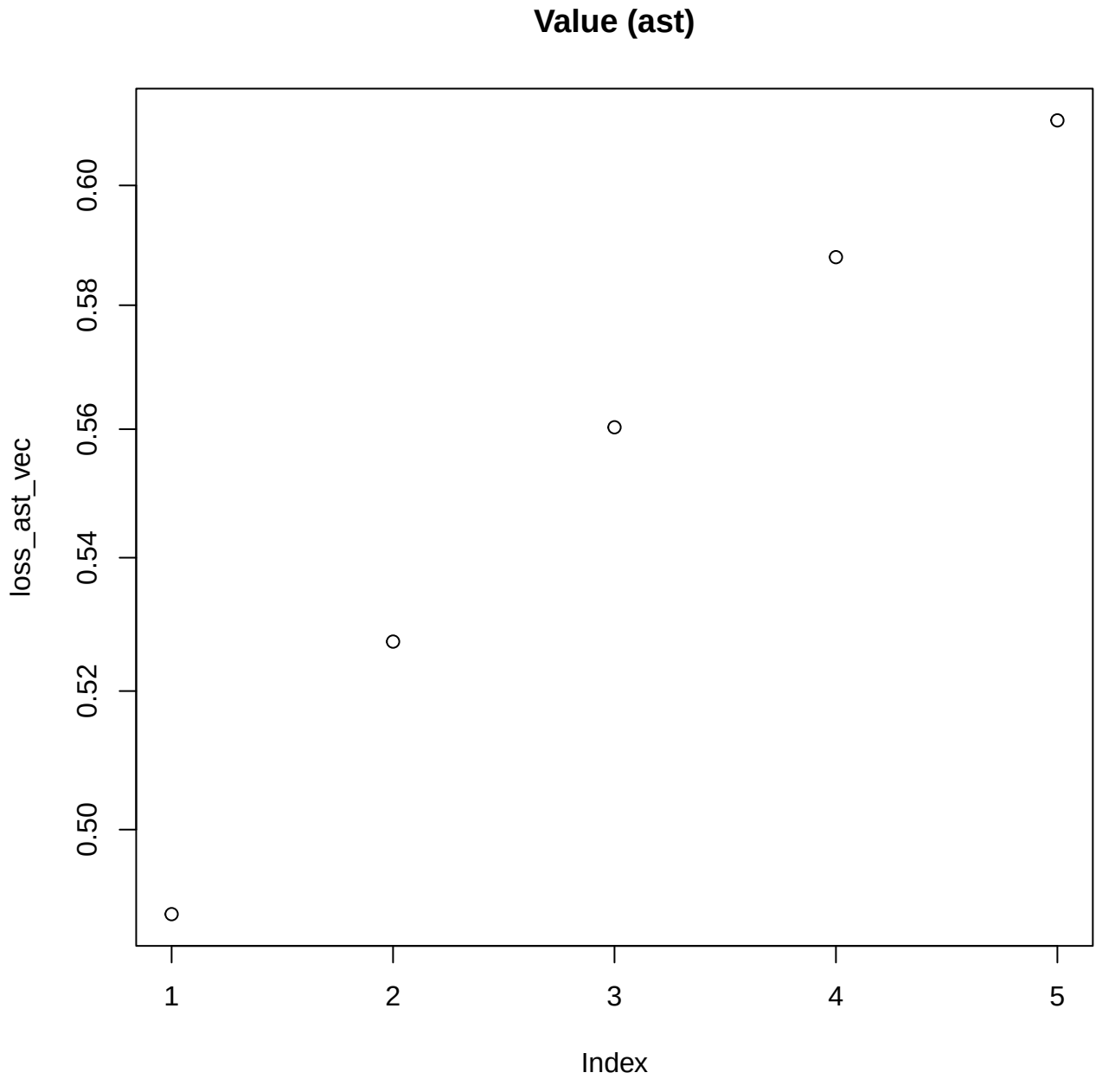

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00058
0.00059
0.00060
0.00061
0.00062
1.0
1.2
1.4
1.6
1.8
2.0
Index

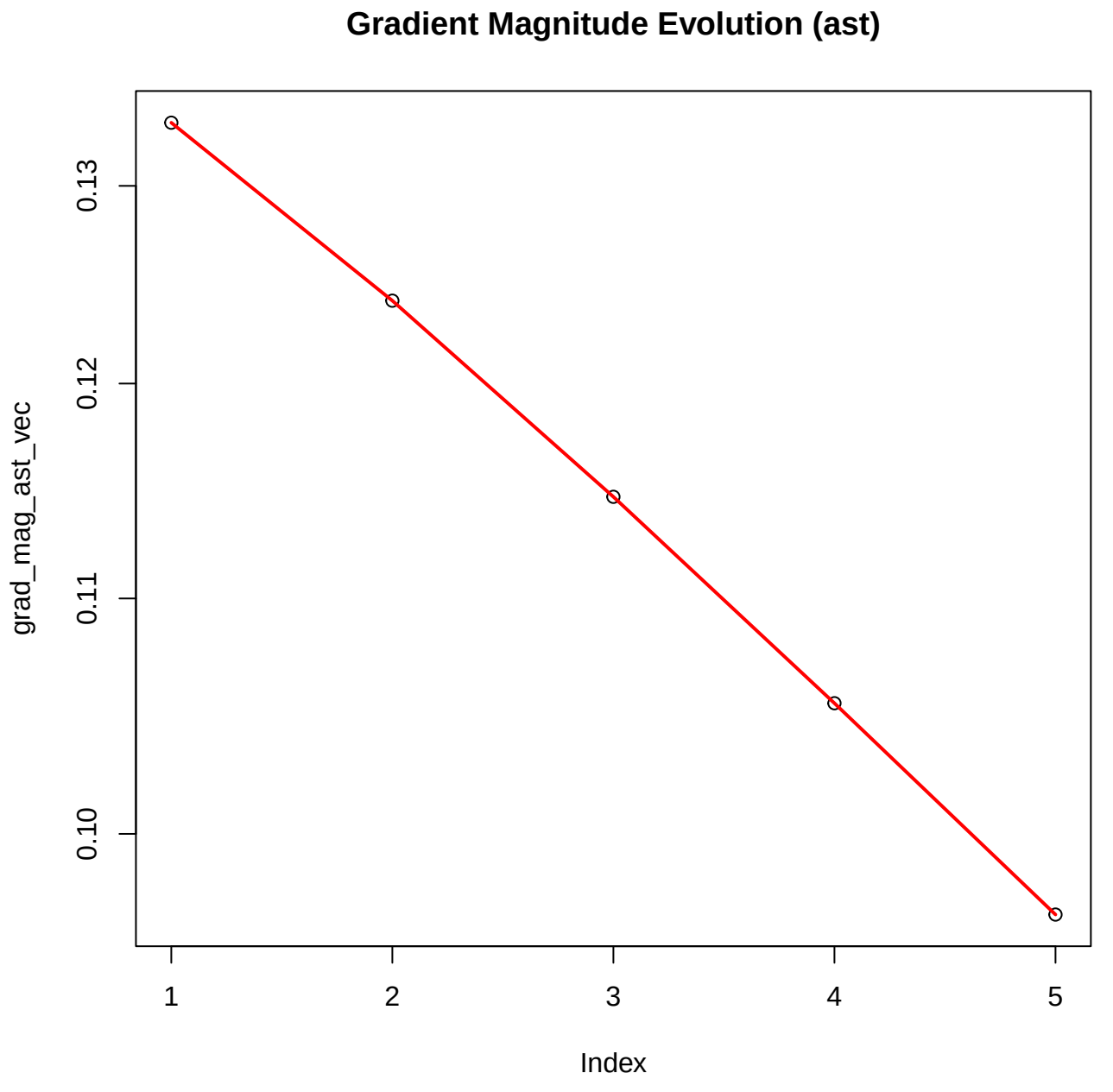

Value (ast)
loss_ast_vec
0.3
0.4
0.5
0.6
0.7
1.0
1.2
1.4
1.6
1.8
2.0
Index

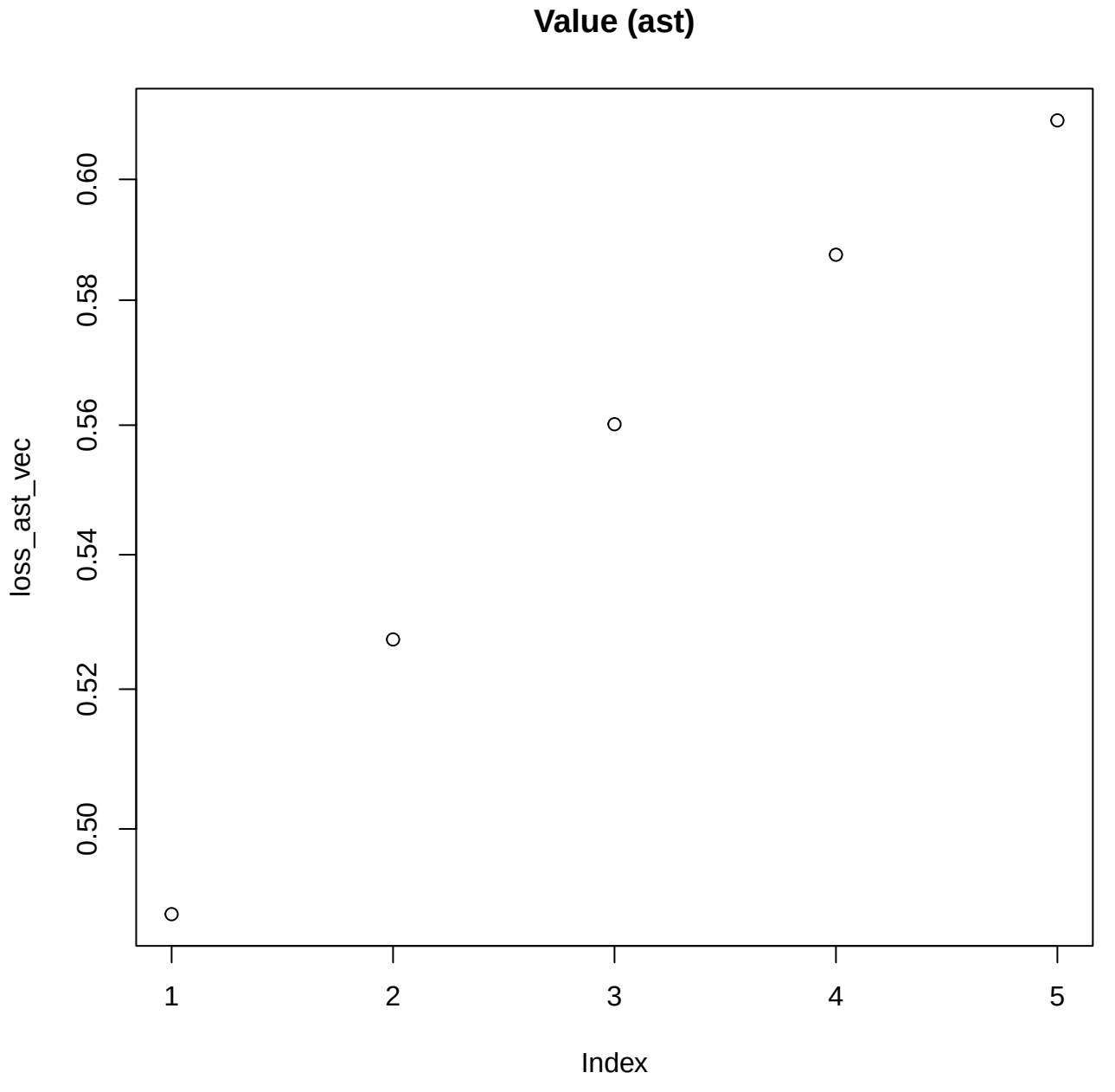

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index

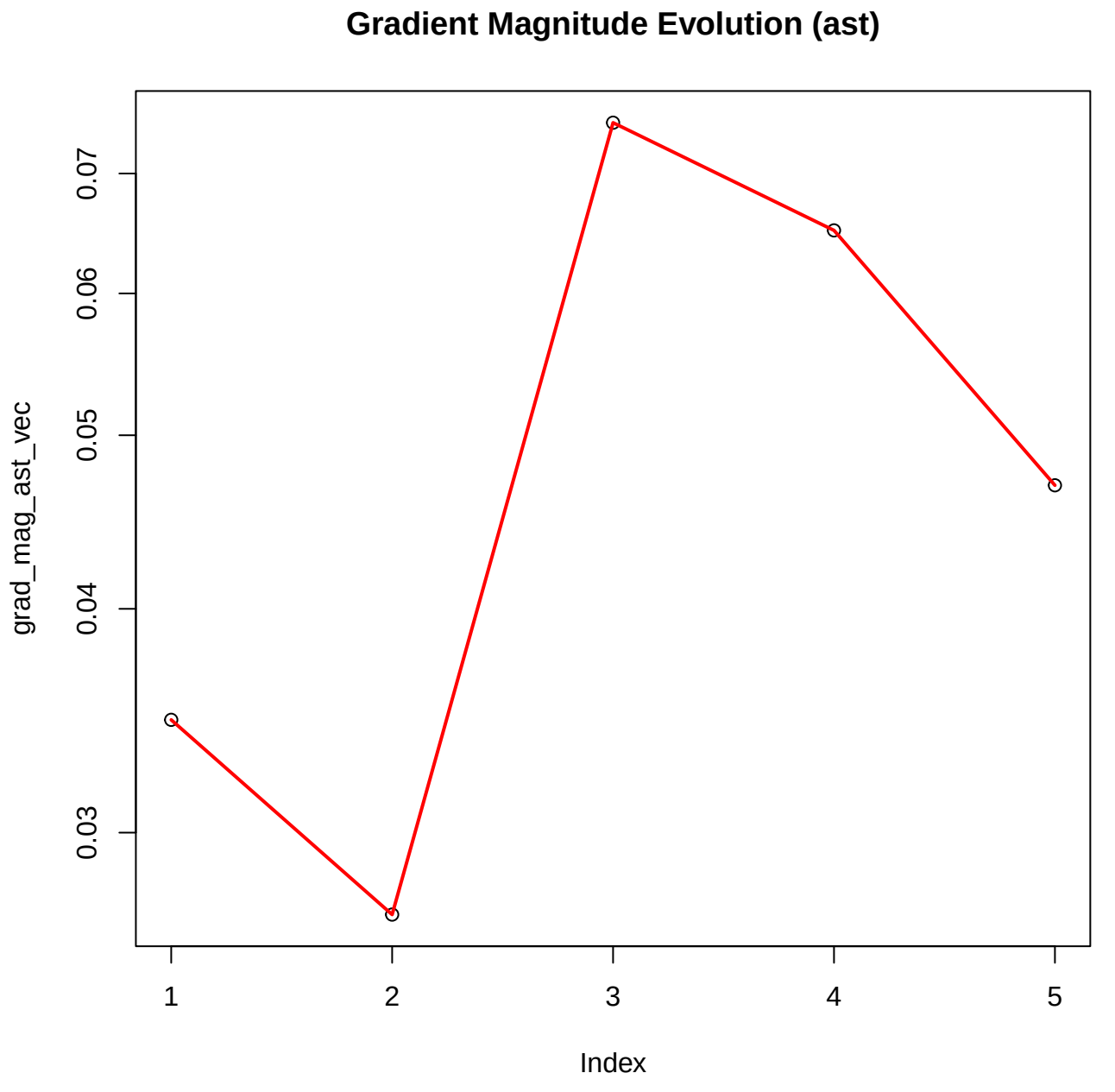

Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index

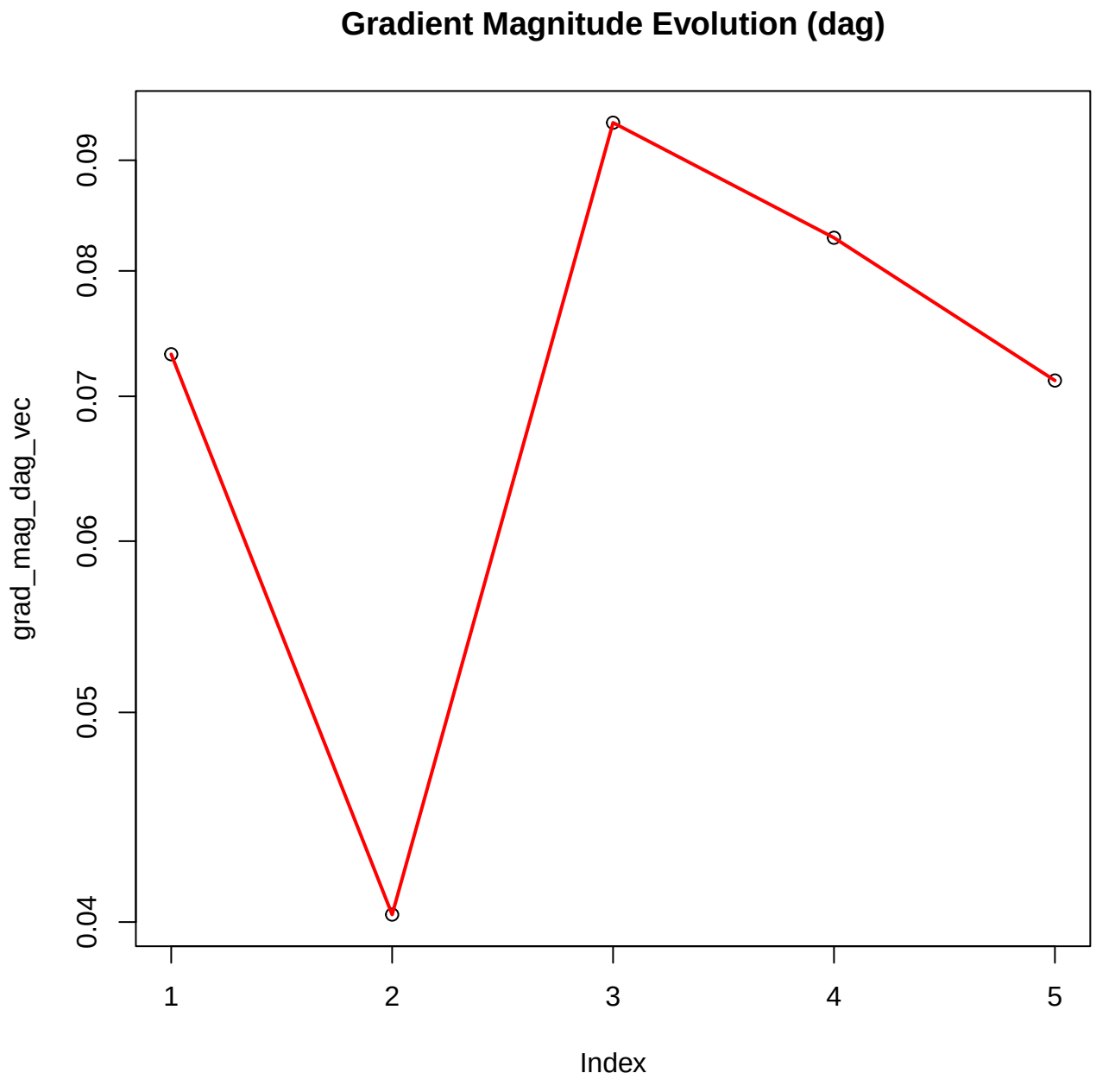

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.100
0.105
0.110
0.115
0.120
0.125
0.130
1
2
3
4
5
Index

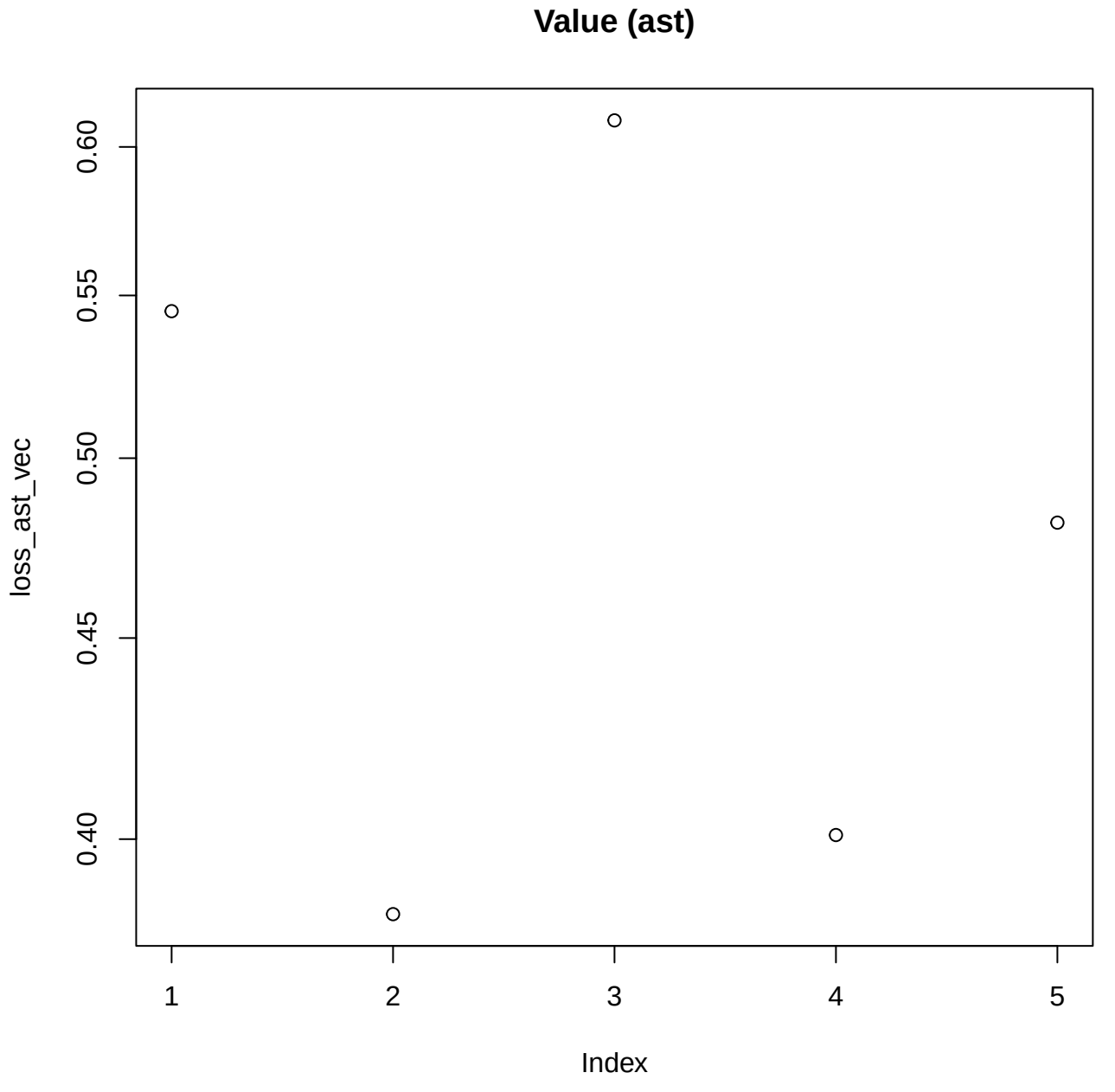

Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index

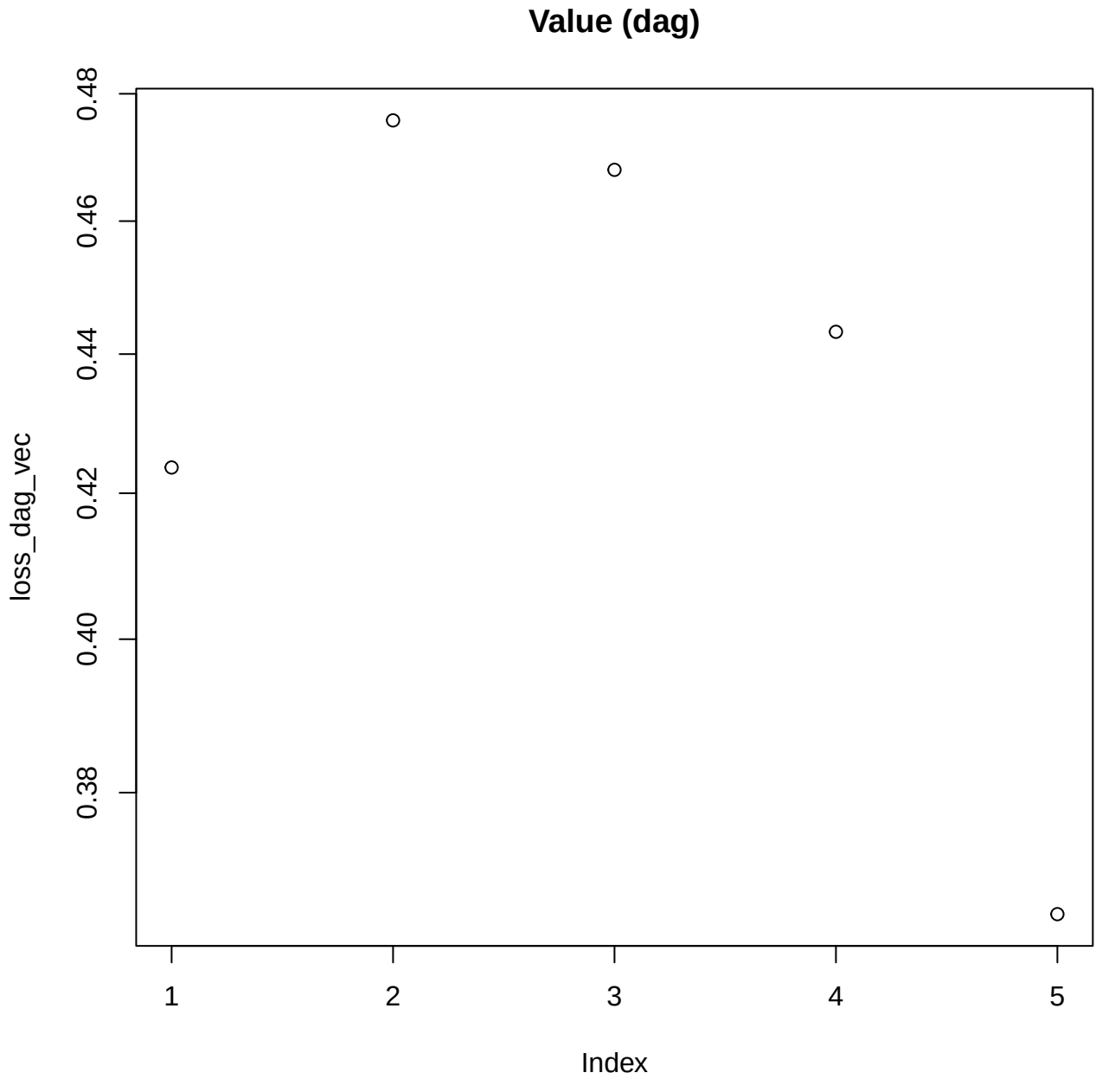

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index

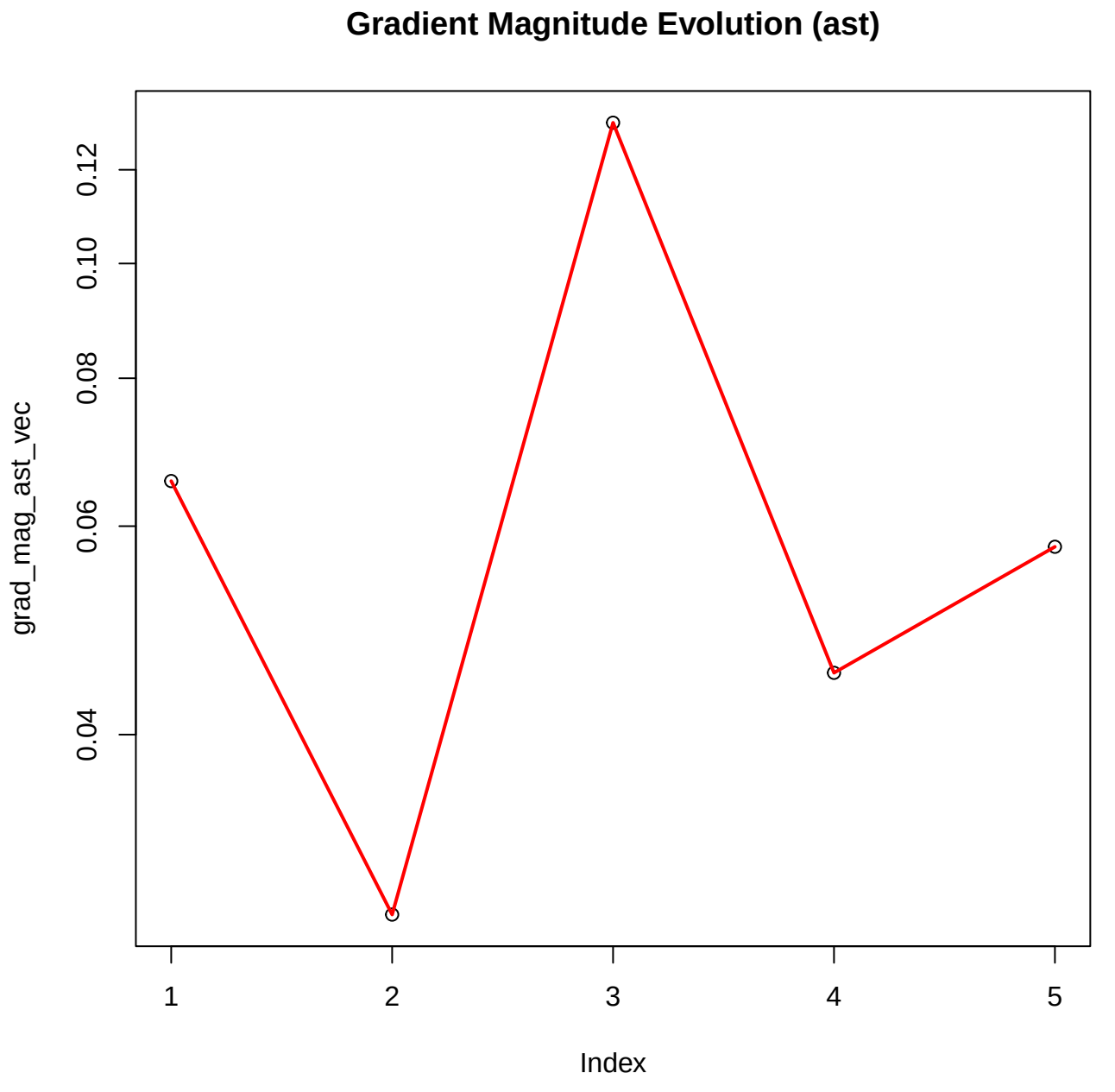

Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index

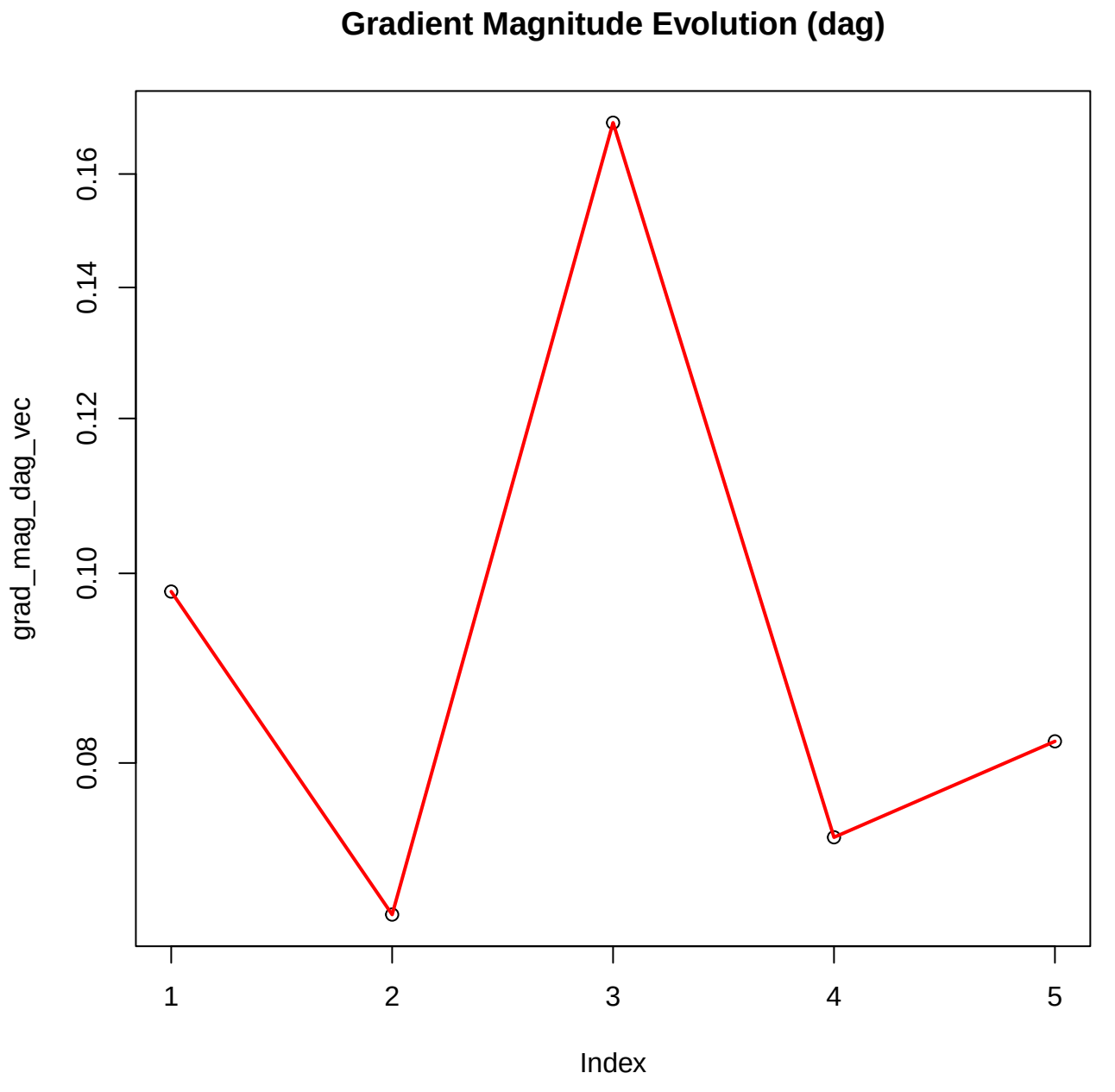

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.03
0.04
0.05
0.06
0.07
1
2
3
4
5
Index

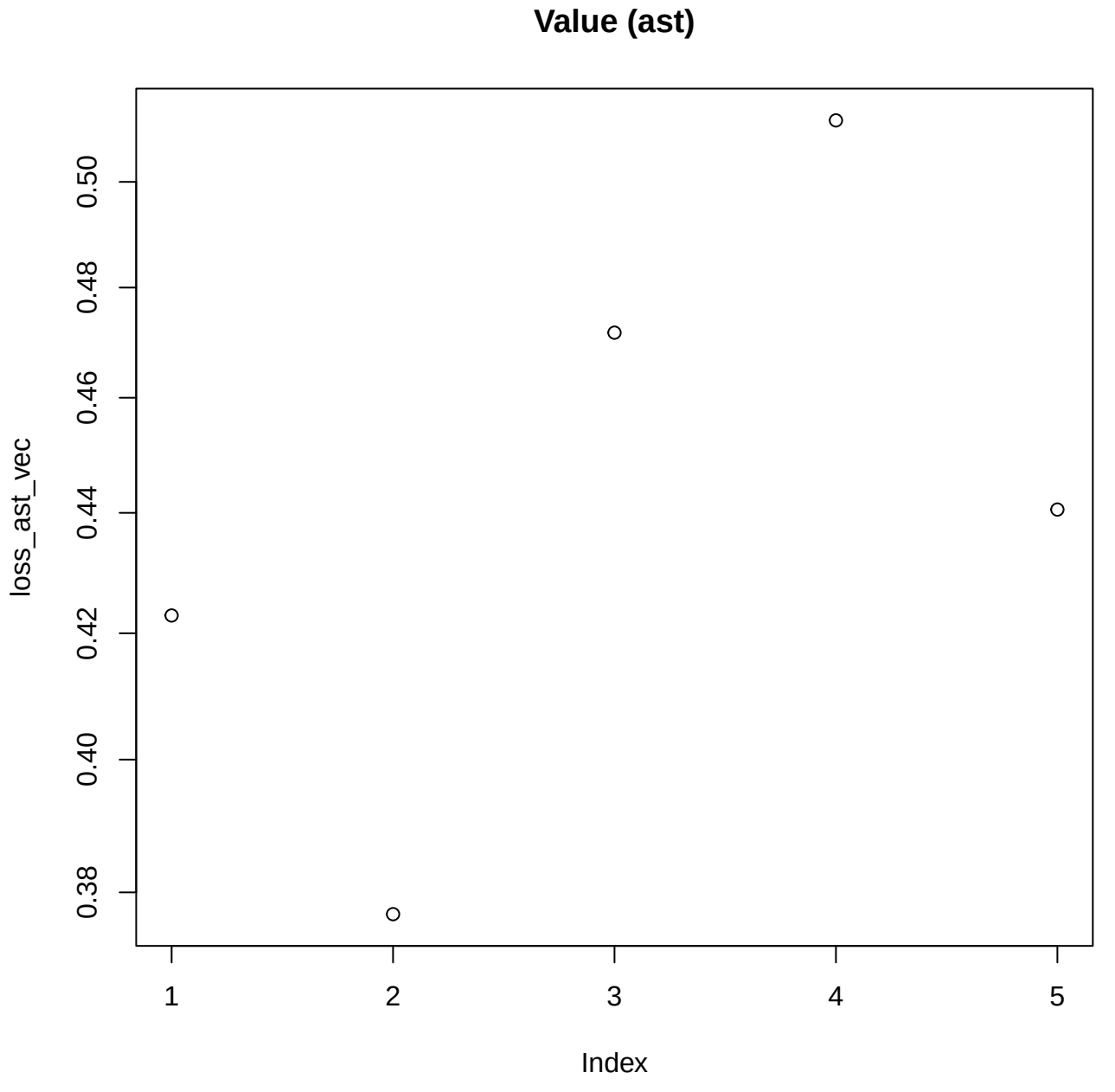

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.04
0.05
0.06
0.07
0.08
0.09
1
2
3
4
5
Index

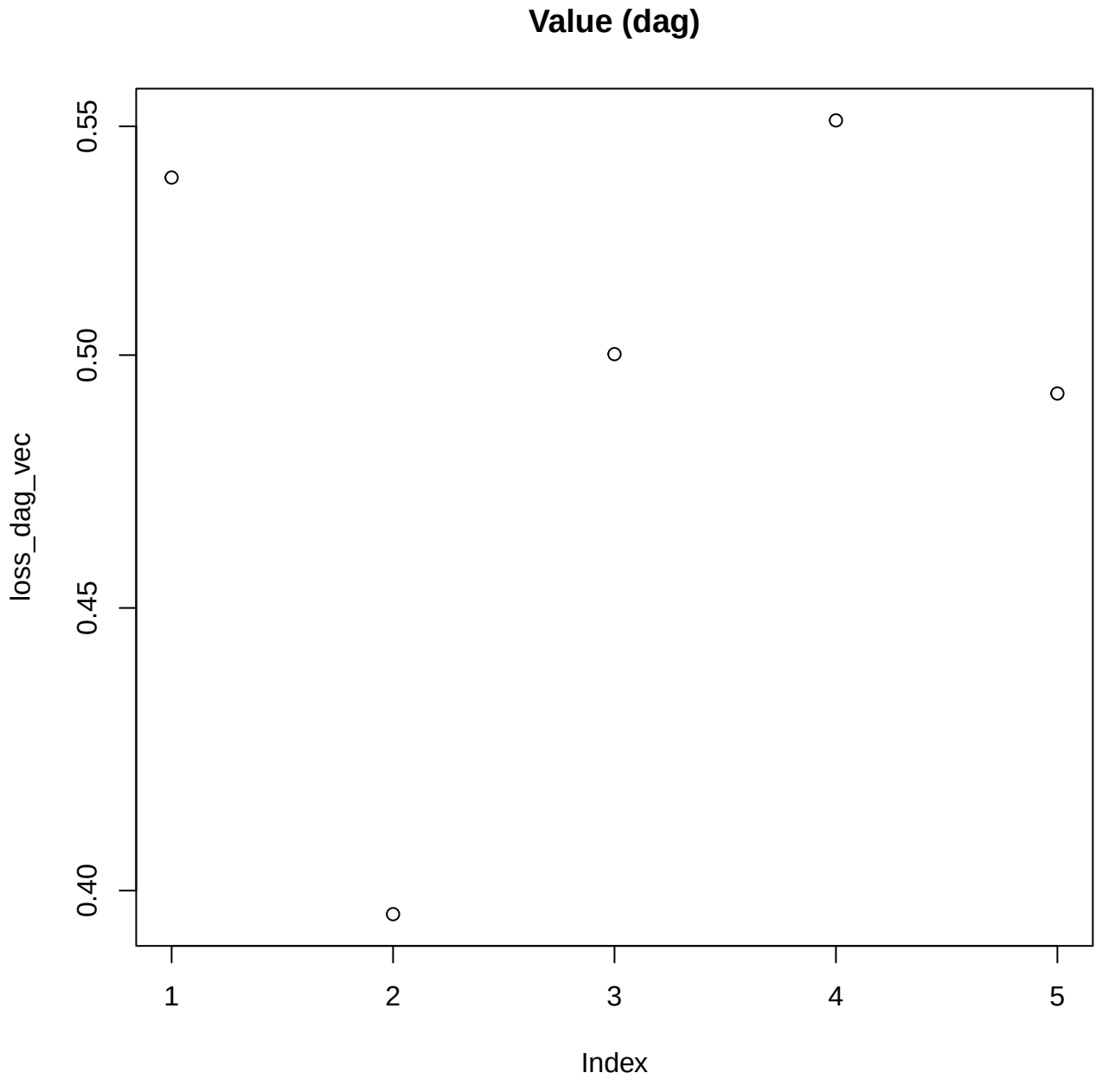

Value (ast)
loss_ast_vec
0.40
0.45
0.50
0.55
0.60
1
2
3
4
5
Index

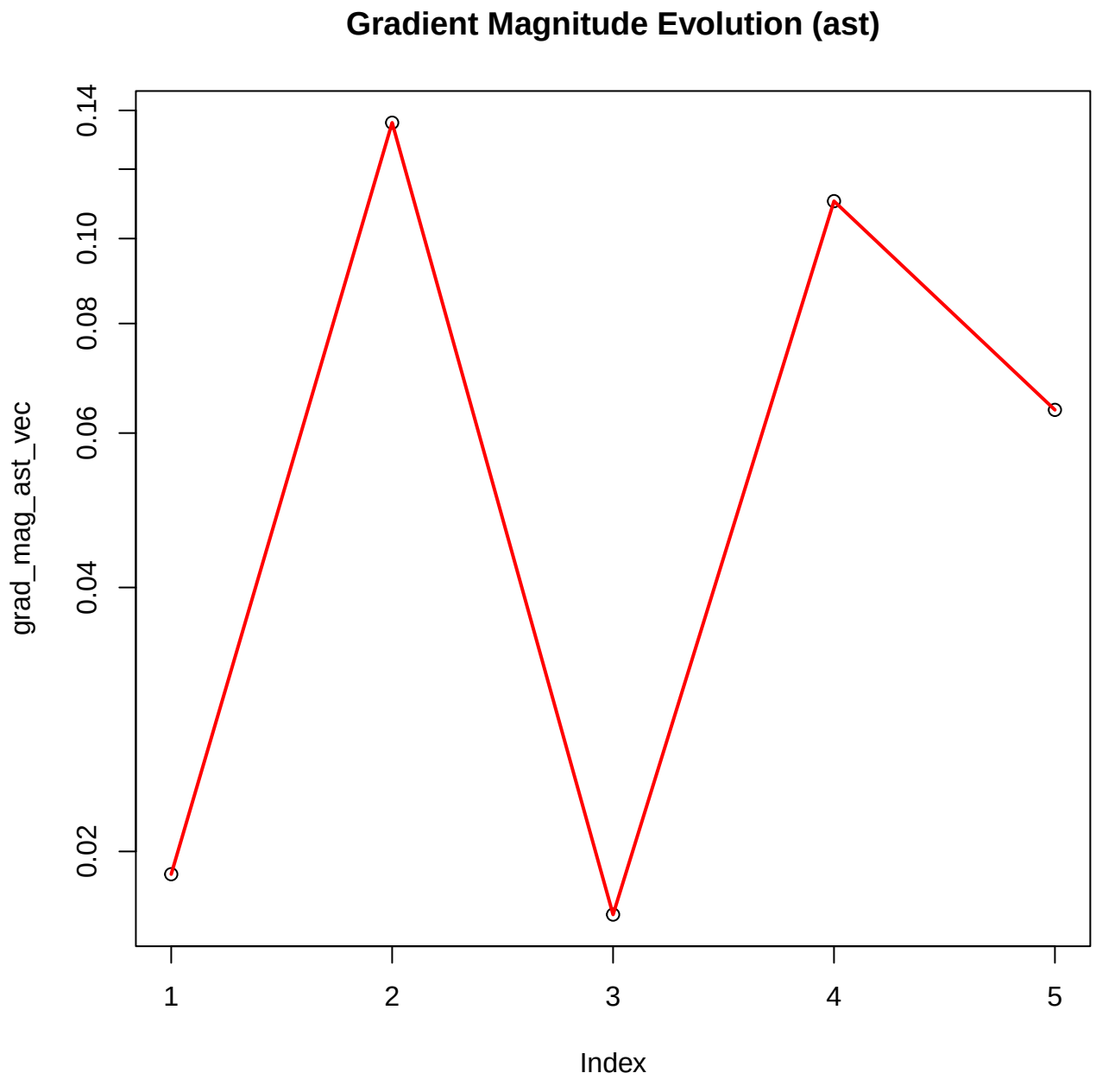

Value (dag)
loss_dag_vec
0.38
0.40
0.42
0.44
0.46
0.48
1
2
3
4
5
Index

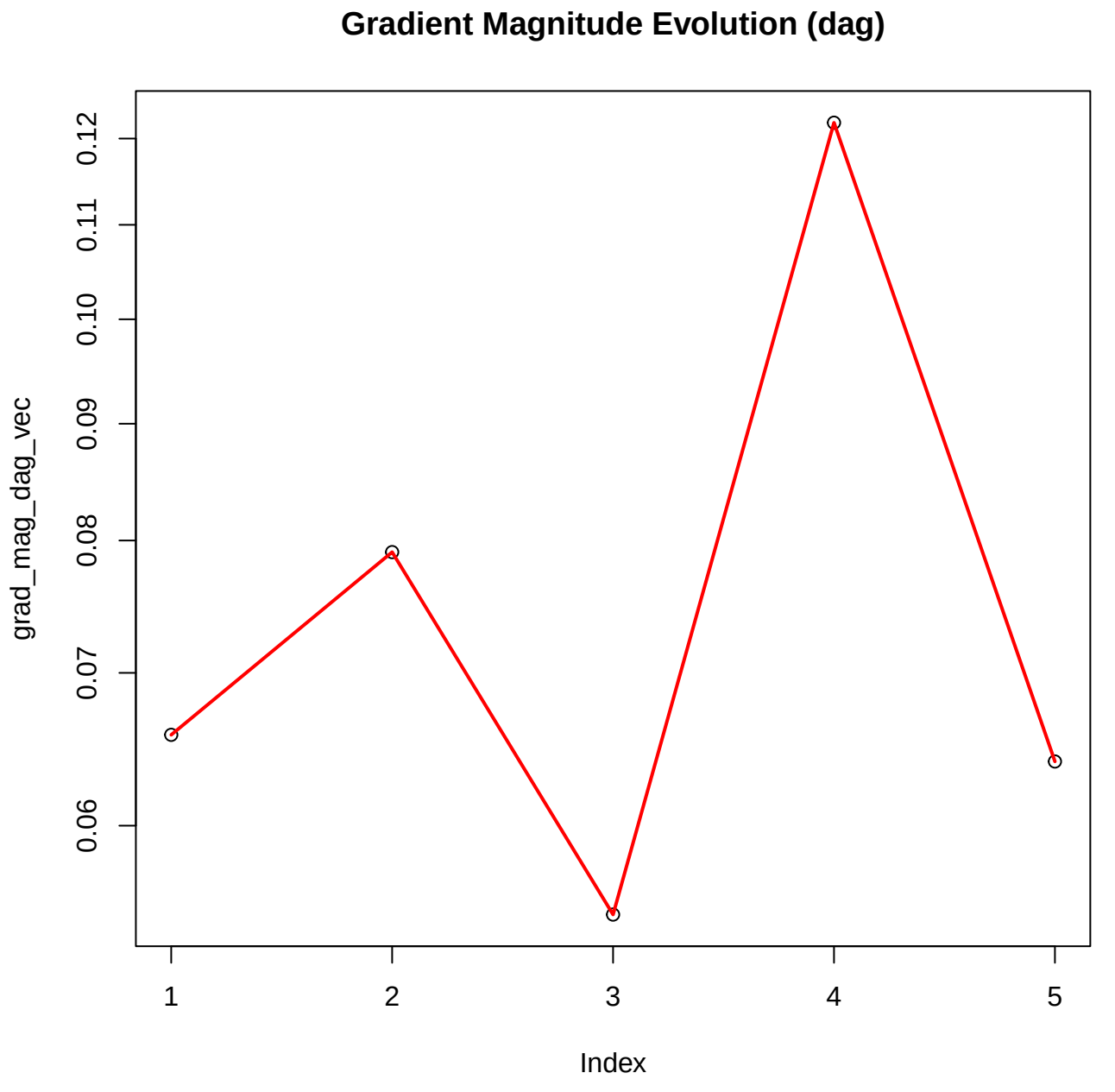

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.04
0.06
0.08
0.10
0.12
1
2
3
4
5
Index

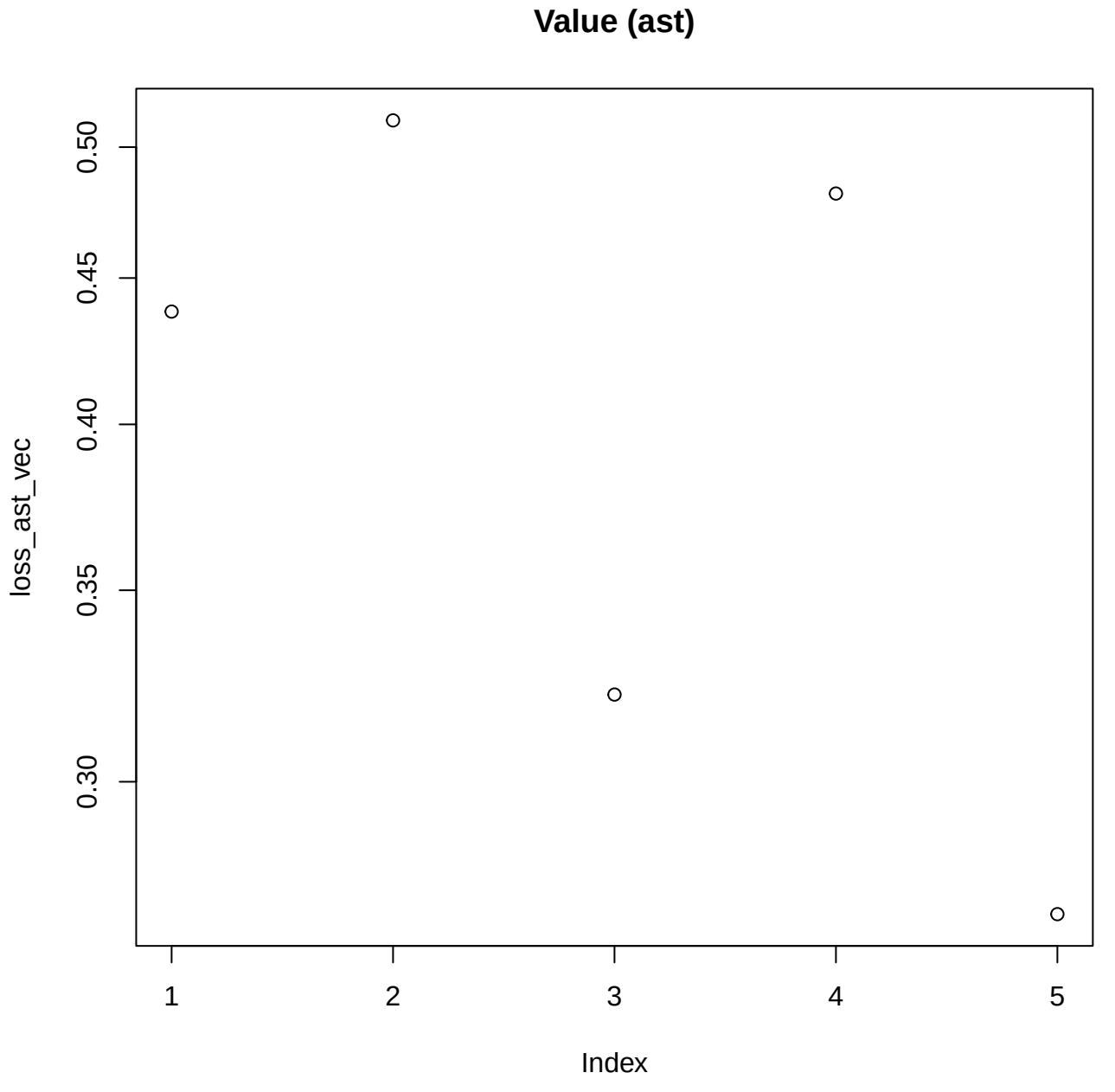

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.08
0.10
0.12
0.14
0.16
1
2
3
4
5
Index

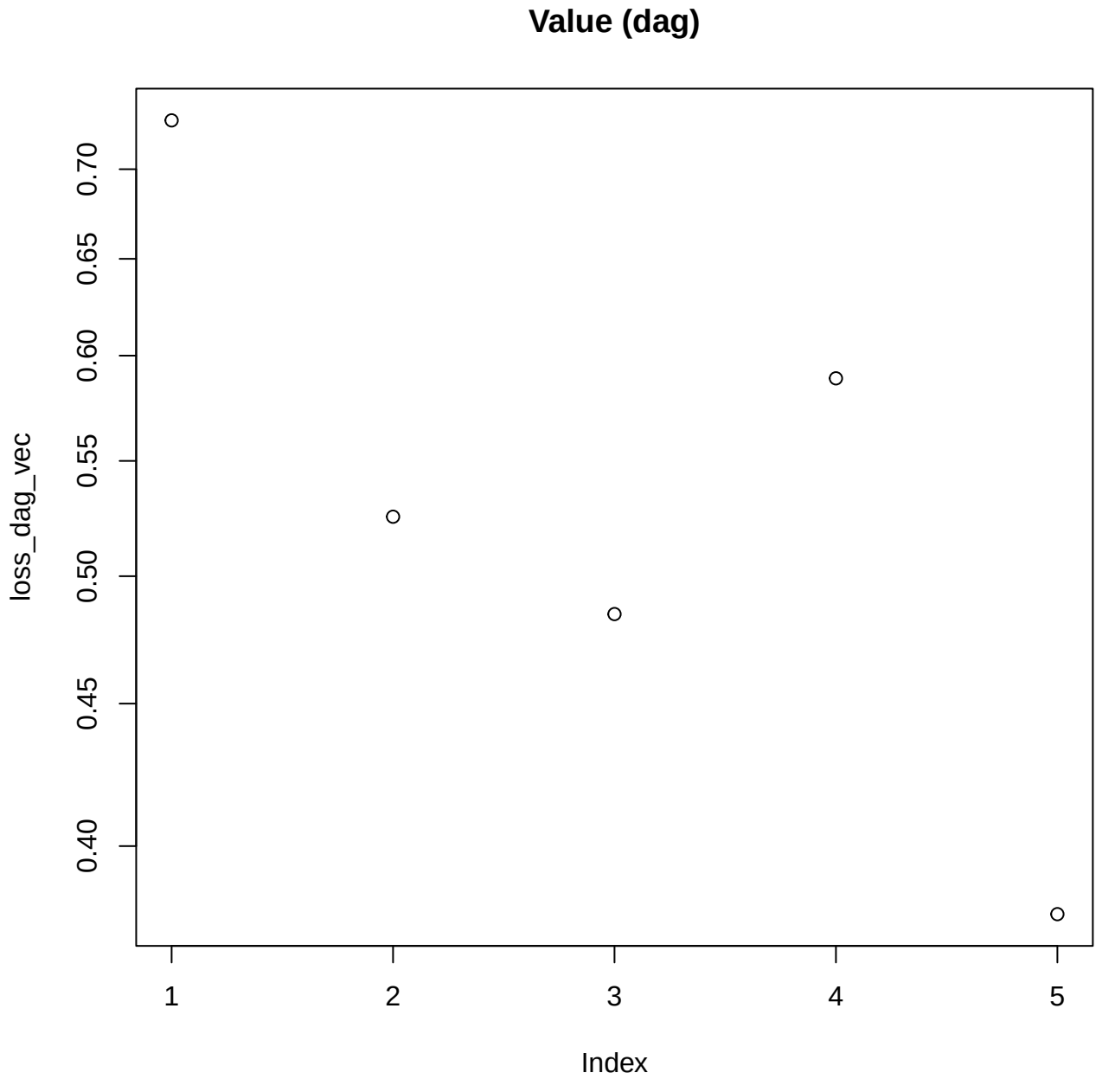

Value (ast)
loss_ast_vec
0.38
0.40
0.42
0.44
0.46
0.48
0.50
1
2
3
4
5
Index

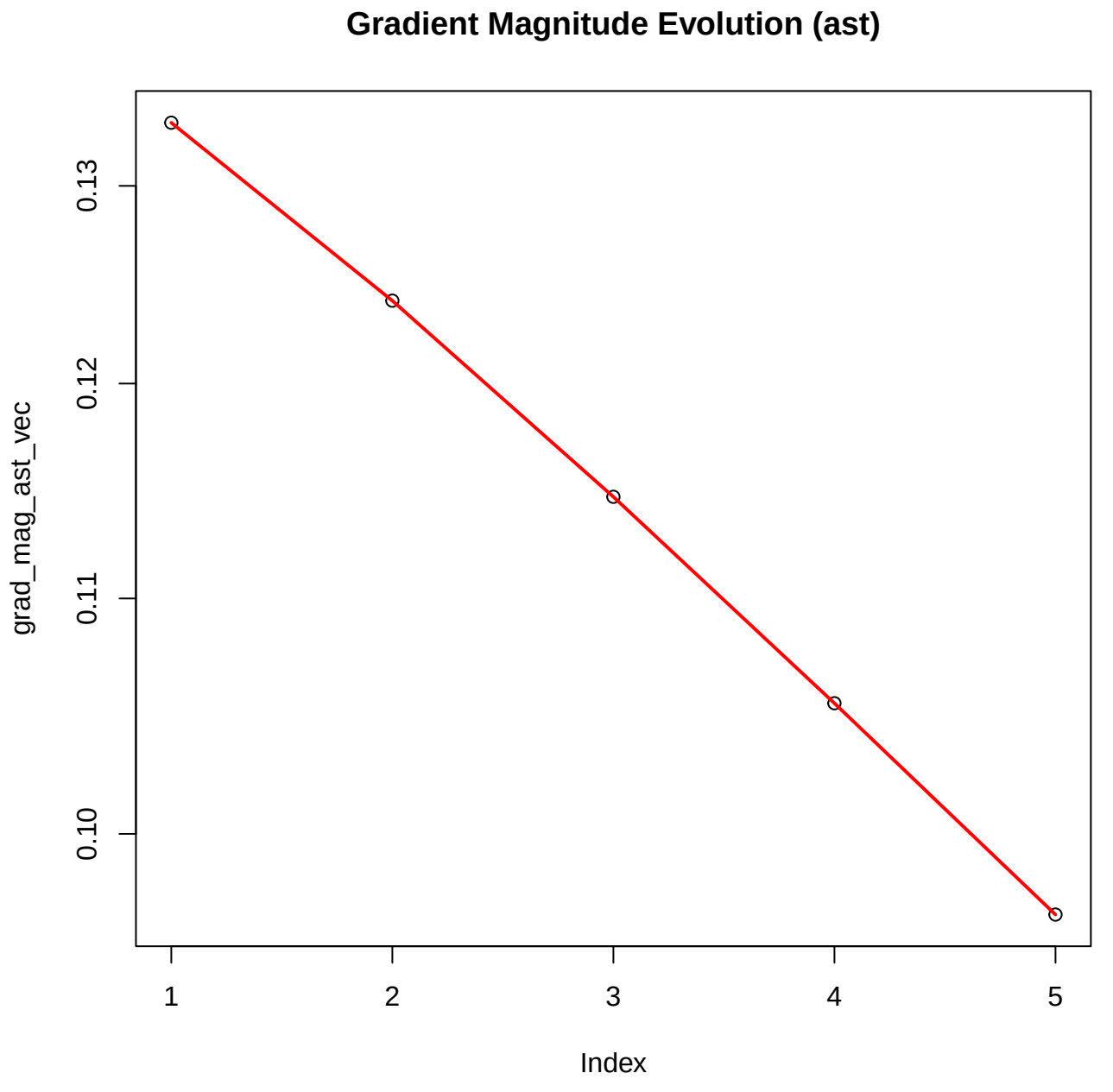

Value (dag)
loss_dag_vec
0.40
0.45
0.50
0.55
1
2
3
4
5
Index

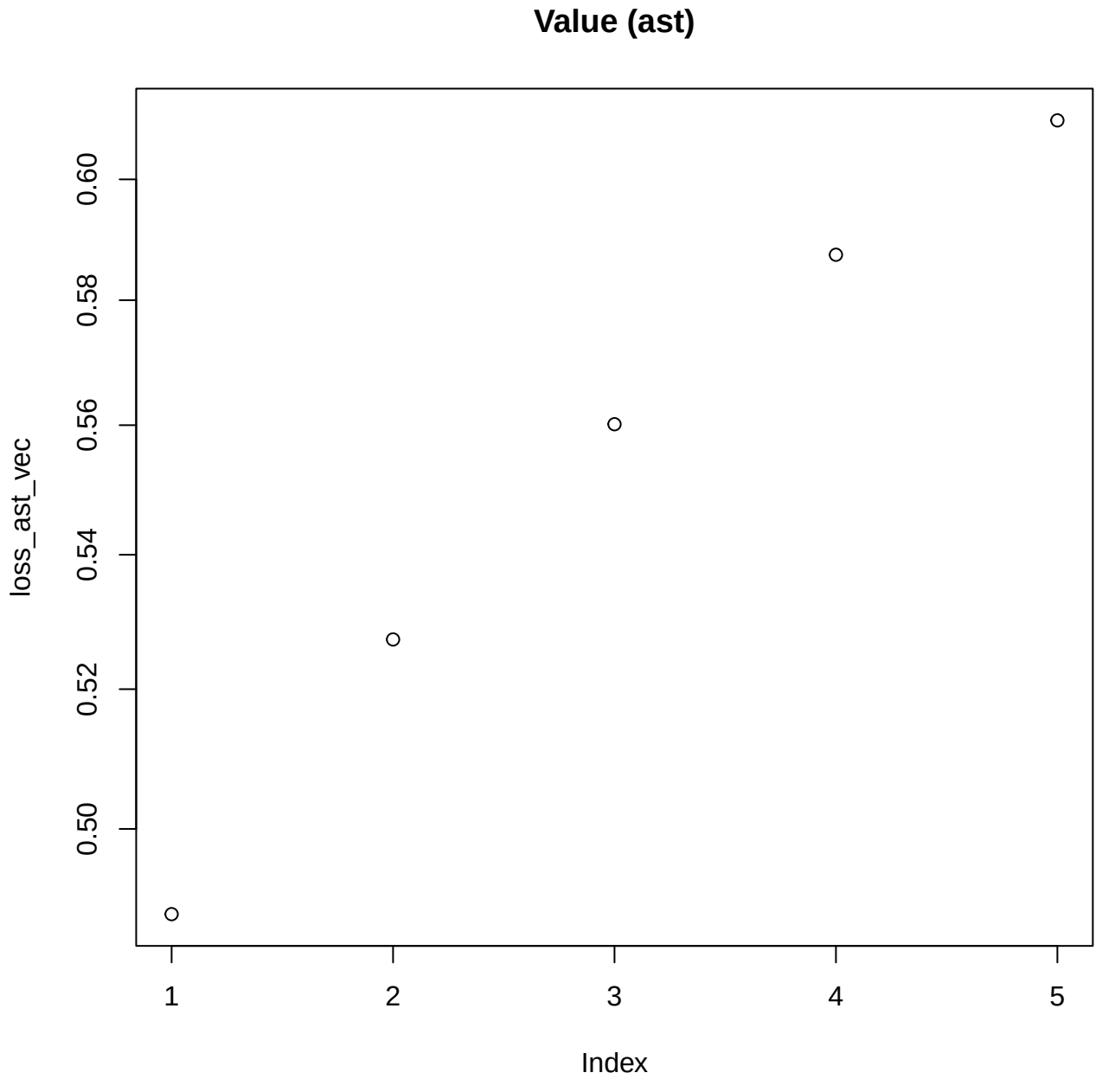

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.02
0.04
0.06
0.08
0.10
0.14
1
2
3
4
5
Index

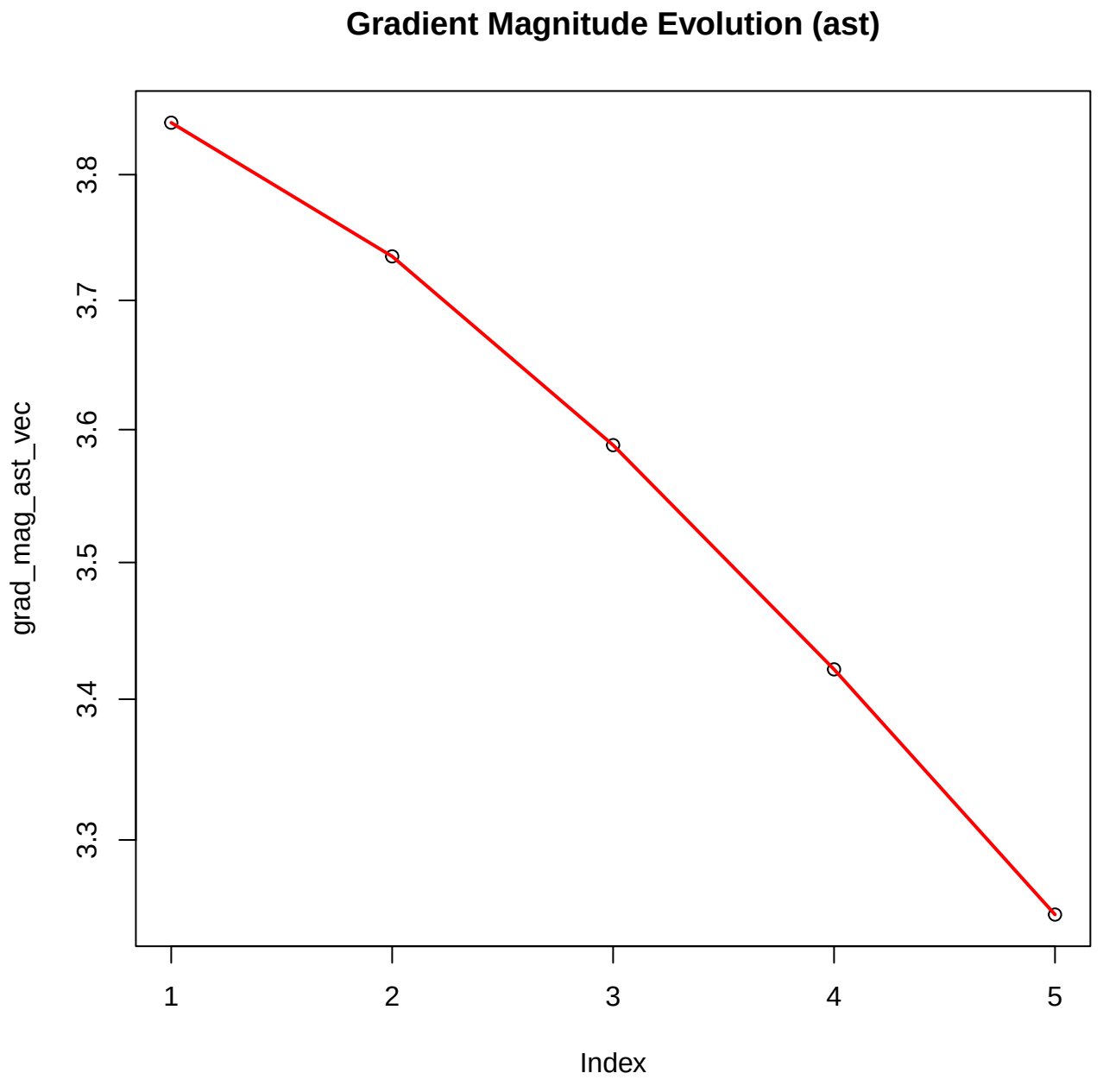

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.06
0.07
0.08
0.09
0.10
0.11
0.12
1
2
3
4
5
Index

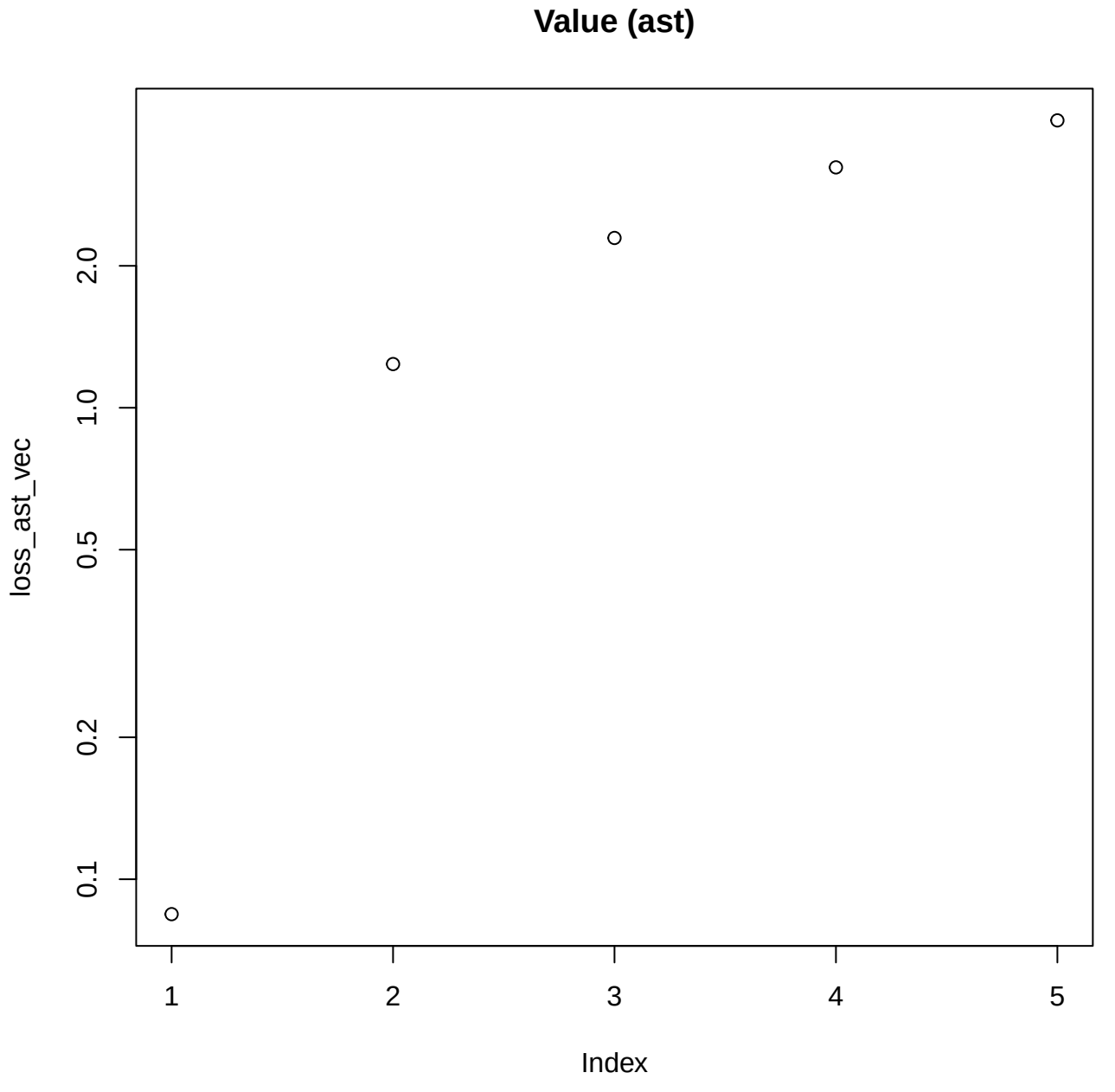

Value (ast)
loss_ast_vec
0.30
0.35
0.40
0.45
0.50
1
2
3
4
5
Index

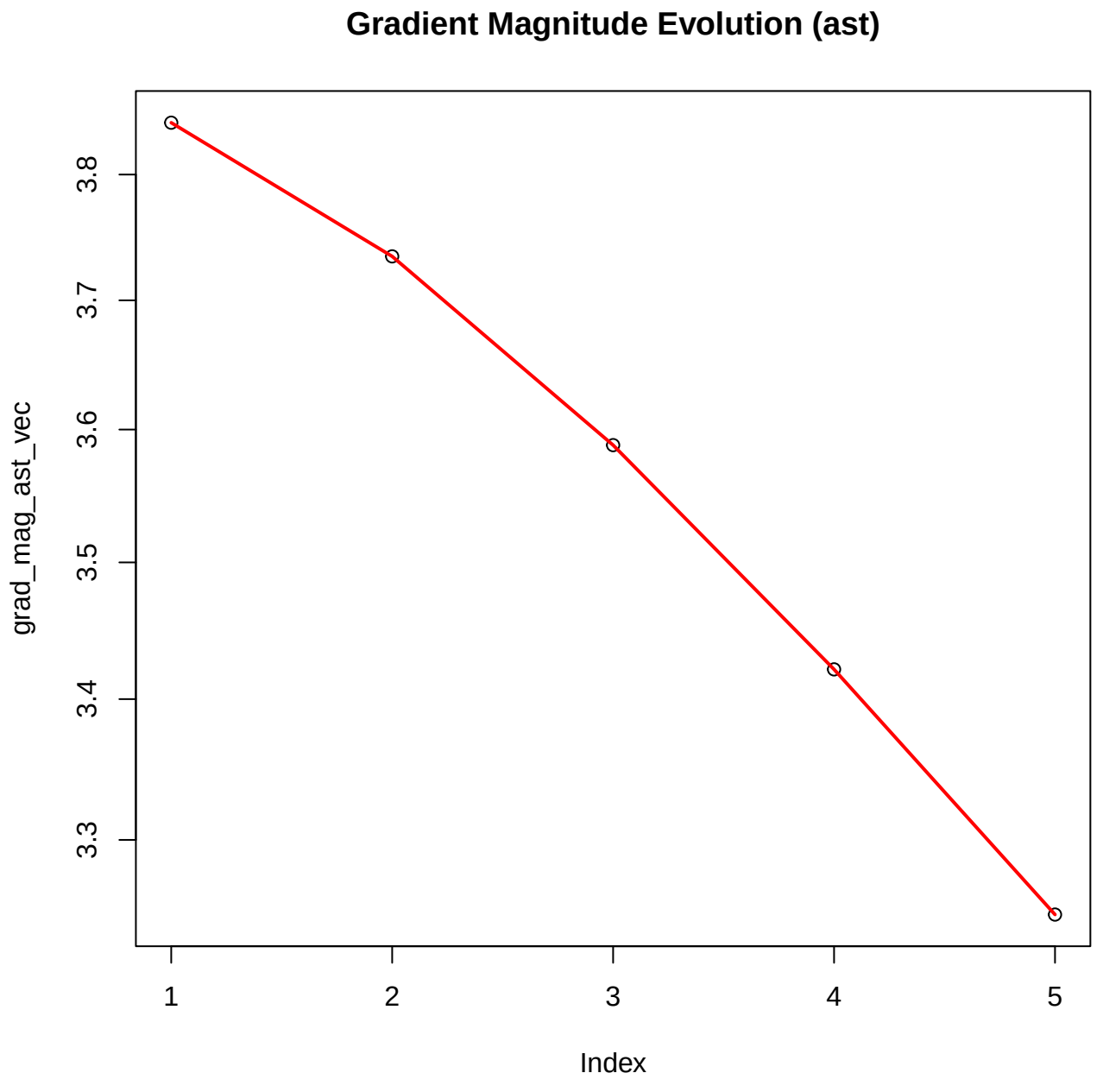

Value (dag)
loss_dag_vec
0.40
0.45
0.50
0.55
0.60
0.65
0.70
1
2
3
4
5
Index

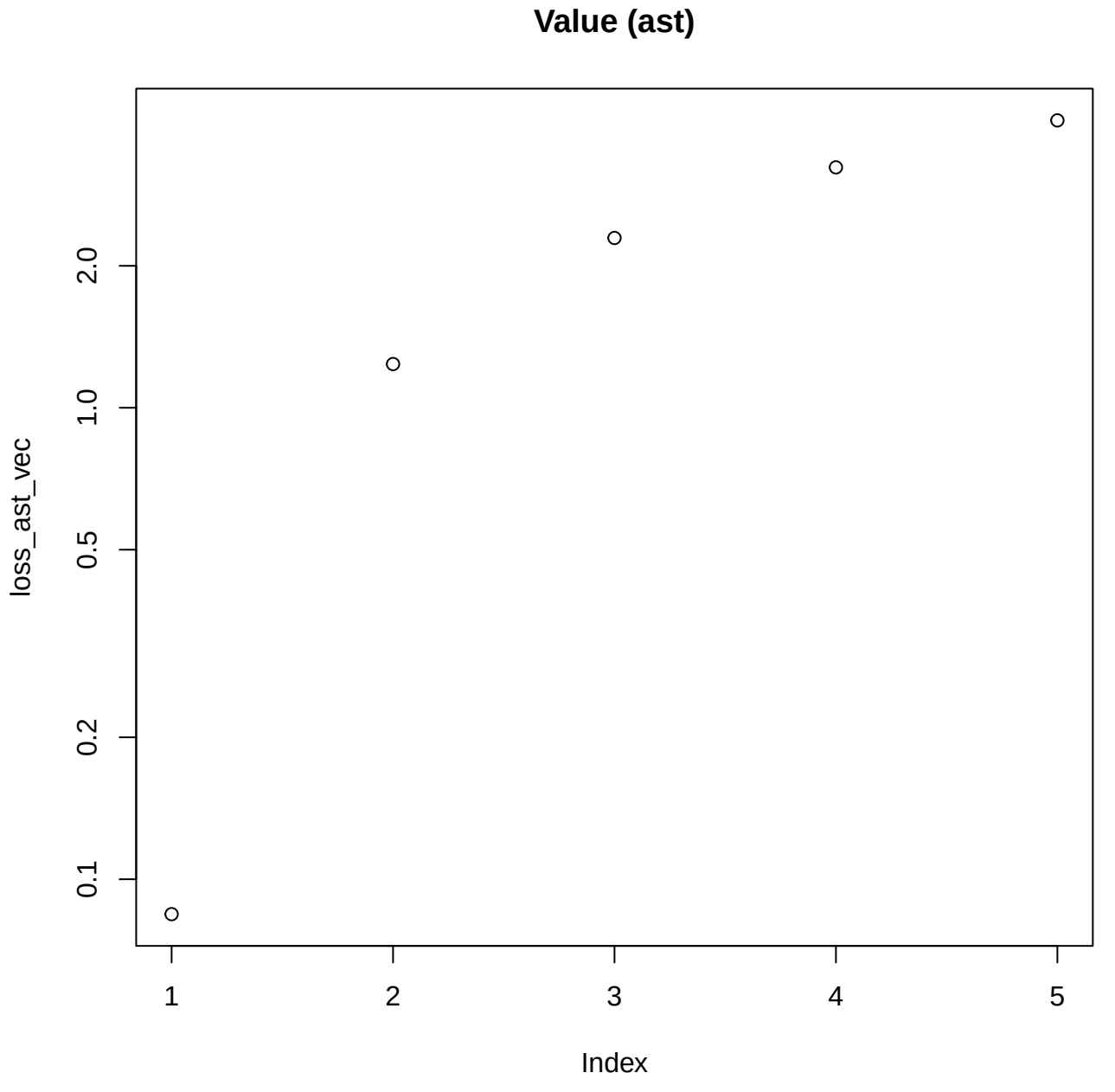

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index

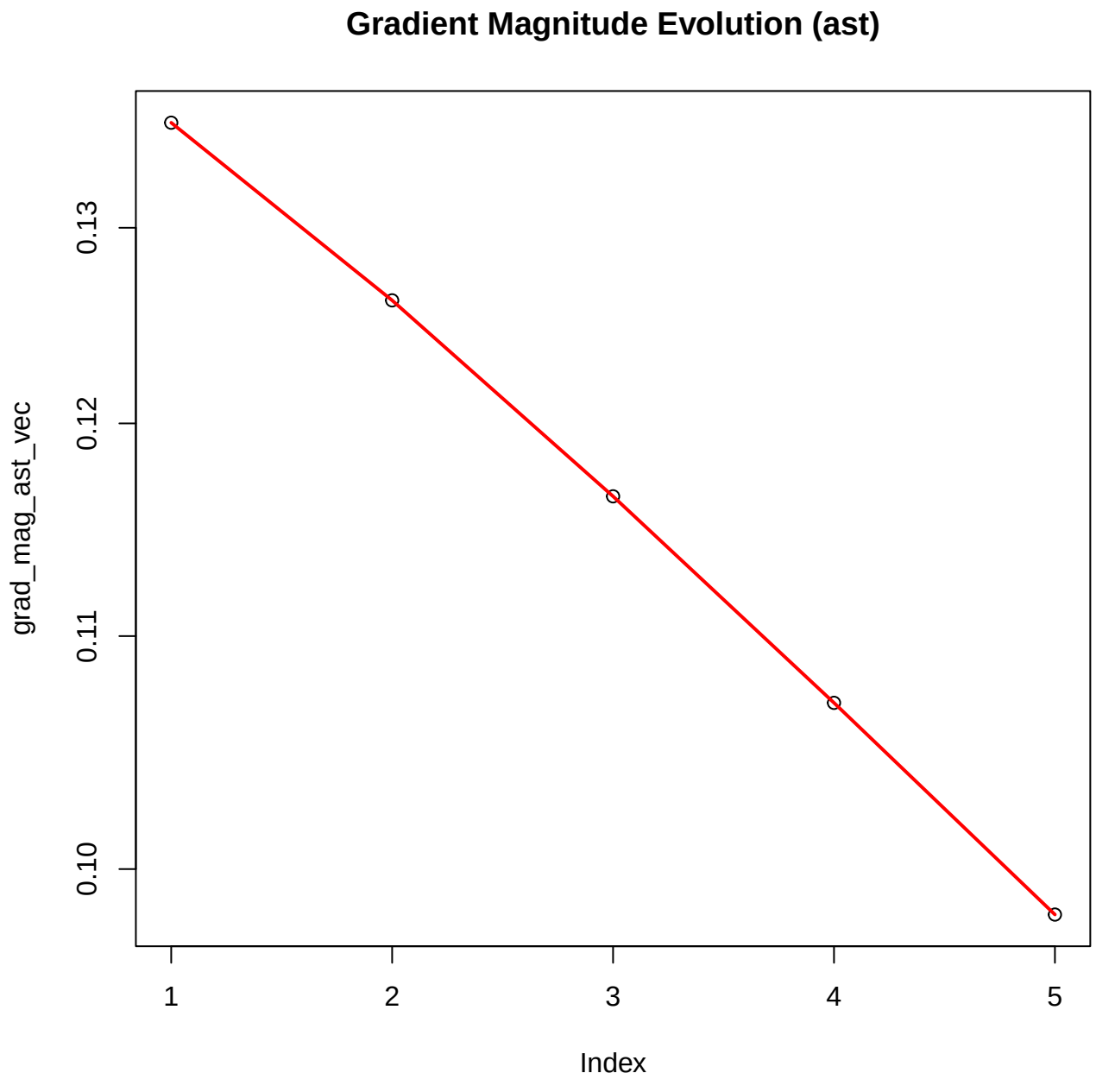

Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index

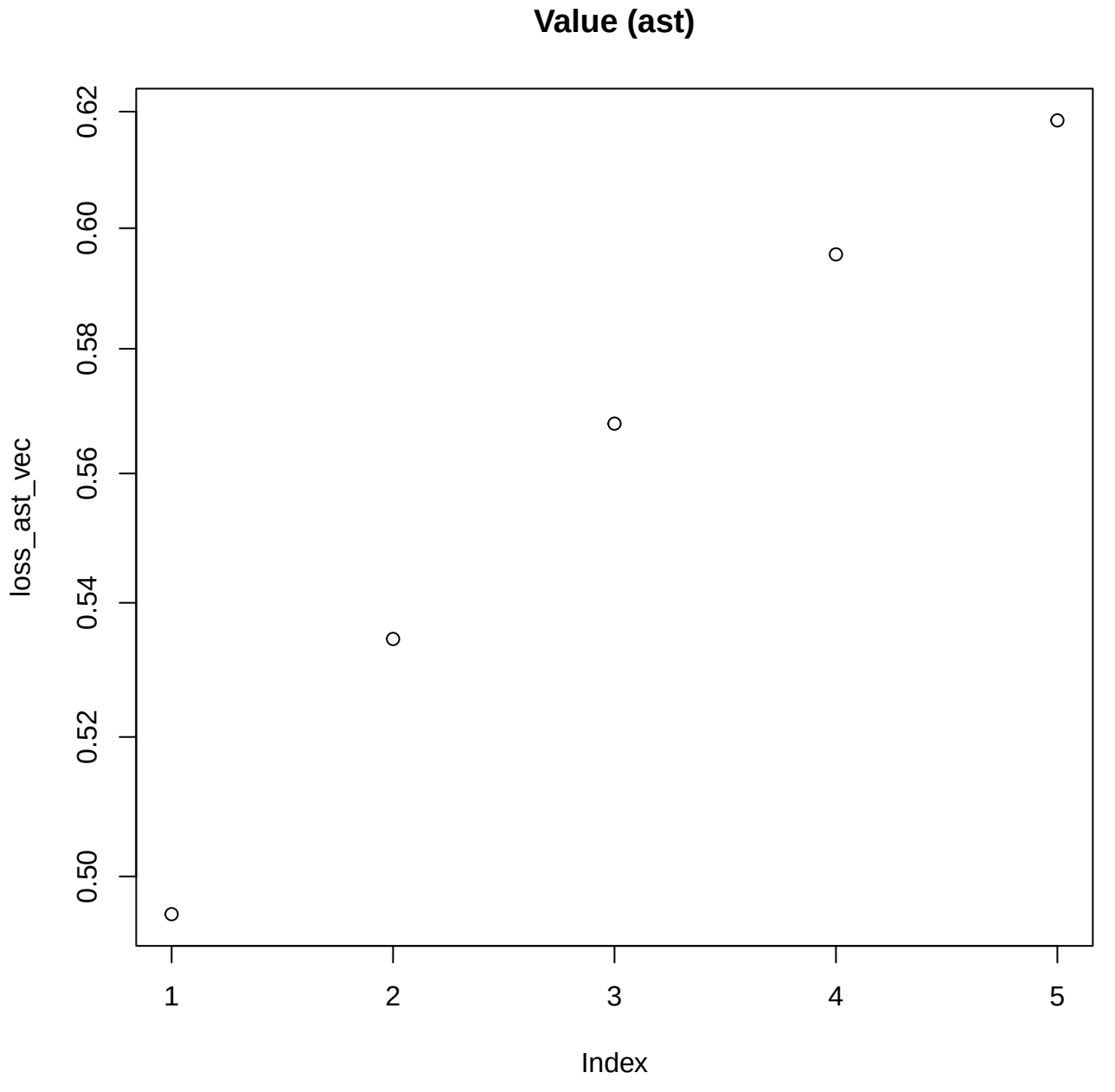

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
3.3
3.4
3.5
3.6
3.7
3.8
1
2
3
4
5
Index

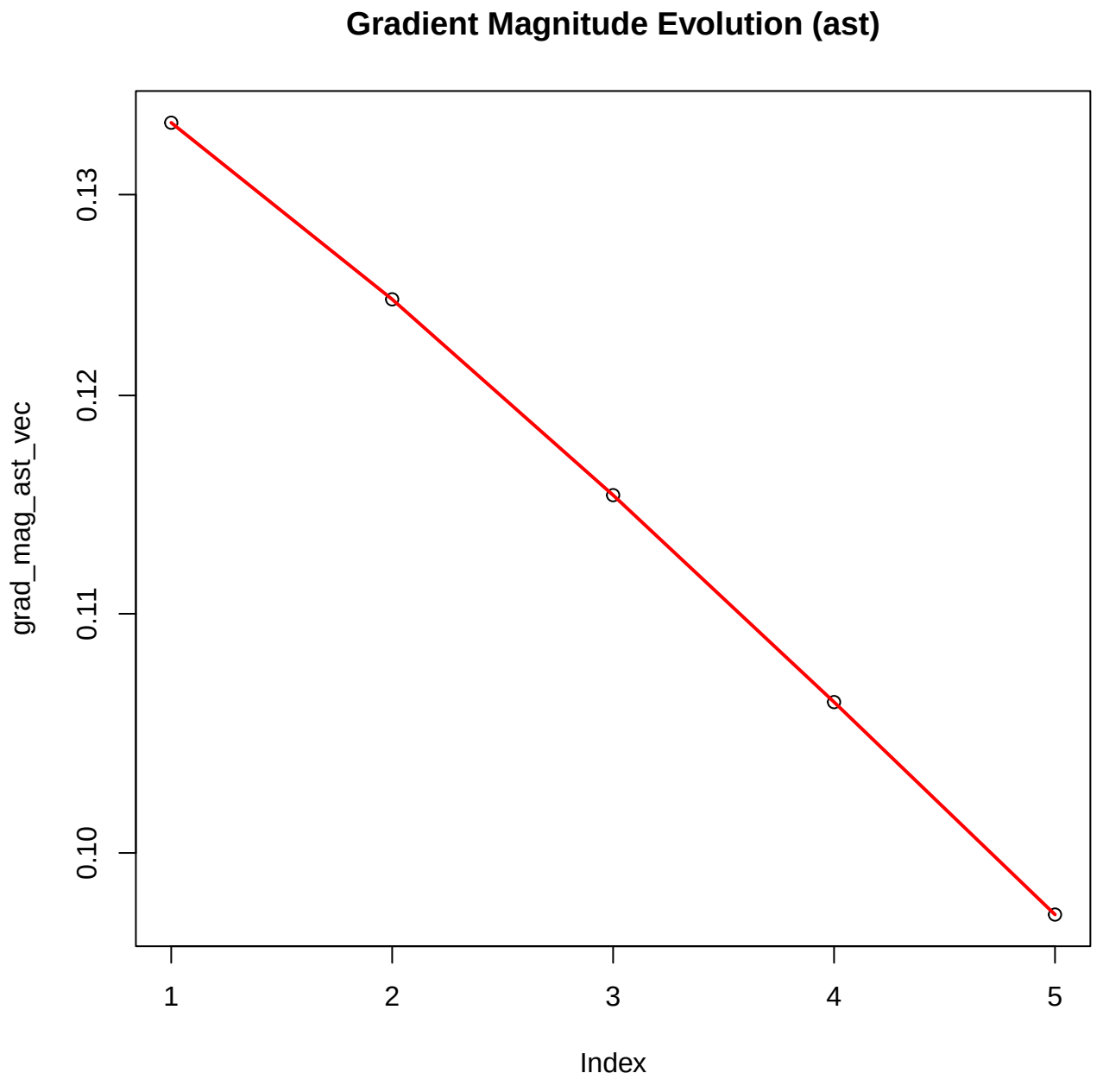

Value (ast)
loss_ast_vec
0.1
0.2
0.5
1.0
2.0
1
2
3
4
5
Index

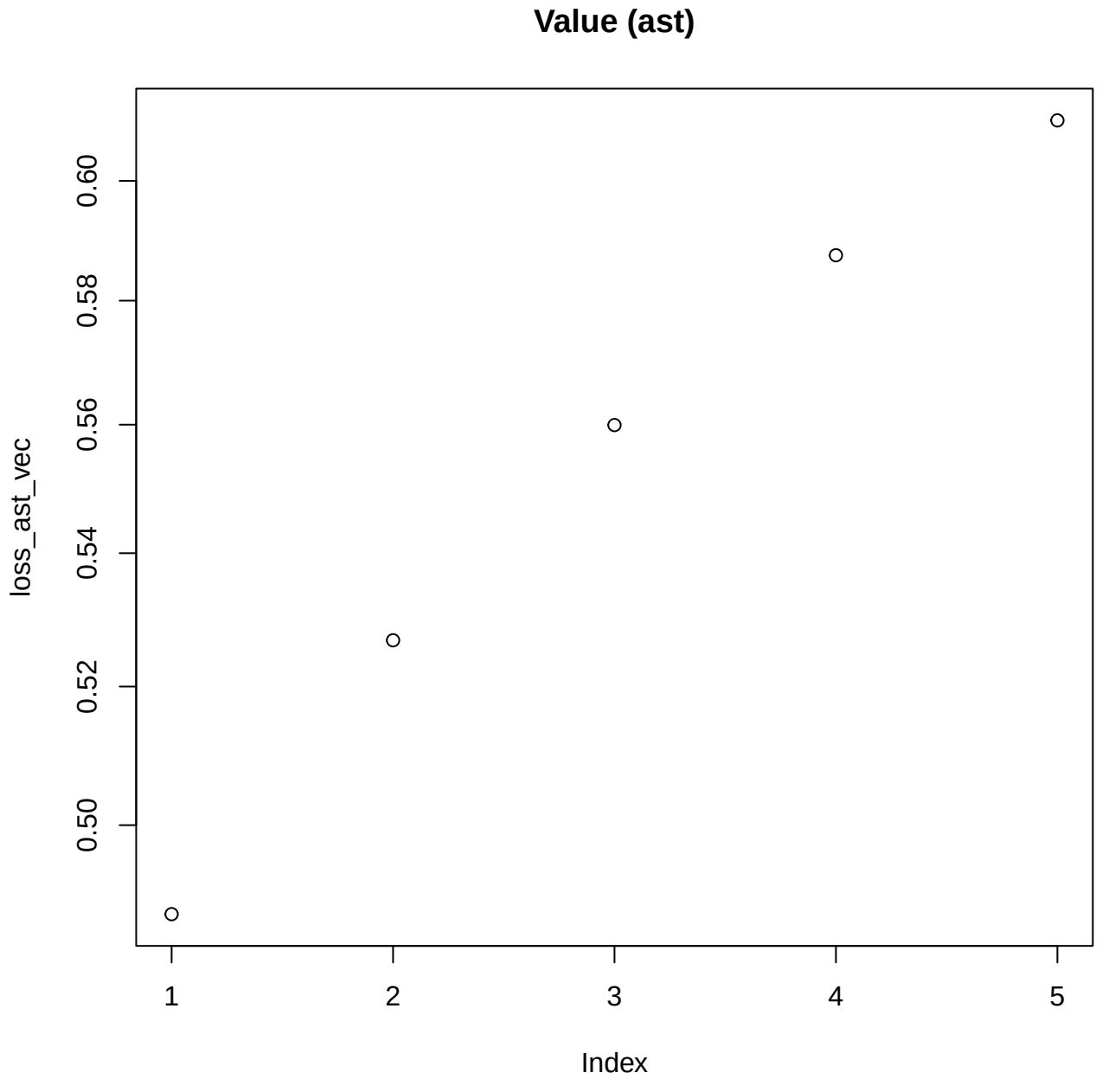

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
3.3
3.4
3.5
3.6
3.7
3.8
1
2
3
4
5
Index

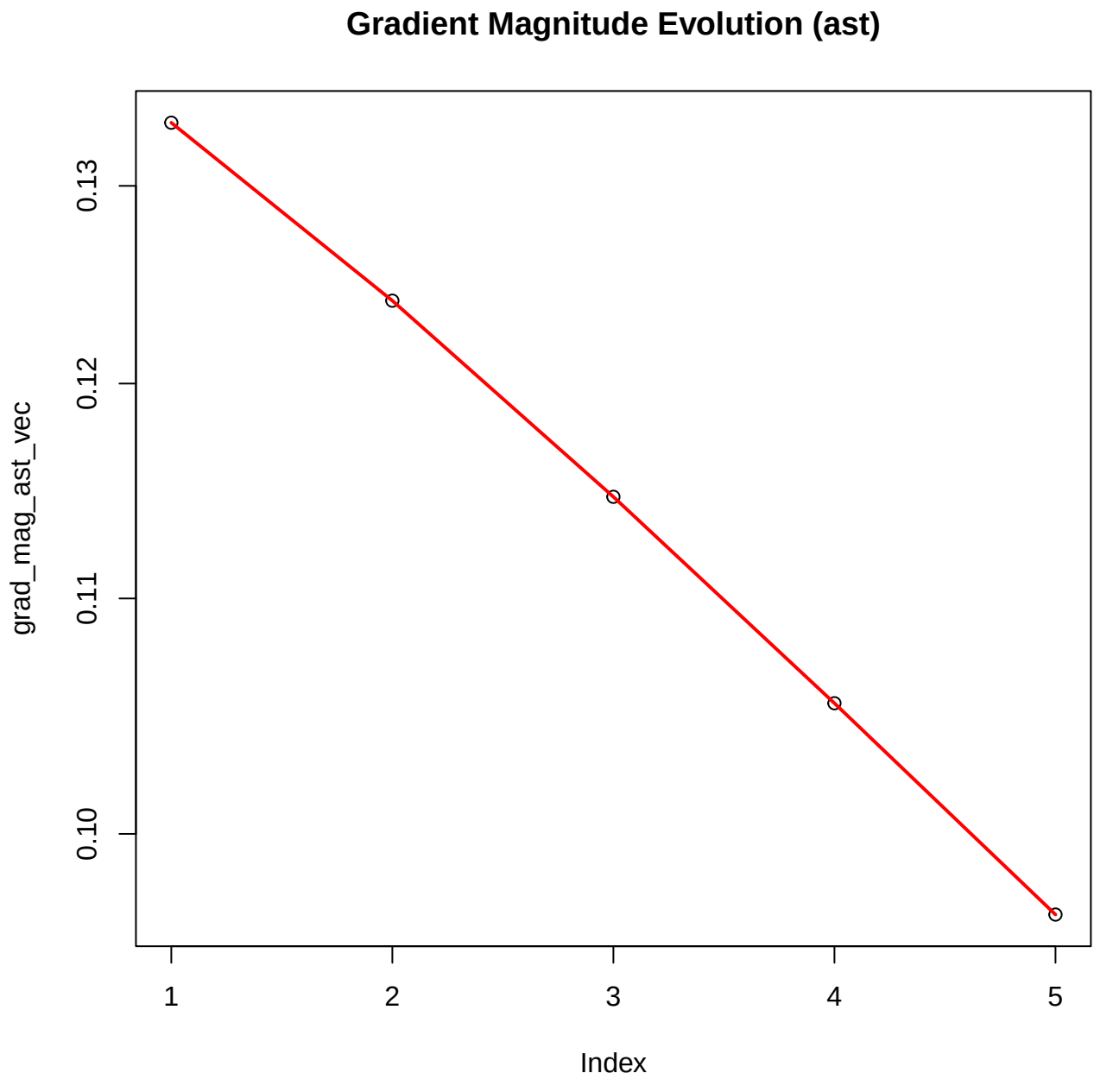

Value (ast)
loss_ast_vec
0.1
0.2
0.5
1.0
2.0
1
2
3
4
5
Index

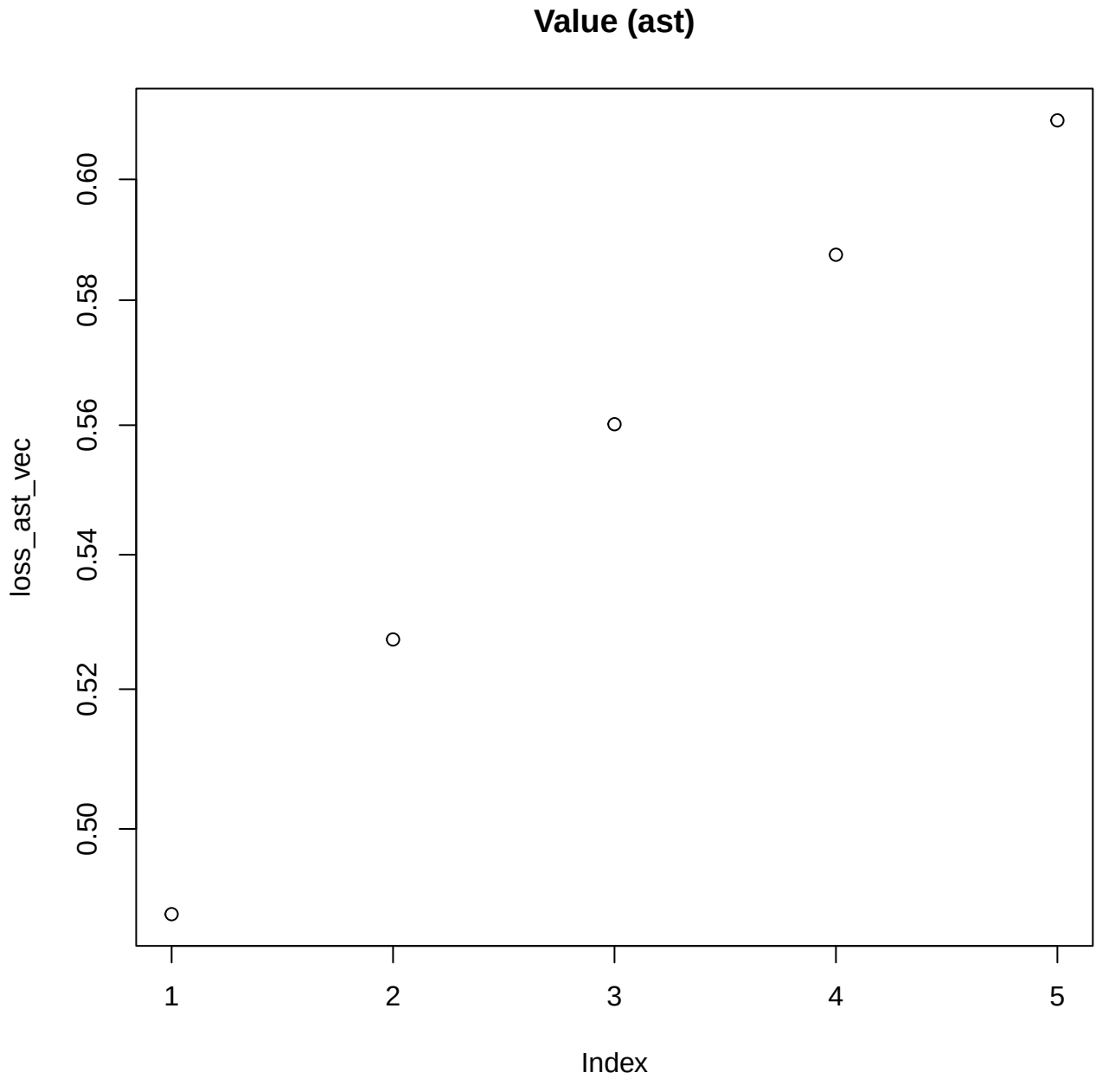

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index

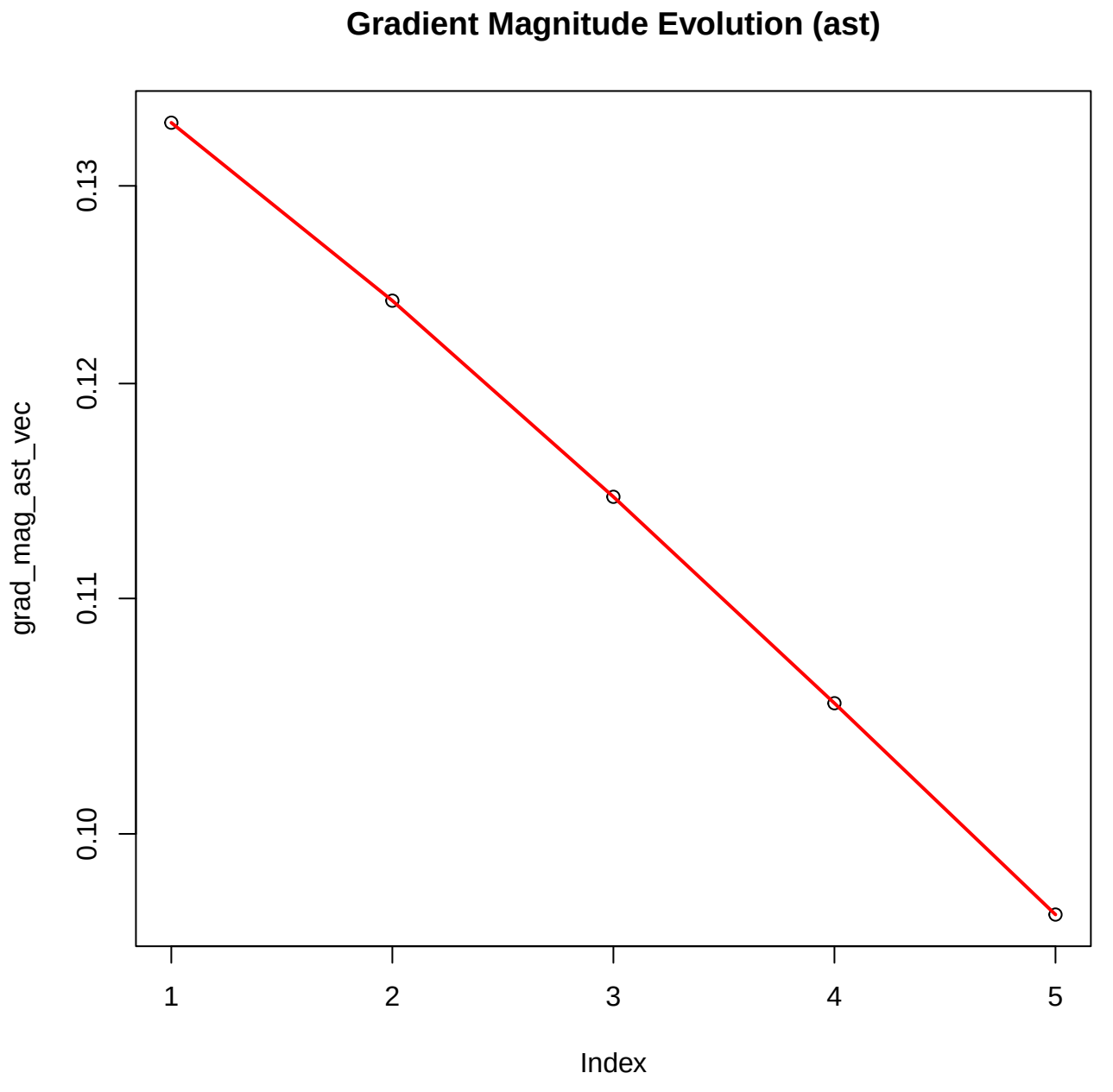

Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
0.62
1
2
3
4
5
Index

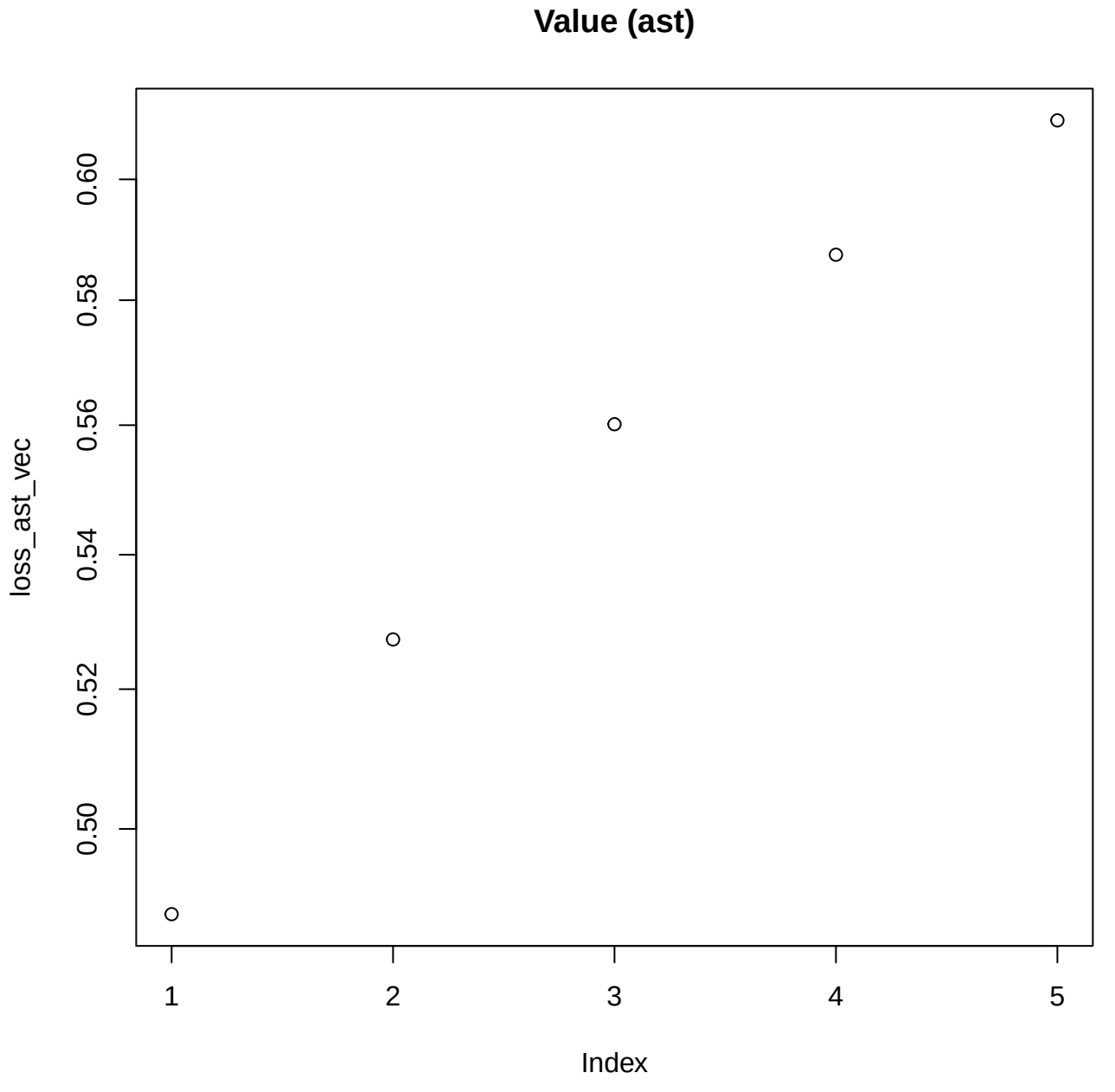

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index

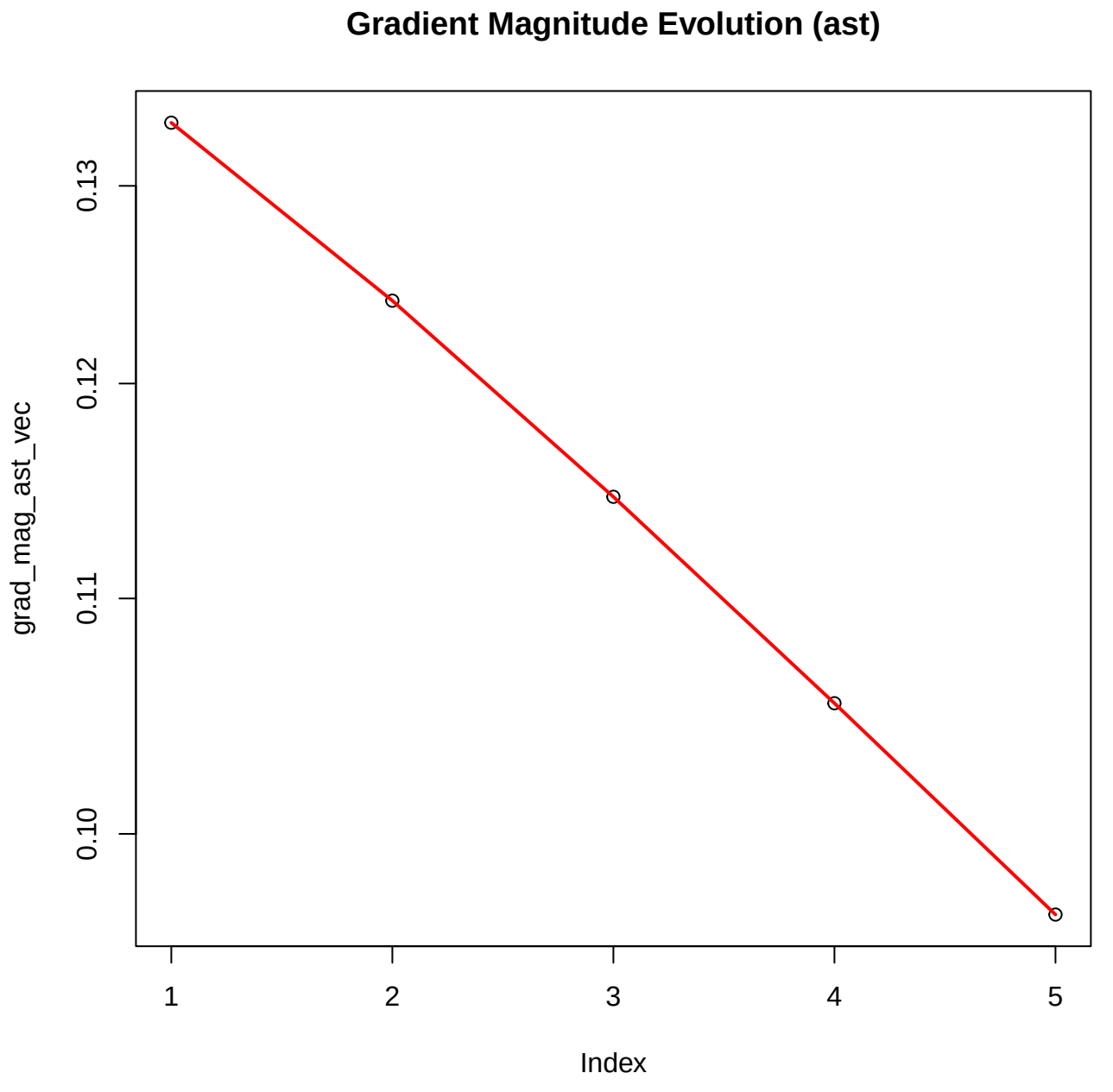

Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index

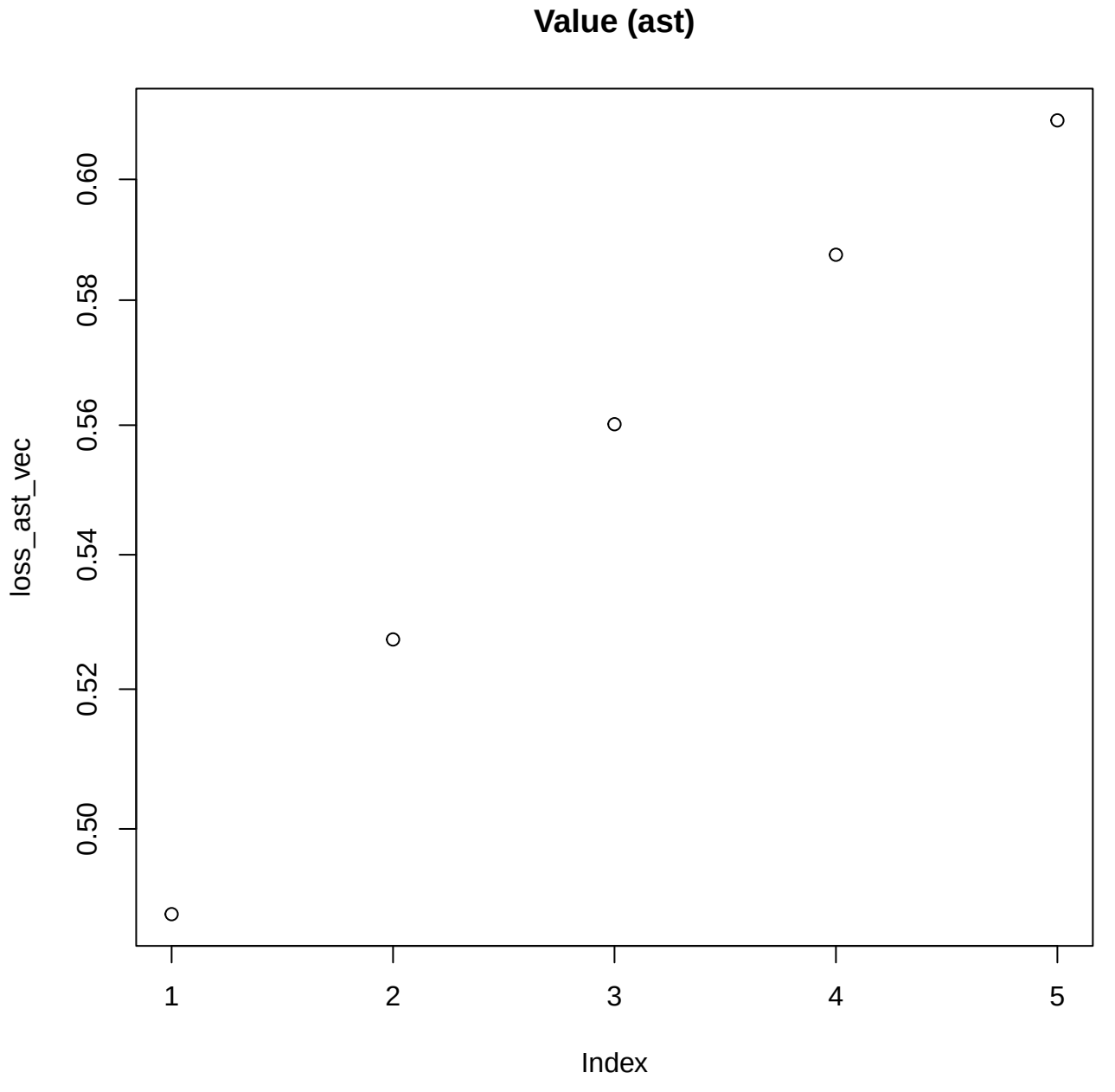

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index

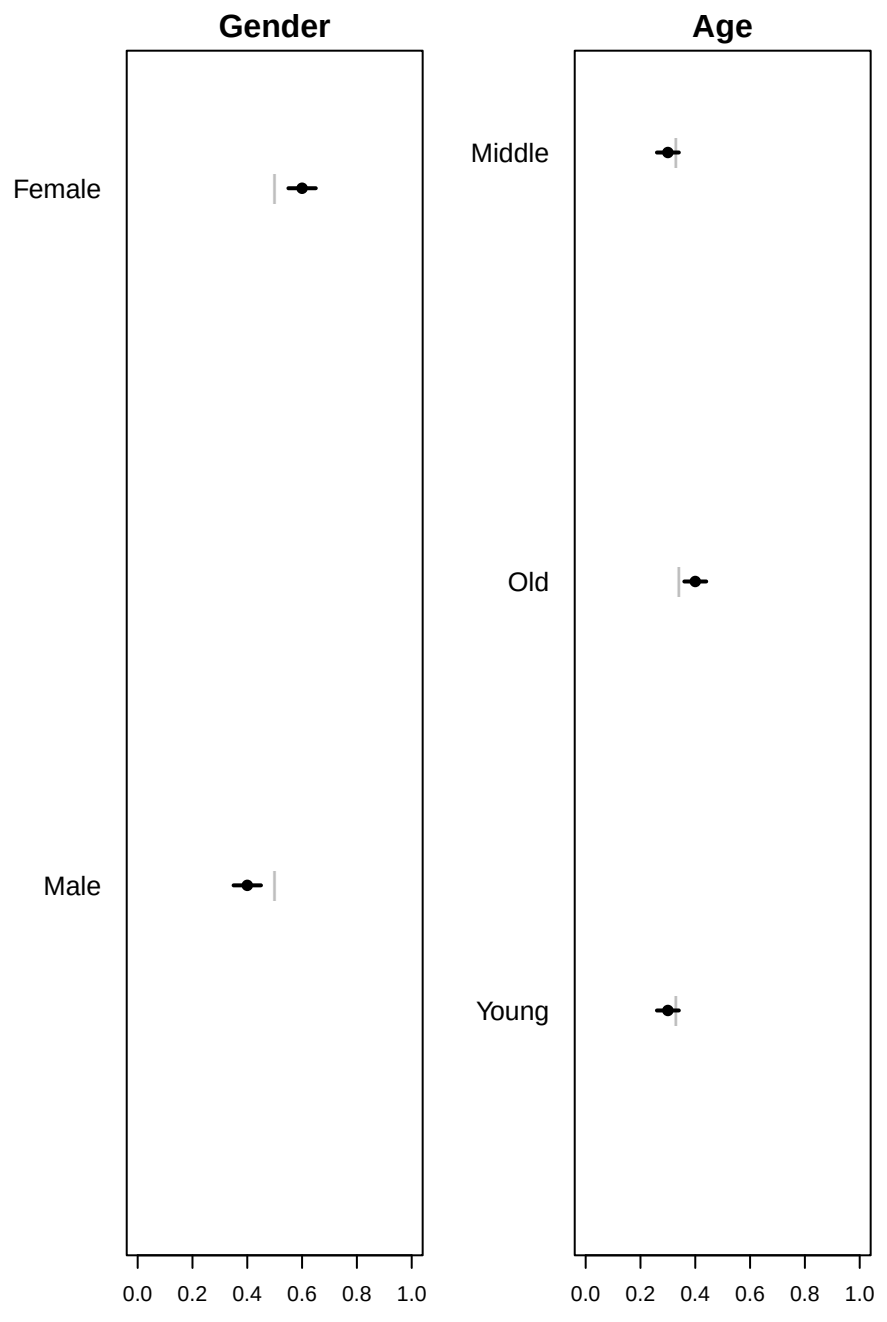

Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index

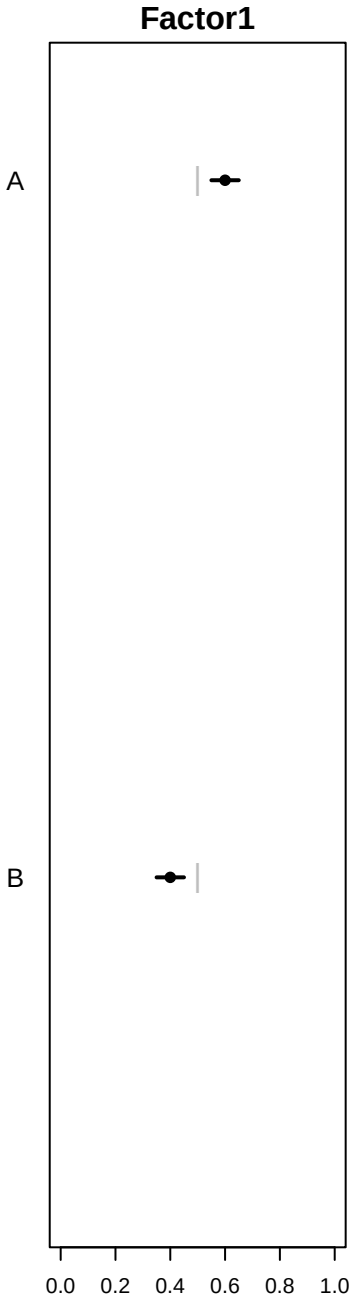

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index

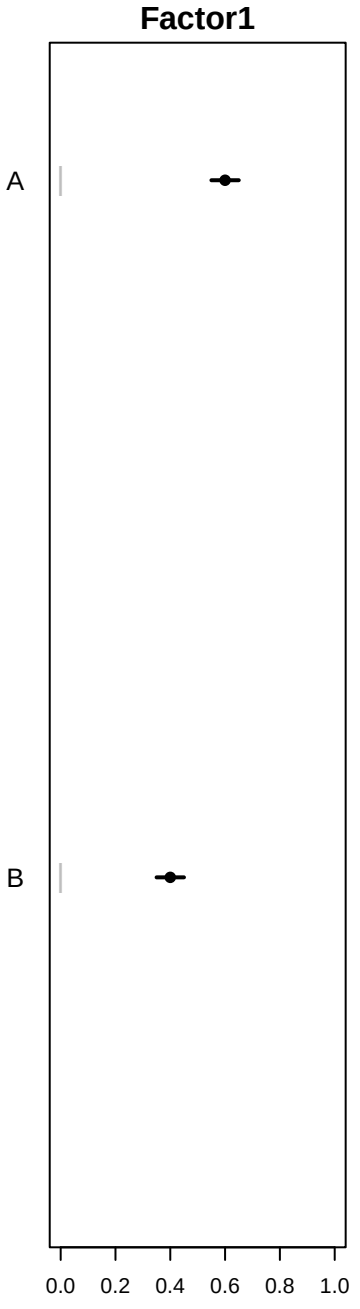

Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index

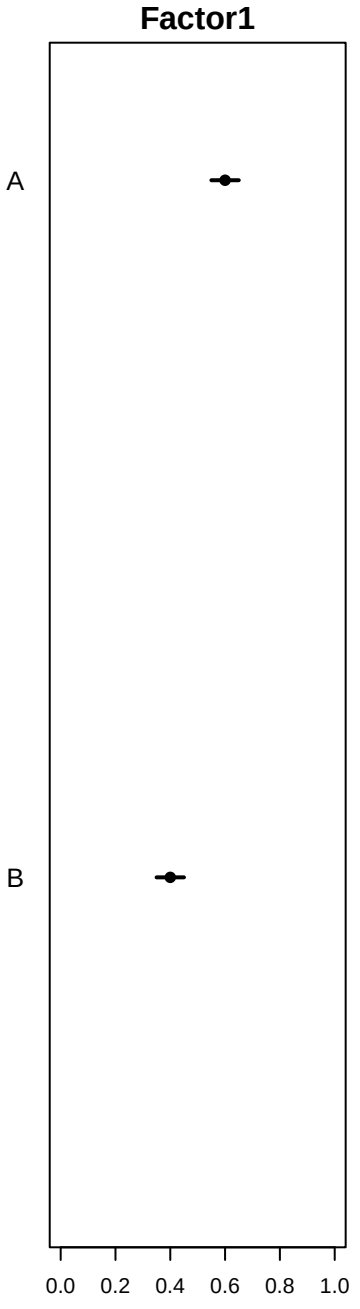

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index

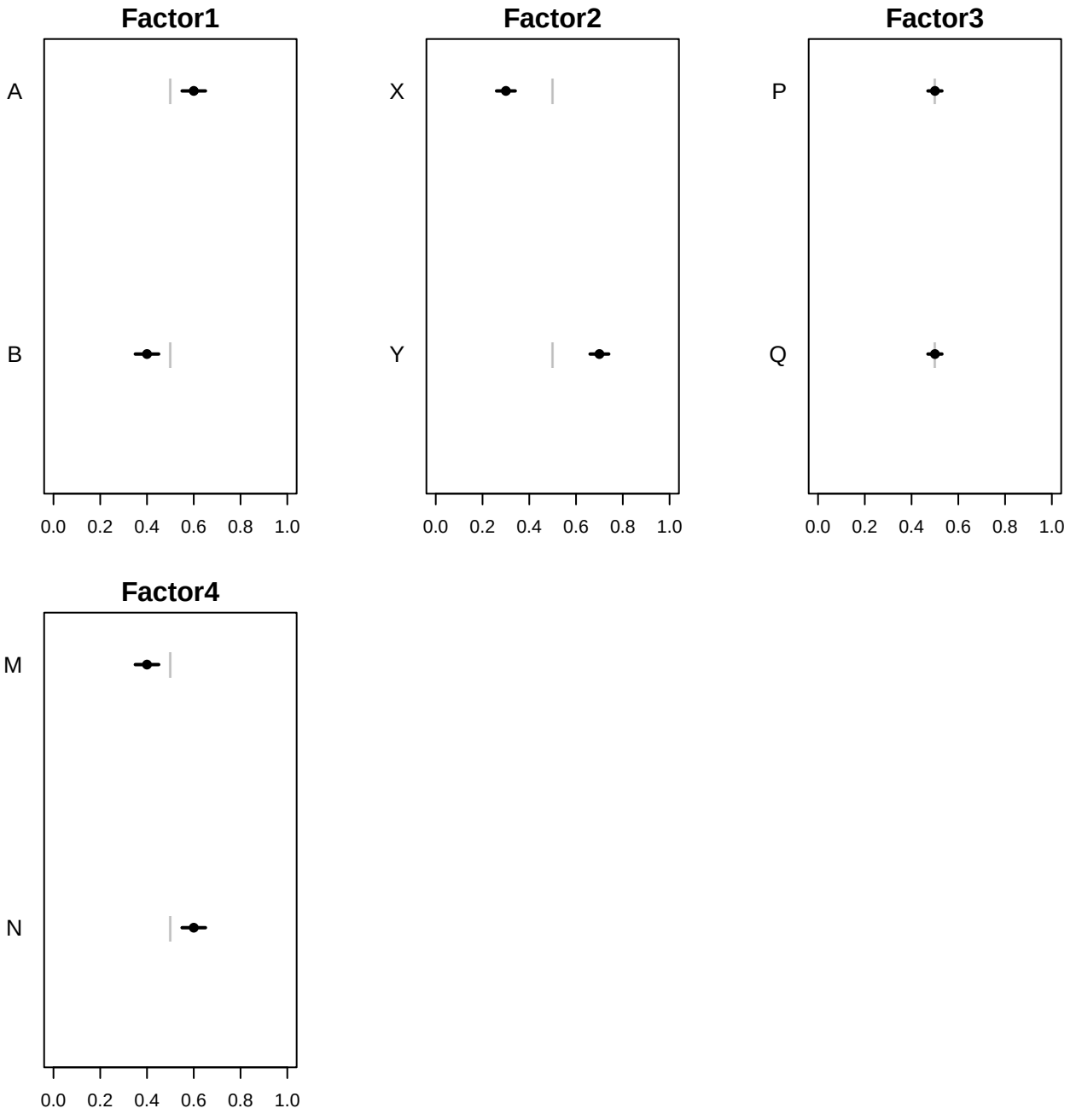

Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index

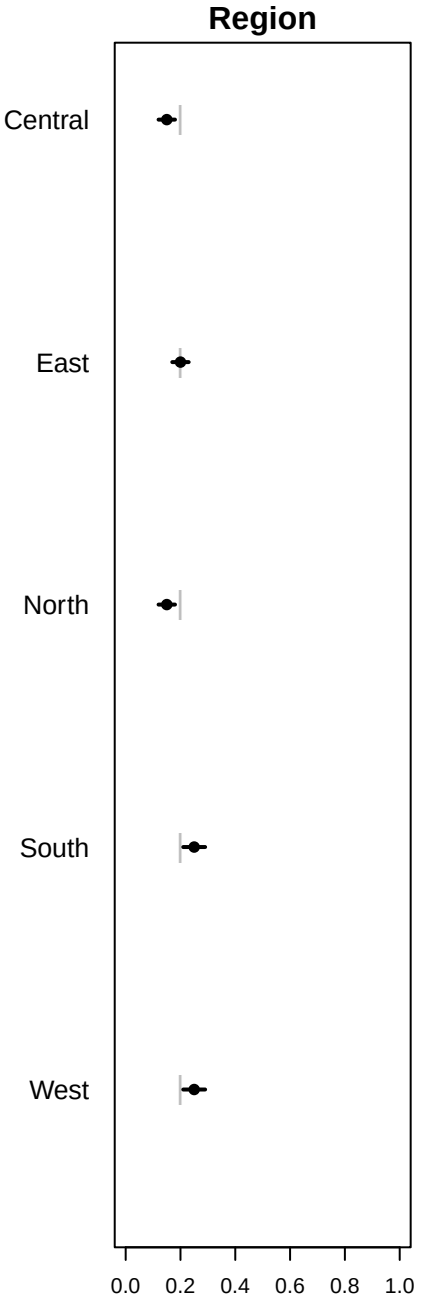

Gender
Female
Male
0.0
0.2
0.4
0.6
0.8
1.0
Age
Middle
Old
Young
0.0
0.2
0.4
0.6
0.8
1.0

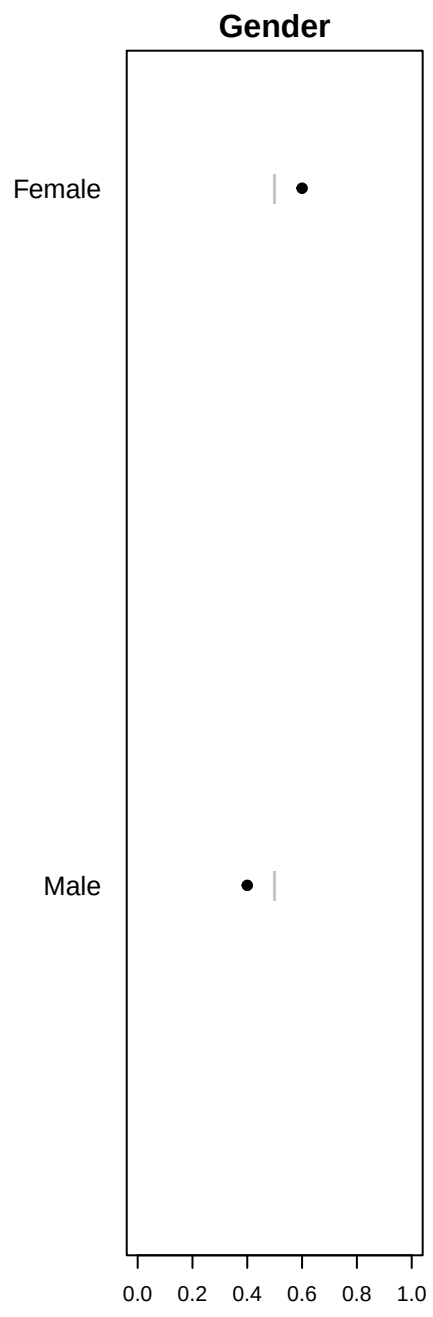

Factor1
A
B
0.0
0.2
0.4
0.6
0.8
1.0

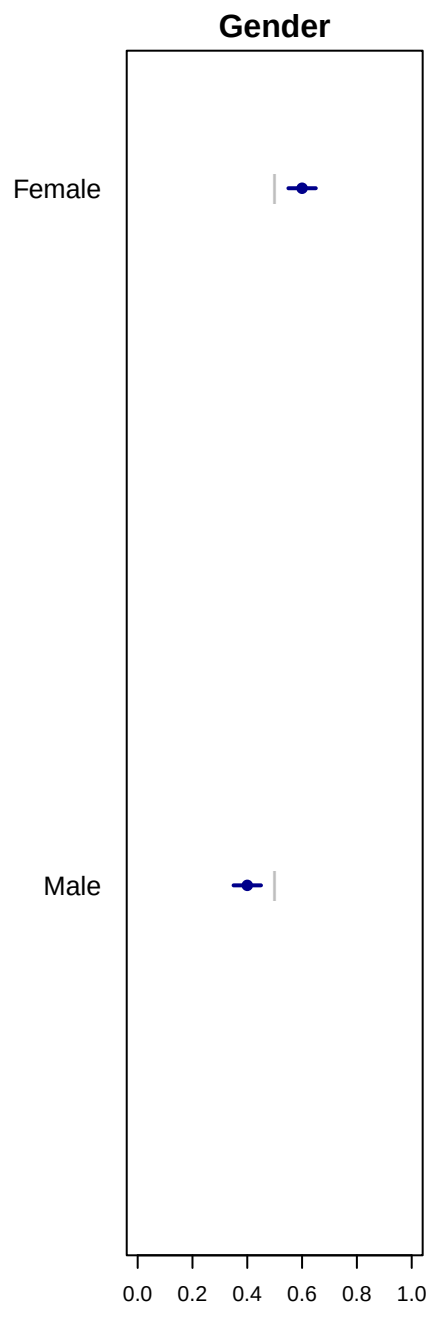

Factor1
A
B
0.0
0.2
0.4
0.6
0.8
1.0

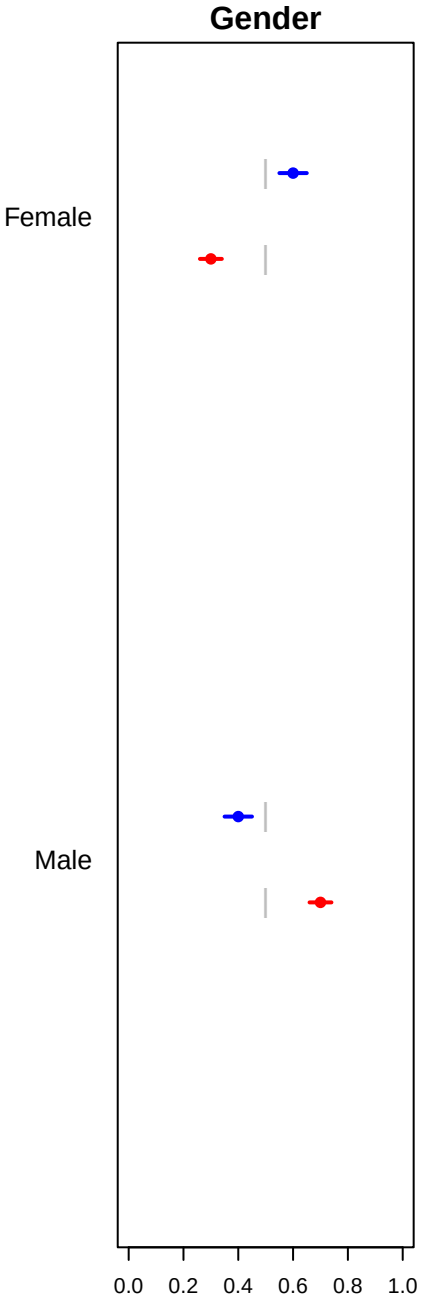

Factor1
A
B
0.0
0.2
0.4
0.6
0.8
1.0

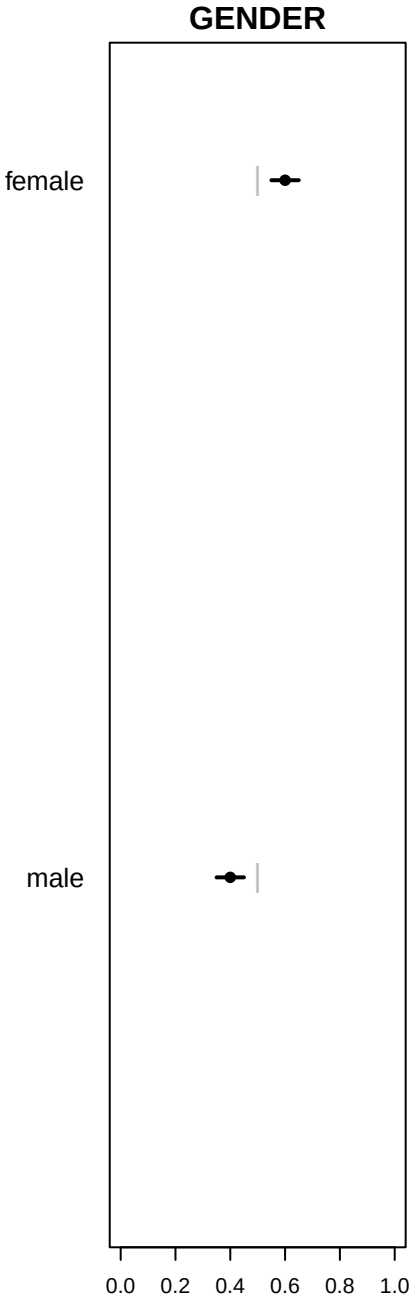

Factor1
A
B
0.0
0.2
0.4
0.6
0.8
1.0
Factor2
X
Y
0.0
0.2
0.4
0.6
0.8
1.0
Factor3
P
Q
0.0
0.2
0.4
0.6
0.8
1.0
Factor4
M
N
0.0
0.2
0.4
0.6
0.8
1.0

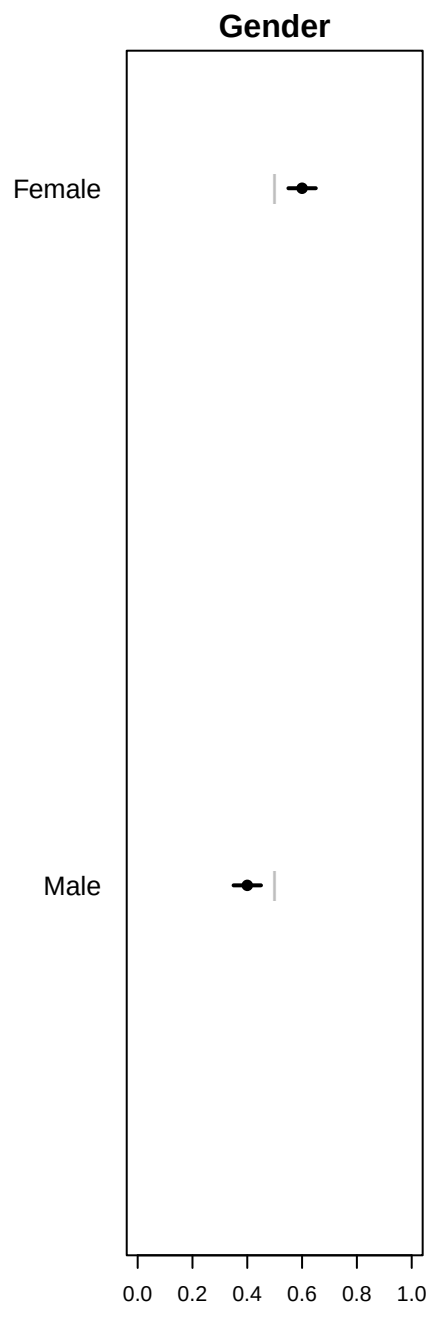

Region
Central
East
North
South
West
0.0
0.2
0.4
0.6
0.8
1.0


Gender
Female
Male
0.0
0.2
0.4
0.6
0.8
1.0


Gender
Female
Male
0.0
0.2
0.4
0.6
0.8
1.0


Gender
Female
Male
0.0
0.2
0.4
0.6
0.8
1.0


GENDER
female
male
0.0
0.2
0.4
0.6
0.8
1.0


Gender
Female
Male
0.0
0.2
0.4
0.6
0.8
1.0